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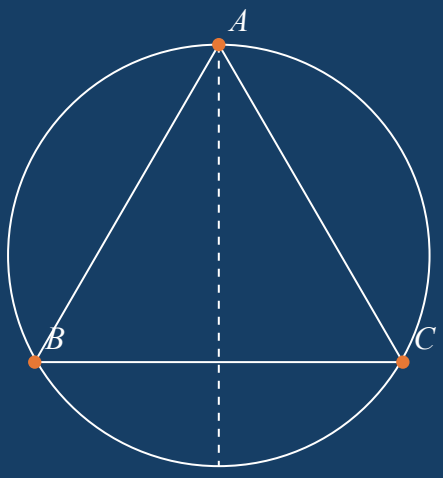
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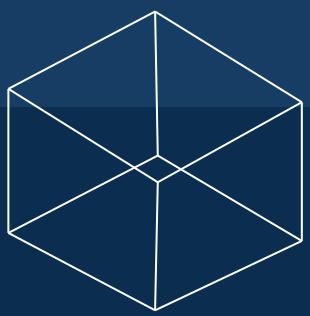
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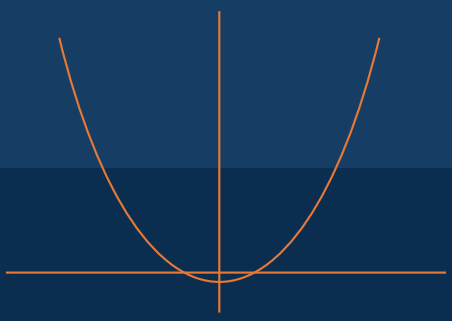
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2000-2026

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The Mathematics Atlas

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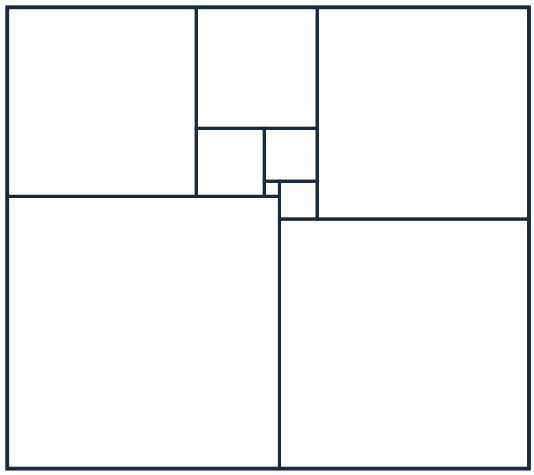
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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q04] | Part A (Q1-Q5)

Answer: 260



1. Label the central smallest square side x and the square immediately lower-right side y. Matching the segment lengths successively gives the other seven sides $x+y, 2x+y, 3x+2y, 5x+3y, 4x+4y, 4x+5y, 8x+4y$.

2. The two vertical boundaries have lengths $(8x+4y)+(5x+3y)$ and $(4x+5y)+(4x+4y)$. Equality gives $5x=2y$.

3. Write $x=2t, y=5t$. The rectangle dimensions are $69t, 61t$. As $\gcd(69,61)=1$, positive integral coprime dimensions force $t=1$. Hence perimeter $2(69+61)=260$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2004-II-Q05] | Part A (Q1-Q5)

Answer: 766

1. Normalize the scheduled time per quarter to one, so the full deadline is four units. The first three quarters use $1 + 1000/900 + 1000/800 = 121/36$ units.

2. The last quarter has $23/36$ time left and needs 1000 worker-time units. Thus the workforce must be at least $\lceil 36000/23 \rceil = 1566$.

3. Additional hires are $1566 - 800 = 766$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2006-I-Q05] | Part A (Q1-Q5)

Answer: 936

1. Square and match the independent radicals: $ab = 52, ac = 234, bc = 72$. Multiplying gives $(abc)^2 = 52 \cdot 234 \cdot 72 = 936^2$.

2. Positivity yields $abc = 936$. Indeed $(a, b, c) = (13, 4, 18)$ also gives constant part $338 + 48 + 1620 = 2006$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2007-I-Q02] | Part A (Q1-Q5)

Answer: 052

1. At time $t \geq 4$, positions are $6t, 10(t-2), 8(t-4)$. Set twice each candidate position equal to the sum of the others.

2. B as midpoint gives $t = 4/3$, inadmissible; C gives an inconsistent equation; A gives $12t = 18t - 52$, so $t = 26/3$. The midpoint position is $6t = 52$, before any finish.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2007-II-Q04] | Part A (Q1-Q5)

Answer: 450

1. Let one X,Y require u, v worker-hours. Equations $300u + 200v = 100$, $240u + 300v = 120$ give $u = 1/7, v = 2/7$.

2. Then $150/7 + 2m/7 = 150$, so $m = 450$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2008-I-Q01] | Part A (Q1-Q5)

Answer: 252

1. If the original total is N, the unchanged girl count gives $.60N = .58(N + 20)$, so $N = 580$.

2. The new dancing count is $.40 \cdot 580 + 20 = 252$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2008-II-Q02] | Part A (Q1-Q5)

Answer: 620

1. If R's speed is r miles/min, its time is $50/r + 49 \cdot 5$. J's time is $200/(3r) + 24 \cdot 5$.

2. Equating gives $125 = 50/(3r)$, hence $r = 2/15$. Total time $375 + 245 = 620$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2009-II-Q01] | Part A (Q1-Q5)

Answer: 114

1. If each stripe uses s, red use beyond its pure stripe is $164 - 130 = 34$ and white extra use is $188 - 130 = 58$. These combine to the pink stripe, so $s = 34 + 58 = 92$.

2. Each leftover is $130 - 92 = 38$, hence total 114.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-I-Q01] | Part A (Q1-Q5)

Answer: 085

1. The combined final volume is ten liters with five liters of acid. Conservation gives $4(0.45) + 5(0.48) + k/100 = 5$, so $k = 80$.

2. If x liters go to A, $1.8 + 0.8x = 0.5(4 + x)$, giving $x = 2/3$. The remaining $1/3$ also makes B fifty-percent.

3. Hence $k + m + n = 80 + 2 + 3 = 85$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-II-Q01] | Part A (Q1-Q5)

Answer: 037

1. Normalize the actual purchase to one and let consumption be r. Then $1/2 - 2r = (2/9)(1 - r)$.

2. Multiplication by eighteen gives $9 - 36r = 4 - 4r$, so $r = 5/32$. It lies below $1/4$, making the hypothetical leftover nonnegative. The answer is 37.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2012-I-Q04] | Part A (Q1-Q5)

Answer: 279

1. Walking one mile takes the faster traveler fifteen minutes and the slower twenty-four; riding takes ten. If the faster traveler walks x of the first n miles, the slower rides those miles, so equality of arrival times is $15x + 10(n - x) = 10x + 24(n - x)$, or $19x = 14n$. Thus n must be a multiple of nineteen.

2. To verify attainability under the stated handoff rule, let Δ be faster minus slower arrival time at successive posts. When $\Delta < 0$, the slower traveler rides and Δ increases by five; when $\Delta > 0$, the faster rides and it decreases by fourteen. Starting at zero, the first mile gives five.

3. This keeps Δ in the integer interval $[-13, 5]$ and gives $\Delta \equiv 5n \pmod{19}$. Hence it first returns to zero exactly at post nineteen. Then $x = 14$.

4. The common time is $15(14) + 10(5) = 260$ minutes, so $n + t = 279$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2012-II-Q04] | Part A (Q1-Q5)

Answer: 061

1. Let those dimensions be L, W, x. Equal times give $(2L + W - x)/8.6 = (W + x)/6.2 = \sqrt{W^2 + x^2}/5$. The first equality gives $31L = 6W + 37x$.

2. The second equality, squared, is $168W^2 - 625Wx + 168x^2 = (24W - 7x)(7W - 24x) = 0$.

3. If $x = 7W/24$, then $L = 13W/24 < W$, impossible. Thus $x = 24W/7, L = 30W/7$, which also has $0 < x < L$.

4. The ratio is $30 : 7 : 24$, giving 61.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-I-Q01] | Part A (Q1-Q5)

Answer: 150

1. For swimming speed s, the three times are $0.5/s, 3/s, 1.6/s$ hours. Their sum is $5.1/s = 4.25$, so $s = 1.2$.

2. Cycling speed is twelve mph, and thirty miles take 2.5 hours, or 150 minutes.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-I-Q03] | Part A (Q1-Q5)

Answer: 018

1. The two square sides are x, y, so $x^2 + y^2 = (9/10)(x + y)^2$.

2. Rearranging gives $x^2 + y^2 = 18xy$. Dividing by the positive product yields the required value 18.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-II-Q01] | Part A (Q1-Q5)

Answer: 275

1. The old time is 396 of the 1440 daily minutes. Multiplying the fraction of a day by 1000 gives $396 \cdot 1000/1440 = 275$ decimal minutes.

2. The display is $2{:}75$, and the required value is 275.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-I-Q04] | Part A (Q1-Q5)

Answer: 049

1. Let east and west train speeds be u, v and length L. The one-minute conditions give $u - 20 = v + 20 = 60L$, hence $u = 60L + 20, v = 60L - 20$.

2. Overtaking the westbound rider requires $v > 20$. The tenfold time ratio gives $u + 20 = 10(v - 20)$, so $60L + 40 = 10(60L - 40)$.

3. Thus $L = 22/27$ miles, giving 49. The derived westbound speed $260/9$ exceeds twenty, verifying the branch.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-II-Q01] | Part A (Q1-Q5)

Answer: 334

1. The per-minute rates are $1/900, 1/600, 1/450$. After ninety minutes, one-tenth of the room is done.

2. The next two-fifths take $360(2/5) = 144$ minutes at combined rate $1/360$. The final half takes one hundred minutes at rate $1/200$.

3. The total is $90 + 144 + 100 = 334$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2017-II-Q05] | Part A (Q1-Q5)

Answer: 791

1. Let the four-number total be s. The six pair sums split into three complementary pairs, each totaling s. If x, y formed one pair, the four given values would split into two equal-sum pairs; none of the three pairings does so ($509 \neq 521$, $476 \neq 554$, $423 \neq 607$).

2. Therefore two given values must themselves be a complementary pair. Since all six sums total $3s$, $x + y = 3s - 1030 \leq 3(320 + 287) - 1030 = 791$.

3. Attainment is given by the four distinct numbers $103/2, 275/2, 365/2, 471/2$. Their remaining pair sums are373 and418, totaling791.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2018-II-Q01] | Part A (Q1-Q5)

Answer: 800

1. Let $AB = x$. Paul covers four-fifths of the initial gap $1800 - x$ before meeting Eve, then retraces this distance. His total is $8(1800 - x)/5$.

2. In the same time Ina covers x, while Paul runs twice as fast, so $8(1800 - x)/5 = 2x$. Solving gives $x = 800$ m.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2022-I-Q05] | Part A (Q1-Q5)

Answer: 550

1. Place the slower start at $(0,0)$ and faster start at $(D,0)$. The target is $(D/2,264)$. For common time t, write $u=D/(2t)$ and $v=264/t$. The water-relative velocity vectors are $(u-14,v)$ and $(-u-14,v)$.

2. Subtract their squared speeds: $(u+14)^2-(u-14)^2=80^2-60^2$, so $56u=2800$ and $u=50$. Then $v^2=60^2-36^2=2304$, giving $v=48$.

3. Thus $t=264/48=11/2$ minutes, and $D=2ut=550$ m.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2023-II-Q04] | Part A (Q1-Q5)

Answer: 273

1. Subtract the first two equations to obtain $(y-4)(x-z)=0$. Thus either $y=4$ or $x=z$.

2. If $y=4$, then $x+z=15$ and $xz=44$, giving $x=4$ or 11.

3. If $x=z$, then $x(y+4)=60$, so $x\neq 0$. Substitution into $x^2+4y=60$ gives $x^3-76x+240=(x-4)(x-6)(x+10)=0$. Every resulting value has a valid real companion $y=60/x-4$.

4. Therefore $S=\{-10,4,6,11\}$ and the square sum is $100+16+36+121=273$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2024-I-Q01] | Part A (Q1-Q5)

Answer: 204

1. Let the stop last u hours. Then $9/s + u = 4$ and $9/(s+2) + u = 12/5$.

2. Eliminating u gives $18/[s(s+2)] = 8/5$, so $s = 5/2$ (the other root is negative). Thus $u = 2/5$ h.

3. The requested speed is 3 km/h. The total is $60(9/3 + 2/5) = 204$ minutes.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2026-I-Q01] | Part A (Q1-Q5)

Answer: 277

1. Let first speed be v and its travel time $T > 2$. Equal distances give $vT = (v+2)(T-1) = (v+9)(T-2)$.

2. The first equality gives $v = 2T - 2$; the last gives $2v = 9T - 18$. Solving yields $T = 14/5, v = 18/5$, so distance $252/25$ and the answer 277.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q07] | Part B (Q6-Q10)

Answer: 005

1. Let $t = z + 1/y$. Because $xyz = 1$, expanding the product of the three cyclic expressions gives $(x + 1/z)(y + 1/x)(z + 1/y) = 2 + (x + 1/z) + (y + 1/x) + (z + 1/y)$.

2. Thus $145t = 36 + t$, yielding $t = 1/4$ and the requested sum 5.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q10] | Part B (Q6-Q10)

Answer: 173

1. Let S be the total sum. Each equation says $2x_k = S - k$. Summing gives $2S = 100S - 5050$, hence $S = 2525/49$.

2. Therefore $x_{50} = (S - 50)/2 = 75/98$, and the requested sum is 173.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2010-I-Q09] | Part B (Q6-Q10)

Answer: 158

1. Put $t = xyz$. Multiplying $x^3 = t + 2, y^3 = t + 6, z^3 = t + 20$ gives $t^3 = (t + 2)(t + 6)(t + 20)$.

2. After cancellation, $7t^2 + 43t + 60 = 0$, whose roots are $-4, -15/7$. Both are achieved by the real cube roots of the three expressions: their product has cube t^3, hence equals t.

3. The desired sum is $28 + 3t$, maximized at $t = -15/7$, giving $151/7$ and answer 158.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-I-Q06] | Part B (Q6-Q10)

Answer: 011

1. Vertex form gives $y = a(x - 1/4)^2 - 9/8$. At $x = 1$, its integer value j is $9(a - 2)/16$.

2. Thus $a = 2 + 16j/9 > 0$, requiring integer $j \geq -1$. The least choice is $j = -1$, giving $a = 2/9$ and answer 11.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2021-II-Q07] | Part B (Q6-Q10)

Answer: 145

1. From the first two equations, $ab = 3c - 4$. Grouping the cubic relation gives $ab(c + d) - 3cd = 14$, hence $d = (3c^2 - 4c - 14)/4$.

2. Substitute in $abcd = 30$ to get $(c + 2)(3c - 10)(3c^2 - 4c + 6) = 0$. The last factor has no real root. For $c = 10/3$, $ab = 6$ and $a + b = -3$ would give a negative discriminant, so this case is impossible.

3. Thus $c = -2, d = 3/2, ab = -10$. The square sum is $9 - 2(-10) + 4 + 9/4 = 141/4$, giving 145.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2001-I-Q11] | Part C (Q11-Q15)

Answer: 149

1. If the selected columns are c_i, old and new numbers are $(i-1)N + c_i$ and $5c_i - 5 + i$. The first two equations yield $c_1 = 5c_2 - 3$ and $N = 24c_2 - 19$.

2. The remaining three are $2N + c_3 = 5c_4 - 1$, $3N + c_4 = 5c_5$, $4N + c_5 = 5c_3 - 2$. Eliminating gives $124c_5 = 89N + 7$.

3. Substituting for N requires $7c_2 \equiv 18 \pmod{31}$, so $c_2 \equiv 7 \pmod{31}$. The smallest positive choice gives $N = 149$. Columns $(32, 7, 141, 88, 107)$ lie in range and verify every equation, proving attainability.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-I-Q14] | Part C (Q11-Q15)

Answer: 263

1. Add one to each fraction and four to the right side, obtaining $x\sum_{k\in\{3,5,17,19\}}1/(x-k)=x(x-11)$. The candidate $x=0$ is valid; otherwise divide by x and set $t=x-11$.

2. Pair symmetric denominators: $2t/(t^2-64)+2t/(t^2-36)=t$. Besides valid $t=0$, divide by t and set $u=t^2$.

3. The resulting equation is $u^2-104u+2504=0$, giving $u=52\pm\sqrt{200}$. Neither equals the excluded pole squares $36,64$.

4. The largest candidate is $11+\sqrt{52+\sqrt{200}}$, exceeding all other sign choices and $x=0,11$. Hence the sum is 263.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2015-II-Q14] | Part C (Q11-Q15)

Answer: 089

1. Put $s = x + y, p = xy$. Factoring gives $p^4 s = 810$ and $p^3(s^3 - 3ps) = 945$. Both s, p are nonzero.

2. Divide the second equation by the first: $(s^2 - 3p)/p = 7/6$, so $s^2 = 25p/6$ and $p > 0$. Squaring $p^4 s = 810$ then gives $(25/6)p^9 = 810^2$, hence $p^3 = 54$.

3. The second equation gives $x^3 + y^3 = 945/54 = 35/2$. Therefore the target is $2(35/2) + 54 = 89$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2001-I-Q03] | Part A (Q1-Q5)

Answer: 500

1. The degree-2001 terms cancel. The resulting degree-2000 polynomial has leading coefficient $2001/2$ and next coefficient $-\binom{2001}{2}/4$.

2. Vieta gives root sum $\binom{2001}{2}/(2 \cdot 2001) = 2000/4 = 500$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2013-I-Q05] | Part A (Q1-Q5)

Answer: 098

1. The equation is $9x^3 = (x+1)^3$. Taking real cube roots gives $\sqrt[3]{9}\,x = x+1$, hence $x = 1/(\sqrt[3]{9}-1)$.

2. Use $(u-1)(u^2+u+1) = u^3-1$ to rationalize: $x = (\sqrt[3]{81}+\sqrt[3]{9}+1)/8$. The linear equation after cube roots also proves uniqueness.

3. The requested sum is $81+9+8 = 98$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-II-Q05] | Part A (Q1-Q5)

Answer: 420

1. Let the third root of p be $t = -r - s$. The other cubic has third root $t - 1$. Equality of their pairwise root-product sums gives $t = 4r - 3s + 13$.

2. The constant-term difference is $rst - (r + 4)(s - 3)(t - 1) = 240$. Using the preceding linear relation reduces this to $(r - s)^2 + 7(r - s) - 8 = 0$. Thus $r - s = 1$ or -8.

3. Combining with $t = -r - s$ gives the respective root triples $(-5, -6, 11)$ and $(1, 9, -10)$. Their constant terms are -330 and 90. Each shifted triple verifies the required second cubic.

4. The absolute-value sum is 420.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2018-I-Q01] | Part A (Q1-Q5)

Answer: 600

1. Write the factors $(x+r)(x+s)$ with $0 \le r \le s$. Since their sum is positive and product nonnegative, this covers every integer-root possibility. The condition is $r+s=a$, with $b=rs$ uniquely determined.

2. For each a, there are $\lfloor a/2 \rfloor + 1$ choices. Pair $a=2k-1, 2k$ for $k=1,\ldots,50$; their contribution is $k+(k+1)$.

3. The total is $2(50 \cdot 51/2) + 50 = 2600$, leaving 600.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2021-II-Q04] | Part A (Q1-Q5)

Answer: 330

1. Both also have root $m - i\sqrt{n}$. The first cubic has zero quadratic coefficient, so $-20 + 2m = 0$ and $m = 10$.

2. The second has zero linear coefficient, so the pairwise root products sum to zero: $m^2 + n - 42m = 0$. Hence $n = 320$, and $m + n = 330$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2022-I-Q01] | Part A (Q1-Q5)

Answer: 116

1. Add the polynomials first. Their quadratic terms cancel, so their sum R is linear, with $R(16) = 108$ and $R(20) = 106$.

2. The slope is $-1/2$. Moving from $x = 16$ back to 0 increases R by 8, giving $R(0) = 116$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2003-II-Q09] | Part B (Q6-Q10)

Answer: 006

1. Division gives $P = (x^2+1)Q + (x^2 - x + 1)$. Therefore at every root, $P(z_i) = z_i^2 - z_i + 1$.

2. Vieta gives $\sum z_i = 1$, $\sum_{i<j} z_i z_j = -1$, so $\sum z_i^2 = 3$. The requested sum is $3 - 1 + 4 = 6$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2004-I-Q07] | Part B (Q6-Q10)

Answer: 588

1. With $u_j = (-1)^j j$, the coefficient is $\sum_{i<j} u_i u_j = [(\sum u_j)^2 - \sum u_j^2]/2$.

2. The alternating sum is -8, and $\sum_{j=1}^{15} j^2 = 1240$. Thus $C = (64 - 1240)/2 = -588$, so $|C| = 588$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2005-II-Q07] | Part B (Q6-Q10)

Answer: 125

1. Put $y = 5^{1/16}$. Multiplying the denominator by $y - 1$ repeatedly uses difference of squares, producing $(y-1)(y+1)(y^2+1)(y^4+1)(y^8+1) = y^{16} - 1 = 4$.

2. Therefore $x = y - 1$, so $(x+1)^{48} = y^{48} = 5^3 = 125$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2007-I-Q08] | Part B (Q6-Q10)

Answer: 030

1. The two quadratics must share a root, since otherwise their degree-four product divides a cubic. Twice the first minus the second is $-15x - 3k$, so a common root is $x = -k/5$.

2. Substitution into the first gives $-4k^2 + 120k = 0$, hence $k = 0$ or 30. For 30, the factors are $(x + 6)(x - 5)$ and $(x + 6)(2x + 5)$, which indeed divide their cubic least common multiple.

3. The maximum is 30.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2008-II-Q07] | Part B (Q6-Q10)

Answer: 753

1. Vieta gives $r+s+t=0$ and $rst=-251$. Each pair sum is the negative remaining root, so the requested sum is $-(r^3+s^3+t^3)$.

2. When the root sum is zero, the cube-sum identity yields $r^3+s^3+t^3=3rst=-753$. The answer is 753.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2010-I-Q06] | Part B (Q6-Q10)

Answer: 406

1. The bounds are $(x-1)^2+1$ and $2(x-1)^2+1$, so $P(1)=1$. The nonnegative quadratic $P(x)-[(x-1)^2+1]$ vanishes at 1 and must have a double root there.

2. Consequently $P(x)=a(x-1)^2+1$. Its value at 11 gives $a=9/5$, within the required interval $[1,2]$.

3. Therefore $P(16)=(9/5)225+1=406$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2010-II-Q06] | Part B (Q6-Q10)

Answer: 008

1. If an integer root r exists, $n = r^3 + 63/r$, with r a positive divisor of 63. For $r = 1, 3$ this gives $64, 48$; for $r \geq 7$ it exceeds 48. Negative r give negative n.

2. Otherwise monicity permits a factorization $(x^2 + px + q)(x^2 - px + r)$ with integers. Matching coefficients gives $qr = 63, q + r = p^2, n = p(q - r)$. Negative q, r cannot have square sum; positive factor pairs have sums $64, 24, 16$.

3. Only pairs $(1, 63), (7, 9)$ have square sums. Their positive possible n are $496, 8$.

4. The minimum is 8, attained by $(x^2 + 4x + 9)(x^2 - 4x + 7)$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-I-Q06] | Part B (Q6-Q10)

Answer: 036

1. The expanded quadratics have root sums $2h$ and root products $2013/3 = 671 = 11 \cdot 61$ and $2014/2 = 1007 = 19 \cdot 53$.

2. Positive factor pairs give possible sums $72, 672$ for the first and $72, 1008$ for the second. The only common sum is seventy-two.

3. Therefore $h = 36$, attained by roots $11, 61$ and $19, 53$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-I-Q09] | Part B (Q6-Q10)

Answer: 002

1. Put $a = \sqrt{2014}$. The cubic factors as $(ax - 1)(x^2 - 2ax - 2)$.

2. The quadratic roots are $a \pm \sqrt{a^2 + 2}$, one negative and one greater than a. Thus the middle root is $1/a$, while the other two sum to $2a$.

3. The product requested is 2.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2015-I-Q10] | Part B (Q6-Q10)

Answer: 072

1. Each cubic $f-12, f+12$ has at most three distinct roots, so the six inputs split into two root triples. Since the cubics differ only by a constant, the two triples have equal sums and equal pairwise-product sums.

2. The total root sum is 24. The only three-element subsets with sum 12 are $\{1,5,6\}$ and $\{2,3,7\}$, complementary and both having pairwise-product sum 41.

3. Change the sign of f if needed. Then $f=c(x-2)(x-3)(x-7)+12=c(x-1)(x-5)(x-6)-12$. At $x=0$, $-42c+12=-30c-12$, so $c=2$ and $f(0)=-72$. The requested absolute value is 72.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2015-II-Q06] | Part B (Q6-Q10)

Answer: 440

1. If the roots are $r \le s \le t$, Vieta gives $r+s+t=a$ and $rs+rt+st=(a^2-81)/2$. Subtraction yields $r^2+s^2+t^2=81$.

2. As $r \le \sqrt{27} < 6$, test $r=1,2,3,4,5$ and $r \le s \le \sqrt{(81-r^2)/2}$. The only square remainders give triples $(1,4,8),(3,6,6),(4,4,7)$.

3. The root sums are respectively $13, 15, 15$. The stated ambiguity therefore means $a=15$. Since $c=2rst$, the two values are 216 and 224, with sum 440.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2016-II-Q06] | Part B (Q6-Q10)

Answer: 275

1. All substituted exponents are odd, so replacing x by $-x$ makes every factor have nonnegative coefficients. Thus the coefficient $(-1)^j a_j$ of $Q(-x)$ is nonnegative, and equals $|a_j|$.

2. Summing the coefficients evaluates at $x = 1$: $\sum |a_j| = Q(-1) = P(-1)^5 = (3/2)^5 = 243/32$. Therefore $m + n = 275$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2018-II-Q06] | Part B (Q6-Q10)

Answer: 037

1. The polynomial factors as $(x-1)(x+2)[x^2+(2a-1)x+1]$. Only the quadratic can have nonreal roots.

2. Its discriminant is nonnegative exactly when $|2a-1| \geq 2$, or $a \leq -1/2$ or $a \geq 3/2$. This excludes an interval of length two from a sample interval of length thirty-eight.

3. The probability is $36/38 = 18/19$, giving 37.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2019-I-Q08] | Part B (Q6-Q10)

Answer: 067

1. Let $u = \sin^2 x, v = \cos^2 x, p = uv$, so $u + v = 1$ and $0 \le p \le 1/4$. The fifth power sum is $u^5 + v^5 = 1 - 5p + 5p^2$, giving $36p^2 - 36p + 5 = 0$.

2. The roots are $1/6, 5/6$; the bound selects $p = 1/6$. The sixth power sum is $1 - 6p + 9p^2 - 2p^3 = 13/54$.

3. The answer is 67.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2019-I-Q10] | Part B (Q6-Q10)

Answer: 352

1. Vieta on roots repeated three times gives $T = \sum z_j = -20/3$. Let $U = \sum_{j<k} z_j z_k$. The next coefficient counts three same-root-copy pairs and nine cross-root-copy pairs, so $19 = 3\sum z_j^2 + 9U$.

2. Substitute $\sum z_j^2 = T^2 - 2U$, obtaining $19 = 400/3 + 3U$. Thus $U = -343/9$, absolute value $343/9$, and the answer is 352.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2023-I-Q09] | Part B (Q6-Q10)

Answer: 738

1. Factor $p(x) - p(2) = (x-2)[x^2 + (a+2)x + 2a + b + 4]$. The constant c is unrestricted, supplying a factor of 41.

2. A monic integer quadratic with one integer root has two integer roots. Thus the bracket must have roots $2, k$ with $k \neq 2$, or repeated root $k \neq 2$.

3. The first case gives $a = -k - 4, b = 4k + 4$ and $-6 \leq k \leq 4$, excluding 2: ten choices. The second gives $a = -2k - 2, b = k^2 + 4k$ and $-6 \leq k \leq 2$, again excluding 2: eight choices.

4. These cases are disjoint, and every listed coefficient is in range. The total is $(10 + 8)41 = 738$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2000-II-Q13] | Part C (Q11-Q15)

Answer: 200

1. Use the cube difference $(10x^2)^3 - 1$ to factor the polynomial as $(100x^4 + 10x^2 + 1)(20x^2 + x - 2)$.

2. The first factor is positive on the real line. The quadratic roots are $(-1 \pm \sqrt{161})/40$, so the root with the positive radical has $(m, n, r) = (-1, 161, 40)$ and the sum is 200.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2008-I-Q13] | Part C (Q11-Q15)

Answer: 040

1. The axial and $(1, \pm 1)$ zeros reduce the polynomial to $u(x - x^3) + v(y - y^3) + wxy(1 - x)$. The zero at $(2, 2)$ imposes $w = -3(u + v)/2$.

2. A universal zero must annihilate both $x - x^3 - 3xy(1 - x)/2$ and $y - y^3 - 3xy(1 - x)/2$. Positive nonintegral x makes the first condition $y = 2(x + 1)/3$.

3. Substitution into the second yields $19x^2 - 43x + 10 = 0$, with roots $2, 5/19$. The new nonintegral point is $(5/19, 16/19)$, so $a + b + c = 40$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2013-II-Q12] | Part C (Q11-Q15)

Answer: 540

1. If all roots are real, they are a multiset of three from $\{-20, -13, 13, 20\}$, giving $\binom{6}{3} = 20$ polynomials.

2. Otherwise the unique real root is one of these four integers, say k, and the remaining roots are a nonreal conjugate pair of squared modulus $R^2 = 169$ or 400. Their quadratic factor is $z^2 - tz + R^2$, where integer coefficients require exactly that t be an integer.

3. Nonreality requires $|t| < 2R$. This allows 51 values when $R = 13$ and 79 when $R = 20$. Every choice gives a distinct integer cubic.

4. The total is $20 + 4(51 + 79) = 540$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2016-I-Q11] | Part C (Q11-Q15)

Answer: 109

1. Substitute $x = 1, 0, -2$ to obtain roots $1, 0, -1$. Thus $P = x(x-1)(x+1)L(x)$ for a polynomial L.

2. The functional identity reduces, as a polynomial identity, to $L(x+1) = L(x)$. A nonconstant polynomial has a nonzero finite difference of degree one lower, so L must be constant $c \neq 0$.

3. The normalization gives $(6c)^2 = 24c$, hence $c = 2/3$. Then $P(7/2) = (2/3)(7/2)(5/2)(9/2) = 105/4$, giving 109.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-I-Q11] | Part C (Q11-Q15)

Answer: 510

1. The inputs $u = f(2)$ and $v = f(4)$ coincide precisely when $a = -6$. Then every b, c works by setting $d = -u^2 - cu$, giving 441 triples.

2. Otherwise u, v are distinct roots, so $c = -u - v = -20 - 6a - 2b$. Thus $-15 \leq 3a + b \leq -5$, with $b \in [-10, 10]$.

3. For $a = -8, -7, -5, -4, -3, -2, -1, 0, 1$ the counts of b are $2, 5, 11, 11, 11, 11, 9, 6, 3$; all other allowed $a \neq -6$ contribute zero. Their sum is 69, and the total is 510.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-I-Q14] | Part C (Q11-Q15)

Answer: 085

1. Write $P = x^2 - sx + q$. The four roots split into two pairs, each satisfying $P(x) = r$ for one root r of P; each pair sums to s.

2. If $3, 4$ are paired, then $s = 7$ and $a + b = 7$, giving 49. This case exists: take $q = 12 + r$ with $r^2 - 6r + 12 = 0$; the other pair is distinct from $3, 4$ and from itself.

3. If they are in different pairs, the other roots are $s - 3, s - 4$. Since $P(3), P(4)$ are the roots of P, their sum gives $q = 4s - 25/2$. Thus $P(3) = s - 7/2, P(4) = 7/2$, and their product condition gives $(7/2)(s - 7/2) = 4s - 25/2$.

4. Solve $s = 1/2$, obtaining $a + b = -6$ and 36. All four roots $3, 4, -5/2, -7/2$ are distinct, so the final sum is $49 + 36 = 85$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-II-Q11] | Part C (Q11-Q15)

Answer: 071

1. Name the shared roots so that $P+Q=2(x-a)(x-b)$, $P+R=2(x-a)(x-c)$, $Q+R=2(x-b)(x-c)$. Combining these expressions recovers $2P$, whose linear coefficient yields $a=3/2$.

2. The constant term of $P+Q$ gives $2ab=-5$, hence $b=-5/3$. If $s=R(0)$, the other constants give $2ac=s-7$ and $2bc=s+2$.

3. Subtract to get $2c(b-a)=9$, so $c=-27/19$. Then $s=2bc-2=52/19$; the three roots are distinct and the answer is 71.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2024-II-Q11] | Part C (Q11-Q15)

Answer: 601

1. Set $q = ab + bc + ca, p = abc$. The degree-three sum is $300q - 3p$, so the condition is $100q - p = 2000000$. Expanding shows this is equivalent to $(100 - a)(100 - b)(100 - c) = 0$.

2. For each specified variable equal to 100, the other two sum to 200, giving 201 choices. Any overlap forces $(100, 100, 100)$, counted three times.

3. Correct the overcount: $3 \cdot 201 - 2 = 601$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2007-I-Q03] | Part A (Q1-Q5)

Answer: 015

1. Expansion gives imaginary parts $18b$ and $243b - b^3$. Equating and dividing by positive b gives $b^2 = 225$.

2. Thus $b = 15$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2009-I-Q02] | Part A (Q1-Q5)

Answer: 697

1. Write $z = u + 164i$ and clear the denominator: $u + 164i = -656 + 4(u+n)i$.

2. Compare components: $u = -656, u + n = 41$. Thus $n = 697$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2018-II-Q05] | Part A (Q1-Q5)

Answer: 074

1. Multiplying all equations gives $(xyz)^2 = 240^2(16+30i) = 240^2(5+3i)^2$. Thus $xyz = \pm 240(5+3i)$.

2. Dividing by the three given products gives matching-sign triples $x = \pm(20+12i), y = \pm(-10-10i), z = \pm(-3+3i)$. Both signs satisfy the products.

3. The sum is $\pm(7+5i)$, whose squared modulus is 74.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2022-I-Q04] | Part A (Q1-Q5)

Answer: 834

1. The arguments of w, z, i are $30^\circ, 120^\circ, 90^\circ$. Equality is equivalent to $r + 3 \equiv 4s \pmod{12}$.

2. For $s \equiv 1, 2, 0 \pmod{3}$, the required r residues modulo 12 are $1, 5, 9$. These occur $9, 8, 8$ times in $1, \ldots, 100$; the three s classes have sizes $34, 33, 33$.

3. Therefore the count is $34 \cdot 9 + 33 \cdot 8 + 33 \cdot 8 = 834$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2026-I-Q05] | Part A (Q1-Q5)

Answer: 065

1. In complex coordinates put $B = 0, A = 1$. Then $A\prime = e^{i\theta}$ and $B\prime = A\prime + e^{-i\theta}(B - A\prime) = e^{i\theta} - 1$.

2. Therefore $AB\prime^2 = |e^{i\theta} - 2|^2 = 5 - 4\cos\theta = 16/9$. Hence $\cos\theta = 29/36$, so $m + n = 65$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2000-II-Q09] | Part B (Q6-Q10)

Answer: 000

1. The quadratic equation for z has roots $e^{\pm 3^\circ i}$. Therefore $z^{2000} + z^{-2000} = 2\cos 6000^\circ = 2\cos 240^\circ = -1$.

2. The least integer strictly exceeding -1 is 0.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2005-I-Q06] | Part B (Q6-Q10)

Answer: 045

1. Adding one yields $(x-1)^4 = 2006$. The nonreal roots are $1 \pm i\sqrt[4]{2006}$.

2. Their product is $1 + \sqrt{2006}$. Since $44^2 < 2006 < 45^2$, this lies between 45 and 46, so the floor is 45.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2005-II-Q09] | Part B (Q6-Q10)

Answer: 250

1. Write $\sin t + i\cos t = ie^{-it}$. The identity becomes $i^n e^{-int} = ie^{-int}$, equivalent to $n \equiv 1 \pmod 4$.

2. The integers $1, 5, \ldots, 997$ number 250.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2008-II-Q09] | Part B (Q6-Q10)

Answer: 019

1. With $\omega = e^{i\pi/4}$, the move is $z \mapsto \omega z + 10$. Since $\omega^8 = 1$ and the eight powers sum to zero, the affine map also has period eight, so only six moves remain.

2. Thus $z_{150} = 5\omega^6 + 10\sum_{j=0}^{5}\omega^j = -5\sqrt{2} + (5 + 5\sqrt{2})i$. The absolute-coordinate sum is $5 + 10\sqrt{2}$, between nineteen and twenty.

3. The floor is 19.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2010-II-Q07] | Part B (Q6-Q10)

Answer: 136

1. Write $w = x + yi$. The root sum is real, forcing $4y + 12 = 0$. The roots become $x, x + 6i, 2x - 4 - 6i$.

2. The two nonreal roots are conjugate, so $x = 2x - 4 = 4$. Hence $P(1) = (-3)[(-3)^2 + 6^2] = -135$.

3. As $P(1) = 1 + a + b + c$, the required absolute value is 136.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2011-II-Q08] | Part B (Q6-Q10)

Answer: 784

1. The roots have radius eight and arguments $j\pi/6$. Each independent choice contributes $8\max(\cos(j\pi/6), -\sin(j\pi/6))$ to the best real part.

2. For $j = 5, \ldots, 10$, choose the negative sine; for $j = 1, 2, 3, 4, 11, 12$, choose the cosine. Each six-term sum of the normalized selected values is $1 + \sqrt{3}$.

3. The maximum is $16 + 16\sqrt{3} = 16 + \sqrt{768}$, so $m + n = 784$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-I-Q06] | Part B (Q6-Q10)

Answer: 071

1. The imaginary part is positive, so $z \neq 0$. Substitution gives $z^{142} = 1$, whose upper-half-plane roots have arguments $k\pi/71$, $1 \leq k \leq 70$.

2. Sine equality between angles strictly between zero and π means $m/n = k/71$ or $1 - k/71$. Since 71 is prime and k is neither zero nor seventy-one, both fractions reduce with denominator 71.

3. Thus $n = 71$; every such upper root also satisfies the original equations with $w = z^{13}$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-II-Q06] | Part B (Q6-Q10)

Answer: 125

1. The distance factors as $125|1+2i-z^2|$, with $|z^2|=25$. The triangle inequality gives maximum $125(25+\sqrt{5})$ when z^2 points opposite to $1+2i$.

2. Thus $z^2=-25(1+2i)/\sqrt{5}=-5\sqrt{5}(1+2i)$. This nonreal point has a square root in the upper half-plane, so the bound is attained under $b>0$.

3. Squaring yields $z^4=125(-3+4i)=-375+500i$. Hence $c+d=125$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-II-Q08] | Part B (Q6-Q10)

Answer: 040

1. Multiply the two equations and put $v = zw$. The result is $v - 240/v = -30 + 14i$, hence $v^2 + (30 - 14i)v - 240 = 0$.

2. The quadratic formula uses $\sqrt{416 - 210i} = \pm(21 - 5i)$, giving $v = 6 + 2i$ or $-36 + 12i$. Their squared moduli are $40, 1440$.

3. The lower one is attained by $z = 1 - i, w = 2 + 4i$, which satisfy both original equations. Thus the minimum is 40.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2013-I-Q10] | Part B (Q6-Q10)

Answer: 080

1. The conjugate is also a root, while the remaining root $q = a - 2r$ is an integer. Their product gives $q(r^2 + s^2) = 65$, so q is positive and $r^2 + s^2$ is $5, 13$, or 65.

2. The unordered positive square representations are $5 = 1^2 + 2^2$, $13 = 2^2 + 3^2$, $65 = 1^2 + 8^2 = 4^2 + 7^2$. For each, r may be either sign of either component, yielding four distinct polynomials; changing only s sign does not change the polynomial.

3. The root sum is $q + 2r$. The four r values cancel, contributing $4q$ for each representation. The coefficients produced are all nonzero.

4. The total is $4(13 + 5 + 1 + 1) = 80$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2014-I-Q07] | Part B (Q6-Q10)

Answer: 100

1. The quotient w/z ranges over the radius-$1/10$ circle at the origin, so $(w-z)/z$ ranges over the circle centered at -1 with that radius.

2. Extreme slopes from the origin are tangent to this circle. The right triangle formed by origin, circle center, and tangency point gives $\sin^2\alpha = 1/100$ for the extreme deviation from the real axis.

3. Hence $\tan^2\theta \le (1/100)/(99/100) = 1/99$, with tangency attaining equality. The sum is 100.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2014-II-Q10] | Part B (Q6-Q10)

Answer: 147

1. The nonzero denominators permit multiplication, yielding $z^2 + zw + w^2 = 0$. Hence w/z is one of the two primitive cube roots of unity; both make all original denominators nonzero.

2. The vertices form an equilateral triangle of circumradius 2014. Its area is $(3\sqrt{3}/4)2014^2 = 3 \cdot 1007^2\sqrt{3}$.

3. The coefficient modulo one thousand is $3 \cdot 7^2 = 147$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2016-I-Q07] | Part B (Q6-Q10)

Answer: 103

1. When $ab + 2016 \geq 0$, the first radical is real. Vanishing imaginary part requires $b = -a$. Then $a^2 \leq 2016$ gives $-44 \leq a \leq 44$, with $a = \pm 10$ excluded by the denominator. This contributes 87.

2. When $ab + 2016 < 0$, the numerator is purely imaginary and must vanish: $-ab - 2016 = |a + b|$. For positive $a + b$, this becomes $(a + 1)(b + 1) = -2015$; for negative sum it becomes $(a - 1)(b - 1) = -2015$. A zero sum cannot satisfy it.

3. The positive divisors of $2015 = 5 \cdot 13 \cdot 31$ are $1, 5, 13, 31, 65, 155, 403, 2015$. In the positive-sum case the positive factor must be one of the last four; either coordinate may carry it, giving 8 pairs. Negating both variables gives another 8 negative-sum pairs. Their denominators are nonzero.

4. Total $87 + 8 + 8 = 103$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2017-I-Q10] | Part B (Q6-Q10)

Answer: 056

1. The real cross-ratio condition equates directed inscribed angles modulo180 degrees. Since the three fixed points are noncollinear, the locus is their circumcircle, excluding the undefined point z_3.

2. The first two points have perpendicular bisector $y = 61$. The second and third have midpoint $(48, 69)$ and slope1, so their bisector is $y - 69 = -(x - 48)$. The center is therefore $(56, 61)$.

3. The highest point is directly above the center, with real coordinate56. Its height exceeds99, so it is not the excluded point z_3.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2018-I-Q06] | Part B (Q6-Q10)

Answer: 440

1. Write $z = e^{i\theta}$ with $0 \le \theta < 2\pi$. The condition is $\sin 720\theta = \sin 120\theta$, so either $600\theta \equiv 0 \pmod{2\pi}$ or $840\theta \equiv \pi \pmod{2\pi}$.

2. The respective families contain $600, 840$ angles. An overlap would require $\pi k/300 = (2j+1)\pi/840$, or $14k = 5(2j+1)$, contradicting parity.

3. The count is 1440, leaving 440.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2019-II-Q08] | Part B (Q6-Q10)

Answer: 053

1. The input is a sixth root of unity ω, so the powers reduce to $\omega^2, \omega, 1$. Therefore the real and imaginary parts are $c + (b - a)/2$ and $(a + b)\sqrt{3}/2$.

2. Imaginary parts give $a + b = 4038$. Since both coefficients are at most 2019, equality forces $a = b = 2019$. Then $c = 2015$, so $f(1) = 6053$, remainder 53.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2020-I-Q08] | Part B (Q6-Q10)

Answer: 103

1. Identify east with the positive real direction. The displacement multiplier is $z = \frac{1}{2}e^{i\pi/3} = (1 + i\sqrt{3})/4$, whose modulus is less than one.

2. The limiting displacement is $5/(1 - z)$. Since $|1 - z|^2 = (3/4)^2 + (\sqrt{3}/4)^2 = 3/4$, its squared modulus is $25/(3/4) = 100/3$.

3. The required integer is $100 + 3 = 103$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2023-II-Q08] | Part B (Q6-Q10)

Answer: 024

1. Put $P(t) = t^3 + t + 1$ and let its roots be $\beta_1, \beta_2, \beta_3$. Interchanging the factors in the product gives $\prod_{k=0}^{6} P(\omega^k) = \prod_{j=1}^{3}(1 - \beta_j^7)$, because $t^7 - 1$ has the seven roots ω^k.

2. Reducing powers using $\beta^3 = -\beta - 1$ gives $\beta^7 = 2\beta^2 - 1$. Thus $1 - \beta^7 = 2(1 - \beta)(1 + \beta)$.

3. The product becomes $8P(1)[-P(-1)] = 8 \cdot 3 \cdot 1 = 24$. The negative sign in the second bracket comes from the odd degree of P.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2024-I-Q07] | Part B (Q6-Q10)

Answer: 540

1. Write $z = 4(\cos\theta + i\sin\theta)$. Since $1/z = (\cos\theta - i\sin\theta)/4$, the real part simplifies to $324\cos\theta - 432\sin\theta$.

2. Cauchy–Schwarz bounds it by $\sqrt{324^2 + 432^2} = 540$. Equality is attained at $(\cos\theta, \sin\theta) = (3/5, -4/5)$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2001-II-Q14] | Part C (Q11-Q15)

Answer: 840

1. The equation is $z^8(z^{20}-1)=1$. On the unit circle this forces $|z^{20}-1|=1$, so $z^{20}=\operatorname{cis}(\pm 60^\circ)$.

2. For the positive sign, $z^8=\operatorname{cis}(-120^\circ)$ and $z^4=z^{20}/(z^8)^2=\operatorname{cis}300^\circ$, giving $\theta\equiv 75^\circ \pmod{90^\circ}$. The negative sign gives $\theta\equiv 15^\circ \pmod{90^\circ}$. Substitution confirms both classes suffice.

3. The ordered arguments are $15, 75, 105, 165, 195, 255, 285, 345$ degrees. The requested sum is $75+165+255+345=840$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2002-I-Q12] | Part C (Q11-Q15)

Answer: 275

1. Direct composition gives $F^2(z) = i(z+1)/(z-1)$ and $F^3(z) = z$ wherever the orbit is defined. The initial orbit avoids the poles, so it repeats every three steps.

2. As $2002 \equiv 1 \pmod 3$, compute $F(1/137 + i) = (1/137 + 2i)/(1/137) = 1 + 274i$. The sum is 275.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2003-II-Q15] | Part C (Q11-Q15)

Answer: 015

1. The triangular coefficient pattern is the convolution of two length-24 all-one lists, so $P(x) = x(1 + x + \cdots + x^{23})^2$.

2. Its distinct zeros are 0 and $e^{15k^\circ i}$ for $k = 1, \ldots, 23$. Thus the required terms are $|\sin 30k^\circ|$; the zero root contributes nothing.

3. Each twelve-step absolute-sine cycle sums to $4 + 2\sqrt{3}$. Over the twenty-three terms the absent twenty-fourth term is zero, so the total is $8 + 4\sqrt{3}$, giving 15.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2004-I-Q13] | Part C (Q11-Q15)

Answer: 482

1. Multiply by $(x-1)^2$: the result is $(x^{18}-1)^2 - x^{17}(x-1)^2 = (x^{19}-1)(x^{17}-1)$. The two roots at one cancel, while $P(1) = 323 \neq 0$.

2. Thus the roots are all nontrivial seventeenth and nineteenth roots of unity. The first five arguments are $1/19, 1/17, 2/19, 2/17, 3/19$.

3. Their sum is $6/19 + 3/17 = 159/323$, giving 482.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2008-II-Q13] | Part C (Q11-Q15)

Answer: 029

1. The exterior is the union of six rotated half-planes such as $\Re z > 1/2$. Under $w = 1/z$, that inequality becomes $u/(u^2 + v^2) > 1/2$, or $|w - 1| < 1$. Thus S is the union of six unit disks centered at the sixth roots of unity.

2. Partition into sixty-degree wedges centered on the disk-center rays. In each wedge its corresponding disk reaches farthest in every direction, with boundary endpoints $(3/2, \pm\sqrt{3}/2)$ for the wedge around the real axis.

3. That wedge piece has a 120° unit sector of area $\pi/3$ plus two triangles of total area $\sqrt{3}/2$. Multiply by six: area $2\pi + 3\sqrt{3} = 2\pi + \sqrt{27}$, giving 29.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2009-II-Q13] | Part C (Q11-Q15)

Answer: 672

1. Use complex coordinates $A=-2, B=2, C_i=2\omega^i$, with $\omega=e^{i\pi/7}$. Then $AC_i\cdot BC_i=4|1-\omega^{2i}|$.

2. The numbers ω^{2i} are the six nontrivial seventh roots. Evaluating $(x^7-1)/(x-1)$ at one gives $\prod_{i=1}^{6}|1-\omega^{2i}|=7$.

3. Thus $N=4^6\cdot 7=28672$, remainder 672.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-I-Q14] | Part C (Q11-Q15)

Answer: 375

1. Vieta gives $a+b+c=0$. Suppose the right angle is at b, with hypotenuse length $h=|a-c|$. Its midpoint $(a+c)/2=-b/2$ is the circumcenter, so $3|b|/2=h/2$, or $h=3|b|$.

2. The parallelogram identity gives $|a|^2+|c|^2=[|a-c|^2+|a+c|^2]/2=(h^2+|b|^2)/2$.

3. Hence $250=h^2/2+3|b|^2/2=(2/3)h^2$, yielding $h^2=375$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2013-I-Q14] | Part C (Q11-Q15)

Answer: 036

1. The absolutely convergent complex series is $Q + iP = \sum_{k \geq 0} (ie^{i\theta}/2)^k = 2/(2 - ie^{i\theta})$.

2. Rationalizing gives $P/Q = \cos\theta/(2 + \sin\theta)$; the denominator is positive. Squaring the given equality yields $49(1 - s^2) = 8(2 + s)^2$ for $s = \sin\theta$.

3. This factors as $(3s - 1)(19s + 17) = 0$. The interval requires $s \leq 0$, so $s = -17/19$. The original positive ratio then chooses $\cos\theta = 6\sqrt{2}/19$, a valid fourth-quadrant value.

4. The answer is $17 + 19 = 36$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2017-II-Q12] | Part C (Q11-Q15)

Answer: 110

1. Use complex coordinates with initial center0 and $A_0 = 1$. Let O_n denote the centers. Induction gives $A_n - O_n = r^n i^n$; internal tangency then gives $O_{n+1} - O_n = (1-r)(ir)^n$.

2. The radii shrink to zero in nested disks, so their unique common point is the center limit $B = (1-r)/(1-ir)$. Its distance from0 is $(1-r)/\sqrt{1+r^2}$.

3. At $r = 11/60$, this equals $49/\sqrt{3600+121} = 49/61$. Thus $m+n = 110$. The center-to-limit distance after stage n is $r^n(1-r)/\sqrt{1+r^2} < r^n$, so the point is inside every disk.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2019-I-Q12] | Part C (Q11-Q15)

Answer: 230

1. The consecutive displacement ratio is $[f(f(z)) - f(z)]/[f(z) - z] = (z - 19)(z + 1)$, using $f(z) - z = z(z - 20)$. Perpendicular nonzero legs require this ratio purely imaginary and nonzero.

2. For $z = x + 11i$, its real part is $x^2 - 18x - 140$. Set zero to obtain $x = 9 \pm \sqrt{221}$. The positive specified form selects $x = 9 + \sqrt{221}$, and its ratio has nonzero imaginary part.

3. Thus $m + n = 9 + 221 = 230$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2023-I-Q15] | Part C (Q11-Q15)

Answer: 349

1. Except for $p = 2$, write $p = a^2 + b^2$ with integers $a > b > 0$. Multiplication by units and conjugation allow signs and order of the real and imaginary cube components to be adjusted; their positive magnitudes are therefore the candidate sides.

2. For $X = a^3 - 3ab^2, Y = 3a^2b - b^3$, factor $X + Y = (a - b)(p + 4ab)$ and $X - Y = (a + b)(p - 4ab)$. Since the first magnitude exceeds p, the triangle condition is exactly $(a + b)|p - 4ab| < p$. In particular $|a^2 + b^2 - 4ab| < \sqrt{p}$.

3. The finite bounds are $1 \le b \le 21$ and $b < a \le 31$. Testing a for each such b in $(a^2 + b^2 - 4ab)^2 < a^2 + b^2 < 1000$ gives possible values
$10, 17, 53, 68, 130, 153, 212, 241, 272, 349, 386, 520, 565, 725, 778, 905, 964$. The primes among them are $17, 53, 241, 349$; their $(a + b)|p - 4ab|$ values are $5, 27, 19, 253$, all smaller than p.

4. The largest is 349. Indeed $z = 18 + 5i$ has $z^3 = 4482 + 4735i$, and $4735 - 4482 = 253 < 349 < 4482 + 4735$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2024-II-Q13] | Part C (Q11-Q15)

Answer: 321

1. Factor $t^2 - 2t + 2 = (t - (1 + i))(t - (1 - i))$. Since 13 is prime, the powers of ω exhaust the roots of $t^{13} - 1$. The product is therefore $[1 - (1 + i)^{13}][1 - (1 - i)^{13}]$.

2. As $(1 + i)^{13} = -64 - 64i$, this becomes $(65 + 64i)(65 - 64i) = 65^2 + 64^2 = 8321$. The remainder is 321.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2026-II-Q11] | Part C (Q11-Q15)

Answer: 132

1. Choose nonzero square roots s_1, s_2, s_3 of the roots. The eight sign sums come in opposite pairs. Exactly seven values therefore requires a zero sign sum; coincidences between nonopposite choices force $s_i = \pm s_j$ and would reduce the count to at most six.

2. The product of the four essentially different sign sums is $4e_2 - e_1^2$, where e_1, e_2 are elementary symmetric functions of s_i^2. Its vanishing gives $4(n-11) - (n/6)^2 = 0$, so $n = 12$ or 132.

3. At $n = 132$, factor the cubic as $(x-16)(x^2 - 6x + 25)$ and choose roots $4, 2+i, 2-i$. The sign sums are zero twice and six distinct values $\pm 8, \pm(4+2i), \pm(4-2i)$. Thus 132 qualifies and is greatest.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2000-II-Q01] | Part A (Q1-Q5)

Answer: 007

1. Reciprocal logarithms convert the expression to $2\log_{2000^6} 4 + 3\log_{2000^6} 5 = \log_{2000^6}(4^2 \cdot 5^3)$.

2. Since $4^2 \cdot 5^3 = 2000$, the value is $1/6$ and the answer is 7.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2003-I-Q04] | Part A (Q1-Q5)

Answer: 012

1. The logarithm domains make both sine and cosine positive, and the first relation gives their product $1/10$.

2. Their squared sum is $1 + 2/10 = 12/10$. Taking the logarithm gives exactly $(\log_{10} 12 - 1)/2$, so $n = 12$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2005-II-Q05] | Part A (Q1-Q5)

Answer: 054

1. Put $t = \log_a b$, so $t + 6/t = 5$. Factoring $t^2 - 5t + 6$ yields $t = 2, 3$, hence $b = a^2$ or a^3.

2. Squares allow $a = 2, \ldots, 44$, giving 43 pairs; cubes allow $2, \ldots, 12$, giving 11. The families do not overlap for $a \geq 2$, so total 54.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2006-II-Q02] | Part A (Q1-Q5)

Answer: 893

1. Triangle inequalities require $\log(75/12) < \log n < \log(12 \cdot 75)$, equivalent to $25/4 < n < 900$. These also ensure positivity and the remaining inequality.

2. Thus $n = 7, \ldots, 899$, giving 893 values.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2009-II-Q02] | Part A (Q1-Q5)

Answer: 469

1. Raise each given equality to its own logarithmic exponent. The terms become $27^{\log_3 7} = 7^3$, $49^{\log_7 11} = 11^2$, and $(\sqrt{11})^{\log_{11} 25} = 5$.

2. The sum is $343 + 121 + 5 = 469$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2010-I-Q03] | Part A (Q1-Q5)

Answer: 529

1. Take the positive xth root after substitution: $x^{3/4} = 3x/4$. Thus $x^{-1/4} = 3/4$ and $x = 256/81$.

2. Then $y = 64/27$, so $x + y = 448/81$ in lowest terms. The answer is 529.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2010-II-Q05] | Part A (Q1-Q5)

Answer: 075

1. Put $a = \log_{10} x, b = \log_{10} y, c = \log_{10} z$. Then $a + b + c = 81$ and $ab + ac + bc = 468$.

2. Thus $a^2 + b^2 + c^2 = 81^2 - 2 \cdot 468 = 5625$, whose positive square root is 75.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2013-II-Q02] | Part A (Q1-Q5)

Answer: 881

1. Undo the logarithms outward to inward: the middle logarithm is one, the innermost logarithm is 2^a, and therefore $b2^a = 1000$.

2. Since $1000 = 2^3 \cdot 125$, positive a is $1, 2, 3$, giving $(a, b) = (1, 500), (2, 250), (3, 125)$.

3. The sums are distinct and total $501 + 252 + 128 = 881$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2016-II-Q03] | Part A (Q1-Q5)

Answer: 265

1. Put $u = \log_5 x, v = \log_5 y, w = \log_5 z$, with sum s. Exponentiating the outer logs gives $5^s + u = 35$, $5^s + v = 84$, and $5^s + w = 259$.

2. Add: $3 \cdot 5^s + s = 378$. The left side is strictly increasing, and $s = 3$ solves it, so it is the only solution.

3. Then $(u, v, w) = (-90, -41, 134)$. These define positive x, y, z and valid original logarithm arguments. The requested absolute sum is $90 + 41 + 134 = 265$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2018-I-Q05] | Part A (Q1-Q5)

Answer: 189

1. The first equation with its positive argument is equivalent to $(2x+y)^2 = x^2 + xy + 7y^2$, factoring as $(x-y)(x+2y) = 0$. Neither branch allows $y = 0$.

2. The second equation similarly gives $K = 6(x/y)^2 + 2(x/y) + 1$. The branches $x/y = 1, -2$ yield $K = 9, 21$.

3. Both occur with all log domains valid: choose $y > 0$ in the first branch, $y < 0$ in the second. Their product is 189.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2020-I-Q02] | Part A (Q1-Q5)

Answer: 017

1. Use $t = \log_2 x$ to rewrite the terms as $(1+t)/3, t/2, t$. Their relation is $t^2/4 = t(1+t)/3$.

2. $t = 0$ produces $1/3, 0, 0$, which has no positive common ratio. For $t \neq 0$, $3t = 4 + 4t$, giving $t = -4$.

3. Therefore $x = 1/16$; the terms are $-1, -2, -4$, so the ratio is indeed 2. The result is 17.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2020-II-Q03] | Part A (Q1-Q5)

Answer: 103

1. Logarithm bases impose $x \neq 0, -3$. Changing to one fixed base and cancelling $\log 3 / \log 2$ gives $20/x = 2020/(x+3)$.

2. Thus $20x + 60 = 2020x$, so $x = 3/100$, which satisfies both base restrictions. The result is 103.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2022-II-Q04] | Part A (Q1-Q5)

Answer: 112

1. Call the common value t. Then $\ln(22x) = t\ln(20x)$ and $\ln(202x) = t\ln(2x)$.

2. Subtract the first equality from the second to eliminate $\ln x$: $\ln(101/11) = t\ln(1/10)$. Thus $t = \log_{10}(11/101)$.

3. The given existence is consistent: $(1-t)\ln x = t\ln 20 - \ln 22$ determines a positive x, and neither excluded value satisfies this equation. The coprime pair is $(11, 101)$, giving 112.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2023-I-Q02] | Part A (Q1-Q5)

Answer: 881

1. Put $t = \log_b n \geq 0$. The first relation becomes $\sqrt{t} = t/2$, whose only possibilities are $t = 0, 4$.

2. The second relation is $bt = 1 + t$, ruling out $t = 0$. Hence $t = 4$ and $b = 5/4$.

3. Therefore $n = (5/4)^4 = 625/256$. The powers of 5 and 2 are coprime, so the sum is 881.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2024-I-Q02] | Part A (Q1-Q5)

Answer: 025

1. Extracting the powers gives $x \log_x y = 10$ and $4y \log_y x = 10$.

2. Multiply: change of base yields $(\log_x y)(\log_y x) = 1$, so $4xy = 100$ and $xy = 25$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2024-II-Q04] | Part A (Q1-Q5)

Answer: 033

1. Put $a = \log_2 x, b = \log_2 y, c = \log_2 z$. The equations become $a - b - c = 1/2, b - a - c = 1/3, c - a - b = 1/4$. Their sum gives $a + b + c = -13/12$.

2. Therefore $(a, b, c) = (-7/24, -3/8, -5/12)$, and $|4a + 3b + 2c| = 25/8$. The required sum is 33.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2025-II-Q04] | Part A (Q1-Q5)

Answer: 106

1. Change base and factor the exponents. The kth factor becomes $\frac{k-1}{k-2}\frac{k+1}{k+2}\frac{\log(k+1)}{\log k}$.

2. The three products telescope to $62/2 = 31$, $5/65 = 1/13$, and $\log 64/\log 4 = 3$. Their product is $93/13$, so the requested sum is 106.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2000-I-Q09] | Part B (Q6-Q10)

Answer: 025

1. Put $a = \log_{10} x, b = \log_{10} y, c = \log_{10} z$, and $L = \log_{10} 2$. The equations become $(1 - a)(1 - b) = L$, $(1 - b)(1 - c) = L$, $(1 - c)(1 - a) = 1$.

2. Since $L \neq 0$, the first two yield $a = c$. Therefore $1 - a = 1 - c = \pm 1$. Correspondingly $1 - b = \pm L$, with matching signs.

3. Thus $y = 10^{1-L} = 5$ or $10^{1+L} = 20$, and their sum is 25.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2002-I-Q06] | Part B (Q6-Q10)

Answer: 012

1. Let $p = \log_{225} x, q = \log_{64} y$, with $p, q \neq 0$. Then $p + q = 4$ and $1/p - 1/q = 1$, yielding $p^2 - 6p + 4 = 0$.

2. The two p-values sum to 6, and the q-values to 2. Hence the product is $225^6 \cdot 64^2 = 30^{12}$, so the required logarithm is 12.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2005-I-Q08] | Part B (Q6-Q10)

Answer: 113

1. Put $y = 2^{111x} > 0$, converting the equation to $y^3 - 8y^2 + 16y - 4 = 0$. The three stated real x roots correspond bijectively to three positive y roots.

2. By Vieta their product is four, so the x-sum is $\log_2(4)/111 = 2/111$. The answer is 113.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2011-I-Q09] | Part B (Q6-Q10)

Answer: 192

1. Positivity permits exponentiation and squaring: $(24\sin x)^3 = (24\cos x)^2$, hence $24\sin^3 x = \cos^2 x$.

2. With $s = \sin x$, this is $24s^3 + s^2 - 1 = (3s-1)(8s^2+3s+1) = 0$. The quadratic has negative discriminant, so $s = 1/3$. It gives valid base $8 \neq 1$ and positive argument.

3. Then $\cot^2 x = (1 - 1/9)/(1/9) = 8$, and the result is 192.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2012-I-Q09] | Part B (Q6-Q10)

Answer: 049

1. Write $x = 2^a, y = 2^b, z = 2^c$, and denote the nonzero common value by t. Then $b + 1 = ta/2, c + 2 = t(a + 1)/2, b + c + 3 = t(4a + 1)$.

2. Add the first two expressions and compare with the third: $t(2a + 1)/2 = t(4a + 1)$. Since $t \neq 0$, one gets $a = -1/6$.

3. The first two relations now imply $c + 2 = -5(b + 1)$, or $5b + c = -7$. Hence $xy^5z = 2^{-43/6}$ and $p + q = 49$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2013-I-Q08] | Part B (Q6-Q10)

Answer: 371

1. The domain is $[1/(mn), m/n]$, so its length condition is $2013(m^2-1) = mn$.

2. As m is coprime to m^2-1, one must have $m \mid 2013 = 3 \cdot 11 \cdot 61$. Conversely every such divisor greater than one yields integer $n = 2013(m - 1/m)$.

3. The sum is $2014m - 2013/m$, strictly increasing for positive m. The least permitted divisor is three, giving $n = 5368, m+n = 5371$ and remainder 371.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2017-II-Q07] | Part B (Q6-Q10)

Answer: 501

1. The domain requires $x > -2$ and $kx > 0$. On this domain the equation is $kx = (x+2)^2$, or $x^2 + (4-k)x + 4 = 0$.

2. For each $k < 0$, the allowed interval is $(-2, 0)$; the line kx strictly decreases while $(x+2)^2$ strictly increases, with opposite endpoint order, giving exactly one crossing. All500 negative values work; $k = 0$ is undefined.

3. For $k > 0$, one needs positive roots. Discriminant $k(k-8)$ is negative below8, zero at8 with double root2, and positive above8 with two positive roots. Only $k = 8$ contributes, so the total is501.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2019-II-Q06] | Part B (Q6-Q10)

Answer: 216

1. Put $y = \log_b x$. Change of base in the second equation gives $\log_b y = y/54$. The first becomes $y/2 + \log_b y = 56/3$.

2. Thus $y/2 + y/54 = 56/3$, so $y = 36$. Then $\log_b 36 = 2/3$, giving $b = 36^{3/2} = 216$. This value fixes a unique y, hence a unique x.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2026-I-Q06] | Part B (Q6-Q10)

Answer: 441

1. Put $t = \log_{2026} x$ and take logarithms. This yields $t^2 - 20t - 20\log_{2026} 26 = 0$, with two distinct real roots and therefore two positive x values. Vieta gives root sum 20.

2. The product is $2026^{20} = 2^{20}1013^{20}$. Trial division by primes up to 31 verifies 1013 prime, so the divisor count is $(20+1)^2 = 441$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2017-I-Q14] | Part C (Q11-Q15)

Answer: 896

1. Put $c = \log_2 a > 0$. Exponentiating the outer first logarithm and multiplying by c gives $\log_2(24/c) = 128c + c2^{128c}$. The left side is strictly decreasing and the right strictly increasing, so there is at most one positive solution.

2. The value $c = 3/64$ works: the left is $\log_2 512 = 9$, and the right is $6 + 3 = 9$. The second equation then gives $x = a^{a^{256}} = (2^{3/64})^{4096} = 2^{192}$, a positive integer.

3. Modulo8 the result is0. Modulo125, Euler gives $2^{192} \equiv 2^{-8} \equiv 6^{-1} \equiv 21$. Solving these two congruences gives $x \equiv 896 \pmod{1000}$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2001-II-Q03] | Part A (Q1-Q5)

Answer: 898

1. Substituting the formula for x_{n-1} into that for x_n cancels all but $-x_{n-5}$, so $x_{n+5} = -x_n$ and $x_{n+10} = x_n$.

2. The indices reduce to $1, 3, 5$ modulo 10. Since $x_5 = 523 - 420 + 375 - 211 = 267$, the sum is $211 + 420 + 267 = 898$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2002-I-Q04] | Part A (Q1-Q5)

Answer: 840

1. Partial fractions give $a_k = 1/k - 1/(k+1)$, hence the sum is $1/m - 1/n = 1/29$.

2. Rearrange as $(29-m)(29+n) = 29^2$. Here $29-m > 0$ and $29+n > 29$, leaving only factors $1, 841$. Thus $m = 28, n = 812$, and the sum is 840.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2004-I-Q02] | Part A (Q1-Q5)

Answer: 201

1. The respective means are 2 and $1/2$. Thus m is odd, the maximum of A is $2 + (m-1)/2$, and that of B is m.

2. The condition is $|3-m|/2 = 99$. Its only positive solution is $m = 201$, which has the required odd parity.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2005-II-Q03] | Part A (Q1-Q5)

Answer: 802

1. For first term a and ratio r, $a/(1-r)=2005$ and $a^2/(1-r^2)=20050$. Dividing gives $a/(1+r)=10$.

2. Thus $2005(1-r)=10(1+r)$, so $r=1995/2015=399/403$. The terms are coprime and sum to 802.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2008-II-Q01] | Part A (Q1-Q5)

Answer: 100

1. Each four-term block $(4j)^2 + (4j-1)^2 - (4j-2)^2 - (4j-3)^2$ simplifies to $32j - 12$.

2. Sum for $j = 1, \ldots, 25$: $N = 32 \cdot 325 - 12 \cdot 25 = 10100$, remainder 100.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q03] | Part A (Q1-Q5)

Answer: 143

1. With first angle a and positive integer step d, the angle sum gives $18a + 153d = 2880$, or $2a + 17d = 320$.

2. Parity forces even $d \geq 2$. Convexity requires $a + 17d = (320 + 17d)/2 < 180$, so $17d < 40$.

3. Therefore $d = 2$ and $a = 143$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q05] | Part A (Q1-Q5)

Answer: 542

1. Group terms into blocks of 2011. Their sums form a geometric sequence, and its first two terms are $200, 180$.

2. The block ratio is $9/10$, so the third block sums to 162. The requested total is $200 + 180 + 162 = 542$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2012-I-Q02] | Part A (Q1-Q5)

Answer: 195

1. If there are n terms, the total increase is the sum of the first n odd numbers, $n^2 = 121$. Thus $n = 11$.

2. The middle term equals the mean, $715/11 = 65$, and the two endpoint terms sum to twice this. The requested sum is 195.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2012-II-Q02] | Part A (Q1-Q5)

Answer: 363

1. For common ratio $r \neq 0$, $27r^{14} = 99r^{10}$ gives $r^4 = 11/3$.

2. Therefore $a_9 = 27r^8 = 27(11/3)^2 = 363$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2013-II-Q03] | Part A (Q1-Q5)

Answer: 350

1. Burning j complete centimeters takes $10\sum_{k=1}^{j} k = 5j(j+1)$ seconds. Thus $T/2 = (5 \cdot 119 \cdot 120)/2 = 35700$.

2. The equation $j(j+1) = 7140 = 84 \cdot 85$ shows exactly eighty-four centimeters have burned. Thus $h = 35$ and $10h = 350$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2015-I-Q01] | Part A (Q1-Q5)

Answer: 722

1. Pair terms with the common factor $2k$. Their difference in $B - A$ is $2k[(2k+1) - (2k-1)] = 4k$.

2. Therefore $B - A = 1 - 39 + 4\sum_{k=1}^{19} k = -38 + 4 \cdot 190 = 722$. This is positive, so its absolute value is 722.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q01] | Part A (Q1-Q5)

Answer: 336

1. Geometric summation gives $S(a)S(-a) = 144/(1-a^2) = 2016$, so $(1-a^2)^{-1} = 14$.

2. The sum is $12/(1-a) + 12/(1+a) = 24/(1-a^2) = 336$. The resulting $a^2 = 13/14$ is consistent with convergence.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q05] | Part A (Q1-Q5)

Answer: 053

1. If there are $k \geq 2$ days, arithmetic summation yields $k(2n + k - 1) = 748$ and $k(2t + k - 1) = 638$.

2. Subtraction gives $k(n - t) = 55$. Also k divides both $748, 638$, so it divides 22. Thus $k \mid \gcd(55, 22) = 11$, forcing $k = 11$.

3. Now $2n + 10 = 68$ and $2t + 10 = 58$, giving $n = 29, t = 24$. Hence $n + t = 53$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-II-Q01] | Part A (Q1-Q5)

Answer: 108

1. The remaining total is 405, so the middle arithmetic term is 135. Thus $B = 144$ initially and $A + C = 300$.

2. Geometric progression gives $AC = B^2 = 20736$. The roots of $t^2 - 300t + 20736 = 0$ are $108, 192$. Therefore $A = 108$, and the remaining counts $103, 135, 167$ check the arithmetic condition.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2023-II-Q01] | Part A (Q1-Q5)

Answer: 220

1. Let the least count be a and greatest be $2a$. The sum of six terms is $6(a+2a)/2 = 9a$.

2. Thus $a = 990/9 = 110$, and the largest count is 220.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2026-II-Q01] | Part A (Q1-Q5)

Answer: 178

1. The common difference must be a positive divisor of both 20 and 30, hence one of $1, 2, 5, 10$. Each such divisor works.

2. The tenth term is $4 + 9d$, so the sum is $4 \cdot 4 + 9(1 + 2 + 5 + 10) = 178$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2002-II-Q06] | Part B (Q6-Q10)

Answer: 521

1. Use $1/(n^2-4) = \frac{1}{4}(1/(n-2) - 1/(n+2))$. The value becomes $250(1 + 1/2 + 1/3 + 1/4) - R = 520 + 5/6 - R$, where $R = 250(1/9999 + 1/10000 + 1/10001 + 1/10002)$.

2. The bound $0 < R < 1000/9999 < 1/3$ puts the value strictly between 520.5 and $520 + 5/6$. Thus the nearest integer is 521.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2003-II-Q08] | Part B (Q6-Q10)

Answer: 348

1. Index from $j = 0$. Its term formula is quadratic, $T_j = aj^2 + bj + c$. The first values give $c = 1440, a + b = 276, 4a + 2b = 408$.

2. Thus $a = -72, b = 348$. The eighth term is $T_7 = -72 \cdot 49 + 348 \cdot 7 + 1440 = 348$.

3. The factorization $T_j = (180 - 24j)(8 + 3j)$ confirms that two real arithmetic sequences realize the formula.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2004-II-Q09] | Part B (Q6-Q10)

Answer: 973

1. Put $a_2 = r$. Induction gives $a_{2k+1} = [1 + k(r-1)]^2$ and $a_{2k+2} = [1 + k(r-1)][1 + (k+1)(r-1)]$.

2. The given sum becomes $(4r-3)(9r-7) = 646$, or $(r-5)(36r+125) = 0$. Positivity gives $r = 5$.

3. Thus the terms are $(4k+1)^2$ and $(4k+1)(4k+5)$, increasing with the index. The boundary terms are $a_{16} = 29 \cdot 33 = 957$, $a_{17} = 1089$. The requested sum is $16 + 957 = 973$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2008-I-Q06] | Part B (Q6-Q10)

Answer: 017

1. Induction gives row-r, position-j entry $2^{r-1}(r+2j-2)$, for $1 \leq j \leq 51-r$.

2. As sixty-seven is odd, divisibility requires $r+2j-2=67$, the only positive multiple of sixty-seven within the possible range up to ninety-nine. It occurs exactly for odd $r \leq 33$.

3. There are seventeen such rows, each with one qualifying entry.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2008-II-Q06] | Part B (Q6-Q10)

Answer: 561

1. Dividing by u_{n-1} shows successive ratios increase by one: $u_n/u_{n-1} = 1 + u_{n-1}/u_{n-2}$.

2. Therefore $a_n = n!$, $b_n = (n+2)!/2$. Their ratio at thirty-two is $34 \cdot 33/2 = 561$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2009-I-Q07] | Part B (Q6-Q10)

Answer: 041

1. Rewrite as $5^{a_{n+1}-a_n} = (3n+5)/(3n+2)$. Multiplication telescopes to $5^{a_k-1} = (3k+2)/5$, hence $a_k = \log_5(3k+2)$.

2. Thus $3k+2 = 5^j$. Modulo three this requires odd j; $j=1$ gives excluded $k=1$, and the next is $j=3$, giving $k=41$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2009-I-Q08] | Part B (Q6-Q10)

Answer: 398

1. Differences are all distinct: $2^j - 2^i = 2^i(2^{j-i} - 1)$ has two-adic valuation i, then the odd part uniquely determines j. Thus summing pairs counts each difference once.

2. The coefficient of 2^j is $j - (10 - j) = 2j - 10$. Hence $N = 2\sum_{j=0}^{10} j2^j - 10\sum_{j=0}^{10} 2^j = 2(18434) - 10(2047) = 16398$.

3. The remainder is 398.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2014-II-Q07] | Part B (Q6-Q10)

Answer: 021

1. At integer k, $f(k) = [(k+1)(k+2)]^{(-1)^k}$. The sum of logs is the log of their telescoping product.

2. For even n, the product is $(n+2)/2$; for odd n, it is $1/[2(n+2)]$. Absolute log one means a product of ten or one-tenth.

3. The even case gives $n = 18$, and the odd case $n = 3$. No other sign choices fit their product ranges. Their sum is 21.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2015-I-Q09] | Part B (Q6-Q10)

Answer: 494

1. An adjacent equal pair forces a zero two steps later. An adjacent pair differing by 1 first creates an equal pair, then a zero. Thus all triples with $|a_2 - a_1| \leq 1$ or $|a_3 - a_2| \leq 1$ qualify.

2. Count the complement by fixing a_2. At either endpoint $1, 10$, each neighbor has 8 choices at distance at least 2; at each other value there are 7 choices. Hence $2 \cdot 8^2 + 8 \cdot 7^2 = 520$ avoid these initial triggers, leaving 480 qualifying triples.

3. Among the 520, if $a_3 \geq 2$, then $a_4 \geq 2a_3$ and the subsequent terms increase with every adjacent gap at least 2, so zero never occurs. If $a_3 = 1$, the only additional trigger is $|a_2 - a_1| = 2$, making $a_4 = 2$ and an adjacent gap 1.

4. For $a_3 = 1$, the previous gap restriction requires $a_2 \geq 3$. Pairs $a_1 = a_2 - 2$ give 8 choices, and $a_1 = a_2 + 2$ give 6. Larger first gaps produce $a_4 \geq 3$ and permanent growth. Therefore the total is $480 + 14 = 494$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q10] | Part B (Q6-Q10)

Answer: 504

1. Write the initial reduced ratio as $b/a > 1$. Integrality gives the first three terms ca^2, cab, cb^2 for a positive integer c. Put $d = b - a \geq 1$. Inducting through the two triple rules yields $a_{2j+1} = c(a + jd)^2$ and $a_{2j+2} = c(a + jd)(a + (j + 1)d)$ for $j \geq 0$.

2. Thus $c(a + 6d)^2 = 2016 = 2^5 \cdot 3^2 \cdot 7$. The integer $N = a + 6d \geq 7$ has square dividing2016, so it divides12. The only divisor of12 at least7 is12.

3. Hence $c = 14$ and $a + 6d = 12$. Positivity forces $d = 1, a = 6$, with $b = 7$ coprime to a. The displayed formulas give an increasing integral sequence for all indices, verifying existence.

4. The requested first term is $14 \cdot 6^2 = 504$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-II-Q09] | Part B (Q6-Q10)

Answer: 262

1. Write $a_n = 1 + (n-1)d$ and $b_n = r^{n-1}$, where $d \geq 1, r \geq 2$ are integers. Since $c_1 = 2$, $k \geq 3$. The conditions become $(k-2)d + r^{k-2} = 99$ and $kd + r^k = 999$.

2. Subtracting gives $2d + r^{k-2}(r^2 - 1) = 900$. The power term is divisible by3, hence $3 \mid d$. The first equation then forces $3 \mid r$.

3. Bounds $r^{k-2} < 99, r^k < 999$ leave $(r,k) = (3,3), (3,4), (3,5), (3,6), (6,3), (9,3)$. Substituting into the two equations, only $(9,3)$ works, with $d = 90$.

4. Thus $c_3 = 1 + 2 \cdot 90 + 9^2 = 262$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2020-II-Q06] | Part B (Q6-Q10)

Answer: 626

1. Rescale by $u_n = 5t_n$. Then $u_n = (u_{n-1} + 1)/u_{n-2}$. Starting from arbitrary positive a, b, the next three values are $(b+1)/a$, $(a+b+1)/(ab)$, $(a+1)/b$, followed by a, b.

2. The recurrence therefore repeats every five indices. Since $2020 \equiv 0 \pmod 5$, use $u_5 = (100+1)/105 = 101/105$.

3. It follows that $t_{2020} = 101/525$ in lowest terms, and the answer is 626.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2002-II-Q11] | Part C (Q11-Q15)

Answer: 518

1. A sum-one geometric series with ratio r begins with $1-r$, so its second term is $r(1-r)$. Distinct ratios giving the same value must be $r, 1-r$.

2. The third-term condition gives $r^2(1-r) = 1/8$. With $t = 2r$, this is $(t-1)(t^2-t-1) = 0$. The root $r = 1/2$ gives identical series; the negative root makes the other ratio exceed one. Thus $r = (1+\sqrt{5})/4$.

3. The common second term is $(\sqrt{5}-1)/8$, so the requested encoding is $500 + 10 + 8 = 518$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2005-II-Q11] | Part C (Q11-Q15)

Answer: 889

1. Multiply by a_k. The adjacent products $b_k = a_k a_{k-1}$ satisfy $b_{k+1} = b_k - 3$, with $b_1 = 2664$. Hence $b_k = 3(889 - k)$.

2. For $k \leq 888$ these products are positive, so all needed denominators remain nonzero. At $k = 889$ the product vanishes and $a_{888} \neq 0$, forcing the first zero $a_{889} = 0$. Thus $m = 889$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2006-I-Q15] | Part C (Q11-Q15)

Answer: 027

1. Squaring gives $x_k^2 - x_{k-1}^2 = 6x_{k-1} + 9$. Summing through $k = 2007$ yields $\sum_{k=1}^{2006} x_k = (x_{2007}^2 - 18063)/6$.

2. Induction shows $x_k/3$ is an integer of parity k. Thus x_{2007} is an odd multiple of three. The nearest such square to 18063 is $135^2 = 18225$; neighboring candidates $129^2, 141^2$ are farther. The resulting absolute sum is 27.

3. Attain it by $x_k = 3k$ through $k = 45$, then alternate -138 at even indices and 135 at odd indices. This obeys the recurrence and finishes at $x_{2007} = 135$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2006-II-Q11] | Part C (Q11-Q15)

Answer: 834

1. For $S_n = \sum_{j=1}^{n} a_j$, induction gives $2S_n = a_{n+2} + a_n$. It starts at $n = 1$, and adding $2a_{n+1}$ uses exactly the tribonacci recurrence.

2. Thus $S_{28} = (20603361 + 6090307)/2 = 13346834$, remainder 834.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2007-I-Q14] | Part C (Q11-Q15)

Answer: 224

1. The invariant $(a_{n+1}+a_{n-1})/a_n$ is constant: subtract consecutive product recurrences and factor to equate its consecutive values. Since $a_3 = 672$, that constant is $(672+3)/3 = 225$.

2. Thus $a_{n+1} = 225a_n - a_{n-1}$ and $a_{n+1}^2 + a_n^2 + 2007 = 225a_n a_{n+1}$. Positive terms increase strictly after a_2, so the late adjacent product exceeds 2007.

3. The target ratio is $225 - 2007/(a_{2006}a_{2007})$, strictly between 224 and 225. Its floor is 224.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2007-II-Q12] | Part C (Q11-Q15)

Answer: 091

1. Write $x_j = 3^{u+vj}$ with integer $v \geq 1$. The logarithm sum gives $2u + 7v = 77$.

2. The positive sum lies strictly between its largest term x_7 and $3x_7$, since its ratio backwards is at most one third. Its logarithm therefore lies strictly between integers $u + 7v$ and $u + 7v + 1$, forcing $u + 7v = 56$.

3. Solve $u = 21, v = 5$, giving $u + 14v = 91$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q11] | Part C (Q11-Q15)

Answer: 073

1. Expand the determinant along its first row, then the first column of the off-diagonal minor. This gives $D_n = 10D_{n-1} - 9D_{n-2}$, using $D_0 = 1, D_1 = 10$.

2. The recurrence has characteristic roots $9, 1$. Its initial values give $D_n = (9^{n+1} - 1)/8$, also verifiable directly by induction.

3. Therefore the sum is $\sum_{n=1}^{\infty} 9^{-(n+1)} = (1/81)/(1 - 1/9) = 1/72$. It converges geometrically, and $p + q = 73$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q05] | Part A (Q1-Q5)

Answer: 539

1. Increasing T by nine raises rounded Celsius by five and the final Fahrenheit by nine. The success pattern is therefore periodic modulo nine.

2. For $T = 32, \ldots, 40$, final values are $32, 34, 34, 36, 36, 37, 37, 39, 39$; five inputs succeed. Through 994 there are 107 full blocks, giving 535. The six remaining temperatures match the first six residues, with four successes.

3. Total 539. No half-integer rounding ties arise in these conversions.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2002-II-Q08] | Part B (Q6-Q10)

Answer: 049

1. For positive k, only positive n matter. Such an n lies in $(2002/(k+1), 2002/k]$. Its interval length is $2002/[k(k+1)]$, at least one for $k \leq 44$, so all these values occur.

2. The choices $n = 44, 43, 42, 41$ give $45, 46, 47, 48$. The next input 40 gives 50, and monotonicity leaves no input for 49. Thus the answer is 49.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q07] | Part B (Q6-Q10)

Answer: 477

1. The ceiling-floor difference is zero exactly for integral logarithms and one otherwise. An integer power of $\sqrt{2}$ that is also an integer is a power of two.

2. Thus subtract $1 + 2 + \cdots + 512 = 1023$ from the sum of all integers one through one thousand: $N = 500500 - 1023 = 499477$. The remainder is 477.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-II-Q07] | Part B (Q6-Q10)

Answer: 553

1. The interval is $k^3 \le n < (k+1)^3 = k^3 + 3k^2 + 3k + 1$. Its multiples of k have quotients $k^2, k^2+1, \ldots, k^2+3k+3$, so there are $3k+4$.

2. From $3k + 4 = 70$, $k = 22$. Maximum quotient $22^2 + 66 + 3 = 553$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2008-I-Q07] | Part B (Q6-Q10)

Answer: 708

1. Gaps between consecutive squares through $50^2 = 2500$ are at most ninety-nine, so every block $S_0, \ldots, S_{25}$ contains a square.

2. After that, consecutive square gaps exceed one hundred, so each later block has at most one. The squares 51^2 through 316^2 occupy 266 distinct later blocks; $317^2 > 99999$.

3. Of the 974 later blocks, $974 - 266 = 708$ are empty of squares.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2009-I-Q06] | Part B (Q6-Q10)

Answer: 412

1. On $k \le x < k+1$ for positive integer k, the image is $[k^k, (k+1)^k)$, containing $(k+1)^k - k^k$ integers. These intervals for different k are disjoint.

2. For $k = 1, 2, 3, 4$ the counts are $1, 5, 37, 369$, all below one thousand; $5^5 > 1000$ excludes larger k. Values $0 < x < 1$ only yield one, already counted. Negative x gives a negative value or a positive value below one, so no additional integer.

3. Total $1 + 5 + 37 + 369 = 412$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2010-I-Q08] | Part B (Q6-Q10)

Answer: 132

1. The integer floor pairs are $(\pm 5, 0), (0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3)$. Each contributes a half-open unit square; any closed enclosing disk also contains their closures.

2. The union of closed squares is centrally symmetric about $Q = (1/2, 1/2)$. An opposite pair $X, 2Q - X$ forces every enclosing radius to be at least $|X - Q|$.

3. The greatest corner distance from Q is attained at $(4, 5)$ and its symmetric counterparts: its square is $(7/2)^2 + (9/2)^2 = 130/4$. All other corners have no greater distance, so the disk centered at Q with radius $\sqrt{130}/2$ attains the bound.

4. Thus $m + n = 130 + 2 = 132$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2012-II-Q10] | Part B (Q6-Q10)

Answer: 496

1. With $a = \lfloor x \rfloor$, positivity of n forces $a \geq 1$. The range $a \leq x < a + 1$ gives $a^2 \leq n < a^2 + a$. Conversely every integer in this interval is attained by $x = n/a$.

2. These integer intervals are disjoint, and each contributes exactly a numbers. All intervals for $a = 1, \ldots, 31$ lie below 1000, ending at 991; the next begins at 1024.

3. The count is $1 + \cdots + 31 = 496$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2021-I-Q08] | Part B (Q6-Q10)

Answer: 057

1. Let $F(x) = |20|x| - x^2|$, an even function. On $x \geq 0$, it rises from 0 to 100, falls to 0 at 20, then rises without bound.

2. Thus the level $F(x) = k$ has 6 roots for $0 < k < 100$, 4 at $k = 100$, 2 for $k > 100$, 3 at $k = 0$, and none for $k < 0$.

3. The two distinct levels are $c - 21, c + 21$. Twelve roots require six from each, so $0 < c - 21 < c + 21 < 100$. The integer range is $22 \leq c \leq 78$, giving 57.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2023-I-Q10] | Part B (Q6-Q10)

Answer: 944

1. Without floors the sum is $S(a) = [\sum n^2 - a\sum n]/5$. It vanishes at $a = (2\cdot 2023+1)/3 = 1349$.

2. Changing a by one changes S by $\sum n/5 = 2023\cdot 1012/5 > 400000$, whereas floors decrease the sum by less than 2023. Thus only $a = 1349$ can work.

3. For this value, $n^2 - na \equiv n(n+1) \pmod 5$, whose residues for $n \equiv 0,1,2,3,4$ are $0,2,1,2,0$. Their total through $2023 = 5\cdot 404 + 3$ is $404\cdot 5 + (2+1+2) = 2025$.

4. Because $S(1349) = 0$, the floor sum is $U = -2025/5 = -405$. This satisfies the bound and gives $a + U = 944$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2004-I-Q12] | Part C (Q11-Q15)

Answer: 014

1. For base b, an even floor $2j$ corresponds to $b^{-2j-1} < x \leq b^{-2j}$. These disjoint intervals have total length $\sum_{j\geq 0}(1-1/b)b^{-2j} = b/(b+1)$.

2. The x- and y-conditions are independent Cartesian factors of lengths $2/3, 5/6$. Area $= (2/3)(5/6) = 5/9$, giving 14.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q11] | Part C (Q11-Q15)

Answer: 955

1. $b(p) = k$ exactly when $k^2 - k + 1 \le p \le k^2 + k$, a block of $2k$ integers. The blocks $k = 1, \ldots, 44$ end at 1980.

2. They contribute $2\sum_{k=1}^{44} k^2 = 58740$. Remaining 27 inputs each give 45, adding 1215. Total 59955, remainder 955.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2008-I-Q12] | Part C (Q11-Q15)

Answer: 375

1. For $k = \lceil v/15 \rceil$, successive fronts are $4(k+1)$ meters apart, giving flow $F = 250v/(k+1) \leq 3750k/(k+1) < 3750$ cars per hour.

2. A finite hour can include an initial car at its starting instant and then one per completed interval. Therefore at most 3750 cars can pass: 3751 would require 3750 gaps within an hour, contradicting $F < 3750$.

3. Choose k large enough that $F > 3749$, and start with a car at the point; then 3750 cars do pass. Thus $M = 3750$, whose quotient by ten is 375.

4. If instead whole cars per hour means the floor of the steady flow rate, its largest possible value is 3749 because $3749 < F < 3750$ is attainable. Under finite-window counting with a free initial phase it is 3750 as above. Both conventions give the requested integer quotient 375.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2009-II-Q11] | Part C (Q11-Q15)

Answer: 125

1. The interval is $m/n < k < mn$. Write $m = nq + r$, $0 \le r < n$, $q \ge 1$. The count is $mn - q - 1$, so $q(n^2 - 1) + nr = 51$; also $n \ge 2$.

2. For $n \ge 8$ the first term already exceeds fifty-one. Testing $2 \le n \le 7$ with $0 \le r < n$ leaves $(n, q, r) = (2, 17, 0), (3, 6, 1)$, giving $(m, n) = (34, 2), (19, 3)$.

3. The distinct products are $68, 57$ and their sum is 125.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2010-I-Q14] | Part C (Q11-Q15)

Answer: 109

1. Every term is nondecreasing in n. It therefore suffices to exhibit adjacent integers where the bound changes.

2. At $n = 109$, the products have floor logarithms 2 for $k = 1, \ldots, 9$, 3 for $10, \ldots, 91$, and 4 for $92, \ldots, 100$. Their sum is $18 + 246 + 36 = 300$.

3. At $n = 110$, the threshold for 10^4 shifts to $k = 91$, so the sum is $18 + 243 + 40 = 301$. Monotonicity proves the maximum is 109.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2011-II-Q15] | Part C (Q11-Q15)

Answer: 850

1. On a unit interval $[n, n+1)$, the right side must be an integer. For $n = 5, \ldots, 14$, the values $P(n)$ are $1, 9, 19, 31, 45, 61, 79, 99, 121, 145$; only $n = 5, 6, 13$ work, with roots $1, 3, 11$.

2. The polynomial increases throughout the sample interval. For each working n with $P(n) = j^2$, success persists from n until the root of $P(x) = (j+1)^2$, namely $[3 + \sqrt{45 + 4(j+1)^2}]/2$. These cutoffs all lie before $n + 1$.

3. The three interval lengths are $(\sqrt{61} - 7)/2, (\sqrt{109} - 9)/2, (\sqrt{621} - 23)/2$. Divide their sum by ten to obtain $(\sqrt{61} + \sqrt{109} + \sqrt{621} - 39)/20$.

4. The endpoint 15 has probability zero. The required sum is $61 + 109 + 621 + 39 + 20 = 850$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2015-I-Q14] | Part C (Q11-Q15)

Answer: 483

1. On a unit strip $j \le x < j+1$, the floor is constant $k = \lfloor \sqrt{j} \rfloor$. Its trapezoid area is $k(2j+1)/2$. Thus the fractional part of cumulative area changes by $1/2$ exactly when k is odd.

2. Block k consists of $k^2 < n \le (k+1)^2$ and has $2k+1$ values. Starting at $A(1) = 0$, the status at block starts cycles modulo 4: integer for $k \equiv 0, 1$, half-integer for $k \equiv 2, 3$.

3. For four successive blocks $k = 4r-3, 4r-2, 4r-1, 4r$, the counts of integer areas are $4r-3, 0, 4r, 8r+1$. For $r = 1, \ldots, 7$, these sum to $\sum(16r-2) = 434$, covering through $n = 841$.

4. Block $k = 29$ adds 29 values through 900; block 30 adds none through 961; the partial block 31 begins with an integer at $n = 962$ and adds 20 through 1000. Total $434 + 29 + 20 = 483$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2020-II-Q14] | Part C (Q11-Q15)

Answer: 010

1. On $[j, j+1)$, $f(j+t) = (j+t)t$ increases bijectively from 0 to the range $[0, j+1)$. In particular integer endpoints cannot solve the target equation.

2. The inverse of 17 has one point in each interval $(n, n+1)$ for $n \geq 17$. Taking another inverse produces one point in every $(k, k+1)$ with $k \geq n$. A third inverse in $(j, j+1)$ exists exactly for $17 \leq n \leq k \leq j$.

3. For fixed j, the count is $\binom{j-15}{2}$. Sum for $17 \leq j \leq 2019$ to obtain $N = \binom{2005}{3} = 1341349010$. The remainder is 10.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2024-I-Q12] | Part C (Q11-Q15)

Answer: 385

1. Every solution lies in $[0,1]^2$. Put $h(u)=4||u-1/2|-1/4|$ on $[0,1]$. Its values at $0,1/4,1/2,3/4,1$ are $1,0,1,0,1$. Thus $y=F(x)=h(|\sin 2\pi x|)$ has sixteen monotone full-height branches, while $x=G(y)=h(|\cos 3\pi y|)$ has twenty-four full-width branches. The only boundary solution is $(1,1)$.

2. To justify branch-intersection uniqueness, we prove $|F\prime(x)G\prime(y)|=96\pi^2|\cos 2\pi x\sin 3\pi y|>1$ at every interior solution. Such solutions avoid all branch endpoints. If $|\cos 2\pi x|\le 1/20$, then $|\sin 2\pi x|\ge 399/400$, hence $y\ge .99$. It follows that $x\ge 1-18\pi^2(.01)^2>.982$, forcing $|\cos 2\pi x|>.99$, a contradiction.

3. If the derivative product were at most one, at least one factor $|\cos 2\pi x|,|\sin 3\pi y|$ would be at most $1/20$, since $96\pi^2/400>1$. The previous argument excludes the first, so $|\sin 3\pi y|\le 1/20$ and $x\ge .99$. Set $a=1-x$, $b=1-y$. Then $0<a\le .01$ and $b=4\sin 2\pi a\le .08\pi<.252$. The sine bound now forces $0<b<.01$ (the only nearby zero of $\sin 3\pi y$ is at $y=1$). The exact equations are $b=4\sin 2\pi a$ and $a=4(1-\cos 3\pi b)$.

4. Define $H(b)=4(1-\cos 3\pi b)-\arcsin(b/4)/(2\pi)$. On $[0,.01]$, $H\prime\prime(b)=36\pi^2\cos 3\pi b-b/[128\pi(1-b^2/16)^{3/2}]>0$. We have $H(0)=H(b)=0$ for the noncorner solution, so strict convexity gives $H\prime(b)>0$. This says $12\pi\sin 3\pi b>1/(8\pi\cos 2\pi a)$, precisely that the derivative product exceeds one, a contradiction. The claimed inequality follows.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2024-I-Q12] | Part C (Q11-Q15) (continued / 续)

5. For oppositely oriented branches uniqueness follows directly by monotonicity. For equally oriented branches, write the first as $x = F^{-1}(y)$: the difference $G(y) - F^{-1}(y)$ has the same nonzero crossing sign at every zero because the derivative product is greater than one. It cannot have two such crossings. Boundary endpoint signs guarantee one interior crossing in each of the $16 \cdot 24$ branch-pair rectangles, except that the final pair shares $(1, 1)$. There is also an interior crossing for that pair: near $b = 0$, $H(b) < 0$ because $H\prime(0) < 0$, while the opposite rectangle endpoint has positive sign. Therefore there are 384 interior intersections plus the corner, giving 385.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2025-I-Q11] | Part C (Q11-Q15)

Answer: 259

1. All intersections have $-1 \le y \le 1$ and $0 \le x \le 34$. Rising pieces give $34y^2 - y - 4k = 0$; falling pieces give $34y^2 + y - (4k+2) = 0$. A root is valid exactly when its y value is in the corresponding piece range.

2. The rising quadratics have both roots valid for $k = 0, \ldots, 8$, each with sum $1/34$. The falling ones have both roots valid for $k = 0, \ldots, 7$, each with sum $-1/34$; for $k = 8$ only $(-1 + 5\sqrt{185})/68$ is valid.

3. The total is $(1 + 5\sqrt{185})/68$, so $a + b + c + d = 1 + 5 + 185 + 68 = 259$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2008-II-Q03] | Part A (Q1-Q5)

Answer: 729

1. Each cut reduces one dimension by one, so the remaining three dimensions sum to $37 - 10 = 27$. By AM–GM their product is at most 9^3.

2. The dimensions $9, 9, 9$ are attained by cuts $1, 4, 5$ in the respective directions. Maximum volume is 729.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2011-II-Q09] | Part B (Q6-Q10)

Answer: 559

1. Call the given two-term sum r and the target sum s. Expansion gives $r + s = (x_1 + x_4)(x_2 + x_5)(x_3 + x_6)$.

2. These three nonnegative factors sum to one. AM-GM gives $s \leq 1/27 - r \leq 19/540$.

3. The bound is attained by $(x_1, \ldots, x_6) = (1/3, 19/60, 1/3, 0, 1/60, 0)$. All opposite-pair sums are $1/3$ and $r = (1/3)^2(1/60) = 1/540$.

4. Thus the maximum is $19/540$, giving 559.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-I-Q09] | Part B (Q6-Q10)

Corrected result: $\dfrac{248(1600-496\sqrt{6})}{2561-992\sqrt{6}} \approx 728.365125470659$

Historical answer key: 744

The historical source key is 744. Its equality configuration does not satisfy B on segment QR. With all original segment conditions retained, the maximum is the exact value shown above, attained when B=R. This is an independently documented source discrepancy; no official acknowledgment of an erratum is claimed.

1. Put $A=(0,0), Q=(w,0), R=(w,h), S=(0,h), B=(w,b), C=(c,h)$ with $0 \le b \le h, 0 \le c \le w$. Let $\beta=\angle BAQ$, $\gamma=\angle CAS$, and $\alpha=\angle BAC$. The triangle lies inside a rectangle corner, so α is acute and $\cos\alpha = 2\sqrt{6}/5$. Also $\beta+\gamma = 90^\circ - \alpha$.

2. The side lengths give $w = 40\cos\beta$, $b = 40\sin\beta$, $c = 31\sin\gamma$, $h = 31\cos\gamma$. Consequently $K = wh = 620[\cos(\beta-\gamma)+1/5] \le 744$. However equality needs $\beta=\gamma$, where $\tan\beta = \sqrt{2/3}$. Then $b \le h$ would require $40\sqrt{2/3} \le 31$, which is false. Thus 744 is an unattainable upper bound for the stated segment problem.

3. Impose $b \le h$: using $\cos\gamma = \sin(\alpha+\beta)$ gives $\tan\beta \le 31/(200-62\sqrt{6})$. The other condition is automatic since $c = 31\cos(\alpha+\beta) \le 31\cos\alpha\cos\beta < 40\cos\beta = w$. The feasible interval is therefore $0 \le \beta \le \arctan[31/(200-62\sqrt{6})] \approx 32.78425^\circ$.

4. The unconstrained area increases until $\beta = (90^\circ-\alpha)/2 \approx 39.23152^\circ$, beyond that feasible interval. Hence the constrained maximum occurs at its right endpoint, where $b=h$ and $B=R$.

5. At this endpoint, $d = BC$ is horizontal and cosine law gives $d^2 = 2561 - 992\sqrt{6}$. The determinant and dot product of AB, AC give $hd = 40\cdot 31\sin\alpha = 248$ and $wd = 40^2 - 40\cdot 31\cos\alpha = 1600 - 496\sqrt{6}$.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-I-Q09] | Part B (Q6-Q10) (continued / 续)

6. Thus $K_{\max} = 248(1600 - 496\sqrt{6})/(2561 - 992\sqrt{6}) \approx 728.365125470659$. It is attained by $h = 248/d, w = (1600 - 496\sqrt{6})/d, C = (w - d, h), B = R$; these satisfy all original lengths and containment conditions. The value 744 remains recorded only as the historical source key.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2022-II-Q06] | Part B (Q6-Q10)

Answer: 841

1. Total positive mass and total negative magnitude are each $1/2$. Sixteen terms no larger than x_{16} imply $x_{16} \geq -1/32$; twenty-five terms no smaller than x_{76} imply $x_{76} \leq 1/50$.

2. Therefore the difference is at most $1/32 + 1/50 = 41/800$. Equality occurs with sixteen copies of $-1/32$, fifty-nine zeros, and twenty-five copies of $1/50$.

3. This ordered tuple satisfies both constraints, so the bound is the maximum and $m + n = 841$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2005-I-Q11] | Part C (Q11-Q15)

Answer: 544

1. For a unit-radius semicircle whose diameter forms angle $\theta \in [0, \pi/2]$ with a square side, its two perpendicular projection widths are $1 + \cos\theta$ and $1 + \sin\theta$. Thus the least square side for that orientation is their maximum.

2. This maximum is minimized at $\theta = \pi/4$, with value $1 + 1/\sqrt{2}$. Scaling to side eight gives $d = 16/(1 + 1/\sqrt{2}) = 32 - 16\sqrt{2} = 32 - \sqrt{512}$.

3. Therefore $m + n = 544$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2008-I-Q14] | Part C (Q11-Q15)

Answer: 432

1. Use circle center $(0,0)$, $A=(-9,0), B=(9,0)$. A tangent has equation $ux+vy=9$ with $u^2+v^2=1$ and $-1<u<0$ because its axial intercept is beyond A. The projection is $P=A+9(1+u)(u,v)$.

2. Expand $BP^2/81=(u^2+u-2)^2+(1+u)^2(1-u^2)=5-2u-3u^2$. Its maximum on $(-1,0)$ occurs at $u=-1/3$, value $16/3$.

3. Thus $m^2=81\cdot 16/3=432$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2009-II-Q15] | Part C (Q11-Q15)

Answer: 014

1. Let intersections be $Q = AC \cap MN$, $P = BC \cap MN$ and $x = CM/CN > 0$. Pythagoras gives $BN = 4/5$, while $AM = AN$. Cyclic sine-area ratios yield $MP/PN = BM \cdot CM/(BN \cdot CN) = 3x/4$ and $MQ/QN = x$.

2. As $MN = 1$, $d = x/(x+1) - 3x/(3x+4) = 1/(3x + 4/x + 7)$. The full opposite semicircle makes x range over all positive values.

3. AM–GM gives $3x + 4/x \geq 4\sqrt{3}$, with equality at $x = 2/\sqrt{3}$, an attainable ratio. Hence maximum $d = 1/(7 + 4\sqrt{3}) = 7 - 4\sqrt{3}$, giving 14.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-II-Q15] | Part C (Q11-Q15)

Answer: 863

1. The square of $\sum x_i = 1$ converts the second condition into $\sum_i x_i^2/(1-a_i) = 1/215$. All denominators are positive.

2. The geometric sum gives $\sum a_i = 1$, so $\sum(1-a_i) = 215$. Weighted Cauchy therefore gives $1 = (\sum x_i)^2 \le [\sum x_i^2/(1-a_i)]\sum(1-a_i) = 1$.

3. Equality forces $x_i/(1-a_i)$ constant, giving the unique positive tuple $x_i = (1-a_i)/215$. It satisfies both original conditions. Thus $x_2 = 3/860$ is its only, hence maximum, value, and $m+n = 863$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2024-I-Q15] | Part C (Q11-Q15)

Answer: 721

1. For dimensions $a, b, c > 0$, $ab + bc + ca = 27, abc = 23$. The required squared radius is one fourth the maximum space-diagonal square, namely $[(a + b + c)^2 - 54]/4$.

2. A maximum exists: each dimension is at least $23/27$, while $27 \geq a(b + c) \geq 2\sqrt{23a}$ gives $a \leq 729/92$, and similarly for the others. The constrained positive dimensions therefore form a compact set.

3. At an extremum two dimensions coincide: otherwise the cubic with roots a, b, c could have its sum perturbed while maintaining three positive roots. Setting $b = c = t$ gives $t^3 - 27t + 46 = (t - 2)(t^2 + 2t - 23) = 0$. The positive candidates are $t = 2$ and $t = 2\sqrt{6} - 1$.

4. For $t = 2$, $(a, b, c) = (23/4, 2, 2)$ and the sum is $39/4$. For the second candidate the sum is $(-21 + 96\sqrt{6})/23 < 39/4$ (use $\sqrt{6} < 5/2$, giving $s < 219/23 < 39/4$). Thus $r^2 = [(39/4)^2 - 54]/4 = 657/64$ and $p + q = 721$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2025-I-Q14] | Part C (Q11-Q15)

Answer: 060

1. Cosine rule yields $AC = 7\sqrt{3}$, $AD = 13\sqrt{3}$, with $AC \perp BC$, $AD \perp DE$ and $\angle BAC = \angle DAE = 30^\circ$. For $\theta = \angle CAD$, $\cos\theta = 1/7, \sin\theta = 4\sqrt{3}/7$. Hence $\cos(\theta + 60^\circ) = -11/14$ and $BE = 38$.

2. For triangle ACD, the Fermat-point rotation construction gives $\min_X(AX + CX + DX) = 19\sqrt{3}$: erect an outward equilateral triangle on AC, then the distance from its new vertex to D has square $147 + 507 + 429 = 1083$. All triangle angles are below 120°.

3. Let F be that Fermat point. Its 120° angles put it on the circles ABC and AED, so $\angle AFB = \angle AFE = 90^\circ$. Convexity places B, F, E in that order. Therefore both lower bounds $BX + EX \geq BE$ and $AX + CX + DX \geq 19\sqrt{3}$ are attained at F. The minimum is $38 + 19\sqrt{3}$, giving 60.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2025-II-Q15] | Part C (Q11-Q15)

Answer: 240

1. If the minimum is c, then $x(f(x)-c)$ is a monic quartic nonnegative for $x>0$ with two distinct positive double roots, so it equals $(x^2-Sx+P)^2$, with $S,P>0$.

2. Coefficient comparison gives $2S=188+k$, $S^2+2P=10116+188k$, and $P^2=127008k$. Put $k=2t^2$, yielding $S=94+t^2$, $P=504t$. The middle equation becomes $(t-2)(t-4)(t-10)(t+16)=0$.

3. The positive values $t=2,4,10$ give $k=8,32,200$. Each has $S^2>4P$, ensuring two distinct positive roots and making the square factorization sufficient as well. Their sum is 240.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2002-II-Q10] | Part B (Q6-Q10)

Answer: 900

1. Equal sines mean the angles differ by $2j\pi$ or sum to $(2j+1)\pi$. Therefore the positive solution families are $360j\pi/(180-\pi)$, $j \geq 1$, and $180(2j+1)\pi/(180+\pi)$, $j \geq 0$.

2. The second family begins near 3.09 and the first near 6.39; the next second-family value is near 9.26. Thus the two least are the first members of these families.

3. The parameters are $360, 180, 180, 180$, summing to 900.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2008-I-Q08] | Part B (Q6-Q10)

Answer: 047

1. Tangent addition combines the first two angles to tangent $7/11$, then the first three to tangent $23/24$. The sums are acute, so no branch adjustment is needed.

2. The last angle has tangent $(1-23/24)/(1+23/24)=1/47$. Therefore $n=47$.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2008-II-Q08] | Part B (Q6-Q10)

Answer: 251

1. Product-to-sum writes each term as $\sin(k(k+1)a) - \sin((k-1)ka)$. The sum telescopes to $\sin(n(n+1)\pi/2008)$.

2. Integer sine values require an argument multiple of $\pi/2$, so $1004 = 4 \cdot 251$ divides $n(n+1)$. Coprimality of consecutive integers forces $n \geq 250$.

3. $250 \cdot 251$ lacks a factor four, while $251 \cdot 252$ qualifies. The minimum is 251.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2012-II-Q09] | Part B (Q6-Q10)

Answer: 107

1. The sine double-angle ratio is $3/2$. Let $t = \cos^2 y$; the two given squared relations sum to $9(1-t) + t/4 = 1$, so $t = 32/35$.

2. Thus $\cos 2y = 29/35$ and $\cos 2x = -19/35$. In particular neither required double-angle denominator vanishes.

3. The sum is $3/2 - 19/29 = 49/58$, giving 107.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2021-I-Q07] | Part B (Q6-Q10)

Answer: 063

1. Both sines must equal 1. Thus $mx = (4a+1)\pi/2$ and $nx = (4b+1)\pi/2$. Eliminating x gives $4(na - mb) = m - n$.

2. Let $g = \gcd(m, n)$, $m = gu, n = gv$. Integer a, b exist exactly when $u \equiv v \pmod 4$; coprimality then forces both odd.

3. For $g = 1, 2, 3, 4, 5, 6$, coprime pairs $u < v \le \lfloor 30/g \rfloor$ in the same odd residue class mod 4 number $44, 11, 4, 2, 1, 1$. For $g \ge 7$ there are none. The total is 63.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2025-II-Q09] | Part B (Q6-Q10)

Answer: 149

1. Zeros require $\sin 5x = k/7$, where $-7 \le k \le 7$ is integral. Over five full periods, $k = 0$ has nine zeros after excluding endpoints; each of twelve nonzero interior levels has ten; the two extrema have five each.

2. Thus $n = 9 + 120 + 10 = 139$. At a zero, $f\prime(x) = 35\pi \cos(k\pi) \cos 5x$ vanishes only at the ten extrema. These are genuine tangencies because the inner sine turns there. Hence $t = 10, n + t = 149$.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2000-II-Q15] | Part C (Q11-Q15)

Answer: 001

1. The identity $\cot k^\circ - \cot(k+1)^\circ = \sin 1^\circ / [\sin k^\circ \sin(k+1)^\circ]$ converts the sum, after multiplication by $\sin 1^\circ$, into an alternating cotangent sum from 45° through 134°.

2. Angles summing to 180° have equal parity and opposite cotangents, so their contributions cancel. Only $\cot 45^\circ = 1$ remains; the central term $\cot 90^\circ$ is zero.

3. Therefore the original sum is $1/\sin 1^\circ$. The positive integer $n = 1$ works and is already the smallest possible.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2006-I-Q12] | Part C (Q11-Q15)

Answer: 906

1. Use $2\cos 4x\cos x=\cos 3x+\cos 5x$. Setting these latter cosines to u,v turns the equation into $u^3+v^3=(u+v)^3$, so $uv(u+v)=0$.

2. Thus zeros come from $\cos 3x=0$, $\cos 5x=0$, $\cos 4x=0$, or $\cos x=0$. In the given interval their distinct solutions are $150;126,162,198;112.5,157.5;$ and none, respectively.

3. The sum is 906.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2013-II-Q15] | Part C (Q11-Q15)

Answer: 222

1. For triangle angles, the sine and cosine laws give $\sin^2 A + \sin^2 B - 2\sin A \sin B \cos C = \sin^2 C$. Thus each displayed expression equals two minus the squared sine of the omitted angle.

2. Therefore $\sin^2 C = 1/8$ and $\sin^2 A = 4/9$. Acuteness fixes $\sin A = 2/3, \cos A = \sqrt{5}/3, \sin C = \sqrt{2}/4, \cos C = \sqrt{14}/4$.

3. Since $B = \pi - A - C$, $\sin B = (2\sqrt{14} + \sqrt{10})/12$, with square $(33 + 4\sqrt{35})/72$. The resulting $A + C$ is acute, consistent with obtuse B.

4. The target is $2 - \sin^2 B = (111 - 4\sqrt{35})/72$. As 115 and 72 are coprime, the requested sum is 222.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2014-II-Q12] | Part C (Q11-Q15)

Answer: 399

1. Using $A+B+C=\pi$ and sum-to-product identities gives $1-(\cos 3A+\cos 3B+\cos 3C)=4\sin(3A/2)\sin(3B/2)\sin(3C/2)$.

2. The product is zero. For a triangle angle $0<X<\pi$, the only possible zero of $\sin(3X/2)$ is $X=120^\circ$, so one angle must be one hundred twenty degrees. Conversely such an angle satisfies the identity.

3. The largest possible third side puts this angle between the known sides, giving squared length $10^2+13^2+130=399$. If the obtuse angle is opposite a known side, the third side is shorter than that side and cannot exceed this value.

4. Thus $m=399$.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2015-I-Q13] | Part C (Q11-Q15)

Answer: 091

1. Let P be the product of sines at odd angles $1^\circ, 3^\circ, \ldots, 89^\circ$, and Q at even angles $2^\circ, 4^\circ, \ldots, 88^\circ$. The product of all sines from 1° to 89° is PQ; replacing every angle by its complement gives the same product of cosines.

2. Thus $(PQ)^2 = \prod_{j=1}^{89} \sin j^\circ \cos j^\circ = 2^{-89} \prod_{j=1}^{89} \sin(2j)^\circ = 2^{-89} Q^2$, because reflection about 90° pairs all even-angle factors and the 90° factor is 1.

3. Cancel the nonzero Q^2 to get $P^{-2} = 2^{89}$. As 89 is prime, the only integers $m, n > 1$ giving this power are $m = 2, n = 89$. Their sum is 91.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2015-II-Q13] | Part C (Q11-Q15)

Answer: 628

1. The identity $2\sin(1/2)\sin k = \cos(k-1/2) - \cos(k+1/2)$ telescopes to $a_n = [\cos(1/2) - \cos(n+1/2)]/[2\sin(1/2)]$.

2. The denominator is positive. Negativity is equivalent to $n+1/2$ lying within $1/2$ of a multiple $2\pi m$, that is $2\pi m - 1 < n < 2\pi m$.

3. For each positive integer m, this interval contains exactly one integer, $\lfloor 2\pi m \rfloor$, because $2\pi m$ is irrational. Their order increases with m, so the 100th is $\lfloor 200\pi \rfloor = 628$.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2022-I-Q15] | Part C (Q11-Q15)

Answer: 033

1. Real radicands imply $0 < x, y, z \leq 2$. Set $x = 2\sin^2 a, y = 2\sin^2 b, z = 2\sin^2 c$, where $a, b, c \in (0^\circ, 90^\circ]$. The equations become $\sin(a+b) = 1/2$, $\sin(b+c) = \sqrt{2}/2$, and $\sin(c+a) = \sqrt{3}/2$.

2. The possible pair-sums are $a+b \in \{30, 150\}$, $b+c \in \{45, 135\}$, $c+a \in \{60, 120\}$ degrees. Of their eight combinations, only $(30, 45, 60)$ and $(150, 135, 120)$ yield all three angles in the stated range. They give $(a, b, c) = (22.5, 7.5, 37.5)$ or $(67.5, 82.5, 52.5)$.

3. In the first case, the product is $\cos 45^\circ \cos 15^\circ \cos 75^\circ = \sqrt{2}/8$; in the second each cosine changes sign, so the product is its negative. The square is always $1/32$, giving 33.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2023-II-Q13] | Part C (Q11-Q15)

Answer: 167

1. Set $s = \sec A, t = \tan A$. Then $st = 2, s^2 - t^2 = 1$, so for $u = s^2, v = t^2$, one has $u + v = \sqrt{17}, uv = 4$.

2. The sums $w_j = u^j + v^j$ satisfy $w_{j+1} = \sqrt{17}w_j - 4w_{j-1}$, with $w_0 = 2, w_1 = \sqrt{17}$. Induction shows that even-index terms are positive integers and odd-index terms are nonzero integer multiples of $\sqrt{17}$. For odd n, $(s^n + t^n)^2 = w_n + 2^{n+1}$ is irrational. Thus integrality occurs exactly when $4 \mid n$.

3. For $n = 4j$, let $x_j = s^{4j} + t^{4j}$. The two bases have sum 9 and product 16, so $x_{j+1} = 9x_j - 16x_{j-1}$ with $x_0 = 2, x_1 = 9$.

4. Modulo 10, the terms for $j \geq 1$ cycle $9, 9, 7$: the pair $(x_1, x_2) = (9, 9)$ returns after three steps. Valid $j \leq 250$ are exactly those not divisible by 3, numbering $250 - 83 = 167$.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2001-II-Q08] | Part B (Q6-Q10)

Answer: 429

1. Since $2001 \in [3^6, 3^7]$, scaling gives $f(2001) = 729 - |2001 - 1458| = 186$.

2. On $[3^k, 3^{k+1}]$, the graph is a tent of height 3^k, specifically $f(x) = 3^k - |x - 2 \cdot 3^k|$. The first tent tall enough has $k = 5$ and height 243.

3. Solve $243 - |x - 486| = 186$: the first hit is $x = 429$, earlier than the other solution 543.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2018-II-Q10] | Part B (Q6-Q10)

Answer: 756

1. Every second iterate must be a fixed point. Partition the elements into fixed points, nonfixed elements mapping directly to fixed points, and the remaining elements, which must map to that middle level.

2. With $i \geq 1$ fixed points and j middle elements, choose levels in $\binom{5}{i}\binom{5-i}{j}$ ways; their images have $i^j j^{5-i-j}$ possibilities. This categorization is intrinsic, preventing double counting.

3. For $i = 1, 2, 3, 4, 5$, the totals are respectively $205, 380, 150, 20, 1$ (the last is the identity). They sum to 756.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2019-II-Q10] | Part B (Q6-Q10)

Answer: 547

1. Reduce 8θ modulo 180°. It is again acute and has the same entire sign itinerary; the stated uniqueness forces it equal θ. Thus 7θ is a multiple of 180°, leaving $\theta = 180k/7$ for $k = 1, 2, 3$.

2. For $k = 1$, the double angle still has positive tangent; for $k = 2$, the quadruple angle does. Both fail. For $k = 3$, the three residues are $540/7, 1080/7, 900/7$ degrees, respectively positive,negative,negative tangent, and repeat after multiplication by eight.

3. Therefore $\theta = 540/7$ and the answer is 547.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2020-II-Q08] | Part B (Q6-Q10)

Answer: 101

1. If $z \geq 0$ is a zero of f_{n-1}, its preimages are $n \pm z$; negative zeros give no preimages. With $T_j = j(j+1)/2$, induction shows that the zero set is the step-2 progression from $n - T_{n-1}$ to T_n.

2. The endpoints sum to $2n$, and the number of terms is $T_{n-1} + 1$. Hence the sum is $S_n = n(n(n-1)/2 + 1)$, an increasing expression.

3. $S_{100} = 495100$, while $S_{101} = 510151$. Therefore the first index is 101.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2000-I-Q12] | Part C (Q11-Q15)

Answer: 177

1. Composing reflections gives periods 1760 and 1056, hence also their gcd 352. The remaining reflection is $x \mapsto 46 - x$ modulo 352.

2. Among integer residues, this involution fixes 23 and 199 and pairs the other 350 residues. Thus there are $2 + 350/2 = 177$ possible classes; the given list covers every residue.

3. The function $f(x) = \cos(2\pi(x - 23)/352)$ satisfies all reflections and separates exactly these integer classes, proving the bound attainable.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2007-II-Q14] | Part C (Q11-Q15)

Answer: 676

1. The polynomial is nonconstant, and comparing leading coefficients makes it monic. If λ is a complex root, so is $2\lambda^3+\lambda$. A root of modulus greater than one would iterate to arbitrarily larger moduli, impossible for a finite root set.

2. The root product has modulus one because $f(0)=1$ and the polynomial is monic. Since every modulus is at most one, all must equal one. The image root must also have modulus one, so $|2\lambda^2+1|=1$ with $|\lambda|=1$, forcing $\lambda^2=-1$.

3. Real coefficients now give $f(x)=(x^2+1)^j$. Its functional identity is indeed satisfied. The value condition $5^j+10^j=125$ uniquely gives $j=2$. Therefore $f(5)=26^2=676$.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2009-I-Q13] | Part C (Q11-Q15)

Answer: 090

1. Subtract successive cleared recurrences to get $a_{n+2} - a_n = (a_{n+2} + 1)(a_{n+3} - a_{n+1})$. If one difference were nonzero, subsequent nonzero integer absolute differences would decrease by a factor of at least two forever, impossible.

2. Therefore the sequence has period two and $a_1 a_2 = 2009 = 7^2 \cdot 41$. Conversely any positive factor pair gives such a sequence.

3. The closest factors are $41, 49$, minimizing their sum at 90.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2009-II-Q14] | Part C (Q11-Q15)

Answer: 983

1. Normalize $u_n = a_n/2^n$. Then $u_{n+1} = (4/5)u_n + (3/5)\sqrt{1-u_n^2}$. Put $\sin\theta = 3/5, \cos\theta = 4/5$, with $30^\circ < \theta < 45^\circ$.

2. Starting at zero, normalized values are $\sin\theta, \sin 2\theta, \sin 3\theta$. Since $\cos 3\theta < 0$, the positive square root reverses the angle change, so subsequent values alternate $\sin 2\theta, \sin 3\theta$.

3. Thus $a_{10} = 2^{10}\sin 2\theta = 1024 \cdot 24/25 = 983.04$, whose floor is 983.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2012-II-Q11] | Part C (Q11-Q15)

Answer: 008

1. Write $f(x) = (6x - 7)/(9x + 3)$. Direct composition gives $f^{\circ 2}(x) = (-3x - 7)/(9x - 6)$ and $f^{\circ 3}(x) = x$ wherever the iterates exist.

2. As $1001 \equiv 2 \pmod 3$, the equation becomes $(-3x - 7)/(9x - 6) = x - 3$. Clearing the nonzero denominator yields $(3x - 5)^2 = 0$.

3. The candidate $5/3$ cycles through $5/3, 1/6, -4/3$, none of which is the pole $-1/3$, so all 1001 iterates exist. The answer is $5 + 3 = 8$.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2013-II-Q11] | Part C (Q11-Q15)

Answer: 399

1. Let the constant be a, and $S = f^{-1}(a)$. The nonempty set S contains a, since for any $t \in S$, $f(f(t)) = f(a) = a$. All members of S map to a, while every point outside S must map into $S \setminus \{a\}$.

2. For $|S| = k$, choose a in seven ways, the other $k-1$ members in $\binom{6}{k-1}$ ways, and outside images in $(k-1)^{7-k}$ ways. This construction is necessary and sufficient and determines S uniquely.

3. Thus $N = 7\sum_{k=1}^{7}\binom{6}{k-1}(k-1)^{7-k} = 7(0+6+240+540+240+30+1) = 7399$. The remainder is 399.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2021-II-Q15] | Part C (Q11-Q15)

Answer: 258

1. Put $k = \lceil \sqrt{n} \rceil$. Iterating to the next square gives $f(n) = k^2 + k - n$. The g iteration preserves parity, so if $n \equiv k \pmod 2$, it reaches the same square and $f(n)/g(n) = 1$. We therefore need opposite parity.

2. Then the first square of the parity of n is $(k+1)^2$, so $g(n) = f(n) + 2k + 2$. The ratio $4/7$ gives $f(n) = 8(k+1)/3$ and $n = k^2 + k - 8(k+1)/3$. Integrality requires $k \equiv 2 \pmod 3$. This formula always makes n even, so the opposite-parity condition requires odd k, hence $k \equiv 5 \pmod 6$.

3. The defining interval $(k-1)^2 < n \leq k^2$ has lower inequality equivalent to $k > 11$; its upper inequality is automatic. The least eligible integer k is therefore 17, yielding $n = 258$. This minimizes n because the square-root intervals are ordered by k.

4. Finally $17^2 - 258 + 17 = 48$ and $18^2 - 258 + 18 = 84$, giving $48/84 = 4/7$. Thus the least positive input is 258.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2025-II-Q13] | Part C (Q11-Q15)

Answer: 248

1. The substitution $y = (2x-1)/(x+1)$ conjugates the recurrence to $y_{j+1} = y_j(y_j - 1)$, with $y_1 = 13/12$. Thus $y_j = c/d$, where $d = 12^{2^{j-1}}$ and c stays coprime to 6.

2. Invert to get $x_j = (d+c)/(2d-c)$. Its numerator and denominator are positive and coprime: a common divisor divides both $3d, 3c$, hence divides 3, but neither term is divisible by 3. Therefore $m+n = 3d$.

3. For $j = 2025$, exponent $2^{2024} \equiv 16 \pmod{100}$. Modulo 125, $d \equiv 12^{16} = 41$; modulo 8, $d \equiv 0$. Chinese remainders give $d \equiv 416 \pmod{1000}$, so $m+n \equiv 248$.

Source / 来源

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

[AIME-2000-I-Q03] | Part A (Q1-Q5)

Answer: 667

1. The binomial coefficients give $\binom{2000}{2}a^2b^{1998} = \binom{2000}{3}a^3b^{1997}$. Cancelling positive factors gives $b/a = 1998/3 = 666$.

2. Coprimality forces $(a, b) = (1, 666)$, so the sum is 667.

Source / 来源

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

[AIME-2002-I-Q07] | Part B (Q6-Q10)

Answer: 428

1. Let $T = 10^{2002}$. Taylor's formula for $(T+1)^{10/7}$ gives $T^{10/7} + (10/7)T^{3/7} + R$, with $0 < R < (15/49)T^{-4/7}$. Thus the first term is the integer 10^{2860} and the second is $10^{859}/7$, while the positive remainder is below 10^{-1000}.

2. Since $859 \equiv 1 \pmod 6$ and 10 has order 6 modulo 7, the second term has fractional part $3/7 = 0.428571\ldots$.

3. The distance from $3/7$ to the next three-decimal boundary 0.429 exceeds 0.0004, vastly larger than R. Therefore the requested digits are 428.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2008-II-Q05] | Part A (Q1-Q5)

Answer: 504

1. Extend the legs to meet at E. Since the two base angles sum to ninety degrees, triangle EAD is right at E. Parallelism makes EBC a smaller homothetic right triangle, so E, M, N are collinear.

2. The median to a right triangle's hypotenuse is half that hypotenuse. Thus $EN = 1004$, $EM = 500$, and $MN = 504$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2009-I-Q05] | Part A (Q1-Q5)

Answer: 072

1. In quadrilateral $APCM$, diagonals AC, PM bisect each other, so it is a parallelogram. Thus $AM \parallel CP$, making BLP similar to BAM.

2. Angle-bisector theorem gives $AL/LB = 450/300 = 3/2$. Therefore $LP/AM = BL/BA = 2/5$ and $LP = 72$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2016-II-Q05] | Part A (Q1-Q5)

Answer: 182

1. Put $a = BC_0, b = AC_0, c = AB$. Similarity makes the successive altitudes geometric, with first term ab/c and ratio $a/c < 1$. Hence their sum is $ab/(c-a) = 6(a+b+c)$.

2. Use $(c-a)(c+a) = b^2$ to simplify this equality to $a = 6(b+c-a)$, or $7a = 6(b+c)$.

3. Squaring $6c = 7a - 6b$ and using $c^2 = a^2 + b^2$ gives $13a = 84b$. Coprimality forces $a = 84, b = 13$, and then $c = 85$. These sides satisfy the sum condition, so $p = 182$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2024-II-Q05] | Part A (Q1-Q5)

Answer: 080

1. Let the triangle side lengths be P, Q, R and the hexagon side length s. The three cut-off corner triangles are similar to the large triangle because each cutting side is parallel to the opposite triangle side.

2. Along the side of length P, the two corner pieces have lengths Ps/Q and Ps/R, leaving a central segment of length s. Thus $1 = s(1/P + 1/Q + 1/R)$.

3. Substituting gives $s = 1/(1/200 + 1/240 + 1/300) = 80$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2001-I-Q07] | Part B (Q6-Q10)

Answer: 923

1. If h is the altitude to BC and r the inradius, similarity gives $DE/20 = (h-r)/h$.

2. Writing area as K, $r = K/s$ and $h = 2K/20$, so $r/h = 20/(2s) = 20/63$. Hence $DE = 20(43/63) = 860/63$, and the sum is 923.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2004-I-Q09] | Part B (Q6-Q10)

Answer: 035

1. The rectangle cut runs from a vertex to its opposite side and makes a right triangle, which must be similar to the 3–4–5 triangle. Its fitting leg pairs are $(6, 9/2)$ and $(7, 21/4)$; the reversed orientations would exceed the other rectangle side.

2. The residual rectangle trapezoid has two right angles, so the triangle cut must be parallel to a leg rather than the hypotenuse. Matching its acute angle selects a cut parallel to the side of length 3.

3. The trapezoid base ratio, hence the small triangle scale, is $(7 - 9/2)/7 = 5/14$ or $(6 - 21/4)/6 = 1/8$. Their areas are $6(5/14)^2 = 75/98$ and $6(1/8)^2 = 3/32$.

4. The smaller is $3/32$, giving 35.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2015-II-Q07] | Part B (Q6-Q10)

Answer: 161

1. The semiperimeter is 27, so Heron gives triangle area $\sqrt{27 \cdot 15 \cdot 10 \cdot 2} = 90$. Its altitude to BC is $h = 180/25 = 36/5$.

2. Similarity of APQ and ABC gives rectangle height $h(1 - w/25)$. Thus its area is $(36/5)w - (36/125)w^2$, and $\beta = 36/125$.

3. The fraction is reduced, so the requested sum is 161.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2018-II-Q07] | Part B (Q6-Q10)

Answer: 020

1. The first k bands form a similar triangle of area ratio $k/2450$. Thus $P_kQ_k = 5\sqrt{3}\sqrt{k/2450} = \sqrt{6k}/14$.

2. An integer has rational square root exactly when it is a perfect square. The condition $6k$ square is equivalent to $k = 6j^2$ for integer $j \geq 1$, by prime-exponent parity.

3. As $6j^2 \leq 2450$, one has $1 \leq j \leq 20$. The count is 20.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2019-I-Q06] | Part B (Q6-Q10)

Answer: 090

1. Let F be the foot on KN. Right triangle KLN gives $KF \cdot KN = KL^2 = 784$.

2. Triangles KFO, KMN are similar by their right angles and common angle at K, so $KO \cdot KM = KF \cdot KN = 784$. Hence $KM = 98$ and $MO = 90$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2019-II-Q07] | Part B (Q6-Q10)

Answer: 715

1. Use barycentric coordinates $\alpha+\beta+\gamma=1$. A parallel chord at level α has length $220(1-\alpha)$, so the three lines are $\alpha=3/4, \beta=3/4, \gamma=7/8$.

2. The new parallel-sided triangle has similarity ratio $|1-(3/4+3/4+7/8)|=11/8$, as seen by subtracting these constant coordinates from a point.

3. Its perimeter is $(11/8)(120+220+180)=715$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2001-II-Q13] | Part C (Q11-Q15)

Answer: 069

1. Extend AB beyond B and DC beyond C to meet at P. The first angle relation gives $PA = PD$. The second gives similarity of PCB and PBD.

2. Hence $PC/PB = PB/PD = BC/BD = 3/5$. Since $PB = PD - 8$, solve $(PD - 8)/PD = 3/5$ to get $PD = 20$, $PB = 12$, $PC = 36/5$.

3. Then $CD = PD - PC = 64/5$, so the answer is 69.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2002-II-Q13] | Part C (Q11-Q15)

Answer: 901

1. Assign masses $5, 6, 15$ at A, B, C, matching both side division ratios. Let $F = CP \cap AB$; its mass is 11, giving $CP : PF = 11 : 15$ and $FP/FC = 15/26$.

2. The homothety centered at F with ratio $15/26$ sends C to P and the other two sides to the stated parallels. Hence it sends CAB to PQR.

3. The area ratio is $(15/26)^2 = 225/676$, giving 901.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2004-II-Q13] | Part C (Q11-Q15)

Answer: 484

1. Cosine law gives $AC = 7$. Let $F = AD \cap CE$; then $ABCF$ is a parallelogram. If B has distance h from AC, F is the same distance on the opposite side.

2. Because $AC \parallel DE$, the triangles with vertex F are similar at ratio $7 : 15$. Convexity puts DE beyond F from AC, so its distance from F is $15h/7$ and from B is $29h/7$.

3. The area ratio is $(7h/2)/[15(29h/7)/2] = 49/435$, giving 484.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2006-II-Q12] | Part C (Q11-Q15)

Answer: 865

1. Triangle ADF has adjacent sides thirteen,eleven and included angle 120°. Its area is $143\sqrt{3}/4$, and cosine rule gives $AF^2 = 433$.

2. Inscribed-angle equalities and the parallel sides give $\triangle CBG \sim \triangle AFD$, with scale ratio $CB/AF = 2\sqrt{3}/\sqrt{433}$.

3. Hence $[CBG] = (12/433)(143\sqrt{3}/4) = 429\sqrt{3}/433$. The requested sum is 865.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2007-I-Q15] | Part C (Q11-Q15)

Answer: 989

1. Write side length s and $AE = t$. Subtract the three corner triangles from ABC: $[DEF] = (\sqrt{3}/4)[5(s-t) + 2t - 10]$. Hence $5(s-t) + 2t = 66$.

2. The sixty-degree angle at E makes AEF similar to CDE by their remaining angles. Thus $t/5 = 2/(s-t)$, so $t(s-t) = 10$.

3. Substitute $s - t = 10/t$ to get $t^2 - 33t + 25 = 0$, giving $t = (33 \pm \sqrt{989})/2$. Then $s = t + 10/t = (231 \pm 3\sqrt{989})/10$. Since $989 = 23 \cdot 43$ is squarefree, the answer is 989.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2013-I-Q13] | Part C (Q11-Q15)

Answer: 961

1. Heron gives the initial area ninety. Let $r = 17/25$. The outer corner triangle $B_0C_1C_0$ has similarity ratio r, while $C_0C_1 = 17r$, so the retained AB_1C_1 has linear ratio $t = 1 - r^2 = 336/625$.

2. The removed target triangle at the first step has area $90(1 - r^2 - t^2) = 90r^2t$. Subsequent target triangles have disjoint interiors and successive area ratio t^2.

3. The total is $90r^2t/(1 - t^2) = 90t/(1 + t) = 30240/961$. Since $961 = 31^2$ is coprime to the numerator, $q = 961$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2002-II-Q04] | Part A (Q1-Q5)

Answer: 803

1. The interior contains a central cell and rings of $6, 12, \ldots, 6(n-2)$ cells. Its cell count is $1 + 3(n-2)(n-1)$.

2. At $n = 202$, this is 120601. Each unit hexagon has area $3\sqrt{3}/2$, so $m = 3 \cdot 120601 = 361803$, whose remainder is 803.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2003-I-Q02] | Part A (Q1-Q5)

Answer: 301

1. The green annuli have radii $(1,2),(3,4),\ldots,(99,100)$. Their areas divided by π are $2^2-1^2, 4^2-3^2, \ldots, 100^2-99^2$.

2. Each difference is the sum of the two radii, so the total is $1+2+\cdots+100=5050$. The area ratio is $5050/10000=101/200$, giving 301.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2004-I-Q04] | Part A (Q1-Q5)

Answer: 086

1. For the common corner A, the endpoint triangle is right and its hypotenuse is 2, so its midpoint has distance 1 from A. Conversely the entire corner quarter-circle arises from such endpoints.

2. The central region is the square minus four radius-one quarter-disks, area $4-\pi \approx 0.858407$. Rounding gives $k=0.86$, hence $100k=86$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2004-II-Q01] | Part A (Q1-Q5)

Answer: 592

1. Scale the radius to two. The chord is distance one from the center and subtends 120°. Its smaller circular segment has area $4\pi/3 - \sqrt{3}$.

2. The complementary area is $8\pi/3 + \sqrt{3}$, so the ratio becomes $(8\pi + 3\sqrt{3})/(4\pi - 3\sqrt{3})$.

3. The normalized parameters are $(8, 3, 3, 4, 3, 3)$, whose product is 2592. The remainder is 592.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-II-Q01] | Part A (Q1-Q5)

Answer: 046

1. The four equal angles are $(720-180)/4 = 135^\circ$. For side length s, chords BF, CE separate two right isosceles triangles, each area $s^2/2$, from a rectangle of sides $s, s\sqrt{2}$.

2. The total area is $s^2(1+\sqrt{2})$, so $s^2 = 2116$ and $s = 46$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2008-I-Q02] | Part A (Q1-Q5)

Answer: 025

1. If height $h \leq 10$, triangle area is at most fifty, so $h > 10$. The far square side cuts off a similar triangle with height $h - 10$ and base $10(h - 10)/h$.

2. Overlap area is $5h - 5(h - 10)^2/h = 100 - 500/h$. Set it to eighty to obtain $h = 25$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2014-II-Q03] | Part A (Q1-Q5)

Answer: 720

1. Each eighteen-unit half-bar spans half of the twenty-four-unit separation. Its outward horizontal displacement is $\sqrt{18^2 - 12^2} = 6\sqrt{5}$.

2. The middle width is therefore $a + 12\sqrt{5}$. Two congruent height-twelve trapezoids give total hexagon area $24a + 144\sqrt{5}$.

3. Equate with $36a$ to get $a = 12\sqrt{5}$, hence $a^2 = 720$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2019-I-Q03] | Part A (Q1-Q5)

Answer: 120

1. The triangle is right at R, with area 150. Its other angle sines are $\sin P = 4/5, \sin Q = 3/5$.

2. Remove corner triangles having adjacent sides five and five: their areas are $25(4/5)/2 = 10$, $25(3/5)/2 = 15/2$, and $25/2$. Remaining area $150 - 30 = 120$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2021-II-Q02] | Part A (Q1-Q5)

Answer: 336

1. Let $s = AF$. Triangle AEF is equilateral, so $BD = h = \sqrt{3}(840 - s)/2$. The horizontal distance from E to BD is $h/\sqrt{3}$, giving $[BED] = h^2/(2\sqrt{3})$.

2. $\angle AFG = 120^\circ$, so it must be the apex of the isosceles triangle; hence $FG = FA = s$ and $[AFG] = \sqrt{3}s^2/4$.

3. The ratio gives $3s^2/(2h^2) = 8/9$, or $s = 4h/(3\sqrt{3}) = 2(840 - s)/3$. Thus $s = 336$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2025-I-Q02] | Part A (Q1-Q5)

Answer: 588

1. The side ratios are $AD/AB = AF/AC = 1/7$ and $AE/AB = AG/AC = 5/7$. Therefore $[DEGF] = (25-1)[ABC]/49$, yielding $[ABC] = 588$.

2. Use affine coordinates $A = (0,0), B = (1,0), C = (0,1)$. Then $F = (0,1/7), E = (5/7,0), M = (-1/7,2/7), N = (10/7,-5/7)$.

3. The shoelace cross-product sum for A, F, N, B, C, E, M is $-10/49 + 5/7 + 1 - 5/7 + 10/49 = 1$, exactly that of ABC. Affine area ratios are preserved, so the requested area is 588.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2000-II-Q06] | Part B (Q6-Q10)

Answer: 181

1. If the shorter base is b, the midsegment is $b+50$. The equal heights of the two half-trapezoids give $(2b+50)/(2b+150)=2/3$, so $b=75$ and the other base is 175.

2. Extending the legs to their apex makes similar triangles with area proportional to the square of a parallel section. Equal trapezoid areas imply $x^2-75^2=175^2-x^2$, so $x^2=18125$.

3. The floor of 181.25 is 181.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2001-I-Q09] | Part B (Q6-Q10)

Answer: 061

1. The three corner area fractions are $p(1-r), q(1-p), r(1-q)$. Subtracting them from one yields $[DEF]/[ABC] = 1-(p+q+r)+(pq+qr+rp)$.

2. The pairwise-product sum is $[(2/3)^2 - 2/5]/2 = 1/45$. Thus the area fraction is $1 - 2/3 + 1/45 = 16/45$, giving 61.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2003-II-Q06] | Part B (Q6-Q10)

Answer: 112

1. A triangle’s centroid is one-third of each altitude above the corresponding side. After the half-turn, the opposite vertex lies one-third of the altitude beyond that side.

2. Each of the three exterior corners is therefore similar to the original triangle with scale $1/3$ and area $1/9$ of the original. They are disjoint, so union area is $K + 3K/9 = 4K/3$.

3. Heron gives $K = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$, hence union area 112.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-I-Q07] | Part B (Q6-Q10)

Answer: 408

1. Let the first-line distance from the vertex be x spacing units. Similar triangles have area proportional to distance squared. Thus A, B, C, D have proportional areas $x^2, 2x+3, 2x+7, 2x+11$.

2. The ratio $(2x+7)/(2x+3) = 11/5$ gives $x = 1/6$. Consequently $[D]/[A] = (2x+11)/x^2 = 408$.

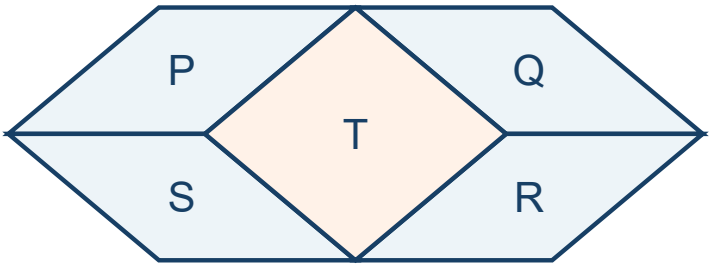
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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-I-Q08] | Part B (Q6-Q10)

Answer: 089



1. Let common side be z and each outer angle incident to the top vertex be α. The central angle is $\pi - 2\alpha$, so $0 < \alpha < \pi/2$ and $z^2 \sin\alpha = \sqrt{2006}$.

2. Central area $K = z^2 \sin 2\alpha = 2\sqrt{2006}\cos\alpha$ takes every value in $(0, \sqrt{8024})$. Since $89^2 < 8024 < 90^2$, the possible integers are 1 through 89.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2011-I-Q08] | Part B (Q6-Q10)

Answer: 318

1. Heron with semiperimeter forty gives area $K = \sqrt{40 \cdot 17 \cdot 13 \cdot 10} = 20\sqrt{221}$. A folded vertex drops by its perpendicular distance to the fold, so a height-h table uses corner triangles of scale factors $ha/(2K), hb/(2K), hc/(2K)$ for opposite sides a, b, c.

2. On any original side, the two endpoint flaps cannot overlap. The three constraints are $h(a+b) \le 2K, h(b+c) \le 2K, h(c+a) \le 2K$. These are also sufficient: the parallel cuts then bound a central tabletop and each folded vertex reaches the floor.

3. The greatest pair sum is $27 + 30 = 57$, so $h_{\max} = 40\sqrt{221}/57$, giving $40 + 221 + 57 = 318$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2013-II-Q10] | Part B (Q6-Q10)

Answer: 146

1. With center O, the points O, B, A are collinear and $AB = 4$. Distances from B, O to any secant through A are in ratio AB/AO, so $[BKL]/[OKL] = 4/(4+\sqrt{13})$.

2. Since $OK = OL = \sqrt{13}$, $[OKL] \leq 13/2$. Equality is possible with a right central angle: the secant need only be at perpendicular distance $\sqrt{13/2}$ from O, which a line through A can attain.

3. Thus maximum $[BKL] = 26/(4+\sqrt{13}) = (104-26\sqrt{13})/3$. The requested sum is 146.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2007-I-Q12] | Part C (Q11-Q15)

Answer: 875

1. The rotation is 15°; denote image vertices $B\prime, C\prime$. Define $D = AB\prime \cap BC$, $E = BC \cap B\prime C\prime$, $F = AC \cap B\prime C\prime$. The overlap is $ADEF = [AB\prime F] - [EB\prime D]$, and $AB\prime \perp BC$.

2. Sine rule in $AB\prime F$ gives $B\prime F = 10\sqrt{6}$, so its area is $50(3+\sqrt{3})$. Right triangle ABD has area fifty. Similarity $EB\prime D \sim ABD$ gives its area as $50[(1-\cos 15^\circ)/\sin 15^\circ]^2 = 50(15 + 8\sqrt{3} - 6\sqrt{6} - 10\sqrt{2})$.

3. Subtract to obtain $500\sqrt{2} - 350\sqrt{3} + 300\sqrt{6} - 600$. Thus the requested expression is $(500+350+300+600)/2 = 875$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2009-I-Q11] | Part C (Q11-Q15)

Answer: 600

1. The fifty points are $P_j = (j, 2009 - 41j)$ for $0 \le j \le 49$. Determinant area is $2009|i - j|/2$.

2. As 2009 is odd, integrality requires equal parity of i, j. There are twenty-five indices of each parity; choose unordered pairs: $2\binom{25}{2} = 600$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2010-I-Q13] | Part C (Q11-Q15)

Answer: 069

1. The radius is 126. With O the midpoint of AB, $AN = AO = ON$, so AON is equilateral. If P is the foot from N to AB, then $AP = 63, NP = 63\sqrt{3}, UP = 21, UO = 42$.

2. Let Q be the foot from U to CD. The region on the A side of the cutting line has one-third of the semicircle minus $[NUO]$, plus one-third of the rectangle plus $[UQT]$. Its rectangle portion is at most two-thirds (its widths are 84 and $DT \leq 252$), and its semicircle portion is less than one-third; hence this side cannot be the larger two-thirds portion.

3. It must be exactly one-third, so $[NUO] = [UQT]$. Relative to $[NUP]$, the first area has ratio $UO/UP = 2$, while the similar triangle UQT has area ratio $(UQ/NP)^2$.

4. Thus $DA = UQ = \sqrt{2}\,NP = 63\sqrt{6}$, giving 69. The resulting T lies on CD, since $DT = 84 + 21\sqrt{2} < 252$.

Source / 来源

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2025-II-Q14] | Part C (Q11-Q15)

Answer: 104

1. First verify the angular order. Since $AK = AL = KL$, the rays AK, AL differ by 60°. If AL lay below AK inside the right angle, both $\angle BAK$ and $\angle LAC$ would exceed 60°, so the equal-radius isosceles relations would give $AB, AC < 14$, contrary to $BC = 38$. Hence the order is AB, AK, AL, AC.

2. AKL is equilateral. Put $\alpha = \angle BAK, \beta = \angle LAC$, so $\alpha + \beta = 30^\circ$. From the two isosceles triangles, $AB = 28\cos\alpha, AC = 28\cos\beta$.

3. Pythagoras gives $\cos^2\alpha + \cos^2\beta = 361/196$, whence $\cos(\alpha - \beta) = 55\sqrt{3}/98$. Therefore $[ABC] = 392\cos\alpha\cos\beta = 208\sqrt{3}$.

4. The removed areas satisfy $[ABK] + [ACL] = 98(\sin 2\alpha + \sin 2\beta) = 55\sqrt{3}$ and $[AKL] = 49\sqrt{3}$. Subtract to obtain $[BKLC] = 104\sqrt{3}$, so $n = 104$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2001-I-Q04] | Part A (Q1-Q5)

Answer: 291

1. $C = 75^\circ$, and in ACT the other angles are $30^\circ, 75^\circ$, so $AC = AT = 24$.

2. Drop $CH \perp AB$. Special triangles give $AH = 12, CH = 12\sqrt{3}, BH = 12\sqrt{3}$, so $AB = 12 + 12\sqrt{3}$.

3. The area is $(12\sqrt{3})(12 + 12\sqrt{3})/2 = 216 + 72\sqrt{3}$, giving 291.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2006-I-Q01] | Part A (Q1-Q5)

Answer: 084

1. Pythagoras in ABC gives $AC^2 = 18^2 + 21^2 = 765$. In ACD, $AD^2 = 765 + 14^2 = 961$, so $AD = 31$.

2. Perimeter $18 + 21 + 14 + 31 = 84$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2007-II-Q03] | Part A (Q1-Q5)

Answer: 578

1. The $5, 12, 13$ triangles AEB, CFD are right at E, F. Extend EA past A and FD past D to meet at G. The complementary acute angles show AGD is right and congruent to BEA, so $GA = 5, GD = 12$.

2. Thus $GE = 17, GF = 17$, with $GE \perp GF$. Pythagoras gives $EF^2 = 2 \cdot 17^2 = 578$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2013-II-Q05] | Part A (Q1-Q5)

Answer: 020

1. Scale the original side to six, making each segment of BC length two. The cosine rule gives $AD = AE = 2\sqrt{7}$.

2. Triangle ADE has one-third of the original area, $3\sqrt{3}$. It also has area $\frac{1}{2}AD\,AE\sin\angle DAE = 14\sin\angle DAE$.

3. The sine is $3\sqrt{3}/14$, giving 20.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2014-I-Q01] | Part A (Q1-Q5)

Answer: 790

1. One level rises $80/3$ mm. Each of the six diagonal crossings has minimum length $\sqrt{50^2 + (80/3)^2} = 170/3$ mm.

2. Adding the bottom width and two ends gives $50 + 6(170/3) + 400 = 790$ mm. Straight taut segments attain the bound.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2015-II-Q04] | Part A (Q1-Q5)

Answer: 018

1. Let $u = \log 2 > 0$. The altitude is $4u$, while half the base difference is $(\log 192 - \log 3)/2 = 3u$.

2. Each equal leg is the hypotenuse of a $3u$ by $4u$ right triangle, so it is $5u$. The perimeter is $\log 3 + \log 192 + 10 \log 2 = \log(2^{16}3^2)$. Therefore $p + q = 18$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2018-I-Q04] | Part A (Q1-Q5)

Answer: 289

1. The original cosine rule gives $\cos A = 7/25$. Put the common length x, making $AE = 10 - x > 0$.

2. The cosine rule in ADE gives $x^2 = x^2 + (10-x)^2 - (14/25)x(10-x)$. Cancel and divide by $10 - x$, obtaining $10 - x = 14x/25$.

3. Thus $x = 250/39$, in the required interval, giving 289.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2020-I-Q01] | Part A (Q1-Q5)

Answer: 547

1. Put $\angle A = \alpha$. Equal sides in AED give $\angle ADE = \alpha$; because EA and EB are opposite rays, $\angle DEB = 2\alpha$.

2. Next $ED = DB$ gives $\angle DBE = 2\alpha$ and $\angle EDB = 180^\circ - 4\alpha$. The straight angle along ADC then gives $\angle BDC = 3\alpha$.

3. From $DB = BC$ and $AB = AC$, both base angles of ABC are 3α. Thus $7\alpha = 180^\circ$, so $\angle B = 540/7$ degrees and the requested sum is 547.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2021-II-Q05] | Part A (Q1-Q5)

Answer: 736

1. If the included angle is θ, area is $20\sin\theta$. Its obtuse branch supplies exactly one triangle for each $0 < s < 20$.

2. For the acute branch, the third side satisfies $c^2 = 116 - 80\cos\theta$. The only possible obtuse angle is opposite side 10, requiring $16 + c^2 < 100$, or $\cos\theta > 2/5$. This branch exists as obtuse exactly for $s < 4\sqrt{21}$.

3. Thus uniqueness among obtuse triangles holds for $4\sqrt{21} \le s < 20$; at the lower endpoint the competing triangle is right. Therefore $a^2 + b^2 = 336 + 400 = 736$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2023-II-Q03] | Part A (Q1-Q5)

Answer: 250

1. Let the common angle be ω and leg length be L. Triangle APC has angles $90^\circ - \omega, \omega, 90^\circ$, so $L = 10/\sin\omega$.

2. In triangle ABP, the angles at A, B, P are $\omega, 45^\circ - \omega, 135^\circ$. The sine rule gives $10\sin 135^\circ = L\sin(45^\circ - \omega)$.

3. Substitution reduces this to $\sin\omega = \cos\omega - \sin\omega$, so $\tan\omega = 1/2$ and $\sin^2\omega = 1/5$.

4. Therefore $L^2 = 500$ and the area is $L^2/2 = 250$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-I-Q10] | Part B (Q6-Q10)

Answer: 083

1. Scale $AC = BC = 1$. In AMC, angle $M = 150^\circ$, so sine law gives $CM = 2\sin 7^\circ$. Also $\angle MCB = 83^\circ$.

2. Cosine law in BMC yields $BM^2 = 1 + 4\sin^2 7^\circ - 4\sin 7^\circ \cos 83^\circ = 1$.

3. Thus $BM = BC$, and its base angles at M, C coincide. The requested angle is 83°.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-II-Q07] | Part B (Q6-Q10)

Answer: 400

1. Let side length be s and $\alpha = A/2$. The diagonals are $2s\cos\alpha, 2s\sin\alpha$. Extended sine law gives $12.5 = s/(2\cos\alpha)$ and $25 = s/(2\sin\alpha)$.

2. Hence $\tan\alpha = 1/2$, with $\sin\alpha = 1/\sqrt{5}, \cos\alpha = 2/\sqrt{5}$, and $s = 10\sqrt{5}$.

3. The area is half the diagonal product, $2s^2\sin\alpha\cos\alpha = 400$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2005-I-Q07] | Part B (Q6-Q10)

Answer: 150

1. Extend AD and BC to meet at P. Triangle ABP is equilateral. Writing $x = AB$ gives $PD = x - 10, PC = x - 8$ and included angle sixty degrees.

2. Cosine rule gives $144 = (x-10)^2 + (x-8)^2 - (x-10)(x-8) = x^2 - 18x + 84$. Thus $x = 9 + \sqrt{141}$, so $p + q = 150$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2008-I-Q10] | Part B (Q6-Q10)

Answer: 032

1. Triangle inequality gives $AD \geq ED - EA = 20\sqrt{7}$. In triangle ACD, sine rule gives $AD = AC \sin \angle ACD / \sin 60^\circ \leq 20\sqrt{7}$. Both inequalities must be equalities.

2. Thus $AD = 20\sqrt{7}$, $AC \perp CD$, and D, A, E are collinear in that order. Pythagoras gives $CD = 10\sqrt{7}$, hence $DF = CD \cos 60^\circ = 5\sqrt{7}$.

3. Therefore $EF = ED - DF = 25\sqrt{7}$, giving 32.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2013-I-Q09] | Part B (Q6-Q10)

Answer: 113

1. The crease is the perpendicular bisector of AT, meeting AB, AC at P, Q. Put $AP = PT = x$, $AQ = QT = y$.

2. Triangle BPT gives $x^2 = (12 - x)^2 + 81 - 9(12 - x)$, hence $x = 39/5$. Similarly at C, $y^2 = (12 - y)^2 + 9 - 3(12 - y)$, giving $y = 39/7$. Both are between zero and twelve, so the crease endpoints are on the sides.

3. With included angle sixty degrees, $PQ^2 = x^2 + y^2 - xy = 39^3/35^2$. Thus $PQ = 39\sqrt{39}/35$, and the answer is 113.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2000-I-Q14] | Part C (Q11-Q15)

Answer: 571

1. Scale the common length to 1 and let $A = \alpha$. In APQ, equal sides give $AQ = 2\cos\alpha$ and $\angle APQ = 180^\circ - 2\alpha$. In ABC, sine law gives $AB = 1/(2\sin(\alpha/2))$.

2. Using $AB = AQ + 1$ gives $4\sin(\alpha/2)\cos\alpha + 2\sin(\alpha/2) = 1$, hence $\sin(3\alpha/2) = 1/2$.

3. The possible $\alpha = 20^\circ, 100^\circ$ reduce to 20° because $AQ > 0$. Thus $r = 80/140 = 4/7$, and the floor is 571.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-I-Q12] | Part C (Q11-Q15)

Answer: 777

1. Let $AD = u, BC = v$ and the common angle be α. Cosine law applied to diagonal BD in both triangles gives $u^2 + 180^2 - 360u\cos\alpha = v^2 + 180^2 - 360v\cos\alpha$.

2. Divide by $u - v \neq 0$ to obtain $\cos\alpha = (u+v)/360 = 280/360 = 7/9$. Thus the floor is $\lfloor 7000/9 \rfloor = 777$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2005-II-Q14] | Part C (Q11-Q15)

Answer: 463

1. The cevians AD, AE are isogonal. Applying sine-area ratios to both and multiplying cancels the sine factors: $(BD/DC)(BE/EC) = AB^2/AC^2$.

2. Thus $BE/EC = (169/196)(6/9) = 169/294$, so $BE = 15 \cdot 169/(169+294) = 2535/463$. The numerator and denominator are coprime, giving $q = 463$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2006-II-Q15] | Part C (Q11-Q15)

Answer: 009

1. The nonnegative radical sums make x, y, z positive and imply strict triangle inequalities. The first equation describes an altitude of length $1/4$ to side x, with the two nonnegative radical lengths as base parts. The other altitudes are $1/5, 1/6$.

2. If triangle area is K, then $(x, y, z) = (8K, 10K, 12K)$. Heron with semiperimeter $15K$ yields $K^2 = 15 \cdot 7 \cdot 5 \cdot 3\, K^4$, so $K = 1/(15\sqrt{7})$. The ratio triangle is acute, confirming all altitude feet are internal.

3. Thus $x + y + z = 30K = 2/\sqrt{7}$ and $m + n = 9$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2007-II-Q11] | Part C (Q11-Q15)

Answer: 179

1. At simultaneous ground/small-cylinder tangency, center separation is 96 and vertical difference 48, so the center line is 30° from horizontal. Crossing sweeps the large center through 120° on radius 96, advancing $2 \cdot 96 \cos 30^\circ = 96\sqrt{3}$.

2. No slip around the stationary small circle makes large-cylinder spin $(96/72)120^\circ = 160^\circ$. The remaining 200° spin is on flat ground, advancing $(200/360)2\pi \cdot 72 = 80\pi$.

3. Total displacement $80\pi + 96\sqrt{3}$ gives 179.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2008-II-Q14] | Part C (Q11-Q15)

Answer: 007

1. Use rectangle vertices $A=(0,0), B=(a,0), C=(a,b), D=(0,b)$ and points $E=(x,0), F=(a,b-y)$. The three equal quantities are exactly squared sides of equilateral triangle DEF.

2. Let $\theta=\angle ADE$. Then $x=b\tan\theta$, $y=a\tan(30^\circ-\theta)$, and equal sides give $a/b=\cos(30^\circ-\theta)/\cos\theta=\sqrt{3}/2+\tan\theta/2$. Nonnegative x,y imply $0\le\theta\le 30^\circ$.

3. The ratio increases with θ and reaches $2/\sqrt{3}$ at $\theta=30^\circ$, with $y=0, x=b/\sqrt{3}<a$. Thus $\rho^2=4/3$ and the answer is 7.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2010-II-Q12] | Part C (Q11-Q15)

Answer: 676

1. Write the bases as $8a, 7a$ for positive integer a, and common perimeter as P. An isosceles triangle with base b and perimeter P has area squared $b^2P(P-2b)/16$.

2. Equal areas yield $64(P-16a) = 49(P-14a)$, hence $P = 338a/15$. The equal legs in the respective triangles are $(P-8a)/2 = 109a/15$ and $(P-7a)/2 = 233a/30$.

3. Because 233 is coprime to 30, integrality requires $30 \mid a$. Taking $a = 30$ gives sides $(218, 218, 240)$ and $(233, 233, 210)$, both nondegenerate and of perimeter 676.

4. Thus the least perimeter is 676.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2010-II-Q14] | Part C (Q11-Q15)

Answer: 007

1. Let $\alpha = \angle ACP$ and $u = AP$. In triangle APC, $\angle A = 180^\circ - 3\alpha$, so $45^\circ < \alpha < 60^\circ$. The sine rule gives $u = \sin\alpha / \sin 3\alpha = 1/(4\cos^2\alpha - 1)$ and $AC = 2u\cos\alpha$.

2. The large right triangle gives $AC = 4\cos A = -4\cos 3\alpha$. Divide the two expressions by $2\cos\alpha$ to obtain $u = 6 - 8\cos^2\alpha$.

3. Substitute $4\cos^2\alpha = 1 + 1/u$, yielding $u^2 - 4u + 2 = 0$. Since $\alpha > 45^\circ$, the first expression for u makes $u > 1$, selecting $u = 2 + \sqrt{2}$.

4. Then $BP = 2 - \sqrt{2}$ and $AP/BP = 3 + 2\sqrt{2}$, giving 7.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2018-II-Q12] | Part C (Q11-Q15)

Answer: 112

1. Let the four diagonal portions be $a = AP, b = BP, c = CP, d = DP$. The common sine of the intersection angle cancels from the area equality, giving $(a - c)(b - d) = 0$. Thus at least one diagonal is bisected.

2. Let u be half the bisected diagonal and v, w the other portions, orienting the angle θ appropriately and putting v next to the fourteen-unit side. Averaging and subtracting the adjacent cosine-law equations gives $u^2 + v^2 = 148, u^2 + w^2 = 180, uv|\cos\theta| = 24, uw|\cos\theta| = 40$. These relations apply whether $a = c$ or $b = d$, with the diagonal names exchanged.

3. Hence $w/v = 5/3$ and $w^2 - v^2 = 32$, so $v = 3\sqrt{2}, w = 5\sqrt{2}, u^2 = 130$. It follows that $\cos^2\theta = 16/65$, and $\sin\theta = 7/\sqrt{65}$.

4. The area is $u(v + w)\sin\theta = \sqrt{130}(8\sqrt{2})(7/\sqrt{65}) = 112$.

Source / 来源

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2025-II-Q12] | Part C (Q11-Q15)

Answer: 019

1. The common sine is $5/13$, so consecutive radial lengths multiply to $26/5$. Hence the ten radial lengths alternate between a, b, with $ab = 26/5$, and the desired sum is $u = a + b$.

2. Cosine rule makes each of the nine intervening polygon sides have square $a^2 + b^2 - 2ab(12/13) = u^2 - 20$. Perimeter gives $u + 9\sqrt{u^2 - 20} = 20$.

3. Squaring with $20 - u > 0$ yields $4u^2 + 2u - 101 = 0$, so $u = (9\sqrt{5} - 1)/4$. The requested sum is $9 + 5 + 1 + 4 = 19$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2025-II-Q05] | Part A (Q1-Q5)

Answer: 336

1. The midpoint circle is the nine-point circle; G, H, J are altitude feet. The medial-triangle angles give $\widehat{DE} = 2\angle DFE = 72^\circ$ and $\widehat{FG} = 2\angle FDG = 72^\circ$.

2. H, J also lie on the circle with diameter BC, centered at D. Isosceles triangles BDJ, CDH give $\angle JDB = 60^\circ, \angle HDC = 108^\circ$. Thus $\angle JDH = 12^\circ$ and, as an inscribed angle of the nine-point circle, it gives $\widehat{HJ} = 24^\circ$.

3. The requested sum is $72 + 2 \cdot 24 + 3 \cdot 72 = 336$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2001-II-Q06] | Part B (Q6-Q10)

Answer: 251

1. Let the original side be s, put its center at zero and CD on $y = s/2$. Equal-height points G, H on the circle force horizontal symmetry. For the smaller square of side t, its circle point is $(t/2, s/2 + t)$ outside the original square.

2. The circle equation gives $t^2/4 + (s/2 + t)^2 = s^2/2$, equivalent to $(5t - s)(t + s) = 0$. Hence $t = s/5$. The inward choice gives $t = s$, excluded by the strict area ratio.

3. The area ratio is $1/25$, giving $10 \cdot 25 + 1 = 251$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2011-II-Q10] | Part B (Q6-Q10)

Answer: 057

1. Let the chord midpoints be M, N. Right triangles give $OM = 20, ON = 24$, while $MN = 12$. The angles OMP, ONP are right, placing all four points on the circle with diameter OP.

2. In triangle OMN, the cosine of the included angle is $(400 + 576 - 144)/960 = 13/15$, hence its sine is $2\sqrt{14}/15$.

3. The extended sine rule in the new circle gives $OP = MN/\sin\angle MON = 90/\sqrt{14}$, so $OP^2 = 4050/7$.

4. The numerator-denominator sum is 4057, leaving remainder 57.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2013-II-Q08] | Part B (Q6-Q10)

Answer: 272

1. Let the minor central angles for chords twenty-two and twenty be α, β. No edge can use its major arc: replacing a repeated edge angle by $2\pi - \alpha$ or $2\pi - \beta$ would already force the sum of all six positive arcs to exceed 2π. Thus $4\alpha + 2\beta = 2\pi$.

2. Then $\cos\alpha = \sin(\beta/2) = 10/R$. The chord formula also gives $\cos\alpha = 1 - 242/R^2$.

3. Equating gives $R^2 - 10R - 242 = 0$, whose positive root is $5 + \sqrt{267}$. Thus $p + q = 272$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2015-I-Q06] | Part B (Q6-Q10)

Answer: 058

1. Let each central interval in the second circle be t. Since $\angle ACE = 5t$, the first circle arc AE avoiding C has measure $10t$. Each of its four short intervals is therefore $q = (360 - 10t)/4$.

2. Inscribed-angle rules give $\angle ABD = (360 - 3q)/2$ and $\angle AHG = (360 - 3t)/2$. Their difference 12 simplifies to $21t/4 - 135 = 12$, hence $t = 28$ and $q = 20$ degrees.

3. Then $\angle BAE = 3q/2 = 30^\circ$ and $\angle EAG = t = 28^\circ$. The two minor-arc orders put AB and AG on opposite sides of ray AE, so $\angle BAG = 30 + 28 = 58^\circ$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2016-II-Q10] | Part B (Q6-Q10)

Answer: 043

1. The intersecting-chord relation at Q gives $QC \cdot QT = QA \cdot QB = 42$. Extend AB past B to R with $BR = 8$, so $QP \cdot QR = 3 \cdot 14 = 42$. The converse chord-product theorem shows C, P, T, R are cyclic.

2. Circle angles now give $BRT = PCT = SAT$, while the original cyclic quadrilateral gives the exterior angle $RBT = AST$. Thus triangles RBT, AST are similar.

3. The corresponding lengths satisfy $ST/BT = AS/RB = 7/8$. Hence $ST = 35/8$ and $m + n = 43$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2023-II-Q09] | Part B (Q6-Q10)

Answer: 033

1. Tangency parallel to a chord places the tangency point over that chord midpoint. Consequently, taking $P=(0,0), X=(-10,0), Y=(14,0)$ gives $A=(-5,h), B=(7,h)$ and $AB=12, XY=24$.

2. The radical axis PQ meets AB at its midpoint $M=(1,h)$, since the powers MA^2, MB^2 agree. Reflection in the line of the two circle centers interchanges P, Q and the two common external tangents. Thus choosing the tangent nearer P makes its distance from P smaller than its distance from Q. Since M, P, Q are collinear and M has positive circle power, their order is M, P, Q, so $MP \cdot MQ = MA^2 = 36$ and $MQ = MP + 5$.

3. Thus $MP = 4$, and $h^2 = MP^2 - 1^2 = 15$. The alternative same-side chord-end placement would give $AB = 2, |M_x| = 6$; then $MP \geq 6$ and $MP \cdot MQ \geq 6(6-5) > 1 = MA^2$, impossible.

4. The area is $(24+12)\sqrt{15}/2 = 18\sqrt{15}$, yielding $m+n=33$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2024-I-Q10] | Part B (Q6-Q10)

Answer: 113

1. The tangent–chord theorem makes DBC isosceles, with base angles A. Cosine rule gives $\cos A = 11/25$, so $DB = (9/2)/(11/25) = 225/22$.

2. Also $\cos B = 1/15$, and $\cos(A+B) = -13/15$. Applying cosine rule in ABD yields $DA^2 = 25 + (225/22)^2 + 2 \cdot 5(225/22)(13/15) = (325/22)^2$.

3. Power of D gives $DP \cdot DA = DB^2$. Thus $AP = DA - DP = (DA^2 - DB^2)/DA = 100/13$ and $m+n = 113$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2025-I-Q06] | Part B (Q6-Q10)

Answer: 504

1. The height is the incircle diameter, 6, so the area gives $r+s=24$. Equal tangent lengths imply the two legs also sum to 24, hence each leg is 12.

2. An end right triangle gives $((r-s)/2)^2=12^2-6^2=108$. Therefore $r^2+s^2=[(r+s)^2+(r-s)^2]/2=(576+432)/2=504$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2026-II-Q10] | Part B (Q6-Q10)

Answer: 340

1. Angle-bisector ratios give $BD : CD = 8 : 9$. As both are integers, write $BD = 8t, CD = 9t$ with positive integer t. Powers of B, C yield $BE = 8t^2/25, CF = 9t^2/25$.

2. Integrality requires $5 \mid t$, say $t = 5s$. Then $AE = 200 - 8s^2 > 0$ and $AF = 225 - 9s^2 > 0$, so $s \leq 4$. Hence $BC = 85s \leq 340$.

3. For $s = 4$, the side lengths $200, 225, 340$ form a triangle and give $BD = 160, CD = 180, AE = 72, AF = 81$, all positive integers. The maximum is 340.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2001-I-Q13] | Part C (Q11-Q15)

Answer: 174

1. Set $t = d/2$ and $c = \cos t$. Chord ratios give the $2d$ and $3d$ lengths as $44c$ and $22(4c^2 - 1)$.

2. Their difference condition reduces to $44c^2 - 22c - 1 = 0$. As $t < 60^\circ$, take $c = (11 + \sqrt{165})/44$.

3. The required chord equals $44c - 20 = -9 + \sqrt{165}$, giving 174.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2002-I-Q13] | Part C (Q11-Q15)

Answer: 063

1. Let P be the centroid. Then $AE = 12$, $AP = 12$, $PE = 9$. The chord power at E gives $EC \cdot EF = EA \cdot EB = 144$, so $EF = 16/3$.

2. In isosceles AEP with equal sides 12 and base 9, the height to PE is $3\sqrt{55}/2$. Hence $[AEP] = 27\sqrt{55}/4$.

3. Collinear-base scaling gives $[AEF] = (EF/EP)[AEP] = 4\sqrt{55}$. Since E bisects AB, $[AFB] = 8\sqrt{55}$, so the answer is 63.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2005-I-Q15] | Part C (Q11-Q15)

Answer: 038

1. Let M be the midpoint of AB, let R be the incircle contact point on AC, and let the incircle cut CM at one-third and two-thirds positions. Its tangent lengths from M and C have equal square $2CM^2/9$. Relabel A, B so the base tangency point T lies in AM. Equal tangent lengths give $AC = AR + CR = AT + MT = AM = 10$.

2. Put $a = BC$. With semiperimeter $15 + a/2$, $MT = (a - 10)/2$ and median formula gives $CM^2 = (a^2 - 100)/2$. Thus $(a - 10)^2/4 = (a^2 - 100)/9$. Since $a > 10$, cancellation yields $9(a - 10) = 4(a + 10)$, so $a = 26$.

3. Heron gives area $\sqrt{28 \cdot 8 \cdot 18 \cdot 2} = 24\sqrt{14}$. The requested sum is 38.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2009-I-Q15] | Part C (Q11-Q15)

Answer: 150

1. Cosine rule gives $\angle A = 60^\circ$. The incenter-angle identities give $\angle BI_BD + \angle CI_CD = 180^\circ + \angle A/2 = 210^\circ$. Cyclic-angle relations put the second intersection below BC and yield $\angle BPC = 150^\circ$, independent of D.

2. Thus P lies on a fixed circular arc through B, C. For fixed chord, maximal perpendicular distance occurs at the arc midpoint, where $BP = PC = x$. This point is attainable: as D varies continuously from B to C, its first circle collapses to B at one end and its second to C at the other, so the distinct second intersection varies continuously from B to C along that arc and must pass its midpoint.

3. Cosine rule at the midpoint gives $196 = (2+\sqrt{3})x^2$. Hence the maximal area is $x^2 \sin 150^\circ/2 = 49(2-\sqrt{3}) = 98 - 49\sqrt{3}$, and the requested sum is 150.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2010-II-Q15] | Part C (Q11-Q15)

Answer: 218

1. Angle bisectors give $AD = 195/29$, $AE = 65/9$. Thus $AD - AM = 13/58$ and $AN - AE = 5/18$.

2. Use unit vectors u, v from A toward C, B and write points as $su + tv$. A circle through A, meeting these rays at distances d, e, has equation $s^2 + t^2 + 2st\cos A - ds - et = 0$.

3. Subtract the equations of the two circles. Their common chord AP has $(AD - AM)s + (AE - AN)t = 0$, so $t/s = (13/58)/(5/18) = 117/145$.

4. On BC, oblique coordinates give $BQ/CQ = (15/13)(s/t) = 725/507$. The factors $725 = 5^2 \cdot 29$ and $507 = 3 \cdot 13^2$ are coprime, so $m - n = 218$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2012-II-Q15] | Part C (Q11-Q15)

Answer: 919

1. Let E' be antipodal to E on ω. Both $\angle DFE$ and $\angle E'FE$ are right, so E', D, F are collinear. This reduces the second-circle intersection to a line-circle intersection.

2. Take $B = (0,0), C = (7,0), A = (65/14, 15\sqrt{3}/14)$. The bisector gives $D = (35/8, 0)$. The circumcenter is $(7/2, -7\sqrt{3}/6)$ with radius squared $49/3$. The second bisector-circle intersection is the midpoint of the arc BC not containing A: it lies on $x = 7/2$ and on ray AD beyond D, hence $E = (7/2, -7\sqrt{3}/2)$. Its antipode is therefore $E' = (7/2, 7\sqrt{3}/6)$.

3. Parameterize line $E'D$ by $E' + t(3, -4\sqrt{3})$. Substitution in the circle equation gives $57t^2 - 56t = 0$. The second point is $t = 56/57$, yielding $F = (245/38, -105\sqrt{3}/38)$.

4. Therefore $AF^2 = (240/133)^2 + 3(510/133)^2 = 900/19$. It is reduced, so $m + n = 919$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2014-I-Q15] | Part C (Q11-Q15)

Answer: 041

1. Take $B=(0,0), A=(0,3), C=(4,0), E=(0,e), D=(d,0)$. Since angle EBD is right, ED is a circle diameter and its equation is $x^2+y^2=dx+ey$. The point equidistant from E, D on the circle and in the positive quadrant is $F=((d+e)/2,(d+e)/2)$. As AC is $3x+4y=12$, this forces $d+e=24/7$.

2. Substitute $y=(12-3x)/4, d=24/7-e$ in the circle equation and remove the known root $x_F=12/7$. The second point is $G=(x,y)=(28(3-e)/25,(12+21e)/25)$. These coordinates satisfy $G-D=(-y/7,y)$ and $G-E=(x,x/7)$.

3. Therefore $DG/EG=y/x=3/4$. Together with $3x+4y=12$, this gives $G=(2,3/2)$. The circle equation now says $2d+(3/2)e=25/4$.

4. Combine with $d+e=24/7$ to obtain $d=31/14, e=17/14$, both within the sides. Thus $DE=\sqrt{d^2+e^2}=25\sqrt{2}/14$, and the sum is 41.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2016-I-Q15] | Part C (Q11-Q15)

Answer: 270

1. Let M be the midpoint of the common tangent segment AB. Its powers to the two circles are $MA^2 = MB^2$, so it lies on their radical axis XY. Both circles, and hence X, Y, lie in the same half-plane of AB. Their near/far order is therefore M, X, Y, with $MX \cdot MY = MA^2$.

2. All angles in the next two steps are directed modulo 180°, so replacing a ray by its opposite does not affect the statements. Cyclicity gives $\angle(DAX) = \angle(DYX)$ and $\angle(CBX) = \angle(CYX)$. Since D, Y, C are collinear, these angles agree. If $AD \parallel BC$, this would force $AX \parallel BX$, contradicting $X \notin AB$. Thus $Z = AD \cap BC$ exists, and the radical-axis theorem places it on XY.

3. The same angle equality now reads $\angle ZAX = \angle ZBX$, proving A, B, X, Z concyclic. From this circle and the tangent-chord theorem on the second original circle, $\angle AZX = \angle ABX = \angle BYX$. Since $ZX \parallel YX$, it follows that $ZA \parallel BY$. Interchanging the two original circles similarly gives $ZB \parallel AY$.

4. Thus $AYBZ$ is a parallelogram. Its diagonals AB, YZ bisect each other at M, so Z is the reflection of Y through M. The full order on their common line is Z, M, X, Y. In particular $XZ = MX + MY$ and $XY = MY - MX$.

5. For the similarity, Z, C, B are collinear and Z, D, A are collinear. Cyclicity of B, C, Y, X gives $\angle XCZ = \angle XYB = \angle XZD$, using $YB \parallel ZA$. Likewise $\angle XZC = \angle XYA = \angle XDZ$, using $ZC \parallel YA$ and cyclicity of A, D, Y, X. Therefore triangles XZC and XDZ are similar, and $XZ/XD = XC/XZ$, hence $XZ^2 = XC \cdot XD$.

6. Finally $AB^2 = 4MX \cdot MY = (MX + MY)^2 - (MY - MX)^2 = XZ^2 - XY^2 = 67 \cdot 37 - 47^2 = 270$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2018-I-Q15] | Part C (Q11-Q15)

Answer: 059

1. Write one cyclic order of side arcs as $\alpha, \beta, \gamma, \delta$, totaling 2π. The other two orders exchange the three possible opposite-side pairings. By the intersecting-chord angle theorem, their diagonal-angle sines are the sines of half the corresponding opposite-arc sums.

2. The diagonals in any one quadrilateral have lengths twice the diagonal-angle sines of the other two orders, using the unit-circle chord formula. Consequently the common area is $\frac{1}{2}d_1 d_2 \sin\phi = 2\sin\phi_A \sin\phi_B \sin\phi_C$. This formula is unchanged if an angle is replaced by its supplement.

3. Substitute to obtain $K = 2(2/3)(3/5)(6/7) = 24/35$, hence $m + n = 59$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2018-II-Q14] | Part C (Q11-Q15)

Answer: 227

1. Let the tangencies on AB, AC be Z, W and their common distance from A be ℓ. Because Y is the first intersection from A, the tangent cuts off the corner: $PY = PZ = \ell - AP$ and $BX = BZ = AB - \ell$.

2. The tangent-chord angle theorem makes the angles opposite AP in triangle APY and opposite AB in triangle ABX supplementary. Their angle at A is common. The sine rule therefore gives $PY/AP = BX/AB$, or $\ell/AP + \ell/AB = 2$.

3. With $AP = 3$, $AB = 7$, this yields $\ell = 21/5$. The same argument on AC gives $\ell/AQ + \ell/AC = 2$, hence $1/AQ = 10/21 - 1/8 = 59/168$.

4. Thus $AQ = 168/59$, within AC, giving 227.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-I-Q13] | Part C (Q11-Q15)

Answer: 032

1. The cyclic angles give $\cos\angle DFE = \pm\cos\angle ACB = \pm 3/4$. With $DF = 2, EF = 7$, the two possible values of $\delta = DE$ are $\sqrt{32}$ and $\sqrt{74}$; the point order must select the valid one.

2. Use $A = (0,0), B = (4,0), C = (27/8, 15\sqrt{7}/8)$, $D = (d,0), E = (d+\delta, 0)$, and $F = (d+w, y)$. Distance subtraction gives $w = (\delta^2 - 45)/(2\delta)$ and $y^2 = 4 - w^2$. The two circle equations are $x^2 + y^2 - dx + (27d - 288)y/(15\sqrt{7}) = 0$ and $x^2 + y^2 - (d+\delta+4)x - (180 + 5d + 5\delta)y/(15\sqrt{7}) + 4(d+\delta) = 0$.

3. Substitute the two signs of y into both equations. For $\delta = \sqrt{32}$ the common solution has $y = -7\sqrt{14}/16$ and $d = (21 + 5\sqrt{2})/4 > 4$. For $\delta = \sqrt{74}$ it has $y = 7\sqrt{518}/148$ and $d = (84 - 5\sqrt{74})/26 < 4$, violating the required order. The opposite sign in each case does not satisfy both circles. Thus only the first branch is allowed.

4. It gives $BE = d + \delta - 4 = (5 + 21\sqrt{2})/4$. The explicitly obtained coordinates satisfy both circle equations and both distances, so the branch is attainable. The requested sum is $5 + 21 + 2 + 4 = 32$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-I-Q15] | Part C (Q11-Q15)

Answer: 065

1. The inner circles touch ω at A, B. Let the tangents there meet at Z. Equal tangent lengths make its powers in the two inner circles equal, so Z lies on PQ. With $s = ZP$, midpoint M of AB gives $ZA^2 - s^2 = MA^2 - MP^2 = 16 - 1 = 15$.

2. Let H be midpoint of XY and $t = ZH$. Power at Z in ω gives $t^2 - (11/2)^2 = s^2 + 15$, so $t^2 - s^2 = 181/4$. Power at P gives $PH^2 = (11/2)^2 - AP \cdot PB = 61/4$. As $t > s$, $PH = t - s$.

3. Subtracting yields $2s(t - s) = 30$, hence $t = s + 15/s$. Power in an inner circle gives $ZQ = ZA^2/ZP = s + 15/s = t$, so $Q = H$. Therefore $PQ^2 = 61/4$, giving 65.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-II-Q15] | Part C (Q11-Q15)

Answer: 574

1. Write $b = AC, c = AB, a = BC, k = \cos A > 0$, $u = AP = bk, v = AQ = ck$. Similar triangles APQ, ACB give $ak = 25$. Since the feet lie inside the circle and $XP, QY < PQ$, the order is X, P, Q, Y. Powers give $u(c - u) = 400$ and $v(b - v) = 525$.

2. Put $w = bck = uv/k$. Then $u^2 = w - 400, v^2 = w - 525$. Cosine rule multiplied by k^2 gives $625 = u^2 + v^2 - 2uvk$, so $wk^2 = w - 775$.

3. Now $(w - 400)(w - 525) = (uv)^2 = w^2k^2 = w(w - 775)$, yielding $w = 1400, k^2 = 25/56$. Therefore $bc = w/k = 560\sqrt{14}$ and the answer is 574.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2020-II-Q15] | Part C (Q11-Q15)

Answer: 717

1. Tangent-chord and sine-law relations yield $TX = 16\sin C$, $TY = 16\sin B$. Since A, X, T, Y are concyclic with diameter AT, $XY = AT\sin A$. For circumradius R, the given sum is $[64(AB^2 + AC^2) + 121AT^2]/R^2 = 1143$.

2. Set $B = (-11, 0), C = (11, 0), T = (0, -\sqrt{135})$. Perpendicular radius and tangent give circumcenter $O = (0, 121/\sqrt{135})$, hence $R^2 = 30976/135$.

3. For $A = (u, v)$ on the circle, $u^2 + v^2 = 242v/\sqrt{135} + 121$. Therefore $AB^2 + AC^2 = 484v/\sqrt{135} + 484$ and $AT^2 = 512v/\sqrt{135} + 256$.

4. The sum condition gives $v = 291/\sqrt{135}$. Thus $AT^2 = 183552/135$ and $XY^2 = 121AT^2/R^2 = 183552/256 = 717$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2021-I-Q11] | Part C (Q11-Q15)

Answer: 301

1. Let P be the diagonal intersection and θ their acute angle. Reflection across an angle bisector swaps the two diagonal lines; multiplying by $\cos\theta$ then sends every vertex to its foot on the other diagonal. Thus all side lengths scale by $\cos\theta$ and the new perimeter is $22\cos\theta$.

2. Ptolemy gives $AC \cdot BD = 4 \cdot 6 + 5 \cdot 7 = 59$. The area is $\sqrt{7 \cdot 6 \cdot 5 \cdot 4} = 2\sqrt{210}$, so $\sin\theta = 4\sqrt{210}/59$.

3. Therefore $\cos\theta = 11/59$, and the perimeter is $242/59$. The requested sum is 301.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2021-II-Q14] | Part C (Q11-Q15)

Answer: 592

1. Let M be the midpoint of BC. The centroid gives A, G, M collinear. Since $OA \perp AX$ and $OG \perp GX$, points O, A, X, G are cyclic. Also $OM \perp BC$ and $OG \perp GY$, so O, G, Y, M are cyclic.

2. The two cyclic quadrilaterals imply the same directed angle modulo 180°: $\angle(OX, OA) = \angle(GX, GA)$ and $\angle(OY, OM) = \angle(GY, GM)$. The right-hand angles are equal modulo 180° because X, G, Y and A, G, M are collinear. Hence ordinary $\angle XOY$ is either $\angle AOM$ or its supplement; this step alone does not fix the rays.

3. Write $B = 13t, C = 2t, A = 180^\circ - 15t$. Acuteness forces $6 < t < 90/13$. Consequently the given $\angle XOY = 17t$ is obtuse. The ray OM points toward the midpoint of the minor arc BC; the chord/arc-angle relations give $\angle AOM = A + 2C = 180^\circ - 11t$, also obtuse throughout this interval. Therefore the supplementary alternative is excluded.

4. We must have $17t = 180^\circ - 11t$, yielding $t = 45/7$. Thus $A = 585/7$ degrees; all three angles are acute and the indicated nondegenerate intersections exist. The reduced numerator-denominator sum is $585 + 7 = 592$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2022-I-Q11] | Part C (Q11-Q15)

Answer: 150

1. The tangent lengths from A, C are $\sqrt{3 \cdot 12} = 6$ and $\sqrt{16 \cdot 25} = 20$. Writing $AB = x$, equal tangents at B give $BC = x + 14$.

2. Set $\angle BAD = 2t$ and circle radius r. The tangent length from A gives $r = 6 \tan t$. The distance between the parallel tangent sides is $x \sin 2t = 2r$, so $x \cos^2 t = 6$.

3. The cosine law along $AC = 28$ gives $784 = x^2 + (x + 14)^2 + 2x(x + 14) \cos 2t$. Substitute $\cos 2t = 12/x - 1$ to obtain $784 = 24x + 532$, hence $x = 21/2$ and $\cos^2 t = 4/7$.

4. Then $\sin 2t = 4\sqrt{3}/7$. The area is $(21/2)(49/2)(4\sqrt{3}/7) = 147\sqrt{3}$, so the requested sum is 150.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2022-II-Q15] | Part C (Q11-Q15)

Answer: 140

1. Let the tangent-circle radii be r, s, so $r + s = 15$; put $t = rs > 0$. Diagonal O_1O_2 splits the hexagon into cyclic quadrilaterals with side lists $(r, 2, s, 15)$ and $(r, 16, s, 15)$. Brahmagupta gives their squared areas $K_2^2 = 14(t + 16)$ and $K_{16}^2 = 56(t + 184)$.

2. For a cyclic quadrilateral with consecutive sides u, v, w, z, its circumradius satisfies $16K^2R^2 = (uv + wz)(uw + vz)(uz + vw)$. Both pieces share the same R. Substituting the two side lists gives $R^2 = (6750 + 169t)(t + 30)/[224(t + 16)]$ and $R^2 = (54000 + t)(t + 240)/[896(t + 184)]$.

3. Equate these positive fractions and clear denominators. The resulting equation factors as $3(t - 40)(225t^2 + 48136t + 486000) = 0$. The quadratic factor is positive for $t > 0$, so $t = 40$.

4. The areas are therefore $\sqrt{14 \cdot 56} = 28$ and $\sqrt{56 \cdot 224} = 112$. Convexity makes their interiors disjoint on opposite sides of the diagonal, so the required total is 140.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2011-I-Q04] | Part A (Q1-Q5)

Answer: 056

1. Reflect C across the B-bisector to P on ray BA, giving $BP = BC = 120$. Reflect it across the A-bisector to Q on ray AB, giving $AQ = AC = 117$. Both points lie within AB.

2. The perpendicular feet M, N are the midpoints of CP, CQ. Hence $MN = PQ/2$ by the midpoint theorem.

3. Along the 125-unit base, $PQ = 120 + 117 - 125 = 112$, so $MN = 56$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2011-II-Q04] | Part A (Q1-Q5)

Answer: 051

1. Angle-bisector division gives $BD/DC = 20/11$. Balance masses $11, 20$ at B, C, giving mass 31 at D.

2. Assign mass 31 to A, so the A, D balance point is M. Grouping A, C first shows their balance point P lies on BM.

3. The lever ratio is $CP/PA = 31/20$, giving 51.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-I-Q03] | Part A (Q1-Q5)

Answer: 242

1. Let the A-bisector reach CD at U. Parallel-line angles imply $DU = AD = 333$. In isosceles triangle ADU, the D-bisector is a median, so P is the midpoint of AU. Similarly Q is the midpoint of BV, where $CV = 333$.

2. Choose $D = (0, 0), C = (650, 0), A = (75, h), B = (575, h)$. Then $U = (333, 0)$ and $V = (317, 0)$, yielding $P = (204, h/2)$ and $Q = (446, h/2)$.

3. The points have equal height, so $PQ = 446 - 204 = 242$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2001-II-Q07] | Part B (Q6-Q10)

Answer: 725

1. Use $P=(0,0), Q=(0,90), R=(120,0)$. The original inradius is $(90+120-150)/2=30$, so its center is $(30,30)$ and the two tangent lines are $x=60, y=60$.

2. Triangles at R, Q have similarity scales $1/2, 1/3$, hence inradii $15, 10$. Their centers are $(75,15)$ and $(10,70)$.

3. The squared separation is $65^2+55^2=7250$, giving $n=725$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2002-I-Q10] | Part B (Q6-Q10)

Answer: 148

1. $AB = 12, AC = 37$, so $BC = 35$ and $[ABC] = 210$. The bisector theorem gives $BD : DC = 12 : 37$, hence $[ADC] = 210 \cdot 37/49 = 1110/7$.

2. In AEF, the same bisector gives $EG : GF = 3 : 10$. Moreover $[AEF] = 210(3/12)(10/37) = 525/37$, so $[AGF] = (10/13)(525/37) = 5250/481$.

3. Subtract to get $[DCFG] = 1110/7 - 5250/481 \approx 147.6567$. The nearest integer is 148.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2007-II-Q09] | Part B (Q6-Q10)

Answer: 259

1. Using a rectangle coordinate system, $BE = DF = \sqrt{63^2 + 84^2} = 105$, $BF = DE = 364$, and $EF = \sqrt{63^2 + 280^2} = 287$. Both triangles have semiperimeter 378.

2. Tangent lengths give $EP = 378 - 364 = 14$ and $FQ = 14$. Therefore $PQ = 287 - 28 = 259$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2016-I-Q06] | Part B (Q6-Q10)

Answer: 013

1. The point order is C, I, L, D. Let half-angles at A, C be u, v. The exterior angle DIA equals $u+v$, while $DAI = DAB + BAI = v+u$ by the equal arcs AD, DB. Thus $DA = DI = 5$.

2. Triangles DAL, DCA are similar: their angle at D agrees and $DAL = DCA$ by the inscribed-angle theorem. Consequently $DA^2 = DL \cdot DC$, giving $DC = 25/3$.

3. Subtract DI along the bisector: $IC = DC - DI = 25/3 - 5 = 10/3$. Hence $p+q = 13$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2024-II-Q10] | Part B (Q6-Q10)

Answer: 468

1. Euler's center identity gives $OI^2 = R^2 - 2Rr$. In right triangle OIA, $IA^2 = R^2 - OI^2 = 2Rr = 156$. Since $IA = r/\sin(A/2)$, $\sin^2(A/2) = 3/13$.

2. It follows that $\sin A = 2\sqrt{30}/13$, $a = 2R\sin A = 4\sqrt{30}$, and $s - a = r\cot(A/2) = 2\sqrt{30}$, hence $s = 6\sqrt{30}$.

3. Equate $rs = bc\sin A/2$: $bc = 2 \cdot 6 \cdot 6\sqrt{30}/(2\sqrt{30}/13) = 468$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2026-II-Q08] | Part B (Q6-Q10)

Answer: 245

1. Put $a = AB = BC, b = AC$ and $u = AI/b = CI/b$. Heron plus the right triangle at the incircle contact point gives $u^2 = a/(2a + b)$, with $u > 0$. The perimeter ratio becomes $125(1 - 2u^2)(2u + 1) = 6$.

2. This factors as $(10u - 7)(50u^2 + 60u + 17) = 0$. The quadratic factor is positive for $u > 0$, so $u = 7/10$, yielding $a/b = 49/2$.

3. Integer sides require $a = 49t, b = 2t$ for positive integer t. Then $AI = 7t/5$ is integral precisely when $5 \mid t$. Taking $t = 5$ gives sides $245, 245, 10$ and $7, 7, 10$, so the minimum is 245.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2009-I-Q12] | Part C (Q11-Q15)

Answer: 011

1. Line AB is tangent at D, so this circle is the incircle of ABI. Let base tangent lengths be $u = AD, v = BD$, and the equal tangent lengths from I be w. Its inradius is $CD/2 = \sqrt{uv}/2$, using the right-triangle altitude identity.

2. The semiperimeter is $s = u + v + w$. Equate area rs to Heron $\sqrt{suvw}$: $s^2uv/4 = suvw$, so $s = 4w$. Thus $AB = u + v = 3w$.

3. The perimeter ratio is $2s/AB = 8/3$, giving 11.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2010-I-Q15] | Part C (Q11-Q15)

Answer: 045

1. Let $k = AM/CM$ and $d = BM$. The area ratio is k; equal area-to-semiperimeter ratios give $(12 + AM + d)/(13 + CM + d) = k$. Since $AM = kCM$, this reduces to $d = (13k - 12)/(1 - k) > 0$, requiring $12/13 < k < 1$.

2. Using $AM = 15k/(1 + k), CM = 15/(1 + k)$, Stewart gives $d^2 = (144 + 169k)/(1 + k) - 225k/(1 + k)^2$.

3. Equate the two formulas for d^2 and clear positive denominators. The result is $4k(69k^2 - 112k + 44) = 0$, with roots $0, 2/3, 22/23$. Only $22/23$ lies in the required interval.

4. It is attainable: $AM = 22/3, CM = 23/3, d = 10$ satisfy Stewart and the perimeter ratio. Hence $p + q = 45$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2012-I-Q12] | Part C (Q11-Q15)

Answer: 018

1. In triangle DCB, CE bisects its sixty-degree angle, so $CD/CB = DE/EB = 8/15$. Scale $CD = 8, CB = 15$.

2. The cosine rule gives $BD^2 = 64 + 225 - 120 = 169$, hence $BD = 13$. Applying it at B gives $\cos B = (169 + 225 - 64)/(2 \cdot 13 \cdot 15) = 11/13$.

3. Thus $\sin B = 4\sqrt{3}/13$, so $\tan B = 4\sqrt{3}/11$. The answer is 18.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2014-II-Q14] | Part C (Q11-Q15)

Answer: 077

1. Let the bisector meet the circumcircle again at E. It is the midpoint of arc BC not containing A, so its perpendicular projection onto line BC is M.

2. Projection takes collinear points A, P, E to H, N, M and preserves affine ratios. Since N is the midpoint, P is the midpoint of AE.

3. The angle at B is 105°. Arc angles give $\angle CBE = 15^\circ$, hence $\angle ABE = 120^\circ$, while $\angle AEB = 45^\circ$. The sine rule gives $AE = 10 \sin 120^\circ / \sin 45^\circ = 5\sqrt{6}$.

4. Thus $AP^2 = AE^2/4 = 75/2$, and the answer is 77.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2015-I-Q11] | Part C (Q11-Q15)

Answer: 108

1. Let $AB = t, BC = c$, and let M be the base midpoint. In the right triangles ABM, IBM, $\cos B = c/(2t)$ and $\cos(B/2) = c/16$. The double-angle identity gives $t = 64c/(c^2 - 128)$.

2. Positivity requires $c > \sqrt{128}$, while nondegeneracy and $BM < BI$ give $c < 16$. Thus only $c = 12, 13, 14, 15$ need testing.

3. The corresponding t values are $48, 832/41, 224/17, 960/97$. Only 48 is integral, and sides $48, 48, 12$ form a valid triangle with the derived half-angle relation. Its perimeter is 108.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2018-I-Q13] | Part C (Q11-Q15)

Answer: 126

1. The angle I_1AI_2 is constant at $A/2$. Put $\alpha = \angle AXB$. Incenter angles and the sine rule give $AI_1 = 30\sin(B/2)/\cos(\alpha/2)$ and $AI_2 = 34\sin(C/2)/\sin(\alpha/2)$.

2. Thus area equals $30 \cdot 34\sin(A/2)\sin(B/2)\sin(C/2)/\sin\alpha$. It is bounded below by its constant numerator. The altitude foot from A lies inside BC because its distance from B is $(30^2 + 32^2 - 34^2)/64 = 12$, so $\alpha = 90^\circ$ attains that bound.

3. For $a = 32, b = 34, c = 30, s = 48$, the half-angle product is $(s-a)(s-b)(s-c)/(abc)$. The minimum is therefore $16 \cdot 14 \cdot 18/32 = 126$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2020-I-Q13] | Part C (Q11-Q15)

Answer: 036

1. Let M be the midpoint of AD. In the circle through A, B, D, the bisector at B meets the midpoint of arc AD away from B; this is E, because it is equidistant from A, D. Thus $\angle EAD = B/2$. Similarly $\angle FAD = C/2$, on the opposite side.

2. Therefore $[AEF] = \frac{1}{2}AM^2(\tan(B/2) + \tan(C/2))$. The bisector theorem gives $BD = 2, DC = 3$, and $AD^2 = AB \cdot AC - BD \cdot DC = 18$, so $AM^2 = 9/2$.

3. Cosine law gives $\cos B = 1/8$, $\cos C = 3/4$. Hence the half-angle tangents are $\sqrt{7}/3$ and $1/\sqrt{7}$, with sum $10\sqrt{7}/21$.

4. The area is $15\sqrt{7}/14$, so $15 + 7 + 14 = 36$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-I-Q14] | Part C (Q11-Q15)

Answer: 459

1. To prove the midpoint-cleaver lemma, put C at the origin and use unit vectors u along CA and v along CB. Write $a = CB, b = CA, c = AB$ and $M = (bu + av)/2$. If $a \geq b$, let $X = M - b(u+v)/2 = ((a-b)/2)v$, a point on CB. Then MX is parallel to $u+v$, the internal bisector direction at C. The boundary route $M - A - C - X$ has length $c/2 + b + (a - b)/2 = (a+b+c)/2$; the other route $M - B - X$ has the same length $c/2 + a - (a-b)/2$. Their common cut segment adds equally to both polygon perimeters. If $b > a$, swap the roles and choose $X = ((b-a)/2)u$. Thus the midpoint perimeter bisector is parallel to the opposite internal angle bisector.

2. Apply the lemma at the midpoints of AB and AC: the two lines are parallel to the internal bisectors at C and B. Their acute angle is $90^\circ - A/2$, so the given 30° forces $A = 120^\circ$.

3. Put $b = AC, c = AB, p = b + c, k = |b - c|$. The cosine law gives $219^2 = b^2 + c^2 + bc$, hence $k^2 = 4 \cdot 219^2 - 3p^2$. Modulo 3, then modulo 9, both k and p are multiples of 3.

4. Write $p = 3r, k = 3u$. Then $u^2 + 3r^2 = 146^2$. Triangle inequality and a nonnegative discriminant restrict r to $74, 75, \ldots, 84$. The values $146^2 - 3r^2$ are $4888, 4441, 3988, 3529, 3064, 2593, 2116, 1633, 1144, 649, 148$; only $2116 = 46^2$ is a square.

5. Thus $r = 80, p = 240, k = 138$, so the side lengths are $189, 51, 219$. They satisfy the strict triangle inequality and the cosine equation. The perimeter is 459.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-II-Q11] | Part C (Q11-Q15)

Answer: 180

1. Put $A=(0,0), D=(7,0)$ and the midpoint $M=(u,h)$ above the axis. Reflection across AM sends B to $(2,0)$; reflection across DM sends C to $(4,0)$. Since $MB=MC$, this implies $u=3$.

2. The half-angle tangents at A, D are $h/3, h/4$. Thus $B=(2(9-h^2)/(9+h^2), 12h/(9+h^2))$ and $C=(7-3(16-h^2)/(16+h^2), 24h/(16+h^2))$.

3. The condition $B_x+C_x=6$ reduces to $2h^2(h^2-5)=0$. Convexity gives $h>0$, hence $h=\sqrt{5}$. The coordinates become $B=(4/7, 6\sqrt{5}/7), C=(38/7, 8\sqrt{5}/7)$, and their midpoint is indeed $(3,\sqrt{5})$.

4. The shoelace formula gives area $6\sqrt{5}$; its square is 180. Both stated angles are acute for these coordinates.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-I-Q02] | Part A (Q1-Q5)

Answer: 154

1. For radius r, the width is $14r$. Nearest centers in adjacent rows form equilateral triangles of side $2r$, so the vertical row gap is $\sqrt{3}r$.

2. The height is $2r + 2\sqrt{3}r$, giving ratio $7/(1+\sqrt{3}) = (\sqrt{147}-7)/2$. Hence $p+q = 154$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2005-I-Q01] | Part A (Q1-Q5)

Answer: 942

1. If the small radius is r, the centers form a regular hexagon of side and circumradius $2r$. Internal tangency gives $2r + r = 30$, so $r = 10$.

2. The area is $K = \pi(900 - 6 \cdot 100) = 300\pi$, between 942 and 943. Its floor is 942.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2009-II-Q05] | Part A (Q1-Q5)

Answer: 032

1. Place the first small-circle center at $(0, 7)$ and the equal pair at $(\pm 4\sqrt{3}, -4)$. Equal distances to that pair force the fourth center to be $(0, y)$.

2. If its radius is r, tangency to the top circle requires $y = 4 - r$ or $10 + r$. The latter is impossible in the equation for either lower circle, so use $y = 4 - r$. Then $48 + (8 - r)^2 = (r + 2)^2$.

3. Solve $r = 27/5$, giving 32.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2000-II-Q10] | Part B (Q6-Q10)

Answer: 647

1. For inradius r, the four half-angle tangents are $r/19, r/26, r/37, r/23$, and the half-angles sum to 180°.

2. Pairing opposite half-angles gives $56r/(703 - r^2) = 49r/(r^2 - 598)$. Thus $105r^2 = 56 \cdot 598 + 49 \cdot 703 = 67935$.

3. So $r^2 = 647$. The corresponding positive half-angles give the required 180° sum; uniqueness follows since that sum increases strictly with r.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2005-II-Q08] | Part B (Q6-Q10)

Answer: 405

1. The small-center separation is 14, so $(R-4)+(R-10)=14$ and the outer radius is $R=14$. Its center lies ten units from the radius-four center and four from the other.

2. Signed distance to the common tangent varies affinely along the center line. Thus the outer-center distance is $4+(10/14)(10-4)=58/7$. Chord length is $2\sqrt{14^2-(58/7)^2}=8\sqrt{390}/7$.

3. Since 390 is squarefree and $8,7$ coprime, the answer is 405.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2006-II-Q09] | Part B (Q6-Q10)

Answer: 027

1. Internal similarity centers divide center distances in radius ratios, giving tangent-line intercepts $(4,0)$ and $(16,0)$. Right triangles yield slopes $1/\sqrt{15}$ and $-1/\sqrt{3}$.

2. Equate $(x-4)/\sqrt{15} = -(x-16)/\sqrt{3}$, so $x = (4+16\sqrt{5})/(1+\sqrt{5}) = 19-3\sqrt{5}$. The requested sum is 27.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2007-I-Q09] | Part B (Q6-Q10)

Answer: 737

1. The hypotenuse is 34. Circle centers share the same perpendicular distance r from it, so their tangency-foot separation is $2r$. At the two ends, tangent distances are $r\tan(A/2)$ and $r\tan(B/2)$.

2. Half-angle formulas give $\tan(A/2) = 16/(34+30) = 1/4$ and $\tan(B/2) = 30/(34+16) = 3/5$. Hence $34 = r(1/4 + 2 + 3/5) = 57r/20$.

3. Thus $r = 680/57$ and $p + q = 737$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2014-II-Q08] | Part B (Q6-Q10)

Answer: 254

1. Let the smaller radius be s, so D has radius $3s$. Set the outer center at $(0,0)$, $A=(-2,0)$, the center of D at $(-2+3s,0)$, and the center of E at (x,s), choosing the upper orientation.

2. Internal tangency gives $x^2=4-4s$. External tangency to D gives $(x+2-3s)^2+s^2=16s^2$. Eliminate x^2 to get $x(2-3s)=3s^2+8s-4$.

3. Squaring this equation and using $x^2=4-4s$ yields $s^2(9s^2+84s-44)=0$. The only positive root is $s=(-14+4\sqrt{15})/3$. It has $0<3s<2$, and the positive value of x satisfies the unsquared equation, so it is valid.

4. Thus the requested radius is $3s=4\sqrt{15}-14=\sqrt{240}-14$, giving 254.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2022-I-Q08] | Part B (Q6-Q10)

Answer: 378

1. For the vertex-A circle, let its center distance from A be d>0. Its radius is d/2 by the 30-degree half-angle. Internal tangency is |18-d|=18-d/2. If d<=18 this forces d=0, impossible; hence d>18 and d-18=18-d/2, giving d=24 and radius12.

2. Put the circumcenter at $(0,0)$ and $A=(0,18)$. The small-circle centers are $(0,-6)$, $(3\sqrt{3},3),(-3\sqrt{3},3)$. The last two circles meet on $x=0$ at $y=3\pm\sqrt{117}$.

3. The smaller triangle has circumradius $\sqrt{117}-3$. Its side equals $\sqrt{3}(\sqrt{117}-3)=\sqrt{351}-\sqrt{27}$, giving $351+27=378$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2022-II-Q07] | Part B (Q6-Q10)

Answer: 192

1. The two external tangent lines meet at the external homothety center P. If P is beyond the smaller circle, its distances to the large and small centers have ratio $4:1$ and difference 30, so they are $40, 10$.

2. The half-angle t at P satisfies $\sin t = 24/40 = 3/5$, hence $\tan t = 3/4$. The common tangent at the circles touch point is perpendicular to the center line; its distance from P is $40 - 24 = 16$.

3. The triangle has height 16 and half-base $16 \cdot 3/4 = 12$. Its area is $16 \cdot 12 = 192$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2024-I-Q08] | Part B (Q6-Q10)

Answer: 197

1. Set $k = \cot(B/2) + \cot(C/2)$. A chain of N circles of radius ρ occupies base length $BC = \rho k + 2\rho(N-1)$, because the end-center offsets are $\rho\cot(B/2)$ and $\rho\cot(C/2)$.

2. Thus $34k + 476 = k + 4046$, whence $k = 1190/11$ and $BC = 45696/11$.

3. The incircle is the case $N = 1$, so $r = BC/k = 192/5$. Therefore $m + n = 197$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2025-II-Q06] | Part B (Q6-Q10)

Answer: 293

1. Place the large center at $(0,0)$ and take $B=(15,0), A=(9,0), C=(-15,0), D=(9,12)$. Rectangle vertices have coordinates $(9\pm a, \pm b)$ with $a^2+b^2=36$, and the specified sides make GF top and GH left.

2. The equal triangle areas are $a(12-b)=b(24-a)$, so $a=2b$ and $5b^2=36$.

3. Rectangle area $4ab=8b^2=288/5$, hence $m+n=293$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-II-Q14] | Part C (Q11-Q15)

Answer: 098

1. The tangency to AM occurs at A, so $AO = 19$. If T is the contact point on PM, equal tangents from M give $MT = MA$.

2. Let $a = MA$. Right triangles POT, PMA are similar with ratio $19/a$. The smaller perimeter is $PO + OT + PT = PA + PT = 152 - 2a$. Therefore $19/a = (152 - 2a)/152$, which gives $(a - 38)^2 = 0$.

3. The ratio is $1/2$, so $PO = PM/2$ and $PA = PO + 19$. Using $PA + PM = 114$ gives $PM = 190/3$, $PO = 95/3$. Thus the answer is 98.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-II-Q15] | Part C (Q11-Q15)

Answer: 282

1. Two circles meeting twice must lie on the same side of each common tangent; the common point above the x-axis puts both centers above it. Their centers lie on one angular bisector. Write them as (x_i, tx_i) with radius $tx_i > 0$.

2. The common point gives $x_i^2 - (18 + 12t)x_i + 117 = 0$. Thus $x_1 x_2 = 117$ and $r_1 r_2 = 117t^2 = 68$. An obtuse upper sector would have $t < -1$, incompatible with $t^2 = 68/117 < 1$, so choose $t > 0$.

3. The line slope is $m = 2t/(1 - t^2) = 12\sqrt{221}/49$. Hence $a + b + c = 12 + 221 + 49 = 282$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2004-II-Q12] | Part C (Q11-Q15)

Answer: 134

1. The trapezoid height is $\sqrt{25-1}=\sqrt{24}$. Equal tangency distances to each equal-radius pair put the inner center on the symmetry axis. If its radius is r, its distances from the two base lines are $\sqrt{r^2+6r}$ and $\sqrt{r^2+4r}$.

2. Their sum is $\sqrt{24}$. Isolating and squaring gives $\sqrt{24(r^2+4r)}=12-r$, then $23r^2+120r-144=0$.

3. The positive root is $(-60+48\sqrt{3})/23$; it is positive, below 12, and satisfies the original radical equation. The parameter sum is 134.

4. The root gives a circle wholly inside the trapezoid: both base distances exceed r, and its distance to either leg is $(3\sqrt{24}-\sqrt{r^2+6r})/5>r$. Internal tangency to one member of a base pair is impossible: the two equal corner circles are externally tangent to each other, so such a circle touching the other member would have to touch the base line, contradicting containment with positive radius. Thus all four distance sums used above are valid.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2007-II-Q15] | Part C (Q11-Q15)

Answer: 389

1. Let original inradius and circumradius be r, R, and the common small radius x. The three corner-circle centers form the triangle homothetic to the original about its incenter, with ratio $(r-x)/r$. Their circumradius is $R(r-x)/r$.

2. The fourth center is that circumcenter, distance $2x$ from all three. Heron gives area 84, so $r=4, R=65/8$. Hence $2x=(65/8)(4-x)/4$.

3. Solve $129x=260$, giving $x=260/129$ and answer 389.

4. The original triangle is acute because $15^2<13^2+14^2$. Its circumcenter, and therefore the fourth center, is inside the homothetic corner-center triangle. Every point of that triangle is at least x from each original side, so the fourth radius-x disk is inside the original triangle; the construction is attainable.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2008-II-Q11] | Part C (Q11-Q15)

Answer: 254

1. Use $B=(0,0), C=(56,0), A=(28,96)$. Half-angle tangent at either base vertex is $(24/25)/(1+7/25)=3/4$. Thus the two centers are $(104/3,16)$ and $(4q/3,q)$, where q is the unknown radius.

2. External tangency gives $((104-4q)/3)^2+(16-q)^2=(16+q)^2$, or $(26-q)^2=36q$. Hence $q=44\pm 6\sqrt{35}$.

3. The large root cannot fit inside the triangle (its diameter exceeds the altitude), so $q=44-6\sqrt{35}$ and $m+nk=44+210=254$.

4. The triangle has area 2688 and semiperimeter 128, hence inradius 21. The smaller root satisfies $0<44-6\sqrt{35}<21$. A base-corner circle of radius below the inradius is contained inside the opposite side too, so this root gives a valid circle.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2015-II-Q15] | Part C (Q11-Q15)

Answer: 129

1. The homothety centered at A with ratio -4 maps D to E. Thus $AE = 4AD$, and equal triangle areas require distances $d(B, \ell) = 4d(C, \ell)$. The same-side condition fixes the signed ratio as positive.

2. Place centers at $(0, 1), (4, 4)$, so $B = (0, 0), C = (4, 0), A = (4/5, 8/5)$. Write the line as $u(x - 4/5) + v(y - 8/5) = 0$. The distance ratio gives $-(4u + 8v) = 4(16u - 8v)$, hence $v/u = 17/6$ and $\ell : 6x + 17y = 32$.

3. The distances from B and the small-circle center to this line are $32/\sqrt{325}$ and $15/\sqrt{325}$. Therefore its small-circle chord is $AD = 2\sqrt{1 - 225/325} = 20/\sqrt{325}$.

4. The common area is $\frac{1}{2}(32/\sqrt{325})(20/\sqrt{325}) = 64/65$. Hence the requested sum is 129.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2019-I-Q11] | Part C (Q11-Q15)

Answer: 020

1. Put base $BC = a$, equal sides b, altitude $h = \sqrt{b^2 - a^2/4}$. The inradius is $r = ah/(a + 2b)$, base-exradius $r_A = ah/(2b - a)$, and other exradii equal h. Coordinates may be chosen with incenter $(0, r)$ and excenters $(0, -r_A)$, $(\pm b, h)$.

2. If the target circle radius is ρ, internal tangency forces $\rho = r + 2r_A$, and external tangency gives $b^2 + (h - r)^2 = (h + r + 2r_A)^2$. Therefore $b^2 = 4(r + r_A)(h + r_A)$. Substitute the radii and $h^2 = (4b^2 - a^2)/4$ to get $b^2 = 8ab^2/(2b - a)$, hence $2b = 9a$.

3. Integer sides are $a = 2t, b = 9t$ for $t \geq 1$, giving perimeter $20t$. At $t = 1$ all tangency formulas hold with positive radii, so the minimum is 20.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2020-II-Q13] | Part C (Q11-Q15)

Answer: 060

1. Equal tangents at each vertex give tangent lengths $3, 2, 4, 2, 4$. If the inradius is r, the half-angle at a vertex of tangent length t has tangent r/t. The five half-angles sum to 270°.

2. Put $\beta = \arctan(r/2)$ and $\gamma = \arctan(r/4)$. Then $r/3 = \cot(2\beta + 2\gamma)$. With $T = \tan(\beta + \gamma) = 6r/(8 - r^2)$, this gives $2rT = 3(1 - T^2)$ and hence $5r^4 - 84r^2 + 64 = 0$.

3. The candidates are $r^2 = 16$ and $4/5$. The latter has every half-angle below 45°, so cannot sum to 270°. For $r = 4$ the half-angles have the required sum (the sum is strictly increasing in r).

4. The semiperimeter is 15, so area $= 15r = 60$.

Source / 来源

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2021-I-Q13] | Part C (Q11-Q15)

Answer: 672

1. Let the third center and radius be O, r, and the required center distance be d. External tangency makes the difference of the powers of O equal to $(r+961)^2-961^2-[(r+625)^2-625^2]=672r$.

2. The power difference is a linear function vanishing on AB, with gradient magnitude $2d$. Hence its absolute value at O is $2d\,\mathrm{dist}(O,AB)$.

3. The 120° chord gives distance $r\cos 60^\circ = r/2$. Thus $672r = dr$, giving $d = 672$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-I-Q02] | Part A (Q1-Q5)

Answer: 021

1. The reflected points are $(v,u),(-v,u),(-v,-u),(v,-u)$. The rectangle $BCDE$ contributes $4uv$ and triangle ABE contributes $u(u-v)$.

2. Thus $u(u+3v)=451=11\cdot 41$. Because $u<u+3v<4u$, only $u=11, u+3v=41$ fits. Hence $v=10$ and $u+v=21$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2001-II-Q04] | Part A (Q1-Q5)

Answer: 067

1. Write $P=(8t,15t), Q=(10u,3u)$. The midpoint gives $8t+10u=16, 15t+3u=12$, so $t=4/7, u=8/7$.

2. The displacement is $(48/7,-36/7)$, whose length is $60/7$. Thus the answer is 67.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2009-I-Q04] | Part A (Q1-Q5)

Answer: 177

1. Use affine coordinates $A=(0,0), B=(1,0), D=(0,1), C=(1,1)$. Then MN has intercept equation $1000x/17+2009y/17=1$.

2. On diagonal $x=y=t$, this gives $3009t=17$. Thus $AC/AP=1/t=3009/17=177$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2009-II-Q03] | Part A (Q1-Q5)

Answer: 141

1. With $A=(0,0), B=(100,0), D=(0,h)$, vectors $AC=(100,h)$ and $BE=(-100,h/2)$ have dot product $-10000+h^2/2$.

2. Perpendicularity yields $h=100\sqrt{2}$, between 141 and 142. The answer is 141.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q02] | Part A (Q1-Q5)

Answer: 036

1. Put $D=(0,0), B=(12,10)$. For a unit vector (u,v) with positive components, the conditions give $E=(12-9u, 10-9v), F=(8u,8v)$.

2. Equal heights imply $17v=10$. Thus $u=\sqrt{1-100/289}=3\sqrt{21}/17$.

3. The horizontal separation is $|12-17u|=3\sqrt{21}-12$. The points are inside the rectangle and the line reaches AD, so the valid value gives $3+21+12=36$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q03] | Part A (Q1-Q5)

Answer: 031

1. The axes have equations $L : 5x - 12y - 132 = 0$ and $M : 12x + 5y - 90 = 0$. New coordinates are signed perpendicular distances to the opposite axes.

2. At the point, the M expression is -123, while it is positive at A, so the new horizontal coordinate is $-123/13$. The L expression is -526, with the same negative sign as at B, so the new vertical coordinate is $526/13$.

3. The sum is $403/13 = 31$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-II-Q02] | Part A (Q1-Q5)

Answer: 810

1. Take $B=(0,0), C=(s,0), A=(0,s), D=(s,s)$ and $E=(a,s), F=(b,0)$. Equal BE, FD gives $a=s-b$.

2. Then $EF=BE$ yields $(2a-s)^2=a^2$, so $a=s/3$ or $a=s$. The latter identifies $E=D, F=B$, excluded by the separately named nonvertex points.

3. Therefore $900=s^2+(s/3)^2=10s^2/9$, and the area is 810.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-II-Q04] | Part A (Q1-Q5)

Answer: 040

1. The midpoint is $(3/2, \sqrt{3})$. Add or subtract $\sqrt{3}/2$ times a ninety-degree rotation of $B - A$ to get the possible third vertices. The first-quadrant one is $C = (9/2, \sqrt{3}/2)$.

2. The centroid, also the center, is $(5/2, 5\sqrt{3}/6)$. Thus $xy = 25\sqrt{3}/12$, giving 40.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2015-I-Q04] | Part A (Q1-Q5)

Answer: 507

1. Choose $B=(0,0), A=(-16,0), C=(4,0)$. Equilateral apices are $D=(-8,8\sqrt{3}), E=(2,2\sqrt{3})$, so $M=(-7,\sqrt{3}), N=(-2,4\sqrt{3})$.

2. Use the determinant area directly: $x=\frac{1}{2}|(-7)(4\sqrt{3})-(-2)(\sqrt{3})|=13\sqrt{3}$. Thus $x^2=507$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2018-II-Q04] | Part A (Q1-Q5)

Answer: 023

1. Use coordinates $C=(0,0), A=(1,1), R=(2,1), O=(3,0), L=(3,-1), I=(2,-2), N=(1,-2), E=(0,-1)$. They give the required alternating lengths and exterior turns of forty-five degrees.

2. Lines AI, RN meet CO at $Y=(4/3,0), Z=(5/3,0)$ and intersect at $W=(3/2,-1/2)$. The region consists of four congruent corner triangles like CAY and two central ones like YZW.

3. The former have area $(4/3)\cdot 1/2=2/3$; the latter $(1/3)(1/2)/2=1/12$. Hence total area $4(2/3)+2(1/12)=17/6$, giving 23.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2019-II-Q01] | Part A (Q1-Q5)

Answer: 059

1. Take $A=(0,0), B=(9,0)$. Distance equations give $C=(15,8)$ and reflection $D=(-6,8)$.

2. The overlap is the triangle with base AB and top $AC \cap BD$. Equating $y=8x/15$ and $y=8(9-x)/15$ gives height $12/5$.

3. Area $9(12/5)/2=54/5$, so the answer is 59.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2020-II-Q04] | Part A (Q1-Q5)

Answer: 108

1. The direction $(0, 12)$ becomes $(12, 0)$, so $m = 90$. A clockwise quarter-turn around (x, y) maps (p, q) to $(x + q - y, y - p + x)$.

2. The origin image gives $x - y = 24, x + y = 18$, hence $(x, y) = (21, -3)$. Substitution also gives both remaining images, so the requested sum is $90 + 21 - 3 = 108$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2021-I-Q02] | Part A (Q1-Q5)

Answer: 109

1. Set $A=(0,3), B=(0,0), C=(11,0), D=(11,3)$. The stated orientation and lengths give $F=(28/5,36/5)$ and $E=(27/5,-21/5)$. These satisfy the distance equations and $A+C=E+F$.

2. The second rectangle is bounded by $9 \leq 4x+3y \leq 44$ and $-12 \leq 3x-4y \leq 33$. Throughout $0 \leq x \leq 11, 0 \leq y \leq 3$, the latter pair already holds.

3. The overlap is therefore a parallelogram of height 3 and horizontal width $(44-9)/4 = 35/4$. Its area is $105/4$, giving 109.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-I-Q05] | Part A (Q1-Q5)

Answer: 106

1. Take the circle radius R and vertices $(R,0),(0,R),(-R,0),(0,-R)$, with $P=(R\cos t, R\sin t)$.

2. Distance formulas give $PA\,PC = 2R^2|\sin t|$ and $PB\,PD = 2R^2|\cos t|$. Squaring and adding yields $(2R^2)^2 = 56^2 + 90^2 = 11236$.

3. The square area is half the squared diagonal, namely $2R^2 = 106$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2024-I-Q05] | Part A (Q1-Q5)

Answer: 104

1. Use $D=(0,0), C=(107,0), A=(0,-16), E=(e,0), H=(e,17), G=(e+184,17)$.

2. The circle center is $(e+92,-8)$ by the perpendicular bisectors of AD and HG. Its distances to D, H give $(e+92)^2+64=92^2+625$.

3. Since $e>0$, $e+92=95$, so $e=3$ and $CE=107-3=104$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2025-II-Q01] | Part A (Q1-Q5)

Answer: 468

1. Use line coordinates $A = 0, C = 26, F = 73, D = 40, B = 18, E = 57$. If $G = (u, v)$, subtract $(u - 26)^2 + v^2 = 1600$ and $(u - 40)^2 + v^2 = 900$ to obtain $u = 58$.

2. Then $v^2 = 576$, so height is 24 and base $BE = 39$. Area $39 \cdot 24/2 = 468$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2026-II-Q03] | Part A (Q1-Q5)

Answer: 503

1. Take $A=(0,0), E=(20,0), B=(0,h)$. The angle conditions and nonconvex order give $C=(14,h-14)$ and $D=(20,h-8)$. Same-side strictness requires $h>14$.

2. Shoelace gives area $20h-164$. Divisibility by sixteen is $20h-164\equiv 0 \pmod{16}$, or $h\equiv 1 \pmod{4}$.

3. The possible integers are $17,21,\ldots,2025$, numbering $(2025-17)/4+1=503$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-II-Q08] | Part B (Q6-Q10)

Answer: 110

1. Use $B=(0,0), A=(\sqrt{11},0), C=(0,h), D=(d,h)$. The diagonal dot product gives $h^2=\sqrt{11}d$.

2. Put $u=h^2$. The leg condition is $(u-11)^2/11+u=1001$, yielding $u^2-11u-10890=0=(u-110)(u+99)$. Since $u>0$, $BC^2=110$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2004-II-Q07] | Part B (Q6-Q10)

Answer: 293

1. Folding preserves $EB' = EB = 17$, so right triangle AEB' gives $AB' = 15$, while $AB = 25$. Set $A = (0,0), B = (25,0), B' = (0,15), E = (8,0)$.

2. The crease is the perpendicular bisector of BB', whose slope is $-3/5$; its own equation is $y = 5(x-8)/3$. At F, $x = 25 - 3 = 22$, giving height $BC = 70/3$.

3. The perimeter is $2(25 + 70/3) = 290/3$, so the sum is 293.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2005-I-Q10] | Part B (Q6-Q10)

Answer: 047

1. The midpoint is $(35/2, 39/2)$, so the median line is $q = -5p + 107$.

2. Signed double area gives $|980 - 56p| = 140$, hence $p = 15$ or 20. Since $p + q = 107 - 4p$, its maximum is 47.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2006-I-Q10] | Part B (Q6-Q10)

Answer: 065

1. The line through $(1, 1/2)$ and $(3/2, 2)$ has slope three. Each point is the symmetry center of one pair of tangent disks, so the line bisects those four disks in pairs.

2. It misses the other four disks, leaving two on each side; their centers have signed line values in opposite pairs of safe magnitudes. Thus it bisects the union. Its equation is $y = 3x - 5/2$, or $6x = 2y + 5$.

3. The answer is $36 + 4 + 25 = 65$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2006-II-Q06] | Part B (Q6-Q10)

Answer: 012

1. Take $A=(0,0), B=(1,0), C=(1,1), D=(0,1)$. Equal lengths $AE=AF$ force $E=(1,t), F=(t,1)$. Then $1+t^2=2(1-t)^2$, giving $t=2-\sqrt{3}$.

2. For smaller side u, its opposite vertex is $(1-u,u)$ and lies on $y=tx$. Thus $u=t/(1+t)=(3-\sqrt{3})/6$, so the requested sum is 12.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2009-II-Q10] | Part B (Q6-Q10)

Answer: 096

1. Take $A=(0,0), B=(5,0), C=(5,12)$. The bisector through A has slope $(12/13)/(1+5/13)=2/3$.

2. At C, the ray toward D is the reflection of CA across vertical line CB, so it lies through $(10,0)$, with equation $5y=120-12x$.

3. Intersection gives $D=(180/23, 120/23)$, hence $AD=60\sqrt{13}/23$ and the answer 96.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2010-II-Q09] | Part B (Q6-Q10)

Answer: 011

1. Normalize the circumradius to 1, taking $A = (1, 0), B = (1/2, \sqrt{3}/2), C = (-1/2, \sqrt{3}/2), F = (1/2, -\sqrt{3}/2)$. Then $H = (0, \sqrt{3}/2), G = (3/4, \sqrt{3}/4)$.

2. A small-hexagon vertex is $AH \cap FG$. Its equations $x + 2y/\sqrt{3} = 1$ and $y = 3\sqrt{3}x - 2\sqrt{3}$ give $(x, y) = (5/7, \sqrt{3}/7)$.

3. Sixty-degree symmetry makes the inner hexagon concentric and regular. The area ratio is the squared circumradius ratio, $25/49 + 3/49 = 4/7$. Hence $m + n = 11$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-I-Q07] | Part B (Q6-Q10)

Answer: 041

1. With height h, those centers can be $(6,8,0),(0,8,h/2),(6,0,h/2)$. Two side vectors are $(-6,0,h/2),(0,-8,h/2)$.

2. Their cross product is $(4h,3h,48)$, so area $\frac{1}{2}\sqrt{25h^2+2304}=30$.

3. Thus $h=36/5>0$, and $m+n=41$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2014-I-Q10] | Part B (Q6-Q10)

Answer: 058

1. If the moving center sweeps angle ϕ around E, no-slip rolling causes rotation 5ϕ relative to the line of centers, plus ϕ from the turning line itself. Hence ground-frame rotation is 6ϕ, and one turn gives $\phi = 60^\circ$.

2. Take $E = (0, 0)$, initial moving center $(6, 0)$, and $A = (5, 0)$. The final center is $(3, 3\sqrt{3})$. A full rotation restores the marked point offset $(-1, 0)$, so $B = (2, 3\sqrt{3})$.

3. Thus the squared sine is $27/(4 + 27) = 27/31$, yielding 58.

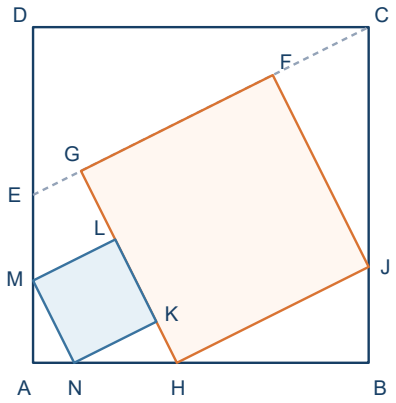
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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2015-I-Q07] | Part B (Q6-Q10)

Answer: 539



1. Normalize the outer square to $A=(0,0), B=(1,0), C=(1,1), D=(0,1)$. Its line CE is $y=x/2+1/2$. Write $H=(h,0)$ and $J=(1,(1-h)/2)$ because $HJ \parallel CE$.

2. Rotating HJ by 90° toward CE gives $G=(h-(1-h)/2, 1-h)$. Requiring G on CE yields $h=3/7$. Consequently $[FGHJ]=(1-h)^2[1+(1/2)^2]=20/49$.

3. Line GH is $y=-2x+6/7$. Put $N=(u,0), M=(0,2u)$; the second square then has $K=(3u,u), L=(2u,3u)$. Its K incidence gives $u=6/49$, hence $[KLMN]=5u^2=180/2401$.

4. The area ratio is $(20/49)/(180/2401)=49/9$. Rescaling the inner area to 99 gives outer inner-square area $99\cdot 49/9=539$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2016-II-Q07] | Part B (Q6-Q10)

Answer: 840

1. Let side lengths be s, t, u, from outer to inner, and rotation angle be θ. Coordinate projections give $s = t(\cos\theta + \sin\theta)$ and $t = u(\cos\theta + \sin\theta)$. The same center follows from the equal extreme coordinates on the four supporting sides.

2. The areas form a geometric progression. If the middle area is T, the inner is $T^2/2016$, and T ranges continuously over $[1008, 2016)$ because $(\cos\theta + \sin\theta)^2$ ranges over $(1, 2]$.

3. For integral T, inner-area integrality requires $2016 \mid T^2$. Since $2016 = 2^5 \cdot 3^2 \cdot 7$, this is equivalent to $168 \mid T$. The allowed extreme multiples are 1008 and 1848, both attainable by rotation. Their difference is 840.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2017-II-Q10] | Part B (Q6-Q10)

Answer: 546

1. Use $A=(0,0), B=(84,0), C=(84,42), D=(0,42)$. Then $CM: y=x/4+21$ and $DN: y=42-3x/2$ intersect at $O=(12,24)$; $N=(28,0)$.

2. Shoelace gives $[BCON]=2184$, with target half1092. Since $[BCO]=1512>1092$, the boundary intersection P lies on CO. From base $BC=42$, $[BCP]=1092$ requires horizontal distance52 from P to BC.

3. Thus $P_x=32, P_y=32/4+21=29$. Finally $[CDP]=(84/2)(42-29)=546$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2018-I-Q08] | Part B (Q6-Q10)

Answer: 147

1. Turn successive edge directions by sixty degrees. Closing the vector chain gives $EF = 2, FA = 16$. Coordinates may be $A = (0, 0), B = (6, 0), C = (10, 4\sqrt{3}), D = (5, 9\sqrt{3}), E = (-7, 9\sqrt{3}), F = (-8, 8\sqrt{3})$.

2. The opposite-side strip widths are $9\sqrt{3}, 11\sqrt{3}, 7\sqrt{3}$. Thus any disk has diameter at most the narrowest width $7\sqrt{3}$.

3. To attain it, take center $(3/2, 11\sqrt{3}/2)$ and radius $7\sqrt{3}/2$. Its distances to CD, FA, DE equal the radius; those to AB, BC, EF are $11\sqrt{3}/2, 5\sqrt{3}, 6\sqrt{3}$, all larger. It lies in all six inward half-planes, hence the disk fits.

4. Therefore $d^2 = 147$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2018-II-Q09] | Part B (Q6-Q10)

Answer: 184

1. For each fan triangle JVW, its centroid is two-thirds of the way from J to the midpoint of VW. Hence the centroid polygon is a homothety of ratio $2/3$ of the polygon of midpoints of $AB, BC, \ldots, GH$.

2. Those midpoints are $(4,3), (27/2,0), (23,3), (27,23/2), (23,20), (27/2,23), (4,20)$. Split their heptagon by vertical lines at $x = 4, 27/2, 23$.

3. Two trapezoids of heights $19/2$ and parallel lengths seventeen, twenty-three contribute 380; the right triangle at the right contributes $17 \cdot 4/2 = 34$. Thus midpoint-polygon area is 414.

4. The target area is $(4/9)414 = 184$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2021-I-Q09] | Part B (Q6-Q10)

Answer: 567

1. The height is 18. Set $D=(0,0), C=(u+2a,0), A=(a,18), B=(u+a,18)$, where $u=AB$. Point-line distances give $18u/\sqrt{324+a^2}=15$ and $18u/\sqrt{324+(u+a)^2}=10$.

2. Squaring and subtracting yields $u+2a=9u/5$, hence $a=2u/5$. Then $324=32u^2/25$, so $u=45\sqrt{2}/4$ and $CD=81\sqrt{2}/4$.

3. $K=9(AB+CD)=567\sqrt{2}/2$, giving $\sqrt{2}K=567$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-I-Q08] | Part B (Q6-Q10)

Answer: 125

1. The parallel-side separation is $9 + 16 = 25$, so the inradius is $25/2$. Put the center at the origin and the parallel sides at $y = \pm 25/2$. Then $P = (\pm 12, 7/2)$.

2. Writing $\alpha = \angle BAD \in (0, 90^\circ)$, orient AB as $x \sin\alpha + y \cos\alpha = -25/2$. Its distance condition rules out $x = 12$ and gives $24 \sin\alpha - 7 \cos\alpha = 15$.

3. Together with the unit-circle identity, this yields $625s^2 - 720s + 176 = 0$, for $s = \sin\alpha$. The roots are $4/5, 44/125$; the latter would make $\cos\alpha < 0$, so $s = 4/5$.

4. The side is height divided by sine, $25/(4/5) = 125/4$, and the perimeter is 125.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2025-I-Q08] | Part B (Q6-Q10)

Answer: 077

1. The first locus is the circle centered at $(25, 20)$ with radius 5. The second is the perpendicular bisector of $(k+4, 0)$ and $(k, 3)$, namely $8x - 6y - 8k - 7 = 0$.

2. Unique intersection means tangency, so the center-to-line distance is $|73 - 8k|/10 = 5$. Thus $k = 23/8, 123/8$, whose sum is $73/4$; the requested sum is 77.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2026-I-Q10] | Part B (Q6-Q10)

Answer: 156

1. Take $B=(0,0), C=(14,0), A=(5,12)$. Equidistance gives circumcenter $O=(7,33/8)$. To turn direction $(3,-4)$ vertical, the rotation has $(\cos\phi,\sin\phi)=(4/5,-3/5)$ or its negative.

2. The opposite-side condition selects $(4/5,-3/5)$: the directed cross products for $A\prime$ and B relative to $B\prime C\prime$ are 168 and $-189/4$. This gives $A\prime=(81/8,93/8), B\prime=(-43/40,201/40), C\prime=(81/8,-27/8)$.

3. Apply the shoelace formula to the six vertices in the stated order. The area is $1557/10=155.7$, whose nearest integer is 156.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-I-Q13] | Part C (Q11-Q15)

Answer: 731

1. An optimal route can be shortened to travel on one road and then straight off-road: any off-road part that returns to a road is no faster than an appropriate road route. Six-minute distances are 5 on-road or $7/5$ off-road.

2. For the positive x-road, the endpoint disks have radius $(7/5)(1-d/5)$ at $(d,0)$. Their union has the straight tangent boundary from $(5,0)$ to the origin-centered radius-$7/5$ disk. Its upper line is $7x+24y=35$; swapping axes gives $24x+7y=35$.

3. The lines meet at $(35/31,35/31)$. Between the x-axis and the diagonal, the first line forms the outer reachable boundary; the other half is its reflection. Thus one quadrant is the quadrilateral $(0,0),(5,0),(35/31,35/31),(0,5)$, of area $175/31$.

4. The full area is $700/31$, giving 731.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-II-Q11] | Part C (Q11-Q15)

Answer: 131

1. Let $v = AD = (1, 7)$ and $w = BC$. Equal legs imply $|w|^2 = 50$, so w ranges over $(\pm 1, \pm 7), (\pm 7, \pm 1), (\pm 5, \pm 5)$. The vectors have the same component normal to the bases and opposite base components; hence $AB \parallel v - w$.

2. Exclude $w = v$ (parallelogram), $w = -v$ (zero height), and $(1, -7), (-1, 7)$ (vertical or horizontal bases). The remaining slopes are $-1, -4/3, 3/4, 1, -1/2, -3, 1/3, 2$.

3. Each is attainable with sufficiently long integer base vectors parallel to $v - w$. The absolute sum is $119/12$, giving 131.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2002-I-Q11] | Part C (Q11-Q15)

Answer: 230

1. Unfold successive reflections into neighboring cubes. Choose coordinates so the initial ray is $(12t, 7t, 5t)$ from the origin.

2. A folded vertex is reached when all three coordinates are multiples of 12. The first coordinate forces t integral, and then $7t$ and $5t$ require $12 \mid t$. The first positive value is $t = 12$.

3. The total distance is $12\sqrt{12^2 + 7^2 + 5^2} = 12\sqrt{218}$, giving 230.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2003-I-Q15] | Part C (Q11-Q15)

Answer: 289

1. Put $B = (0, 0)$ and its bisector on the x-axis. With $c = AB, a = BC$, write $A = (c\cos t, c\sin t)$ and $C = (a\cos t, -a\sin t)$. The bisector theorem gives $D = (2ac\cos t/(a + c), 0)$.

2. The line DF is vertical, giving $F_y = -2ac\sin t/(a + c)$. The midpoint is $M = ((a + c)\cos t/2, (c - a)\sin t/2)$; intersecting ray BM with that vertical gives $E_y = 2ac(c - a)\sin t/(a + c)^2$.

3. Therefore $DE/EF = (a - c)/(2c) = 147/720 = 49/240$. The requested sum is 289.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2003-II-Q11] | Part C (Q11-Q15)

Answer: 578

1. Use $A = (0,0), B = (25,0)$. From $AC = 7, BC = 24$, the x-coordinate of C is $(49 + 625 - 576)/50 = 49/25$. The midpoint has x-coordinate $25/2$.

2. The isosceles point D is vertically above M by $\sqrt{225 - 625/4} = 5\sqrt{11}/2$. Thus take DM as base: the distance of C from its vertical line is $25/2 - 49/25 = 527/50$.

3. The area is $\frac{1}{2}(5\sqrt{11}/2)(527/50) = 527\sqrt{11}/40$, giving 578.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2003-II-Q14] | Part C (Q11-Q15)

Answer: 051

1. Equal parallel opposite sides make the hexagon centrally symmetric, so opposite vertex heights sum to 10. Thus $D_y = 10, E_y = 8$. Reflect horizontally so $b > 0$. If $C_y = 4, F_y = 6$, the first two equal edge vectors are $(b, 2), (-b, 2)$, while the third is $(q, 6)$ with $|q| < b$. Their consecutive cross products are $4b > 0$ and $-6b - 2q < 0$, contradicting convexity. Hence $C_y = 6, F_y = 4$.

2. Reflect horizontally if needed to take $b > 0$. Put $AB = (b, 2), BC = (p, 4), CD = (q, 4)$. Equal lengths and noncollinearity give $p = -q$ and $q^2 = b^2 - 12$. The angle at A gives $bq + 8 = -(b^2 + 4)/2$.

3. Thus $q < 0$ and $b\sqrt{b^2 - 12} = (b^2 + 20)/2$, yielding $b^2 = 100/3$, $q = -8/\sqrt{3}$, $p = 8/\sqrt{3}$.

4. The parallelogram $ABDE$ has area $8b$, and the two remaining triangles each area $4p$. Total $8(b + p) = 48\sqrt{3}$, so the sum is 51.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2004-II-Q15] | Part C (Q11-Q15)

Answer: 593

1. Track strip length L, thickness H, current horizontal position p and bottom-up layer index b. Initially $(L, H, p, b) = (1024, 1, 942, 1)$. A left-half square keeps p, b; a right-half square changes to $p \leftarrow L + 1 - p$, $b \leftarrow 2H + 1 - b$. Then halve L and double H.

2. The successive (p, b) states are $(83, 2), (83, 2), (83, 2), (46, 15), (19, 18), (14, 47), (3, 82), (3, 82), (2, 431), (1, 594)$.

3. The final square is layer 594 from the bottom, so 593 squares lie below it.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2005-I-Q14] | Part C (Q11-Q15)

Answer: 936

1. The convex order forces A, C on opposite sides and B, D on opposite sides. If the side through B has slope m, its line is $mx - y + 9 - 10m = 0$; a perpendicular side through C has line $x + my - 8 = 0$.

2. Equal widths give $|2 - 14m| = |12m - 8|$, so $m = 5/13$ or -3. The first places A, C on opposite sides of the line through B, invalid for a supporting side. For $m = -3$ all four points lie on their required side segments.

3. The side length is $44/\sqrt{10}$, so $10K = 1936$ and the remainder is 936.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2005-II-Q12] | Part C (Q11-Q15)

Answer: 307

1. Put the side midpoint at horizontal coordinate zero and O height 450. Let signed coordinates of E, F be e, f, with $f - e = 400$. The angle tangent formula gives $1 = 450(f - e)/(450^2 + ef)$, so $ef = -22500$.

2. The roots with difference 400 are $e = -200 - 50\sqrt{7}, f = 200 - 50\sqrt{7}$; this orientation follows from $AE < BF$. Thus $BF = 450 - f = 250 + 50\sqrt{7}$.

3. The requested sum is 307.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2006-I-Q14] | Part C (Q11-Q15)

Answer: 183

1. Use top $T = (0, 0, 4)$ and feet $A = (3, 0, 0)$, $B = (-3/2, 3\sqrt{3}/2, 0)$, $C = (-3/2, -3\sqrt{3}/2, 0)$. Shortening TA by one gives $A\prime = (12/5, 0, 4/5)$.

2. The new supporting plane contains the vertical-y line BC. In the xz cross-section its equation through $(-3/2, 0), (12/5, 4/5)$ is $8x - 39z + 12 = 0$. The height is the distance from T to this plane, $144/\sqrt{1585}$.

3. $1585 = 5 \cdot 317$ is squarefree and $39 < \sqrt{1585} < 40$, so the requested floor is 183.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q14] | Part C (Q11-Q15)

Answer: 037

1. Scale the side to 2 and set $r = 1 + \sqrt{2}$, the apothem. Rotate coordinates so $M_1 = (0, r)$ and $M_3 = (-r, 0)$. An inward direction of R_3 is $(\cos\phi, \sin\phi)$; perpendicularity and the inward condition give direction $(\sin\phi, -\cos\phi)$ for R_1. The other rays are the successive quarter-turns.

2. Writing $B_1 = M_1 + a(\sin\phi, -\cos\phi) = M_3 + b(\cos\phi, \sin\phi)$ gives $a = r(\cos\phi - \sin\phi)$ and $b = r(\cos\phi + \sin\phi)$. Both are positive because the intersections lie on the rays. Rotation makes $M_3B_3 = a$, so $B_1B_3 = |a - b| = 2r|\sin\phi|$.

3. The required side length is 2, hence $\sin^2\phi = 1/r^2$. The vector M_3A_3 is vertical, so for $\beta = \angle A_3M_3B_1$ we have $\cos 2\beta = -\cos 2\phi = 2/r^2 - 1 = 5 - 4\sqrt{2} = 5 - \sqrt{32}$.

4. The choices $|\sin\phi| = 1/r < 1/\sqrt{2}$ give positive a, b and valid inward intersections, so this value is attained. Therefore $m + n = 37$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-II-Q13] | Part C (Q11-Q15)

Answer: 096

1. Set $A=(0,0), B=(12,0), C=(12,12), D=(0,12)$ and $P=(p,p)$, with $6<p<12$. Perpendicular-bisector equations give $O_1=(6,p-6), O_2=(6,p+6)$.

2. The vectors from P are $(6-p,-6),(6-p,6)$. The cosine condition is $[(p-6)^2-36]/[(p-6)^2+36]=-1/2$, giving $(p-6)^2=12$.

3. The stated inequality chooses $p=6+2\sqrt{3}$. Hence $AP=p\sqrt{2}=\sqrt{72}+\sqrt{24}$, and $a+b=96$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2012-I-Q13] | Part C (Q11-Q15)

Answer: 041

1. Let $OA = 3, OB = 4, OC = 5$. Rotate sixty degrees about A in the direction taking B to C, and call the image of O point P. Then $AP = OP = 3$ and $PC = 4$, so triangle OPC is a right 3-4-5 triangle at P.

2. The rays PA, PO make sixty degrees and PO, PC make ninety. Consequently $\angle APC$ is either thirty or one hundred fifty degrees. The cosine rule gives the complete possibilities $s^2 = 25 \pm 12\sqrt{3}$. Both occur by constructing the equilateral AOP on either side of OP and undoing the rotation to obtain B.

3. The larger area is $(\sqrt{3}/4)(25 + 12\sqrt{3}) = 9 + (25/4)\sqrt{3}$. Hence $a + b + c + d = 41$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2012-II-Q13] | Part C (Q11-Q15)

Answer: 677

1. Put $s^2 = 111, r^2 = 11$. The two rays AD_i are at signed angles $\pm\theta$ to AB. Rotating each by $\pm 60^\circ$ produces the four AE_k rays, all of length s.

2. Relative to ray AC at 60°, the four angle cosines sum to $2\cos\theta + \cos(120^\circ - \theta) + \cos(120^\circ + \theta) = \cos\theta$.

3. Thus the squared-distance sum is $2s^2(4 - \cos\theta)$. In triangle ABD_i, $r^2 = 2s^2(1 - \cos\theta)$, so this simplifies to $6s^2 + r^2 = 677$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-I-Q12] | Part C (Q11-Q15)

Answer: 021

1. The corner triangle BQC is equilateral, so take $Q=(0,0), C=(1,0), D=(2,0)$. The other vertices are $B=(1/2,\sqrt{3}/2), A=(1,\sqrt{3}), F=(2,\sqrt{3}), E=(5/2,\sqrt{3}/2)$.

2. The angle at R is forty-five degrees, so RP is a line $x+y=t$ supporting the hexagon. The greatest value of $x+y$ at a hexagon vertex occurs at F, hence $t=2+\sqrt{3}$; this also rules out touching only E.

3. Thus $QR=2+\sqrt{3}$. Intersecting with $QP: y=\sqrt{3}x$ gives height $(3+\sqrt{3})/2$. The area is $(9+5\sqrt{3})/4$.

4. The parameter sum is 21.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-II-Q13] | Part C (Q11-Q15)

Answer: 010

1. Set $B=(0,0), C=(t,0), A=(u,v)$, taking $v>0$. Equal sides give $u^2+v^2=2tu$. As $D=(t/4,0)$, the midpoint is $E=((u+t/4)/2, v/2)$.

2. The equation $BE^2=9$ becomes $(5/2)tu+t^2/16=36$. Subtracting BE^2 from CE^2 gives $tu=3t^2/4+2$.

3. Combining yields $t^2=16$, then $tu=14, u=7/2, v^2=63/4$. Thus area $tv/2=3\sqrt{7}$, and $m+n=10$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2014-I-Q13] | Part C (Q11-Q15)

Answer: 850

1. Use $B=(0,0), C=(s,0), D=(s,s), A=(0,s)$ and $E=(0,e), F=(f,0), G=(s,g), H=(h,s)$. The top two areas sum to half the square, so $e+g=s$. The left two sum to $2s^2/5$, so $f+h=4s/5$. Perpendicularity gives $h-f=e-g$.

2. Put $\delta=g-e$. Then $e=(s-\delta)/2, g=(s+\delta)/2, f=2s/5+\delta/2, h=2s/5-\delta/2$, and $s^2+\delta^2=1156$. Intersecting the lines gives $P=(s/2-s^3/11560, s/2-s^2\delta/11560)$.

3. The shoelace formula yields $[AEPH]=s^2/5-s^3\delta/231200$. Equate this with $269s^2/1360$ to obtain $s\delta=510$.

4. Thus s^2, δ^2 are roots of $t^2-1156t+260100=0$, namely $850, 306$. Since $|g-e|<s$, the square area is the larger value 850. The endpoint coordinates from these values lie on their required sides.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2014-II-Q11] | Part C (Q11-Q15)

Answer: 056

1. Take $D = (0, 0), R = (1/2, \sqrt{3}/2)$. The sine rule gives $E = ((1 + \sqrt{3})/2, 0)$ and $M = (1/4, \sqrt{3}/4)$.

2. Perpendicular slopes give the horizontal coordinate of C as $u = (7 - 3\sqrt{3})/11$, lying between D and E.

3. For $A = (t, 0)$, equate $(t - u)^2 = (t - 1/2)^2 + 3/4$. This gives $t = (1 - u^2)/(1 - 2u) = (9 + 4\sqrt{3})/11$, which lies beyond E.

4. Therefore $AE = (7 - 3\sqrt{3})/22 = (7 - \sqrt{27})/22$, giving 56.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2015-II-Q11] | Part C (Q11-Q15)

Answer: 023

1. Place B at the vector origin, and write vectors to A, C, O as a, c, o. Equal circumradii give $2o \cdot a = |a|^2 = 25$ and $2o \cdot c = |c|^2 = 16$.

2. The specified line has equation $o \cdot x = |o|^2$. If $P = ta, Q = sc$, then $25t = 16s = 2|o|^2$, so both parameters are positive and $t/s = 16/25$.

3. It follows that $BP/BQ = (5t)/(4s) = 4/5$. Thus $BP = (4/5)(9/2) = 18/5$, giving $m + n = 23$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2016-I-Q14] | Part C (Q11-Q15)

Answer: 574

1. The segment direction reduces to $(7, 3)$, repeated143 times, because $\gcd(1001, 429) = 143$. Its supporting line is $3x - 7y = 0$.

2. For square center (u, v), the linear form ranges within $|3|/10 + |7|/10 = 1$ of $3u - 7v$, so intersection requires $|3u - 7v| \leq 1$. On one primitive segment the centers are $(0, 0), (2, 1), (5, 2), (7, 3)$, with the middle two touched at corners.

3. A circle can be met only if $|3u - 7v|/\sqrt{58} \leq 1/10$. Since $\sqrt{58} < 10$, this forces $3u - 7v = 0$, so only the two primitive endpoints qualify. No centers beyond the full segment endpoints can intersect it, by the $1/10$ coordinate extents. Every center on the segment contributes a true circumference intersection, including the two full-segment endpoints, because the segment extends at least one radius into each corresponding disk.

4. There are144 lattice centers on the full segment and two additional square-only centers per primitive copy. Hence $m = 144 + 2 \cdot 143 = 430$, $n = 144$, and $m + n = 574$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2017-I-Q15] | Part C (Q11-Q15)

Answer: 145

1. Use outer vertices $(0,0),(5,0),(0,2\sqrt{3})$. If the equilateral side joining the legs has endpoints $(s\cos t,0),(0,s\sin t)$, its third vertex in the positive quadrant is $s(\cos t+\sqrt{3}\sin t,\sin t+\sqrt{3}\cos t)/2$. The other orientation has a negative coordinate and cannot lie in the triangle.

2. The hypotenuse equation is $2\sqrt{3}x+5y=10\sqrt{3}$. Substitution yields $s=20\sqrt{3}/(7\sqrt{3}\cos t+11\sin t)$. Cauchy bounds the denominator by $\sqrt{147+121}=2\sqrt{67}$.

3. Equality has $\cos t=7\sqrt{3}/(2\sqrt{67}),\sin t=11/(2\sqrt{67})$. The leg intercepts are $105/67,55\sqrt{3}/67$, within the side segments, and the third vertex is $(135/67,80\sqrt{3}/67)$ on the hypotenuse segment. Thus the bound is attained.

4. The minimum area is $(\sqrt{3}/4)(300/67)=75\sqrt{3}/67$, so $m+n+p=145$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2017-II-Q15] | Part C (Q11-Q15)

Answer: 682

1. Embed the tetrahedron at $A=(a,b,c), B=(a,-b,-c), C=(-a,b,-c), D=(-a,-b,c)$. The edge equations give $a^2+b^2=196$, $a^2+c^2=484$, $b^2+c^2=676$, so $(a^2,b^2,c^2)=(2,194,482)$. All are positive, and these coordinates realize every given edge.

2. Each vertex vector v has length $R=\sqrt{678}$, and the four vectors sum to0. For any position vector x, Cauchy gives $|v-x|\geq(v-x)\cdot(v/R)=R-v\cdot x/R$.

3. Sum this inequality over the four vertices to obtain $f(X)\geq 4R$. Equality is attained at $x=0$, so the minimum is $4\sqrt{678}$. As $678=2\cdot 3\cdot 113$ is squarefree, $m+n=682$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2019-II-Q11] | Part C (Q11-Q15)

Answer: 011

1. Take $A=(0,0), B=(7,0), C=(33/7, 24\sqrt{5}/7)$, using cosine rule. Tangency and passage through the third point give circle equations $x^2+y^2-7x+77y/(8\sqrt{5})=0$ and $x^2+y^2-189y/(8\sqrt{5})=0$.

2. Subtract to get the radical axis $y=(4\sqrt{5}/19)x$. Its positive unit direction is $(19/21, 4\sqrt{5}/21)$. Put $K=t$ times this vector in the second circle equation: $t^2=(189/(8\sqrt{5}))t(4\sqrt{5}/21)$.

3. The nonzero solution is $t=9/2$, hence $m+n=11$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2019-II-Q13] | Part C (Q11-Q15)

Answer: 504

1. Let side length be s, and u_j the outward unit normal toward midpoint of side A_jA_{j+1}. Signed triangle-area subtraction from the one-eighth sector gives region area $F_j = 1/8 - (s/2)P \cdot u_j$, valid even if P lies beyond the chord.

2. The given values imply $(s/2)P \cdot u_1 = -1/56$, $(s/2)P \cdot u_3 = 1/72$. Normals differ by forty-five-degree steps, so $u_6 = -(u_1 + u_3)/\sqrt{2}$.

3. Thus $(s/2)P \cdot u_6 = -(-1/56 + 1/72)/\sqrt{2} = \sqrt{2}/504$, giving $n = 504$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2020-I-Q15] | Part C (Q11-Q15)

Answer: 058

1. With circumcenter O as vector origin, $H = A + B + C$. The circle HBC is the reflection of ω across BC, so its center is $B + C = H - A$. Its radius toward H is therefore parallel to OA, and $XY \perp OA$.

2. Set $A = (0, R)$ and $H = (h, k)$ with chord XY horizontal. Since H is inside the circle and HX, HY are $2, 6$, the half-chord is 4 and $h^2 = 4$. Thus $R^2 - k^2 = 16$, while $HA^2 = 4 + (R - k)^2 = 9$. Hence $R - k = \sqrt{5}$ and $R = 21/(2\sqrt{5})$.

3. The midpoint M of BC is $(H - A)/2 = (h/2, -\sqrt{5}/2)$, giving $OM = 3/2$. Hence $BC = 2\sqrt{R^2 - 9/4} = 2\sqrt{99/5}$.

4. The distance from A to the chord line BC, whose normal is OM, is $|A \cdot M - |M|^2|/|M| = (21/4 + 9/4)/(3/2) = 5$. Thus the area is $5\sqrt{99/5} = 3\sqrt{55}$, and the answer is 58.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2020-II-Q12] | Part C (Q11-Q15)

Answer: 248

1. Use cell centers $(\text{column}, \text{row})$. The endpoints are $(200, 1)$ and (b, m), where $b = 2000 - (m-1)n$. Since m, n are odd, b is even. Their midpoint is the center of cell 1100; cell 1099 is immediately to its left.

2. The line reaches the left neighbor interior exactly when its absolute slope is below 1: $m - 1 < |b - 200| = |1800 - (m-1)n|$. Also $n \geq 201$ and $(m-1)n < 2000 \leq mn$, so $m = 3, 5, 7, 9$.

3. The odd-column ranges are respectively $667 \leq n \leq 999$, $401 \leq n \leq 499$, $287 \leq n \leq 333$, $223 \leq n \leq 249$. Their sizes are $167, 50, 24, 14$.

4. The slope test excludes respectively $\{899, 901\}, \{449, 451\}, \{299, 301\}, \{225\}$. The valid total is $165 + 48 + 22 + 13 = 248$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2021-II-Q12] | Part C (Q11-Q15)

Answer: 047

1. Let the diagonal vectors be $u = C - A$ and $v = D - B$. Expanding squared side lengths gives $BC^2 + DA^2 - AB^2 - CD^2 = 2u \cdot v = -21$. Hence $|u||v| \cos\theta = 21/2$.

2. The quadrilateral area is $|u \times v|/2$, so $|u||v| \sin\theta = 60$. Dividing gives $\tan\theta = 60/(21/2) = 40/7$, and the sum is 47.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-I-Q12] | Part C (Q11-Q15)

Answer: 075

1. Set $B=(0,0), C=(55,0), A=(55/2, 55\sqrt{3}/2)$. Then $D=(7,0), E=(40,15\sqrt{3}), F=(15/2, 15\sqrt{3}/2)$. The unit directions toward C, A, B from D, E, F are $u_3=(1,0), u_1=(-1/2, \sqrt{3}/2), u_2=(-1/2, -\sqrt{3}/2)$, with sum zero.

2. If d_1, d_2, d_3 are the perpendicular distances from P to CA, AB, BC, projection gives $(P-E)\cdot u_1 = d_1\cot\theta$ and its two analogues. Area decomposition gives $d_1+d_2+d_3 = 55\sqrt{3}/2$.

3. Add the projection equations. The terms involving P cancel, while $E\cdot u_1 = 5/2, F\cdot u_2 = -15, D\cdot u_3 = 7$. Therefore $(55\sqrt{3}/2)\cot\theta = -(5/2-15+7) = 11/2$.

4. Hence $\cot\theta = 1/(5\sqrt{3})$ and $\tan^2\theta = 75$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-II-Q12] | Part C (Q11-Q15)

Answer: 247

1. Take $B=(0,0), C=(14,0), A=(5,12), M=(7,0)$, so $AM=2\sqrt{37}$. The power relation $MA\,MP=49$ gives $P=(567/74,-147/37)$. Directions of BP, CP may be represented by $(27,-14),(-67,-42)$.

2. Write $Q=(5+2t,12-12t)$ with $0\le t\le 1$. Cross-product and dot-product formulas give the angle tangents $(394-296t)/(222t-33)$ at B, and $(1182-888t)/(370t+99)$ at C.

3. The angle at C is acute on this segment, so equality requires $222t-33>0$. The numerators have ratio 3, hence equality is equivalent to $370t+99=3(222t-33)$. It has the single valid root $t=99/148$.

4. Therefore $AQ=t\,AM=99/\sqrt{148}$. Since 99 and 148 are coprime, the answer is 247.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2025-I-Q12] | Part C (Q11-Q15)

Answer: 510

1. Factor the comparisons as $(x-y)(1+z)<0$ and $(y-z)(1+x)<0$. Of the four sign combinations, the bounded one is $-1<x<y<z$; the combination $x>y>z$ with $x,z<-1$ contradicts the plane, and the other two are unbounded.

2. The bounded-region closure has vertices $A=(-1,-1,77), B=(-1,38,38), C=(25,25,25)$, obtained by intersecting boundary lines pairwise.

3. Its area is half the norm of $(B-A)\times(C-A)$, namely $1014\sqrt{3}/2=507\sqrt{3}$. Thus $a+b=510$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2026-I-Q14] | Part C (Q11-Q15)

Answer: 681

1. Let side lengths be s_i in directions differing by 72°. Define $Q = \sum s_i^2 = 308$, $P_1 = \sum s_i s_{i+1}$, $P_2 = \sum s_i s_{i+2}$ cyclically. Diagonals are sums of successive side vectors, so $800 = 2Q + 2\cos 72^\circ P_1$.

2. Vector closure also gives $0 = Q + 2\cos 72^\circ P_1 + 2\cos 144^\circ P_2$. Thus $2P_1 = 184(\sqrt{5}+1)$ and $2P_2 = 492(\sqrt{5}-1)$.

3. The squared perimeter is $Q + 2P_1 + 2P_2 = 676\sqrt{5}$, hence $m + n = 681$.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2001-I-Q05] | Part A (Q1-Q5)

Answer: 937

1. The other vertices are $(u, v), (-u, v)$ with $u > 0$. Equilateral geometry gives $v = 1 - \sqrt{3}u$.

2. Substitution in the ellipse yields $13u^2 - 8\sqrt{3}u = 0$, so $u = 8\sqrt{3}/13$. The side length is $2u$, whose square is $768/169$. Thus the sum is 937.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2024-I-Q09] | Part B (Q6-Q10)

Answer: 480

1. Opposite vertices are negatives, and diagonals are perpendicular. Put $B = (x, mx)$; the hyperbola gives $x^2 = 120/(6 - 5m^2)$. The perpendicular diagonal meets the hyperbola only if $5/6 < m^2 < 6/5$.

2. Consequently $BD^2 = 480(1 + m^2)/(6 - 5m^2)$, an increasing function of m^2 on that interval. Its unattained lower limit as $m^2 \to 5/6$ is 480.

3. Every value is larger than 480 and values approach it arbitrarily closely, so the requested number is 480.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2025-I-Q09] | Part B (Q6-Q10)

Answer: 062

1. An inverse rotation sends (x, y) to $((x + \sqrt{3}y)/2, (-\sqrt{3}x + y)/2)$. Seek a point whose inverse rotation is its reflection $(-x, y)$ across the symmetry axis; both then lie on the original parabola.

2. This requires $y = -\sqrt{3}x$. With $y = x^2 - 4$, the positive root is $x = (-\sqrt{3} + \sqrt{19})/2$, giving $y = (3 - \sqrt{57})/2 < 0$.

3. The stated uniqueness identifies this fourth-quadrant intersection. Thus $a + b + c = 3 + 57 + 2 = 62$.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2026-II-Q06] | Part B (Q6-Q10)

Answer: 050

1. Parameterize the parabola by $(4+u, u^2/2-2)$. At tangency the radius is normal to the parabola, equivalently a critical point of squared distance $D(u) = u^2 + (u^2/2 - 41)^2$.

2. $D\prime(u) = u(u^2 - 80)$, so $u = 0, \pm\sqrt{80}$. Their squared distances are 1681 and 81, giving radii $41, 9$. Their sum is 50.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2005-II-Q15] | Part C (Q11-Q15)

Answer: 169

1. The centers are $F_1 = (-5, 12), F_2 = (5, 12)$ and radii $16, 4$. A desired center P has $PF_1 + PF_2 = 20$, giving ellipse $x^2/100 + (y-12)^2/75 = 1$. Every ellipse point is admissible since its PF_1 is between five and fifteen, making circle radius $16 - PF_1$ positive.

2. Substitute $y = ax$: $(3 + 4a^2)x^2 - 96ax + 276 = 0$. Real intersection requires discriminant $4800a^2 - 3312 \geq 0$.

3. The least positive slope is tangent, with $a^2 = 69/100$, so $p + q = 169$.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2021-I-Q15] | Part C (Q11-Q15)

Answer: 285

1. Add the first parabola equation to half the second to obtain the circle $x^2+y^2-x/2-41y+400-3k/2=0$. Its squared radius is $325/16+3k/2$, so the radius condition gives $k\le 280$.

2. Let $c=k+20$ and $F(x)=2(x^2-c)^2-x-k$. For $1\le k\le 4$, $F(x)>0$ for $x\le -k$; for $-k<x\le 0$, $x^2<16<c$ gives $F(x)>50-k>0$. Thus there are no negative roots. Its derivative has only one positive zero, so there cannot be four roots.

3. For $k\ge 5$, $F(-\sqrt{c})=\sqrt{c}-k\le 0$, $F(0)>0$, $F(\sqrt{c})<0$, and both tails are positive. For $k>5$, the four sign intervals give four distinct roots. At $k=5$, the root $-\sqrt{c}=-5$ has derivative -1, so the second negative root still exists.

4. Thus the allowed integers run from 5 to 280, and their extreme sum is 285.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2022-II-Q12] | Part C (Q11-Q15)

Answer: 023

1. The first ellipse has foci $F=(-4,0), G=(4,0)$ and focal-distance sum $2a$. The second has foci $H=(20,10), I=(20,12)$ and focal-distance sum $2b$.

2. For a common point P, group the four distances as $2(a+b)=(PF+PH)+(PG+PI)\geq FH+GI=26+20=46$. Thus $a+b\geq 23$.

3. Segments FH and GI cross at $P=(14,15/2)$. Its distances to F,G,H,I are $39/2, 25/2, 13/2, 15/2$, so $a=16, b=7$ satisfy the strict parameter restrictions and attain 23.

Source / 来源

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2024-II-Q12] | Part C (Q11-Q15)

Answer: 023

1. Parameterize each segment by intercepts $(\cos\theta, 0), (0, \sin\theta)$, for $0 < \theta < \pi/2$. A positive point (x, y) is on it exactly when $F(\theta) = x\sec\theta + y\csc\theta = 1$. The segment AB corresponds to $\theta = \pi/3$.

2. Both secant and cosecant are strictly convex here, so F is strictly convex and tends to infinity at both endpoints. A point belongs to just one segment precisely when $F(\pi/3) = 1$ is its minimum, requiring $F\prime(\pi/3) = 0$, or $y = 3\sqrt{3}x$.

3. Intersect with AB, whose equation is $y = \sqrt{3}/2 - \sqrt{3}x$: $C = (1/8, 3\sqrt{3}/8)$. Thus $OC^2 = 7/16$ and $p + q = 23$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2022-II-Q03] | Part A (Q1-Q5)

Answer: 021

1. The pyramid height is $h = 3 \cdot 54/36 = 9/2$. Its sphere center lies on the symmetry axis, at signed height z above the base.

2. The squared distance to a base vertex is $z^2 + 18$; to the apex it is $(9/2 - z)^2$. Equating yields $z = 1/4$.

3. The radius is $9/2 - 1/4 = 17/4$, hence $m + n = 21$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-I-Q03] | Part A (Q1-Q5)

Answer: 079

1. If P is distance d from the disk center, the small-sphere center is height 42, at distance $\sqrt{d^2+42^2}$ from the hemisphere center. Containment requires this distance plus 42 at most 200.

2. Thus $d^2 \le 158^2 - 42^2 = 23200$. Allowed points form a disk, so the area ratio is $23200/40000 = 29/50$, giving 79.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2020-I-Q06] | Part B (Q6-Q10)

Answer: 173

1. The centers have heights $\sqrt{r^2-1}$ and $\sqrt{r^2-4}$. Hence $49+(\sqrt{r^2-1}-\sqrt{r^2-4})^2=4r^2$.

2. Put $u=r^2$. Rearrangement gives $\sqrt{(u-1)(u-4)}=22-u$. Squaring cancels u^2 and yields $39u=480$.

3. Thus $u=160/13>4$ and $22-u>0$, so the unsquared equation is satisfied. The requested sum is 173.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2020-II-Q07] | Part B (Q6-Q10)

Answer: 298

1. Orient each axis from its base toward its apex, with origin O. In an axial half-plane, the side is $3u + 8\rho = 15$, since the apex is at $u = 5$ and the base edge at $(u, \rho) = (-3, 3)$.

2. For sphere center axial coordinates u, v, its distances to the axes are at least $|v|, |u|$. Thus $\sqrt{73}r \le 15 - 3u - 8|v|$ and $\sqrt{73}r \le 15 - 3v - 8|u|$. Adding yields $2\sqrt{73}r \le 30$.

3. At O, the sphere with $r = 15/\sqrt{73}$ touches each lateral surface, and $r < 3$ keeps it above both base planes. Hence the maximum square is $225/73$, giving 298.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2021-II-Q10] | Part B (Q6-Q10)

Answer: 335

1. All three spheres lie on the same side of each tangent plane. If two were on opposite sides, external tangency would force the same contact point; tangency of the third to the opposite-side sphere would force that contact point again, making its center lie on the same normal line as the same-side sphere and preventing their external tangency. Let the sector angle be 2θ. A radius-r center has transverse distance $r/\sin\theta$ from ℓ.

2. The two equal centers differ by 72 along ℓ. The small center is midway along that direction, so tangency gives $(23/\sin\theta)^2 + 36^2 = 49^2$. Thus $\sin\theta = 23/\sqrt{1105}$ and $\cos\theta = 24/\sqrt{1105}$.

3. The contact-point distance is $13\cot\theta = 312/23$, giving 335.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2022-I-Q10] | Part B (Q6-Q10)

Answer: 756

1. Let the positive sphere-center heights be u, v, w and common section radius be r. Then $u^2 = 121 - r^2, v^2 = 169 - r^2, w^2 = 361 - r^2$.

2. The first center distance is 24, so $(v-u)^2 = 24^2 - 560 = 16$. Since $v > u$, $v - u = 4$; also $(v-u)(v+u) = 48$, hence $u = 4, v = 8$. Thus $r^2 = 105$ and $w = 16$.

3. Project the 30-unit first-to-third center segment onto the plane: $AC^2 = 30^2 - (16-4)^2 = 756$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2024-II-Q08] | Part B (Q6-Q10)

Answer: 127

1. Take an axial cross-section. The generating-circle center lies 6 units from the axis, and its distance from the sphere center is $11-3=8$ for internal tangency or $11+3=14$ for external tangency.

2. The contact point lies on the line joining these centers, at distance 11 from the sphere center. Similar triangles therefore give $r_i=6\cdot 11/8=33/4$ and $r_o=6\cdot 11/14=33/7$.

3. The difference is $99/28$, so $m+n=127$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2000-II-Q12] | Part C (Q11-Q15)

Answer: 118

1. The perpendicular projection of O to the triangle plane is its circumcenter. The triangle is acute since $15^2 < 13^2 + 14^2$, so this foot is inside and gives the distance to the triangle itself.

2. Heron gives area 84, hence circumradius $R = 13 \cdot 14 \cdot 15/(4 \cdot 84) = 65/8$.

3. The plane distance is $\sqrt{20^2 - R^2} = 15\sqrt{95}/8$. Thus $m + n + k = 15 + 95 + 8 = 118$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2001-I-Q12] | Part C (Q11-Q15)

Answer: 005

1. Tangency to the three coordinate planes puts the sphere center at (r, r, r). The fourth face lies on $x/6 + y/4 + z/2 = 1$, or $2x + 3y + 6z = 12$.

2. The center-to-face distance is $(12 - 11r)/\sqrt{4 + 9 + 36} = (12 - 11r)/7$. Equating it to r gives $18r = 12$, so $r = 2/3$ and the answer is 5.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2011-I-Q13] | Part C (Q11-Q15)

Answer: 330

1. Let the nearest height be h. Projections of the three orthogonal ten-unit edges onto the vertical direction are $10-h, 11-h, 12-h$. Their squared sum is 10^2, because the squared components of a unit vertical vector sum to one.

2. Thus $(10-h)^2+(11-h)^2+(12-h)^2=100$, or $3h^2-66h+265=0$.

3. The solutions are $(33\pm\sqrt{294})/3$. Since the nearest vertex has $h<10$, choose the minus sign. The sum is $33+294+3=330$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2016-II-Q14] | Part C (Q11-Q15)

Answer: 450

1. Let H be the triangle center and M the midpoint of AB. Both P, Q and sphere center O lie on the perpendicular axis through H. Since $OP = OQ = d$, O is the midpoint of PQ and $PQ = 2d$. Also $MH = 100\sqrt{3}$, $CH = 200\sqrt{3}$.

2. The perpendicular section through M, H, P, Q gives angles $u = PMH, v = QMH$ with $u + v = 120^\circ$. Because PQ is a sphere diameter and C lies on the sphere, PCQ is right; its altitude gives $PH \cdot QH = CH^2$. Thus $\tan u \tan v = 4$.

3. Tangent addition gives $-\sqrt{3} = (\tan u + \tan v)/(1 - 4)$, hence the tangent sum is $3\sqrt{3}$. Therefore $2d = PH + QH = 100\sqrt{3} \cdot 3\sqrt{3} = 900$, so $d = 450$. The two positive tangents exist since their sum squared exceeds four times their product.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-I-Q12] | Part C (Q11-Q15)

Answer: 161

1. Let a common tangent plane have unit normal (u, v, h) and equation $ux + vy + hz = c$. A radius-ρ sphere above $z = 0$ has center (x, y, ρ); choose the normal so tangency reads $ux + vy - c = (1 - h)\rho$. Since the second plane is distinct from $z = 0$, $h \neq 1$, making radius an affine function of its contact point.

2. Take $A = (0, 0), B = (6, 0), C = (0, 4)$. The affine radius function is $g(x, y) = 1 + x/6 + y/2$. The centroid $(2, 4/3)$ reflects across $2x + 3y = 12$ to $D = (42/13, 124/39)$.

3. Thus $r = g(D) = 122/39$, with coprime terms, giving 161. The normal can indeed be normalized with $h = -13/23$, so the distinct tangent plane exists.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-II-Q12] | Part C (Q11-Q15)

Answer: 223

1. The exceptional edges $10, 18$ must be opposite; all other edges have square 250. Realize the tetrahedron as $A = (-5, 0, 12), B = (5, 0, 12), C = (0, -9, 0), D = (0, 9, 0)$. Reflection symmetries put both centers on the z axis.

2. For $S = (0, 0, s)$, equal squared distances give $25 + (12 - s)^2 = 81 + s^2$, hence $s = 11/3$. For $R = (0, 0, t)$, face planes $4y - 3z + 36 = 0$ and $12x + 5z = 0$ give $(36 - 3t)/5 = 5t/13$, so $t = 117/16$.

3. Thus $RS = 117/16 - 11/3 = 175/48$, and $m + n = 223$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2002-II-Q02] | Part A (Q1-Q5)

Answer: 294

1. Each of the three pairwise squared distances is 98, so the vertices form an equilateral triangle.

2. In a cube, three mutually equidistant vertices cannot all be joined by edges (the cube graph has no triangles) or by space diagonals (each vertex has only one opposite). They must be joined by face diagonals. Hence $2s^2 = 98$ and surface area $6s^2 = 294$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-I-Q05] | Part A (Q1-Q5)

Answer: 505

1. Decompose by the nearest box feature. The box contributes 60, and the six face slabs contribute $2(12+15+20)=94$.

2. The twelve edge quarter-cylinders combine into cylinders of total axial length $3+4+5=12$, contributing 12π. The eight corner octants form one sphere of volume $4\pi/3$.

3. Total volume is $154+40\pi/3=(462+40\pi)/3$, giving 505.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-II-Q04] | Part A (Q1-Q5)

Answer: 028

1. With centroid $G = (A + B + C + D)/4$, the center opposite A is $(B + C + D)/3 = G - (A - G)/3$. The same relation holds at every vertex.

2. Thus the face-center tetrahedron is a homothetic image with scale $-1/3$, giving volume ratio $1/27$. The requested sum is 28.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-II-Q05] | Part A (Q1-Q5)

Answer: 216

1. Over the radius-six disk, the wedge height is an affine function, zero at the tangent contact point and twelve at its diameter-opposite point.

2. Opposite disk points have heights summing to twelve, so the average height is six. Volume $= \pi \cdot 6^2 \cdot 6 = 216\pi$, giving 216.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2008-I-Q05] | Part A (Q1-Q5)

Answer: 014

1. The base contact travels a circle of radius equal to the slant height $\ell = \sqrt{r^2 + h^2}$. One circuit rolls distance $2\pi\ell$, equal to seventeen base circumferences $17 \cdot 2\pi r$.

2. Thus $\ell = 17r$ and $h^2 = 288r^2$, yielding $h/r = 12\sqrt{2}$ and answer 14.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2012-II-Q05] | Part A (Q1-Q5)

Answer: 750

1. Each flat triangular face has altitude $20 - 15/2 = 25/2$, which becomes the pyramid slant height. The base apothem is $15/2$.

2. The vertical height is $\sqrt{(25/2)^2 - (15/2)^2} = 10$. Thus the volume is $15^2 \cdot 10/3 = 750$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2016-I-Q04] | Part A (Q1-Q5)

Answer: 108

1. Let U, V be the two base neighbors of A, and W its vertical neighbor. The two relevant planes meet along UV. In isosceles triangle AUV, $AU = AV = 12$ and angle $UAV = 120^\circ$.

2. The midpoint M of UV satisfies $AM = 12\cos 60^\circ = 6$. Both AM and WM are perpendicular to UV, so angle AMW is the given dihedral angle.

3. Right triangle AMW gives $h = 6\tan 60^\circ = 6\sqrt{3}$, and $h^2 = 108$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2017-I-Q04] | Part A (Q1-Q5)

Answer: 803

1. Equal apex distances put its perpendicular projection at the base circumcenter. The base altitude is $\sqrt{20^2 - 12^2} = 16$, so area $K = 192$ and circumradius $R = 20 \cdot 20 \cdot 24/(4K) = 25/2$.

2. The pyramid height is $\sqrt{25^2 - R^2} = 25\sqrt{3}/2$. Its volume is $Kh/3 = 800\sqrt{3}$, giving $m + n = 803$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2000-I-Q08] | Part B (Q6-Q10)

Answer: 052

1. The initial liquid occupies $(9/12)^3 = 27/64$ of the total volume. On reversal, the empty apex cone occupies $37/64$.

2. Its height is $12\sqrt[3]{37/64} = 3\sqrt[3]{37}$. Subtracting from 12 gives the liquid depth, so the requested sum is $12 + 3 + 37 = 52$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-I-Q06] | Part B (Q6-Q10)

Answer: 348

1. Triangles have only three squared-side patterns: $(1,1,2)$, $(2,2,2)$, and $(1,2,3)$. The first occurs four times per face, or 24 times, with area $1/2$.

2. The equilateral pattern occurs 8 times with area $\sqrt{3}/2$. The last occurs once for each space diagonal and each of the other six vertices, or 24 times, with area $\sqrt{2}/2$. The counts sum to $56 = \binom{8}{3}$.

3. Total area is $12 + 4\sqrt{3} + 12\sqrt{2} = 12 + \sqrt{48} + \sqrt{288}$, giving 348.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2005-II-Q10] | Part B (Q6-Q10)

Answer: 011

1. Take octahedron vertices $(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)$. Two square pyramids of base area two and height one give volume $4/3$.

2. Face centers are $(\pm 1/3, \pm 1/3, \pm 1/3)$, so cube volume is $(2/3)^3 = 8/27$. Their ratio is $9/2$, giving 11.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2012-I-Q08] | Part B (Q6-Q10)

Answer: 089

1. The cutting plane through $(0,1,0),(1/2,0,0),(1,1,1/2)$ has equation $2x+y-4z=1$. Its smaller portion is near the bottom edge BC.

2. Extend DM to $K=(1,-1,0)$, so $CK=2$ and B is the midpoint of CK. The tetrahedron $KDCN$ has base DCN of area $1/4$ and perpendicular height two, so volume $1/6$.

3. The part beyond the cube is a similar tetrahedron at K of linear scale $1/2$, volume $1/48$. Hence the smaller cube portion is $1/6-1/48=7/48$.

4. The larger portion is $41/48$, giving $p+q=89$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2015-II-Q09] | Part B (Q6-Q10)

Answer: 384

1. Near its lowest vertex, a plane perpendicular to the cube body diagonal cuts an equilateral triangle. At the rim, its circumradius is 4, so its side length is $4\sqrt{3}$.

2. If the equal intercepts along the three mutually perpendicular cube edges are t, the section side is $t\sqrt{2}$, giving $t = 2\sqrt{6} < 8$. Thus the submerged part is exactly a corner tetrahedron, of volume $t^3/6 = 8\sqrt{6}$.

3. Its depth is $t/\sqrt{3} = 2\sqrt{2} < 10$, so it does not reach the barrel bottom. The displaced volume is the tetrahedron volume, and $v^2 = 384$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2021-I-Q06] | Part B (Q6-Q10)

Answer: 192

1. Set $A = 0$, with B, C, D on perpendicular coordinate axes at distance s, and $G = (s, s, s)$. Expanding distances to $P = (x, y, z)$ gives $PB^2 + PC^2 + PD^2 - PG^2 = 2PA^2$; both linear coordinate terms and s^2 cancel.

2. Therefore $2PA^2 = 36000 + 18000 + 28800 - 9072 = 73728$. Thus $PA = \sqrt{36864} = 192$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2001-II-Q12] | Part C (Q11-Q15)

Answer: 101

1. Each face becomes three corner triangles and three outer tetrahedron faces, all of half side length. Thus at stage i there are $4 \cdot 6^i$ faces, with side scale 2^{-i}.

2. The attached tetrahedra in the next step have scale $2^{-(i+1)}$, hence volume $8^{-(i+1)}$ each. The increment is $\frac{1}{2}(3/4)^i$.

3. The first three additions give total volume $1 + 1/2 + 3/8 + 9/32 = 69/32$. The requested sum is 101.

4. The required non-overlap through three stages can be checked exactly: start with vertices $(1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)$. Each outward apex and face midpoint remains rational. Separating-axis tests on the resulting 173 tetrahedra and coplanar-triangle tests on the 864 final boundary faces show only boundary contact, with no positive-volume or hidden positive-area overlap. The accompanying executable certificate uses exact integer projections, so the stated face count and additive volume are valid through stage three.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2001-II-Q15] | Part C (Q11-Q15)

Answer: 417

1. Take $A=(0,0,0), E=(8,8,8)$ and $I=(6,8,8), J=(8,6,8), K=(8,8,6)$. The parallel tunnel lines from I, J, K reach $L=(0,2,2), M=(2,0,2), N=(2,2,0)$.

2. The wall through IJ has plane $2z=x+y+2$ and meets the z-axis at $O=(0,0,1)$. Its pentagon is rectangle $ILMJ$ plus triangle LOM. Their areas are $(2\sqrt{2})(6\sqrt{3})=12\sqrt{6}$ and $(2\sqrt{2})(\sqrt{3})/2=\sqrt{6}$, totaling $13\sqrt{6}$.

3. Each of the three faces at E loses a legs-2 right triangle, area 2. Each face at A loses a quadrilateral, for example $(0,0), (1,0), (2,2), (0,1)$, also area 2.

4. The total is $6(64-2)+3(13\sqrt{6})=372+39\sqrt{6}$. Thus $m+n+p=417$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2002-I-Q15] | Part C (Q11-Q15)

Answer: 163

1. Use $D=(0,0,0), C=(12,0,0), B=(12,12,0), A=(0,12,0)$. Equal distances to C, D put E in $x=6$. Its distance to the filled square forces its perpendicular foot to be within the square; otherwise its squared distance to D and the nearest edge contradict $DE=14$ and distance 12. Hence, choosing the upper side, $E=(6,4,12)$.

2. Isosceles trapezoid symmetry gives $G=(3,y,z), F=(9,y,z)$. The face $ADEG$ lies in $z=2x$, so $z=6$. Then $AG=8$ yields $(y-12)^2=19$.

3. Therefore $EG^2=3^2+6^2+(8\pm\sqrt{19})^2=128\pm16\sqrt{19}$. The specified minus form selects $128-16\sqrt{19}$, so the required sum is 163.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-I-Q11] | Part C (Q11-Q15)

Answer: 512

1. The full cone has slant height 5, lateral area 15π and total painted area 24π. At similarity scale t, the small cone retains painted area $15\pi t^2$, while its new cut face is unpainted.

2. Equate $15t^2/(24-15t^2) = t^3/(1-t^3)$. Cancellation gives $15 = 24t$, so $t = 5/8$.

3. The common ratio is $125/(512-125) = 125/387$, giving 512.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-I-Q14] | Part C (Q11-Q15)

Answer: 813

1. Unroll the wall and extend the tangent free portion in that development. Tautness makes the combined path straight, rising 4 over length 20, so its total horizontal projection is $\sqrt{400-16}=8\sqrt{6}$.

2. The endpoint is at horizontal radius 12. Its tangent to the radius-eight circle has projected length $\sqrt{12^2-8^2}=4\sqrt{5}$. The constant length-to-projection ratio makes the free rope length $20(4\sqrt{5})/(8\sqrt{6})=5\sqrt{30}/3$.

3. The contact length is $(60-\sqrt{750})/3$, giving $60+750+3=813$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-II-Q11] | Part C (Q11-Q15)

Answer: 625

1. The slant height is 800, so its lateral surface develops to a radius-800 sector with angle $360^\circ(600/800) = 270^\circ$. Opposite generators are separated by 135° in the development.

2. The straight segment in that sector is the shortest path. Its squared length is $125^2 + (375\sqrt{2})^2 - 2(125)(375\sqrt{2})\cos 135^\circ = 390625$.

3. The distance is 625.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2007-I-Q13] | Part C (Q11-Q15)

Answer: 080

1. Take base $A=(0,0,0), B=(4,0,0), C=(4,4,0), D=(0,4,0)$ and apex $E=(2,2,2\sqrt{2})$. The cutting plane is $x+y+2\sqrt{2}z=6$.

2. Its section has vertices $(1,1,\sqrt{2}), (7/2,1/2,\sqrt{2}/2), (4,2,0), (2,4,0), (1/2,7/2,\sqrt{2}/2)$. Their xy projections have shoelace area eight.

3. Projection onto xy scales area by $1/\sqrt{1+(1/(2\sqrt{2}))^2+(1/(2\sqrt{2}))^2}=2/\sqrt{5}$. Thus the section area is $8\sqrt{5}/2=\sqrt{80}$, giving $p=80$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2008-I-Q15] | Part C (Q11-Q15)

Answer: 871

1. At one corner use $O=(0,0)$ and $P=(a,0)$ with $a=\sqrt{17}$. The inner cut vertex is $R=(t,t)$. The angle condition makes triangle OPR have angles $45^\circ, 105^\circ, 30^\circ$, so $PR=a\sqrt{2}$ and $t=PR\sin 75^\circ=a(\sqrt{3}+1)/2$.

2. The fold through R is horizontal. Its perpendicular foot above P is $S=(a,t)$, so the panel width is $SP=t$. After folding and taping, symmetry puts P vertically over (a,a), while its distance from S remains t. The horizontal separation is $t-a$.

3. Therefore height squared is $t^2-(t-a)^2=a^2\sqrt{3}=17\sqrt{3}$. Height is $\sqrt[4]{867}$; $867=3\cdot 17^2$ has no fourth-power factor. The answer is 871.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2010-I-Q11] | Part C (Q11-Q15)

Answer: 365

1. The region is a triangle with apex $C = (8, 10)$ and base endpoints $A = (9/2, 13/2), B = (39/4, 33/4)$ on the rotation axis.

2. The perpendicular foot of C is $(87/10, 79/10)$, lying within AB. Therefore rotation gives two cones sharing a base, with total axial height $AB = 7\sqrt{10}/4$ and base radius $7/\sqrt{10}$.

3. The volume is $\pi(49/10)(7\sqrt{10}/4)/3 = 343\pi/(12\sqrt{10})$. Hence the answer is $343 + 12 + 10 = 365$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2015-I-Q15] | Part C (Q11-Q15)

Answer: 053

1. The base chord lies at distance $6\cos 60^\circ = 3$ from the base center. Central symmetry sends it to a parallel chord on the other base at opposite signed offset. The section projects onto the width-6 central strip of the radius-6 disk.

2. Each excluded circular cap has area $12\pi - 9\sqrt{3}$, from its 120° sector minus the central triangle. The projected strip area is $36\pi - 2(12\pi - 9\sqrt{3}) = 12\pi + 18\sqrt{3}$.

3. The plane rises 8 across horizontal distance 6, so the area enlargement factor is $\sqrt{6^2 + 8^2}/6 = 5/3$. Hence section area $20\pi + 30\sqrt{3}$, giving $a + b + c = 53$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2023-I-Q13] | Part C (Q11-Q15)

Answer: 125

1. For three generating edge vectors u, v, w, the rhombus diagonal relations give $|u|^2 = |v|^2 = |w|^2 = (21 + 31)/4 = 13$. Every pairwise dot product is $\pm(31 - 21)/4 = \pm 5/2$.

2. With $x = u \cdot v, y = u \cdot w, z = v \cdot w$, the Gram determinant is $V^2 = 13^3 - 13(x^2 + y^2 + z^2) + 2xyz$. The squared terms are fixed; only the sign of xyz changes.

3. The two squared volumes are $2197 - 975/4 \pm 125/4$, namely $7938/4$ and $7688/4$. Their ratio gives $V_L/V_S = \sqrt{3969/3844} = 63/62$.

4. The fraction is reduced, so $m + n = 125$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2023-II-Q14] | Part C (Q11-Q15)

Answer: 751

1. Use cube-edge coordinates $A=(0,0,0), B=(s,0,0), C=(0,s,s), D=(s,s,s)$ and unit vertical vector u. The vertical rectangle has normal $(0,1,-1)$, so $u_2=u_3$. Heights give $su_1=2, su_2=su_3=4$, whence $s^2=2^2+4^2+4^2=36$.

2. Thus the cube is $[0,6]^3$ and height is $(x+2y+2z)/3$. Reflect all coordinates about 3: the empty region above height 7 has the same volume as $x+2y+2z<9$ inside the cube.

3. The full first-octant tetrahedron has intercepts $9, 9/2, 9/2$ and volume $243/8$. Only the constraint $x\le 6$ truncates it; the removed tetrahedron has intercepts $3, 3/2, 3/2$ and volume $9/8$. Thus air occupies $234/8=117/4$.

4. The water volume is $6^3-117/4=747/4$, giving $m+n=751$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2024-I-Q14] | Part C (Q11-Q15)

Answer: 104

1. Embed the tetrahedron in a rectangular box using alternate vertices. Box dimensions a, b, c satisfy $a^2 + b^2 = 41, a^2 + c^2 = 80, b^2 + c^2 = 89$, giving $(a, b, c) = (4, 5, 8)$. Its volume is $abc/3 = 160/3$.

2. Every face has squared sides $41, 80, 89$. The squared-area identity gives $16F^2 = 2(41 \cdot 80 + 80 \cdot 89 + 89 \cdot 41) - (41^2 + 80^2 + 89^2) = 12096$, so $F = 6\sqrt{21}$.

3. Decomposing into four pyramids gives $V = r(4F)/3$. Hence $r = 20\sqrt{21}/63$, so $m + n + p = 104$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2000-I-Q01] | Part A (Q1-Q5)

Answer: 008

1. Any factor containing both primes $2, 5$ ends in zero. Consequently the only possible factorization avoiding zeros is $2^n \cdot 5^n$; a factor 1 would leave the zero-ending number 10^n.

2. For $1 \leq n \leq 7$, neither 2^n nor 5^n has a zero digit. At $n = 8$, $5^8 = 390625$ contains one. Thus the answer is 8.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2002-II-Q05] | Part A (Q1-Q5)

Answer: 042

1. Prime-exponent comparison shows failure precisely when $a < 6n$ or $a < 6m$. If $n, m \geq 1$, then $a \geq 3 \cdot 2^n \geq 6n$ and $a \geq 2 \cdot 3^m \geq 6m$, so there are no failures.

2. For $m = 0$, $2^n < 6n$ holds exactly at $n = 1, 2, 3, 4$; after $n = 5$, induction preserves the reverse inequality. For $n = 0$, $3^m < 6m$ holds only at $m = 1, 2$.

3. The integers are $2, 4, 8, 16, 3, 9$, summing to 42.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q04] | Part A (Q1-Q5)

Answer: 124

1. The five-adic valuation controls trailing zeros. An integer j occurs in $101 - j$ of the factorials.

2. Multiples $5, 10, \ldots, 100$ contribute $96 + 91 + \cdots + 1 = 970$. Multiples $25, 50, 75, 100$ contribute an additional $76 + 51 + 26 + 1 = 154$. No factor 125 occurs.

3. Thus $N = 1124$, remainder 124.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-II-Q03] | Part A (Q1-Q5)

Answer: 049

1. Count odd multiples of $3, 9, 27, 81$ separately, each adding a valuation layer. Their counts are $33, 11, 4, 1$, with no multiple of 243.

2. The exponent is $33 + 11 + 4 + 1 = 49$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2007-I-Q01] | Part A (Q1-Q5)

Answer: 083

1. For u^2 divisible by $2^3 \cdot 3$, the integer u must be divisible by $2^2 \cdot 3 = 12$, and this condition is sufficient.

2. The positive multiples of twelve below one thousand are $12, 24, \ldots, 996$, numbering 83.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2010-II-Q03] | Part A (Q1-Q5)

Answer: 150

1. A difference d occurs $20-d$ times. Count one contribution for each factor of 2, another for each factor of 4, and similarly for $8, 16$.

2. The contributions are $\sum_{j=1}^{9}(20-2j)=90$, $\sum_{j=1}^{4}(20-4j)=40$, $(20-8)+(20-16)=16$, and $20-16=4$.

3. The total valuation is $90+40+16+4=150$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2015-I-Q03] | Part A (Q1-Q5)

Answer: 307

1. Write $16p = n^3 - 1 = (n-1)(n^2+n+1)$. The integer n is odd, making the second factor odd; thus 16 divides $n-1$.

2. Put $n - 1 = 16k$. Then $p = k(n^2+n+1)$, so primality forces $k = 1$, since $n^2+n+1 > 1$. Hence $n = 17$ and $p = 307$.

3. No prime at most $\sqrt{307} < 18$ divides 307: test $2, 3, 5, 7, 11, 13, 17$. Also $16 \cdot 307 + 1 = 17^3$, verifying the answer.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2019-II-Q04] | Part A (Q1-Q5)

Answer: 187

1. Classify non-five faces by exponent parities of two and three: neutral $1, 4$ (two choices), and single classes $2, 3, 6$. A square product requires an even number of fives, and counts of $2, 3, 6$ either all even or all odd.

2. With zero fives, all-even counts contribute $16 + 72 + 18 + 3 = 109$ sequences and all-odd counts contribute $4! \cdot 2 = 48$, total 157. With two fives, choose positions in six ways and the remaining equal parity classes in $2^2 + 1 + 1 + 1 = 7$ ways, total 42. Four fives gives one.

3. Favorable count 200 of 1296, probability $25/162$, so the answer is 187.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2020-II-Q01] | Part A (Q1-Q5)

Answer: 231

1. Factor $20^{20} = 2^{40}5^{20}$. Necessarily $m = 2^a5^b$, and $n = 2^{40-2a}5^{20-2b}$.

2. Nonnegative remaining exponents require $0 \le a \le 20$ and $0 \le b \le 10$. Each exponent pair determines exactly one n, giving $21 \cdot 11 = 231$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2021-I-Q03] | Part A (Q1-Q5)

Answer: 050

1. A positive integral difference must be $2^b(2^c-1)$ with integers $b \geq 0, c \geq 1$. Negative lower exponents would leave an odd numerator over a power of two, so cannot give an integer. Unique odd parts ensure no double counting.

2. For $c = 1, \ldots, 9$, the admissible counts of b are $10, 9, 8, 7, 6, 4, 3, 2, 1$. For $c \geq 10$ even the smallest value exceeds 999.

3. These counts sum to 50.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2023-I-Q04] | Part A (Q1-Q5)

Answer: 012

1. The prime exponents of $13!$ are $(10, 5, 2, 1, 1, 1)$ for primes $(2, 3, 5, 7, 11, 13)$. Each exponent in m must have the same parity and lie between zero and the corresponding exponent.

2. Independent exponent choices factor the sum into $(1 + 4 + \cdots + 4^5)(3 + 27 + 243)(1 + 25) \cdot 7 \cdot 11 \cdot 13$.

3. These factors are $1365 = 3 \cdot 5 \cdot 7 \cdot 13$, $273 = 3 \cdot 7 \cdot 13$, $26 = 2 \cdot 13$, and $1001 = 7 \cdot 11 \cdot 13$. The resulting exponents are $(1, 2, 1, 3, 1, 4)$, summing to 12.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q09] | Part B (Q6-Q10)

Answer: 046

1. The logarithm sum gives $a^{12}r^{66} = 2^{6018}$, hence $a^2r^{11} = 2^{1003}$. Thus $a = 2^u, r = 2^v$ with nonnegative exponents satisfying $2u + 11v = 1003$.

2. v is odd and $1 \leq v \leq 91$, giving forty-six choices. Each determines a nonnegative integer u, so the count is 46.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2009-II-Q07] | Part B (Q6-Q10)

Answer: 401

1. The summand equals $\binom{2i}{i}/4^i$, so its reduced denominator is purely a power of two. Its original numerator is odd and $(2i)!! = 2^i i!$, hence its reduced denominator exponent is $e_i = i + v_2(i!)$.

2. These exponents strictly increase. With common denominator $2^{e_{2009}}$, only the final summand contributes an odd numerator; all others contribute even numerators. Thus no final two-factor cancellation occurs, and $b = 1$.

3. Legendre gives $v_2(2009!) = 1004 + 502 + 251 + 125 + 62 + 31 + 15 + 7 + 3 + 1 = 2001$. Hence $a = 4010$ and $ab/10 = 401$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2016-II-Q08] | Part B (Q6-Q10)

Answer: 728

1. The product N factors as $3^2 \cdot 7 \cdot 11 \cdot 17 \cdot 31 \cdot 41 \cdot 61$. For ordered triples, each of the six exponent-one primes has three destinations; the exponent of3 splits into three nonnegative parts in $\binom{4}{2} = 6$ ways. This gives $6 \cdot 3^6 = 4374$.

2. If two entries equal v, then $v^2 \mid N$, so $v = 1$ or3. The two multisets $(1, 1, N)$ and $(3, 3, N/9)$ each have three orderings. An all-equal triple is impossible.

3. Remove these six ordered triples and divide by six for the distinct-entry permutations: $(4374 - 6)/6 = 728$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2019-I-Q07] | Part B (Q6-Q10)

Answer: 880

1. Exponentiating shows only primes two and five occur. For either prime, let its exponents be a, b. Then $a + 2\min(a, b) = 60$, $b + 2\max(a, b) = 570$.

2. If $a > b$, solving yields negative b, impossible. Thus $a \leq b$, giving $3a = 60, 3b = 570$, or $a = 20, b = 190$.

3. Both primes have these exponents, so $m = 40, n = 380$ and $3m + 2n = 880$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2020-I-Q10] | Part B (Q6-Q10)

Answer: 407

1. Every prime of n also divides m, so it divides $m+n$ and is at least 11. Since $n \nmid m$, some such prime has exponents $a = v_p(n) > b = v_p(m) \geq 1$. Divisibility requires $bm \geq an$.

2. A candidate is $n = 121, m = 286$: $v_{11}(n^n) = 242 \leq 286 = v_{11}(m^m)$ and $407 = 11 \cdot 37$ is coprime to 210.

3. To rule out smaller sums, assume $m + n < 407$. An exponent $a \geq 3$ forces $n \geq 1331$, impossible. Hence $a = 2, b = 1$, so $m \geq 2n$ and $n \leq 135$. Only $n = 121$ can contain a square of a prime at least 11 in this range.

4. For $242 \leq m < 286$, the multiples of 11 are $242, 253, 264, 275$. The first is divisible by 121; the others give sums divisible respectively by $2, 5, 2$. None works. The minimum is 407.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2000-I-Q11] | Part C (Q11-Q15)

Answer: 248

1. Since $1000 = 2^3 5^3$, coprimality lets each prime appear in at most one of a, b. Thus the exponent differences range independently from -3 to 3, each once.

2. Consequently $S = (\sum_{j=-3}^{3} 2^j)(\sum_{j=-3}^{3} 5^j) = (127/8)(19531/125) = 2480437/1000$. Hence $S/10 = 248.0437$, whose floor is 248.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q13] | Part C (Q11-Q15)

Answer: 899

1. For $1 \leq j < n$, exactly 2^{n-1-j} summands have value 2^j, while the endpoint contributes 2^n. Thus $S_n = (n-1)2^{n-1} + 2^n = (n+1)2^{n-1}$.

2. Even n gives odd two-adic valuation and fails. For odd n, the power of two is already square, so $n+1$ must be square. The largest even square at most one thousand is 900, giving $n = 899$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-II-Q14] | Part C (Q11-Q15)

Answer: 063

1. Among zero-padded strings from 0 to $10^n - 1$, each digit occurs $n10^{n-1}$ times. The final integer 10^n contributes one. Hence $S_n = 1 + n10^{n-1}H_9$, where $H_9 = 7129/2520$ is reduced.

2. As $2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$, integrality requires $63 \mid n$, because powers of ten supply no factors three or seven. The first such $n = 63$ also supplies all factors two and five, so it works and is minimal.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2016-I-Q12] | Part C (Q11-Q15)

Answer: 132

1. The expression is odd. Its residue sets modulo $3, 5, 7$ are respectively $\{1, 2\}, \{1, 2, 3\}, \{2, 3, 4, 6\}$, none containing zero. Thus every prime factor is at least 11.

2. The first possible products of at least four such primes are $11^4 = 14641$ and $11^3 \cdot 13 = 17303$. The first would require discriminant 58521, strictly between 241^2 and 242^2, so is impossible.

3. The expression increases for positive integral m, and at $m = 132$ it equals $132 \cdot 131 + 11 = 17303 = 11^3 \cdot 13$. No smaller value can be a qualifying prime product except the excluded 11^4. Hence the minimum is 132.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2003-I-Q01] | Part A (Q1-Q5)

Answer: 839

1. The number is $720!/6 = 120 \cdot 719!$. No $n \geq 720$ works, since then $kn! \geq 720! > 720!/6$.

2. Thus $n = 719, k = 120$, giving 839.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2007-II-Q02] | Part A (Q1-Q5)

Answer: 200

1. Write $b = au, c = av$, where $u, v \geq 1$. Then $a(1 + u + v) = 100$, so a is a divisor of one hundred with $100/a \geq 3$.

2. The possible a values are $1, 2, 4, 5, 10, 20, 25$, each contributing $100/a - 2$ positive ordered pairs. Their sum is $100 + 50 + 25 + 20 + 10 + 5 + 4 - 14 = 200$.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2015-II-Q01] | Part A (Q1-Q5)

Answer: 131

1. The ratios reduce to $39/50$ and $29/25$. Therefore the integral value N must be a multiple of both 39 and 29.

2. These numbers are coprime, so the smallest value is $39 \cdot 29 = 1131$. It is attained using the original integers $1450, 975$, and its remainder is 131.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2017-I-Q02] | Part A (Q1-Q5)

Answer: 062

1. Equal remainders force $m \mid \gcd(787-702, 855-787) = \gcd(85, 68) = 17$. The divisor1 cannot leave a positive remainder, so $m = 17, r = 5$.

2. Similarly $n \mid \gcd(310, 93) = 31$, and positivity forces $n = 31, s = 9$. The sum is $17 + 31 + 5 + 9 = 62$.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2019-II-Q03] | Part A (Q1-Q5)

Answer: 096

1. As $c \mid 70$, $e \mid 72$ and $cde = 71$ prime, neither c nor e can be seventy-one. Hence $c = e = 1, d = 71$.

2. Then $ab = 70$ has eight ordered factor pairs and $fg = 72$ has twelve. Multiply to obtain 96.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2020-I-Q04] | Part A (Q1-Q5)

Answer: 093

1. If the prefix is k, the number is $10000k + 2020$. Its divisibility by k is equivalent to $k \mid 2020$.

2. As $2020 = 2^2 \cdot 5 \cdot 101$, its positive divisors are
$1, 2, 4, 5, 10, 20, 101, 202, 404, 505, 1010, 2020$. Their digit sums total 45.

3. Each of the 12 suffixes contributes 4, so the total is $45 + 48 = 93$.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2022-II-Q01] | Part A (Q1-Q5)

Answer: 154

1. Write the initial counts as $12k$ people and $5k$ adults, with k a positive integer. The final adult count is $11(12k+50)/25$.

2. Integrality requires $25 \mid 12k$, so $25 \mid k$. The smallest choice $k=25$ gives 350 total and 154 adults.

3. This is feasible: initially there are 125 adults; adding 29 adults and 21 children produces the required final counts. Hence 154 is attained.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2025-II-Q02] | Part A (Q1-Q5)

Answer: 049

1. Modulo $n+2$, substitute $n=-2$: the product becomes $3 \cdot 1 \cdot 13 = 39$. Hence the condition is exactly $(n+2) \mid 39$.

2. The divisors at least 3 are $3, 13, 39$, yielding $n = 1, 11, 37$. Their sum is 49.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2021-I-Q10] | Part B (Q6-Q10)

Answer: 059

1. For each reduced pair define $d = n - m$ and $I = 19m - 18n$. Before reduction the shift preserves I and changes the difference to $d + 1$. The reduction factor is $g = \gcd(d + 1, I)$, so the updated quantities are $(d + 1)/g, I/g$.

2. Initially $(d, I) = (1, 2002)$ with $2002 = 2 \cdot 7 \cdot 11 \cdot 13$. Cancellation successively uses factors $2, 7, 11, 13$, returning d to 1 at indices $2, 8, 18, 30$. Including the initial term gives indices $1, 2, 8, 18, 30$.

3. After I becomes 1, no reduction is possible and d grows forever. Hence the sum is 59.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2025-II-Q07] | Part B (Q6-Q10)

Answer: 237

1. $2025 = 3^4 5^2$ has 15 divisors. The LCM condition requires a selected divisor with full 3-exponent and one with full 5-exponent.

2. Of the divisors, 12 lack full 3-exponent, 10 lack full 5-exponent, and 8 lack both. Inclusion–exclusion gives $2^{15} - 2^{12} - 2^{10} + 2^8 = 27904$ successful subsets.

3. The probability is $27904/32768 = 109/128$, giving 237.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2007-I-Q04] | Part A (Q1-Q5)

Answer: 105

1. Collinearity allows either side of the star, so relative angular displacement must be a half-integer number of turns. The successive relative rates are both $1/60 - 1/84 = 1/84 - 1/140 = 1/210$.

2. Thus the first positive time making relative displacement a multiple of $1/2$ is $210/2 = 105$ years; the third pair then differs by one full turn.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2010-I-Q02] | Part A (Q1-Q5)

Answer: 109

1. For every $j \geq 3$, the factor is -1 modulo 1000. There are 997 such factors.

2. The product is therefore $9 \cdot 99(-1)^{997} = -891 \equiv 109 \pmod{1000}$.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2025-I-Q05] | Part A (Q1-Q5)

Answer: 279

1. Digit sum is 36. The divisibility-by-11 criterion forces each parity class of four positions to have digit sum 18: the alternating difference lies between -16 and 16 and is even, so its only possible multiple of 11 is zero.

2. There are eight four-subsets summing to 18: $1278, 1368, 1458, 1467, 2358, 2367, 2457, 3456$. Across these subsets, the total number of even-digit incidences is 16, since complements pair to all four even digits.

3. Choose the even-position subset, its even units digit, and then order the remaining positions: $N = 16 \cdot 3! \cdot 4! = 2304$. Thus $N - 2025 = 279$.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-I-Q09] | Part B (Q6-Q10)

Answer: 757

1. $h = 1, 2$ cause overlap, so $h \geq 3$. If $h \geq 5$, picket 4 forces $t = 2$: the third painter could reach it only with $u = 1$, which would itself cause overlaps. Then picket 5 forces $u = 2$, after which the second and third painters cover every picket from 2 on, overlapping the first painter's next picket. Thus $h = 3$ or 4.

2. For $h = 3$, picket 5 forces $t = 3$, because $u = 2$ would overlap at 7; the remaining residue class forces $u = 3$. This gives 333.

3. For $h = 4$, picket 4 forces $t = 2$ and the remaining class $3 \pmod 4$ forces $u = 4$, giving 424. These two partitions work, and their sum is 757.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-II-Q07] | Part B (Q6-Q10)

Answer: 112

1. The square-sum formula requires $1200 \mid k(k+1)(2k+1)$. A factor 3 is automatic. All factors of 2 lie in the even one of $k, k+1$, requiring $k \equiv 0, 15 \pmod{16}$.

2. For the prime 5, the three factors are pairwise nonoverlapping modulo 5, so 25 must divide one of them. Thus $k \equiv 0, 24, 12 \pmod{25}$.

3. Combining the six residue pairs gives least positive representatives $112, 175, 224, 287, 399, 400$. The least is 112, which also verifies the original square-sum condition.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2011-I-Q07] | Part B (Q6-Q10)

Answer: 016

1. $m = 1$ is impossible. For $m \geq 2$, reduction modulo $m - 1$ forces $m - 1 \mid 2010$.

2. Conversely, if $2010 = (m - 1)q$, the identity $m^q = m + (m - 1)(m + m^2 + \cdots + m^{q-1})$ expresses one power as $m + (m - 1)(q - 1) = 2011$ powers. Here the initial m denotes m copies of m^0, and the empty sum when $q = 1$ causes no problem.

3. There are $2^4 = 16$ positive divisors of $2010 = 2 \cdot 3 \cdot 5 \cdot 67$, and hence 16 values of m.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-I-Q10] | Part B (Q6-Q10)

Answer: 170

1. A square $u^2 \equiv 256 \pmod{1000}$ requires $4 \mid u$ and $u \equiv \pm 16 \pmod{125}$: the two factors $u-16, u+16$ differ by thirty-two, so only one can contain factors of five.

2. Combining gives $u \equiv \pm 16 \pmod{500}$. The positive square roots run $16, 484, 516, 984, 1016, \ldots$, with the tenth equal to 2484.

3. The tenth truncated square is $(2484^2 - 256)/1000 = (2468 \cdot 2500)/1000 = 6170$, leaving remainder 170.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2014-I-Q08] | Part B (Q6-Q10)

Answer: 937

1. The ending is an idempotent modulo 10000: $N(N-1)$ is divisible by $16 \cdot 625$. Since consecutive integers are coprime, each prime-power factor lies entirely in one of them.

2. The four Chinese-remainder choices give endings $0000, 0001, 0625, 9376$. Only the last has a nonzero thousands digit.

3. Thus the requested three-digit number is 937.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2017-I-Q09] | Part B (Q6-Q10)

Answer: 045

1. Modulo99, multiplying by100 does nothing. Hence $a_n \equiv 10 + 11 + \cdots + n = (n + 10)(n - 9)/2$. Since2 is invertible modulo99, the condition is $99 \mid (n + 10)(n - 9)$.

2. The two factors differ by19, coprime to99, so each of the prime powers9 and11 must divide a single factor. The four resulting CRT classes modulo99 are $n \equiv 9, 45, 53, 89$.

3. The least representative greater than10 is45; its sum is $55 \cdot 36/2 = 990$, divisible by99.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2017-II-Q08] | Part B (Q6-Q10)

Answer: 134

1. After multiplying by720 and removing integral contributions from $1+n$, the condition is divisibility of $F=n^6+6n^5+30n^4+120n^3+360n^2$ by720. Modulo2 and3, only n^6 can obstruct divisibility, so $6 \mid n$.

2. Write $n=6t$. The last three terms are now multiples of720, while the first two combine to $6^6t^5(t+1)$. Since $720=2^4\cdot 3^2\cdot 5$, the only remaining condition is $t\equiv 0$ or4 modulo5.

3. The range is $1\le t\le 336$. Each allowed residue occurs67 times, giving134 integers.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2020-II-Q10] | Part B (Q6-Q10)

Answer: 239

1. Put $K = n(n+1)/2$, so the sum is K^2. Modulo $d = n+5$, $2K \equiv 20$ and therefore $4K^2 \equiv 400$. The desired condition implies $d \mid 332$, with $d > 17$.

2. Thus $d = 83, 166, 332$, corresponding to $n = 78, 161, 327$. Since multiplication by 4 may lose information, test the original congruence: the respective residues $K^2 \bmod d$ are $17, 17, 100$.

3. Only $78, 161$ work, and their sum is 239.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2022-II-Q08] | Part B (Q6-Q10)

Answer: 080

1. Each triple corresponds to a consecutive block of integers. A singleton block requires a jump immediately before and after n: some member of $\{4, 5, 6\}$ divides n and some member divides $n + 1$. The endpoint $n = 1$ is not unique and adds no exception.

2. The even divisors $4, 6$ cannot divide consecutive integers on opposite sides. Thus the divisor pairs are $(4, 5), (5, 4), (6, 5), (5, 6)$. Solving modulo 60 gives their union $\{4, 5, 15, 24, 35, 44, 54, 55\}$.

3. Each of these eight residues occurs ten times in $1, \ldots, 600$, so the count is 80. Uniqueness is checked globally, including potential matches above 600.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2023-I-Q07] | Part B (Q6-Q10)

Answer: 049

1. Write the remainders as $r_2, \ldots, r_6$. For even n, distinctness forces $r_2 = 0, r_4 = 2, r_6 = 4$, then $r_3 = 1, r_5 = 3$. These are exactly $n \equiv 58 \pmod{60}$.

2. For odd n, one must have $r_2 = 1, r_4 = 3, r_6 = 5, r_3 = 2$, leaving $r_5 = 0$ or 4. The resulting classes modulo 60 are 35 and 59.

3. The classes $35, 58, 59$ have respectively $17, 16, 16$ representatives in $1, \ldots, 999$. Their sum is 49.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2024-II-Q07] | Part B (Q6-Q10)

Answer: 699

1. Let the digits be a, b, c, d and $k \equiv N \pmod 7$. Replacement conditions imply $k \equiv 6(a-1) \equiv 2(b-1) \equiv 3(c-1) \equiv d-1$. Thus $(a, b, c, d) \equiv (1-k, 1+4k, 1+5k, 1+k) \pmod 7$.

2. Substituting into $k \equiv 6a + 2b + 3c + d$ gives $k \equiv 5 + 4k$, so $k = 3 \pmod 7$. The digit residues are $(5, 6, 2, 4)$.

3. Choose the largest allowable digit in each class: $N = 5694$. Replacements yield $1694, 5194, 5614, 5691$, all divisible by 7. Hence $Q + R = 5 + 694 = 699$.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2026-I-Q08] | Part B (Q6-Q10)

Answer: 244

1. Factor $17017 = 7 \cdot 11 \cdot 13 \cdot 17$. A divisor is $7^a 11^b 13^c 17^d$, each exponent from zero to seventeen. Modulo twelve, only the parities of a, b, d matter, since the residues are $7, 11, 1, 5$.

2. Of eight parity patterns, exactly $(a, b, d) \equiv (0, 0, 1), (1, 1, 0) \pmod 2$ produce residue five. Each parity allows nine exponent choices and c has eighteen choices. Thus $N = 2 \cdot 9^3 \cdot 18 = 26244$, remainder 244.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-I-Q14] | Part C (Q11-Q15)

Answer: 030

1. If the size is n and sum S, each deletion requires $S - x \equiv 0 \pmod{n-1}$. Thus all elements are $1 \pmod{n-1}$, and $n-1 \mid 2001 = 3 \cdot 23 \cdot 29$.

2. Distinctness gives $2002 \geq 1 + (n-1)^2$, so $n - 1 \leq 44$. The largest divisor of 2001 in that range is 29, yielding $n \leq 30$.

3. Take $1, 30, 59, \ldots, 813$ (twenty-nine terms) together with 2002. All thirty terms are $1 \pmod{29}$, making their total $1 \pmod{29}$ and every deletion divisible by 29. Thus 30 is attainable.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2011-II-Q14] | Part C (Q11-Q15)

Answer: 440

1. For each modulus m, all positions of one residue class receive values from a single residue class. Every position class and every value class has $30/m$ members.

2. Two position classes cannot map to the same value class, as that would require twice as many distinct values as available. Thus the induced residue map is a permutation σ_m.

3. Conversely, any three residue permutations uniquely specify every value by the Chinese remainder theorem modulo 30, and map distinct position triples to distinct value triples.

4. The count is $2!3!5! = 1440$, leaving remainder 440.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-I-Q15] | Part C (Q11-Q15)

Answer: 332

1. Permutation of the seats requires $\gcd(a, n) = 1$. A nonzero seat difference d must also obey $ad \not\equiv d, -d \pmod{n}$, equivalent to multiplication by $a - 1$ and $a + 1$ having no nonzero kernel.

2. Thus all of $a - 1, a, a + 1$ must be coprime to n. Three consecutive integers always contain multiples of both two and three, so n can have neither prime factor.

3. Conversely, if $\gcd(n, 6) = 1$, choose $a = 3$. The consecutive product $2 \cdot 3 \cdot 4$ is coprime to n, proving sufficiency.

4. The integers in range are $6k \pm 1$, $1 \leq k \leq 166$, numbering 332.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-II-Q12] | Part C (Q11-Q15)

Answer: 958

1. The allowed residues modulo p are $3, \ldots, p-3$. There are $2, 6, 8$ choices for the three primes, so the Chinese remainder theorem gives 96 allowed classes modulo 1001.

2. Ten complete blocks, 1 through 10010, contain 960 safe numbers. Among the excess last ten numbers, only $10006, 10007$ are seven-safe; their residues modulo eleven are $7, 8$ and modulo thirteen $9, 10$, so both are fully safe.

3. Subtracting them gives 958.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-I-Q11] | Part C (Q11-Q15)

Answer: 148

1. Write $N = 1680k$. Below fourteen, only $9, 11, 13$ fail to divide 1680, so they must be the three remainder-three divisors.

2. The congruences reduce to $k \equiv 2 \pmod 3, k \equiv 10 \pmod{11}, k \equiv 1 \pmod{13}$. Combining gives $k \equiv 131 \pmod{429}$; for example 131 satisfies all, and the Chinese remainder theorem makes the class unique.

3. The least positive k is 131, a prime. The distinct primes of N are $2, 3, 5, 7, 131$, whose sum is 148.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-I-Q15] | Part C (Q11-Q15)

Answer: 272

1. Write $a = b + d, c = b + 2d$, where $0 < 2d < p$. If the original progression has difference D, then $D \equiv -d$ and $D \equiv 2d \pmod{p}$. Thus $p \mid 3d$ and $p \nmid d$, forcing $p = 3$.

2. Then $d = 1$ and $(b, a, c) = (0, 1, 2)$. Exactly the progressions $A = 1 + 3j, D = 2 + 3k$ qualify, with nonnegative integers j, k.

3. The upper bound is $C = 5 + 3j + 6k \leq 99$, or $j + 2k \leq 31$. Summing $32 - 2k$ over $k = 0, \ldots, 15$ gives 272.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-II-Q14] | Part C (Q11-Q15)

Answer: 512

1. A remainder is at most both $k-1$ and $n-2k$, since the quotient is at least two. Write $n=3m+r$, $r=0,1,2$; the choice $k=m+1$ gives remainder $m+r-2$ for this range of n.

2. For $k \geq m+1$, the bound $n-2k$ proves optimality. For smaller k, $k-1$ does so, except when $r=0, k=m$, which instead has zero remainder. Thus $F(3m)=m-2, F(3m+1)=m-1, F(3m+2)=m$.

3. Group $20,\ldots,100$ into triples $3m-1, 3m, 3m+1$ for $m=7,\ldots,33$. Each contributes $3m-4$, so the sum is $3(7+33)27/2-4\cdot 27=1512$.

4. The remainder is 512.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2019-II-Q14] | Part C (Q11-Q15)

Answer: 071

1. For each residue r modulo five, let M_r be its least representable amount. Adding fives then represents every larger amount in that residue, so the largest missing amount is $\max_r M_r - 5$. Thus the maximum residue minimum must be 96, and in particular $M_1 = 96$.

2. Using k stamps of values $n, n+1$ produces $kn + c$ for $0 \le c \le k$. According as $n \equiv 0, 1, 2, 3, 4 \pmod 5$, the least residue-one amounts are $n+1, n, 2n+2, 2n, 4n$. Setting each to ninety-six gives candidates $95, 96, 47, 48, 24$.

3. For 95 a residue-four minimum exceeds ninety-six; for 96 a residue-three minimum does; for 48 the residue-two minimum is ninety-seven. The survivors $24, 47$ have all residue minima at most ninety-six, with the required maximum. Their sum is 71.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2021-II-Q13] | Part C (Q11-Q15)

Answer: 797

1. $n = 1, 2$ fail. For $n \geq 3$, modulo 8 forces n odd and $n \equiv 5 \pmod 8$; modulo 125 the condition is $n \equiv 2^n \pmod{125}$.

2. First $n \equiv 1 \pmod 4$ yields $2^n \equiv 2 \pmod 5$, so $n \equiv 17 \pmod{20}$. Then $2^{17} \equiv 22 \pmod{25}$ implies $n \equiv 97 \pmod{100}$.

3. Finally $2^{97} \equiv 47 \pmod{125}$, giving $n \equiv 47 \pmod{125}$. Together with $n \equiv 5 \pmod 8$, this forces $n \equiv 797 \pmod{1000}$. Direct modular substitution verifies 797, so it is least.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2017-I-Q03] | Part A (Q1-Q5)

Answer: 069

1. Triangular numbers at indices $n+20, n$ differ by $20n+210$, divisible by10. Thus the last digits have period20.

2. One period is $(1,3,6,0,5,1,8,6,5,5,6,8,1,5,0,6,3,1,0,0)$, with sum70. Since $2017 = 100 \cdot 20 + 17$, the remaining first17 terms sum69.

3. Total 7069 has remainder69, or AIME format069.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2018-II-Q02] | Part A (Q1-Q5)

Answer: 112

1. The initial terms are $2, 5, 8, 5, 6, 10, 7, 4, 7, 6, 2, 5, 8$. The consecutive triple at indices ten through twelve repeats the initial triple.

2. Because three terms determine every future term, this proves period ten. The requested terms are $a_8 = 7, a_0 = 2, a_2 = 8$, whose product is 112.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2001-II-Q10] | Part B (Q6-Q10)

Answer: 784

1. Since 10 is coprime to 1001, divisibility is equivalent to $10^{j-i} \equiv 1 \pmod{1001}$. The orders modulo $7, 11, 13$ are $6, 2, 6$, so $j - i$ must be a multiple of 6.

2. Different pairs give different integers, because the number has i trailing zeros followed by exactly $j - i$ nines. For gap $6d$, there are $100 - 6d$ choices, $1 \leq d \leq 16$.

3. The total is $\sum_{d=1}^{16}(100 - 6d) = 784$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2021-II-Q09] | Part B (Q6-Q10)

Answer: 295

1. If an odd prime divides both numbers, the order of 2 modulo that prime has 2-adic valuation exactly $v_2(m)+1$ and divides n. Necessarily $v_2(n) > v_2(m)$.

2. Conversely, under this inequality let $g = \gcd(m,n)$. Then m/g is odd and n/g even, so 2^g+1 divides both numbers. Thus the criterion is sufficient too.

3. The counts of $v_2 = 0,1,2,3,4$ in 1 through 30 are $15, 8, 4, 2, 1$. Therefore the pair count is $15\cdot 15 + 8\cdot 7 + 4\cdot 3 + 2\cdot 1 = 295$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2011-I-Q11] | Part C (Q11-Q15)

Answer: 007

1. The first three residues are $1, 2, 4$; all later ones are divisible by 8 and not by 5, so these classes are disjoint.

2. Modulo 125, $2^{10} = 24, 2^{20} = 76, 2^{40} = 26$, so $2^{50} = 26 \cdot 24 \equiv -1$. Hence for $n \geq 3$, $2^{n+50} \equiv -2^n \pmod{1000}$, using divisibility by 8 as well.

3. The later residue set is therefore closed under $r \mapsto 1000 - r$. There is no fixed point: 0 and 500 cannot be powers of two modulo 1000. Pairing distinct residues makes their total a multiple of 1000.

4. Only $1 + 2 + 4$ remains, so the answer is 7.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2018-I-Q11] | Part C (Q11-Q15)

Answer: 195

1. The condition is $3^n \equiv 1 \pmod{143^2}$, equivalent to congruence one modulo 121 and 169.

2. Modulo 121, $3^5 = 243 \equiv 1$ and $3 \not\equiv 1$, so the order is five. Modulo thirteen, the order is three; for $n = 3k$, binomial expansion gives $3^{3k} = (1+26)^k \equiv 1 + 26k \pmod{169}$, since 26^2 is divisible by 169.

3. This is one exactly when $13 \mid k$, so the order modulo 169 is thirty-nine. The least simultaneous exponent is $\operatorname{lcm}(5, 39) = 195$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2019-I-Q14] | Part C (Q11-Q15)

Answer: 097

1. If odd prime p divides it, 2019 has multiplicative order exactly sixteen modulo p: its sixteenth power is one, but its eighth is minus one. Hence $p \equiv 1 \pmod{16}$. The first such prime candidates are seventeen and ninety-seven; intermediate numbers $33, 49, 65, 81$ are composite.

2. Modulo seventeen, $2019 \equiv 13$ and $13^2 \equiv -1$, so its eighth power is one, not minus one. Modulo ninety-seven, successive squared residues are $33, 22, -1$, so divisibility holds.

3. The least odd prime factor is 97.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2020-I-Q12] | Part C (Q11-Q15)

Answer: 270

1. Because $149 - 2 = 3 \cdot 7^2$, the odd-prime lifting formula gives $v_3 = 1 + v_3(n)$ and $v_7 = 2 + v_7(n)$. Thus $3^2 7^5 \mid n$.

2. Modulo 5, $(149/2)^n \equiv 2^n$, whose order is 4. Write $n = 4k$. Since $v_5(149^4 - 2^4) = 1$, lifting gives $v_5(149^n - 2^n) = 1 + v_5(k)$, so $4 \cdot 5^4 \mid n$.

3. These conditions are necessary and sufficient. The least exponent is $2^2 3^2 5^4 7^5$, which has $(2+1)(2+1)(4+1)(5+1) = 270$ divisors.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2023-II-Q15] | Part C (Q11-Q15)

Answer: 363

1. Write $a_n = 1 + k_n 2^n$. Minimality makes k_n the representative in $\{1, \ldots, 22\}$ of $-2^{-n} \pmod{23}$. Equality with a_{n+1} holds exactly when k_n is even.

2. Since $2^{11} \equiv 1 \pmod{23}$ and the order is 11 (a prime, and $2 \not\equiv 1$), the sequence k_n repeats every eleven terms.

3. For $n = 1, \ldots, 11$, its values are $11, 17, 20, 10, 5, 14, 7, 15, 19, 21, 22$. These follow from $k_{n+1} = k_n/2$ for even k_n, or $(k_n + 23)/2$ for odd k_n. Even entries occur at residues $3, 4, 6, 0 \pmod{11}$.

4. In $1, \ldots, 1000 = 90 \cdot 11 + 10$, there are ninety full periods and three extra even entries. The count is $90 \cdot 4 + 3 = 363$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2024-I-Q13] | Part C (Q11-Q15)

Answer: 110

1. $p = 2$ fails modulo 4. For odd p, a root has multiplicative order 8, so $8 \mid p - 1$. The first candidate is 17, whose roots of $x^4 = -1$ are $2, 8, 9, 15$.

2. For each root a, write $m = a + 17t$. Modulo 289, the condition is $(a^4 + 1)/17 + 4a^3 t \equiv 0 \pmod{17}$. The derivative coefficient is nonzero, so each root lifts uniquely.

3. The four lifted residues are $155, 110, 179, 134$, respectively. Therefore the least positive root is 110; indeed $110^4 + 1 = 289 \cdot 506609$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2025-I-Q15] | Part C (Q11-Q15)

Answer: 735

1. Modulo 9, a unit cube is 1 or -1. Thus a solution has exactly two units or no units. Cubing maps the 486 units modulo 3^6 bijectively onto the unit cube classes modulo 3^7, which are precisely those congruent to $\pm 1 \pmod 9$; this follows from the kernel of cubing on units having size 3.

2. With two units, choose the divisible variable in three ways, its value in 243 ways, and one unit in 486 ways. The remaining unit is forced by the bijection. This contributes 354294.

3. If all variables are divisible by 3, divide them by 3 and reduce the cube-sum condition modulo 81. Only their residues modulo 27 matter, with 9^3 lifts. At this lower level the two-unit count is $3 \cdot 18 \cdot 9 = 486$; all-divisible triples reduce to a sum-zero condition modulo 3, giving $9^3/3 = 243$. Thus this case contributes $9^3(486 + 243) = 531441$. Total 885735, remainder 735.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2000-II-Q02] | Part A (Q1-Q5)

Answer: 098

1. The two factors $x-y, x+y$ have the same parity, and their product is even, so both are even. Set $x-y=2a, x+y=2b$ to get $ab=10^6=2^6 5^6$.

2. There are 49 ordered positive factor pairs. Each yields $x=a+b, y=b-a$; their simultaneous negatives give 49 more. Opposite signs cannot produce the positive product, so the total is 98.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2002-II-Q03] | Part A (Q1-Q5)

Answer: 111

1. The logarithms give $abc = 6^6$, and geometricity gives $abc = b^3$. Hence $b = 36$ and $ac = 1296$.

2. Write $a = 36 - k^2 > 0$ with $1 \le k \le 5$. The candidates are $35, 32, 27, 20, 11$; only 27 divides 1296.

3. Thus $(a, b, c) = (27, 36, 48)$, and the sum is 111.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-II-Q01] | Part A (Q1-Q5)

Answer: 336

1. Write the numbers as $a, b, c = a + b$. Then $abc = 12c$, so $ab = 12$ and $N = 12(a + b)$.

2. The unordered factor pairs $(1, 12), (2, 6), (3, 4)$ give $N = 156, 96, 84$. Their sum is 336.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q03] | Part A (Q1-Q5)

Answer: 384

1. For integer dimensions p, q, r, the hidden count is $(p-1)(q-1)(r-1) = 231 = 3 \cdot 7 \cdot 11$.

2. Up to order, the hidden dimensions are $(3, 7, 11), (1, 3, 77), (1, 7, 33), (1, 11, 21), (1, 1, 231)$. Adding one to each dimension gives total counts $384, 624, 544, 528, 928$.

3. The least is 384, from a $4 \times 8 \times 12$ block.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2005-I-Q04] | Part A (Q1-Q5)

Answer: 294

1. Write $N = s^2 + 5 = x(x+7)$. Completing the square gives $(2x+7-2s)(2x+7+2s) = 69$.

2. The positive factor pairs are $(1, 69)$ and $(3, 23)$. They give $(x, s) = (14, 17)$ or $(3, 5)$, hence sizes $294, 30$. The maximum is 294.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-I-Q03] | Part A (Q1-Q5)

Answer: 314

1. Eliminating b gives $4j + 5s = 57$. Its positive integer solutions are $(j, s) = (13, 1), (8, 5), (3, 9)$.

2. The respective $2b$ values are $31, 30, 29$, so integrality selects $(b, j, s) = (15, 8, 5)$. Squared sum $225 + 64 + 25 = 314$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-I-Q04] | Part A (Q1-Q5)

Answer: 080

1. Complete the square: $[y-(x+42)][y+(x+42)]=244$. Both factors are positive and of equal parity.

2. The only equal-parity positive factor pair is $2, 122$. Thus $y=62, x=18$, giving 80.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-II-Q04] | Part A (Q1-Q5)

Answer: 089

1. For c children, the arithmetic-series sum is $c(n+1-c)=2009$. Hence $c \mid 2009$ and $n=2009/c+c-1$. Last-place nonnegativity imposes $c(c-1) \leq 2009$.

2. Since $2009=7^2 \cdot 41$, allowed divisors are $c=1,7,41$; larger divisors already exceed the inequality. They give $n=2009,293,89$.

3. The minimum is 89, attained by forty-one children ending at nine grapes.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2010-I-Q05] | Part A (Q1-Q5)

Answer: 501

1. Factor the squared differences. Since $a-b, c-d \geq 1$, one has $(a-b)(a+b)+(c-d)(c+d) \geq a+b+c+d$, with equality only when both differences are 1.

2. Thus $b=a-1, d=c-1$ and $a+c=1006$. The strict middle inequality gives $a \geq 504$, while $d \geq 1$ gives $a \leq 1004$.

3. Every integer in that interval gives $(a, a-1, 1006-a, 1005-a)$, so the count is $1004-504+1=501$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2012-II-Q01] | Part A (Q1-Q5)

Answer: 034

1. Divide by four to get $5m + 3n = 503$. Modulo three, $m \equiv 1$, so write $m = 3k + 1, n = 166 - 5k$.

2. Positivity is exactly $0 \leq k \leq 33$, giving 34 pairs.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2021-I-Q05] | Part A (Q1-Q5)

Answer: 031

1. Write the terms as $a-d, a, a+d$ with $d \geq 1$. The equation is $d^2(a-2)=3a^2$, forcing $a > 2$.

2. With $t=a-2$, obtain $d^2=3t+12+12/t$, so t is a positive divisor of 12. For $t = 1,2,3,4,6,12$ the squared differences are $27,24,25,27,32,49$.

3. Only $(a,d)=(5,5),(14,7)$ work. Their last terms are $10, 21$, summing to 31.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2025-I-Q04] | Part A (Q1-Q5)

Answer: 117

1. Factor as $(3x+2y)(4x-3y)=0$. The two integer families are $(x,y)=(2t,-3t)$ and $(3t,4t)$.

2. The first permits $-33 \le t \le 33$, giving 67 pairs; the second permits $-25 \le t \le 25$, giving 51. Only the origin is shared, so total $67+51-1=117$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2026-I-Q04] | Part A (Q1-Q5)

Answer: 070

1. For $n = a + b + ab$, the condition is $n + 1 = (a + 1)(b + 1)$ with distinct factors at least two. Thus $n + 1$ must be composite but not a prime square.

2. Among $2, \ldots, 101$ there are 100 integers, 26 primes, and four prime squares $4, 9, 25, 49$. Every other composite has unequal factors obtained from its smallest prime factor. Count $100 - 26 - 4 = 70$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2000-I-Q06] | Part B (Q6-Q10)

Answer: 997

1. The equation gives $(\sqrt{y}-\sqrt{x})^2=4$, so $\sqrt{y}-\sqrt{x}=2$. Also $\sqrt{y}+\sqrt{x}=(y-x)/2$ is rational, making both square roots rational and hence integral.

2. Therefore $x=a^2, y=(a+2)^2$. The strict upper bound requires $a+2\leq 999$, so $1\leq a\leq 997$ and there are 997 pairs.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2002-I-Q08] | Part B (Q6-Q10)

Answer: 748

1. With first terms $x \leq y$, iteration gives $a_9 = 13x + 21y$. Suppose two distinct pairs have $x < u$; equality of ninth terms gives $13(u - x) = 21(y - v)$.

2. Coprimality implies $u - x \geq 21$, hence $u \geq 22$ and $v \geq u \geq 22$. Thus the common value is at least $34 \cdot 22 = 748$.

3. The pairs $(1, 35)$ and $(22, 22)$ both give 748, proving this lower bound attainable.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-I-Q07] | Part B (Q6-Q10)

Answer: 380

1. The altitude from D meets AC at its midpoint, distance 15 from A and 6 from B. If $a = AD, b = BD$, subtracting Pythagorean equations gives $(a+b)(a-b) = 225 - 36 = 189$.

2. The positive factor pairs yield $(a, b) = (95, 94), (33, 30), (17, 10), (15, 6)$. The last has zero altitude and is excluded.

3. The valid perimeters are $220, 96, 64$, summing to 380.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-I-Q08] | Part B (Q6-Q10)

Answer: 129

1. Write them as $a, a+d, a+2d, a+30$. Geometricity gives $(a+d)(a+30)=(a+2d)^2$, hence $3a(10-d)=2d(2d-15)$.

2. Positivity makes the two parenthesized factors have the same sign, so $7.5<d<10$. For $d=8$, $a=8/3$ is not integral; for $d=9$, $a=18$.

3. The sequence is $18, 27, 36, 48$, whose sum is 129.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-II-Q10] | Part B (Q6-Q10)

Answer: 156

1. Write them $x, x+60$. Squaring shows $x(x+60)=z^2$ for an integer z, hence $(x+30-z)(x+30+z)=900$.

2. The factors must both be even. Positive factor pairs below the equal pair give $x=196, 48, 20, 4$; the equal pair gives $x=0$, excluded.

3. At $x=196$, both original integers are squares and the resulting radicand is 900, forbidden. At $x=48$, their square roots sum to $10\sqrt{3}=\sqrt{300}$, allowed. Therefore the maximum sum is $48+108=156$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q06] | Part B (Q6-Q10)

Answer: 408

1. Integral splits force the three portions to be $8x, 8y, 24z$. The final totals are $6x + 3y + 11z$, $x + 2y + 11z$, $x + 3y + 2z$.

2. Equating these to ratio $3 : 2 : 1$ gives $3x + 5z = 6y$ and $x + 4y = 7z$. Elimination yields $9y = 13z$, hence $(x, y, z) = (11t, 13t, 9t)$ for positive integer t.

3. The initial total is $8x + 8y + 24z = 408t$, whose minimum is 408.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-II-Q09] | Part B (Q6-Q10)

Answer: 000

1. Subtract one from each variable in the first equation. It bijects to nonnegative solutions of the second. Thus the difference counts its solutions with at least one zero variable.

2. For $x = 0$, even $y = 0, 2, \ldots, 666$ gives 334; for $y = 0$, $x = 0, \ldots, 500$ gives 501; for $z = 0$, $y = 0, 4, \ldots, 664$ gives 167. Only the two axial solutions $(0, 0, 1000), (500, 0, 0)$ are double counted.

3. Difference $334 + 501 + 167 - 2 = 1000$, so remainder 0.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2013-II-Q06] | Part B (Q6-Q10)

Answer: 282

1. Before $500^2 = 250000$, successive squares differ by at most 999, so every earlier thousand-number block contains a square. It remains to check blocks starting with index 250.

2. For $x = 500 + a$, $x^2 = 1000(250 + a) + a^2$. For $0 \leq a \leq 31$, the final term is below one thousand, so blocks 250 through 281 all contain these squares.

3. The next squares are $531^2 = 281961$ and $532^2 = 283024$. They leave the entire block 282000 through 282999 empty of squares.

4. Therefore the least N is 282.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2013-II-Q07] | Part B (Q6-Q10)

Answer: 945

1. Let the starting staff be n and each removed group $k > 0$. Work balance gives $30n + 30(n-k) + 30(n-2k) + 5(n-3k) = 1775$, hence $19n - 21k = 355$.

2. Modulo nineteen, $k \equiv 3$, so $k = 3 + 19t, n = 22 + 21t$ with integer $t \geq 0$. The final staff is $n - 3k = 13 - 36t > 0$, forcing $t = 0$.

3. The first ninety-minute output is $30(22) + 15(19) = 945$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2015-II-Q08] | Part B (Q6-Q10)

Answer: 036

1. If either variable is 1, the target is 1. Otherwise both are at least 2, and the condition rearranges to $(2a-3)(2b-3)<5$.

2. The two factors are positive odd integers. Their only possibilities, apart from swapping, are $(1,1),(1,3)$, giving $(a,b)=(2,2),(2,3)$.

3. The target values are $65/16$ and $217/35=31/5$. The maximum is $31/5$, larger than 1, and $p+q=36$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2017-II-Q06] | Part B (Q6-Q10)

Answer: 195

1. Write the square as m^2. Completing the square gives $(2m-(2n+85))(2m+(2n+85))=843=3\cdot 281$. The first factor is positive because $4m^2>(2n+85)^2$.

2. The increasing positive odd factor pairs are $(1,843)$ and $(3,281)$. If the factors are u,v, then $n=(v-u-170)/4$, giving168 and27, respectively; the corresponding m are211 and71.

3. Both satisfy the original equation and positivity. The requested sum is195.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q14] | Part C (Q11-Q15)

Answer: 108

1. Divide by seven. Only summands $1, 11, 111$ can occur, since $1111 > 1000$. If their multiplicities are x, y, z, then $x + 11y + 111z = 1000$, while $n = x + 2y + 3z = 1000 - 9(y + 12z)$.

2. For $z = 0, \ldots, 9$, the possible values $v = y + 12z$ lie in integer intervals with endpoints $(0, 90), (12, 92), (24, 94), (36, 96), (48, 98), (60, 100), (72, 102), (84, 104), (96, 106)$, and the single value 108.

3. The union is $\{0, 1, \ldots, 106, 108\}$, of size 108. Each gives a distinct positive n, so the answer is 108.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2005-II-Q13] | Part C (Q11-Q15)

Answer: 418

1. Put $S(x) = P(x) - x - 3$. Since $S(17) = S(24) = -10$, division by the two monic linear factors gives $S(x) = -10 + (x-17)(x-24)Q(x)$ with integer-coefficient Q.

2. An integer zero requires $(n-17)(n-24) \mid 10$. The factors differ by seven; testing signed factor pairs of divisors of ten leaves only $(2,-5)$ and $(5,-2)$, giving $n = 19, 22$.

3. The stipulated two distinct solutions must be these, so their product is 418.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-II-Q15] | Part C (Q11-Q15)

Answer: 181

1. For a positive candidate, write $n^2 = 3u^2 + 3u + 1$. Then $(2n-1)(2n+1) = 3(2u+1)^2$. These coprime odd factors force one square and the other three times a square. The lower factor cannot be three times a square, since the upper would be two modulo three. Hence $2n - 1 = k^2$ with odd $k > 0$.

2. Write $2n + 79 = d^2$. Subtracting yields $(d-k)(d+k) = 80$. The positive even factor pairs $2, 40$; $4, 20$; $8, 10$ give $k = 19, 8, 1$. Oddness leaves $k = 19, 1$, giving $n = 181, 1$.

3. The larger works: $181^2 = 105^3 - 104^3$ and $2 \cdot 181 + 79 = 21^2$. Thus the maximum is 181.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-I-Q14] | Part C (Q11-Q15)

Answer: 905

1. Let u_j count entries equal to j. Subtract the count equation from the first- and fourth-power equations to get $u_2 + 2u_3 + 3u_4 = 163$ and $15u_2 + 80u_3 + 255u_4 = 4395$.

2. Eliminating u_2 gives $5u_3 + 21u_4 = 195$. Hence u_4 is a multiple of five and $0 \le u_4 \le 9$, leaving $u_4 = 0, 5$. The corresponding count vectors are $(226, 85, 39, 0), (215, 112, 18, 5)$.

3. Their square sums are $917, 905$, so the minimum is 905.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2011-I-Q15] | Part C (Q11-Q15)

Answer: 098

1. Vieta gives $a+b+c=0$ and $ab+bc+ca=-2011$. After changing all signs if needed, choose $a \geq b \geq 0$, with $c=-(a+b)$. Then $a^2+ab+b^2=2011$.

2. Complete the square: $(2a+b)^2=8044-3b^2$. Since $3b^2 \leq 2011$, it suffices to test $0 \leq b \leq 25$. In this interval the right side is a square only for $b=10$, where it is $7744=88^2$.

3. Thus $a=39, c=-49$, giving $39+10+49=98$. These values satisfy both Vieta relations, so they are achieved by an integer m.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2017-I-Q13] | Part C (Q11-Q15)

Answer: 059

1. If $k^3 \le n < (k+1)^3$, the next candidate cube is $(k+1)^3$. The worst ratio in that block is $(1+1/k)^3$, decreasing with k.

2. For $m \ge 8$, every block works from $n = 1$, so $Q = 1$. For $4 \le m \le 7$, $n = 1$ fails, $2 \le n \le 7$ is covered by8, and blocks $k \ge 2$ have ratio at most $27/8 < 4$, so $Q = 2$.

3. For $m = 3$, $n = 8$ fails; $9 \le n \le 26$ is covered by27, and later ratios are at most $64/27 < 3$, so $Q = 9$. For $m = 2$, $n = 31$ fails; $32 \le n \le 63$ is covered by64, and later ratios are at most $125/64 < 2$, so $Q = 32$.

4. The sum is $32 + 9 + 4 \cdot 2 + 2010 = 2059$, with remainder59.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-I-Q01] | Part A (Q1-Q5)

Answer: 630

1. Write the number as $10a + b$. Neither digit can be zero. Divisibility by a gives $b = ka$; divisibility by b then requires $k \mid 10$. Since $ka \leq 9$, only $k = 1, 2, 5$ occur.

2. For $k = 1$, the numbers $11, 22, \ldots, 99$ sum to 495; for $k = 2$, $12, 24, 36, 48$ sum to 120; for $k = 5$, only 15 works. Total 630.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-II-Q01] | Part A (Q1-Q5)

Answer: 816

1. The allowable blocks are $16, 25, 36, 49, 64, 81$. They give directed digit chains $1 \to 6 \to 4 \to 9$, $2 \to 5$, $3 \to 6 \to 4 \to 9$, and $8 \to 1 \to 6 \to 4 \to 9$.

2. There is no directed cycle, so no arbitrarily long number. The unique longest chain gives $N = 81649$, whose first three digits are 816.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2002-I-Q03] | Part A (Q1-Q5)

Answer: 025

1. Write Jane's future age as $10a + b \geq 26$. Dick's is $10b + a$, which is larger exactly when $b > a$. Each such digit pair uniquely gives $n = 10a + b - 25$ and $d = 25 + 9(b - a)$.

2. Tens digit 2 allows $26, 27, 28, 29$, giving four. Tens digits 3 through 8 allow $6, 5, 4, 3, 2, 1$ choices. Their sum is 25.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2002-II-Q01] | Part A (Q1-Q5)

Answer: 009

1. Write $x = 100a + 10b + c$, so $y = 100c + 10b + a$. Their absolute difference is $99|a - c|$.

2. Both end digits a, c must be in 1 through 9, so $|a - c|$ takes exactly $0, 1, \ldots, 8$. Thus there are 9 distinct values.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2003-II-Q02] | Part A (Q1-Q5)

Answer: 120

1. At most ten digits can be used. A feasible ten-digit number beats every shorter one, so maximize the prefix lexicographically as 9876543 if the remaining digits $0, 1, 2$ admit a suffix divisible by eight.

2. The six suffix permutations are $012, 021, 102, 120, 201, 210$; only 120 is divisible by eight. Thus $N = 9876543120$, and the remainder is 120.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2004-I-Q01] | Part A (Q1-Q5)

Answer: 217

1. For leading digit $a = 3, \ldots, 9$, the number is $1111a - 123$. As $1111 \equiv 1$ and $123 \equiv 12 \pmod{37}$, its remainder is $a + 25$.

2. The remainders are $28, 29, \ldots, 34$, summing to $7 \cdot 31 = 217$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2006-I-Q03] | Part A (Q1-Q5)

Answer: 725

1. Write the number $d10^k + u$ with leading digit d. Then $d10^k = 28u$. Since seven does not divide a power of ten, it divides d, forcing $d = 7$.

2. Thus $u = 10^k/4$, requiring $k \geq 2$. The least case has $u = 25$, giving 725.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2008-II-Q04] | Part A (Q1-Q5)

Answer: 021

1. The ordinary ternary representation is 2202101_3. Replace each digit two by minus one with a carry to the next position, giving $2008 = 3^7 - 3^5 + 3^4 - 3^3 + 3^2 + 3^0$.

2. Balanced ternary digits are unique because reduction modulo three determines the final digit and then induction determines the rest. Exponent sum $7 + 5 + 4 + 3 + 2 + 0 = 21$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2009-I-Q01] | Part A (Q1-Q5)

Answer: 840

1. The smallest possible leading digit is one; integral later digits force positive integer ratio at least two, giving minimum 124. Zero cannot occur, since it would force repeated zeroes.

2. For maximum leading digit nine, the middle digit b requires $b^2/9$ to be an integer digit distinct from nine and b. Thus $b = 3, 6$ give $931, 964$, with maximum 964. Difference $964 - 124 = 840$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2010-II-Q01] | Part A (Q1-Q5)

Answer: 640

1. The available digits $0, 2, 4, 6, 8$ sum to 20. A positive multiple of 9 using them must have even digit sum 18; thus the used nonzero digits are exactly $4, 6, 8$, optionally with 0.

2. The largest arrangement is 8640. Its last two digits are divisible by 4, so it is divisible by both 4 and 9, hence by 36.

3. The remainder is 640.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-I-Q01] | Part A (Q1-Q5)

Answer: 040

1. Divisibility requires $2b + a \equiv 2b + c \equiv 0 \pmod 4$. Thus a, c lie together in either $\{2, 6\}$ or $\{4, 8\}$.

2. Each class gives four ordered pairs. The first requires odd b and the second even b, each with five choices. Hence the count is $2 \cdot 4 \cdot 5 = 40$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-I-Q05] | Part A (Q1-Q5)

Answer: 330

1. Adding one replaces a terminal zero followed by k ones with a one followed by k zeros, changing the one count by $1-k$. It preserves the required weight exactly when $k=1$.

2. Thus the smaller number ends in 01, the larger in 10, and the shared eleven-digit prefix has seven ones and four zeros. There are $\binom{11}{4}=330$ such prefixes and exactly one successful ordered subtraction for each.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2013-I-Q02] | Part A (Q1-Q5)

Answer: 200

1. The last digit is zero or five, but cannot be zero because it equals the first. Thus both endpoint digits are five.

2. Choose the next two digits arbitrarily. The final middle digit must have one specified residue modulo five, with two available decimal digits. Hence the count is $10 \cdot 10 \cdot 2 = 200$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2015-II-Q03] | Part A (Q1-Q5)

Answer: 476

1. Write the number as $17k$. Congruence to its digit sum modulo 9 forces $17k \equiv 17 \pmod 9$, hence $k \equiv 1 \pmod 9$.

2. The first candidates are $k = 1, 10, 19, 28$, giving $17, 170, 323, 476$. The first three have digit sum 8 and the last has digit sum 17. Thus the least answer is 476.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2017-I-Q05] | Part A (Q1-Q5)

Answer: 321

1. Fractional parts are below1 in both bases, so integer parts agree: $8a + b = 13b$, or $2a = 3b$. The base-eight bounds allow only $(a, b) = (3, 2), (6, 4)$.

2. The fractional equality is $(8c + d)/64 = (12b + a)/144$. The two cases respectively give $8c + d = 12$ and24. The latter requires a zero units digit, forbidden; the former yields $c = 1, d = 4$.

3. Thus the requested decimal numeral is321.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2017-II-Q04] | Part A (Q1-Q5)

Answer: 222

1. All one- through six-digit ternary strings qualify under the bound, contributing $\sum_{k=1}^{6} 2^k = 126$, since the largest is $222222_3 = 728$.

2. Convert the bound: $2017 = 2202201_3$. A seven-digit nonzero-digit string is allowed precisely when its first two digits are11,12 or21; prefix22 would exceed the bound at the third digit. This gives $3 \cdot 2^5 = 96$.

3. Total $126 + 96 = 222$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2018-I-Q02] | Part A (Q1-Q5)

Answer: 925

1. Place values give $196a + 14b + c = 225a + 15c + b = 222a + 37c$. The shared base-six digits satisfy $1 \le a \le 5, 0 \le c \le 5$.

2. Equating the last two forms gives $b = 22c - 3a$. The first two give $13b = 29a + 14c$. Combining yields $a = 4c$.

3. The only permitted digits are $c = 1, a = 4, b = 10$, and the value is $222(4) + 37 = 925$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2018-II-Q03] | Part A (Q1-Q5)

Answer: 371

1. We need $b \geq 8$, $3b + 6$ a square, and $2b + 7$ a cube. The last expression is odd and below 2007, so its positive odd cube possibilities give $b = 10, 59, 168, 361, 662$ after discarding the invalid negative case.

2. The corresponding square candidates are $36, 183, 510, 1089, 1992$. Only $36 = 6^2$ and $1089 = 33^2$ qualify, at bases ten and 361.

3. The sum is 371.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2019-I-Q01] | Part A (Q1-Q5)

Answer: 342

1. The geometric sum is the numeral consisting of 321 ones followed by zero, minus 321. Its final four-digit block changes from 1110 to 0789.

2. There are 318 preceding ones. Digit sum $318+7+8+9=342$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2020-I-Q03] | Part A (Q1-Q5)

Answer: 621

1. Place-value expansion gives $121a + 11b + c = 512 + 64b + 8c + a$, or $120a = 512 + 53b + 7c$, where $1 \leq a \leq 7$ and $0 \leq b, c \leq 7$.

2. Necessarily $a \geq 5$. At $a = 5$, solve $53b + 7c = 88$: $b = 0$ fails divisibility, $b \geq 2$ is too large, and $b = 1$ gives $c = 5$.

3. The value is $515_{11} = 621 = 1155_8$. Any $a \geq 6$ gives at least $726 > 621$, proving minimality.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2020-II-Q05] | Part A (Q1-Q5)

Answer: 151

1. The first hexadecimal value needing a digit beyond 9 is 10. For an integer $k < 31$, its base-8 digit sum is at most 9, while $31 = 37_8$ has digit sum 10.

2. The least base-4 number with digit sum at least 31 uses eleven digits, with leading digit 1 and ten trailing 3 digits. This greedy placement puts as much digit mass as possible in low places; thus it also proves minimality.

3. $N = 13333333333_4 = 2 \cdot 4^{10} - 1 = 2097151$, so the remainder is 151.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2022-I-Q02] | Part A (Q1-Q5)

Answer: 227

1. Place value gives $100a + 10b + c = 81b + 9c + a$, or $99a = 71b + 8c$. Here $1 \leq a, b \leq 8$ and $0 \leq c \leq 8$.

2. Modulo 8, $b \equiv 5a$. For $a = 1, \ldots, 8$, the possible positive digits b are respectively $5, 2, 7, 4, 1, 6, 3, 8$. The corresponding $99a - 71b$ are $-256, 56, -200, 112, 424, 168, 480, 224$.

3. Only 56 is in $[0, 64]$ and divisible by 8. Thus $(a, b, c) = (2, 2, 7)$, and the answer is 227.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2023-II-Q02] | Part A (Q1-Q5)

Answer: 585

1. A base-eight number of at most three digits is at most 511. Any larger candidate below 1000 has four octal digits, beginning and ending with 1.

2. Its form is $\overline{1bb1}_8 = 513 + 72b$, with $b = 0, \ldots, 6$. The candidates are $513, 585, 657, 729, 801, 873, 945$.

3. Only 585 is a decimal palindrome; it exceeds all three-digit octal numbers and is therefore the maximum. Indeed $585 = 1111_8$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2024-II-Q03] | Part A (Q1-Q5)

Answer: 045

1. Write rows as a, b, c and d, e, f. The row sum forces $a + d = b + e = c + f = 9$: the units sum cannot be 19, so no carries occur, and the same argument repeats.

2. Let $S = a + b + c$. The column sum is $10S + (27 - S) = 99$, hence $S = 8$. Each nonnegative triple summing to 8 is valid and fixes the bottom digits.

3. No digit upper bound is active, so stars and bars gives $\binom{10}{2} = 45$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2025-I-Q01] | Part A (Q1-Q5)

Answer: 070

1. The divisibility is $(b+7) \mid (9b+7)$, equivalent to $(b+7) \mid 56$ by subtraction of $9(b+7)$.

2. Since $b+7 > 16$, it is 28 or 56. Thus $b = 21, 49$, whose sum is 70.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2026-I-Q02] | Part A (Q1-Q5)

Answer: 062

1. An even-length palindrome has even digit sum, so the middle digit is one of $1, 3, 5, 7, 9$. The left half then has digit sum $s = 6, 5, 4, 3, 2$, respectively, and determines the whole palindrome.

2. Since $s \leq 6$, nonzero-digit strings of sum s are exactly compositions of s, numbering 2^{s-1}. The sum is $32 + 16 + 8 + 4 + 2 = 62$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2026-II-Q04] | Part A (Q1-Q5)

Answer: 279

1. Every one-digit number works. For a number with at least two digits, base $b < 10$ strictly lowers its value because the positive leading-digit weight decreases. Thus a multidigit number works exactly when it contains digit nine.

2. There are nine one-digit numbers, $90 - 8 \cdot 9 = 18$ two-digit numbers containing nine, and $900 - 8 \cdot 9^2 = 252$ three-digit ones. Total $9 + 18 + 252 = 279$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-I-Q08] | Part B (Q6-Q10)

Answer: 315

1. For digits $d_i \in \{0, \ldots, 6\}$, the condition is $\sum d_i(10^i - 2 \cdot 7^i) = 0$. The coefficients for $i = 0, 1, 2, 3$ are $-1, -4, 2, 314$; subsequent coefficients are even larger positive numbers.

2. The total negative contribution is at most 30, so no position $i \geq 3$ can be nonzero. Thus $2d_2 = 4d_1 + d_0$.

3. To maximize $49d_2 + 7d_1 + d_0$, choose $d_2 = 6$ and then $d_1 = 3, d_0 = 0$. The resulting $630_7 = 315$ indeed reads as $630 = 2N$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2006-II-Q07] | Part B (Q6-Q10)

Answer: 738

1. There are 999 positive pairs. Exactly 99 have both units digits zero: $a = 10u, b = 10v, u + v = 100$.

2. If units digits are nonzero, a forbidden number must have form $h0u$ with $h, u = 1, \ldots, 9$. There are 81 choices for either member; its complement has tens digit nine and cannot also have this form. Hence these exclude 162 additional pairs.

3. Valid count $999 - 99 - 162 = 738$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-II-Q07] | Part B (Q6-Q10)

Answer: 032

1. Below 2^{12} are $\binom{12}{8} = 495$ such numbers, and below 2^{13} are $\binom{13}{8} = 1287$. The desired one is thirteen-bit. There are $\binom{11}{6} = 462$ beginning 11, the largest group, so its rank within that group is $462 - (1287 - 1000) = 175$.

2. The initial groups 1100, 11010, 110110 have counts $\binom{9}{6} = 84$, $\binom{8}{5} = 56$, $\binom{7}{4} = 35$, whose sum is 175. Thus take the greatest member of the final group, placing four remaining ones as far left as possible.

3. The binary string is $1101101111000_2 = 7032$, leaving remainder 32.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2015-I-Q08] | Part B (Q6-Q10)

Answer: 695

1. Numbers with fewer than three digits have digit sum at most 18. In addition, $s(864) = 18$, and each carry reduces the naive sum by 9, so exactly two carries must occur.

2. For a three-digit candidate $n = 100a + 10b + c$, $a + b + c = 20$ forces $a \geq 2$, guaranteeing a hundreds carry. If units carry but tens do not, $c \geq 6, b \leq 2$, so the minimum possible candidate is 929.

3. If tens carry but units do not, $c \leq 5$, so $a \geq 20 - 9 - 5 = 6$. The smallest realization is $(a, b, c) = (6, 9, 5)$, giving 695. Indeed $695 + 864 = 1559$ and both digit sums are 20.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2016-I-Q08] | Part B (Q6-Q10)

Answer: 162

1. Let the hundreds-column sum be H and units-column sum be U. Since all digits total 45, the overall sum is $100H + 10(45 - H - U) + U = 90H + 450 - 9U$.

2. Three distinct units digits sum between 6 and 24; ending in 0 requires $U = 10$ or 20. Since $H \geq 6$, the respective lower bounds are 900 and 810. The latter is attainable only with hundreds digits $\{1, 2, 3\}$ and $U = 20$.

3. From remaining digits $4, 5, 6, 7, 8, 9$, the units triples summing to 20 are $\{4, 7, 9\}, \{5, 6, 9\}, \{5, 7, 8\}$. Each permits $(3!)^3$ assignments to the three columns, so $n = 3 \cdot 216 = 648$.

4. Thus $m = 810$ and $|m - n| = 162$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2000-II-Q14] | Part C (Q11-Q15)

Answer: 495

1. The identity $(i+1)! - i! = i \cdot i!$ gives $a! - b! = \sum_{i=b}^{a-1} i \cdot i!$. Group k as $16! + \sum_{h=1}^{62}[(32h+16)! - (32h)!]$.

2. Thus $f_{16} = 1$ and, in each interval $32h \le i \le 32h+15$, $f_i = i$; other digits are zero. These are legal digits on disjoint intervals, so no carrying is needed.

3. Each sixteen-digit interval starts at an even index and contributes eight pairs $-i + (i+1) = 1$. The initial digit contributes -1. Total $62 \cdot 8 - 1 = 495$.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2003-I-Q13] | Part C (Q11-Q15)

Answer: 155

1. A length-$(d+1)$ binary number starts with one. If its remaining d digits contain j ones, the required inequality is $j \geq d/2$. The count through 2047 is therefore the upper halves of binomial rows 0 through 10.

2. By symmetry this count is $[2047 + (1 + 2 + 6 + 20 + 70 + 252)]/2 = 1199$.

3. The 44 integers 2004 through 2047 all have prefix 11111 and at least one more one, so all were included. Thus $N = 1155$, with remainder 155.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2014-II-Q15] | Part C (Q11-Q15)

Answer: 149

1. Index primes as $\rho_0 = 2, \rho_1 = 3, \rho_2 = 5, \ldots$. The invariant is $x_n = \prod_i \rho_i^{d_i}$, where d_i are the binary digits of n. It holds at zero, giving the empty product.

2. For the first missing prime ρ_j, all preceding primes occur once. The update removes them and inserts ρ_j, leaving higher primes unchanged. This is exactly a binary carry through the trailing one-bits, so induction proves the invariant and squarefreeness.

3. Now $2090 = 2 \cdot 5 \cdot 11 \cdot 19 = \rho_0\rho_2\rho_4\rho_7$, so its unique binary index is $2^0 + 2^2 + 2^4 + 2^7 = 149$. Uniqueness of prime and binary representations proves this is the first occurrence.

Source / 来源

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2024-II-Q14] | Part C (Q11-Q15)

Answer: 211

1. Write the digits as x, y and their sum as s. Then $s^2 = bx + y = (b-1)x + s$, so $(b-1) \mid s(s-1)$. Conversely every solution $2 \le s \le b-1$ gives digits $x = s(s-1)/(b-1)$ and $y = s(b-s)/(b-1)$, which satisfy the digit bounds.

2. Put $M = b - 1$. Since consecutive integers are coprime, each prime-power factor of M must divide either s or $s-1$. The Chinese remainder theorem gives $2^{\omega(M)}$ residue classes; excluding $s = 1$ leaves $2^{\omega(M)} - 1$ special integers.

3. This exceeds ten exactly when $\omega(M) \ge 4$. The least such M is $2 \cdot 3 \cdot 5 \cdot 7 = 210$, hence $b = 211$.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2014-II-Q04] | Part A (Q1-Q5)

Answer: 447

1. Let the digit integers be $u = 10a + b, v = 100a + 10b + c$. Then $u/99 + v/999 = 33/37$, or $111u + 11v = 9801$.

2. Modulo eleven, u is divisible by eleven, forcing $a = b$. Divide by eleven and substitute $u = 11a, v = 110a + c$ to get $221a + c = 891$.

3. With $0 \leq c \leq 9$, only $a = 4, c = 7$ work, so the answer is 447.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2023-II-Q05] | Part A (Q1-Q5)

Answer: 719

1. Write $r = a/b$ with $\gcd(a, b) = 1$, and put $g = \gcd(55, b) \in \{1, 5, 11, 55\}$. The reduced form of $55r$ is $(55a/g)/(b/g)$.

2. The equal-sum condition becomes $(g - 1)b = (55 - g)a$. The values $g = 1, 55$ are impossible for positive a, b.

3. For $g = 5$, $2b = 25a$, forcing $(a, b) = (2, 25)$. For $g = 11$, $5b = 22a$, forcing $(a, b) = (5, 22)$. Each has the required gcd and indeed equal sums.

4. The sum is $2/25 + 5/22 = 169/550$, which is reduced, and $p + q = 719$.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2006-I-Q06] | Part B (Q6-Q10)

Answer: 360

1. Each equals $(100a + 10b + c)/999$. There are $10 \cdot 9 \cdot 8 = 720$ triples, and each digit appears in each position 72 times.

2. The sum is $72(0 + 1 + \cdots + 9)(100 + 10 + 1)/999 = 72 \cdot 45/9 = 360$.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2003-I-Q14] | Part C (Q11-Q15)

Answer: 127

1. If 251 occurs later, multiply by a suitable power of ten and take the fractional part. The new reduced denominator divides the original one and the decimal starts $0.251\ldots$. Thus it suffices to minimize the denominator of a fraction in $[251/1000, 252/1000)$.

2. Such a fraction exceeds $1/4$. Let $h = 4m - n \geq 1$. The strict upper endpoint gives $h/(4n) < 1/500$, hence $n > 125h \geq 125$.

3. For $n = 126$, the only possible h is 1, but $4m = 127$ is impossible. At $n = 127$, $m = 32$ gives $32/127 \in [0.251, 0.252)$ and is reduced. Therefore the least denominator is 127.

Source / 来源

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2022-I-Q13] | Part C (Q11-Q15)

Answer: 392

1. These decimals are $k/9999$ for $1 \le k \le 9999$. A positive integer m occurs as a reduced numerator iff $m \le D$ and $\gcd(m, D) = 1$ for some divisor D of $9999 = 9 \cdot 11 \cdot 101$.

2. If m avoids all three primes $3, 11, 101$, choose $D = 9999$, giving $\varphi(9999) = 6000$ numerators. If m is divisible by 3 only, the largest allowed denominator is 1111, giving $\lfloor 1111/3 \rfloor - \lfloor 1111/33 \rfloor - \lfloor 1111/303 \rfloor = 334$.

3. If 11 alone divides m, use $D = 909$, yielding $\lfloor 909/11 \rfloor - \lfloor 909/33 \rfloor = 55$. If both $3, 11$ divide m but not 101, $D = 101$ permits $33, 66, 99$. Any divisibility by 101 forces the maximum coprime denominator below m, so contributes nothing.

4. Thus $N = 6000 + 334 + 55 + 3 = 6392$, with remainder 392.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2000-II-Q04] | Part A (Q1-Q5)

Answer: 180

1. Write $N = 2^a m$ with m odd. Odd divisors number $\tau(m) = 6$, while even divisors number $a\tau(m) = 12$, so $a = 2$.

2. The odd factor with six divisors has form p^5 or $p^2 q$ for distinct odd primes. Their minima are 243 and 45, respectively. Hence the least N is $4 \cdot 45 = 180$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q02] | Part A (Q1-Q5)

Answer: 012

1. The condition is exactly $k \mid 2004$, since the target requires $1 + jk = 2005$ for an integer $j \geq 1$.

2. $2004 = 2^2 \cdot 3 \cdot 167$, so the divisor count is $3 \cdot 2 \cdot 2 = 12$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q03] | Part A (Q1-Q5)

Answer: 109

1. Four total divisors forces the form p^3 or pq for distinct primes. The latter has proper divisors $1, p, q$, so choose two of the fifteen primes below fifty: $\binom{15}{2} = 105$.

2. For p^3, the largest proper divisor is $p^2 < 50$, giving $p = 2, 3, 5, 7$. Total $105 + 4 = 109$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-II-Q04] | Part A (Q1-Q5)

Answer: 435

1. The factorizations give divisor counts $11^2 = 121$, $8^2 = 64$, and $12 \cdot 23 = 276$. Pairwise intersections are divisors of gcds $5^7, 2^{10}, 3^7$, with counts $8, 11, 8$.

2. The triple intersection contains only one. Inclusion–exclusion gives $121 + 64 + 276 - 8 - 11 - 8 + 1 = 435$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2014-I-Q03] | Part A (Q1-Q5)

Answer: 200

1. Write the fraction $a/(1000-a)$ with $1 \leq a < 500$. Reduction is equivalent to $\gcd(a, 1000) = 1$.

2. There are $\varphi(1000) = 1000(1-1/2)(1-1/5) = 400$ coprime integers from one to 999. Pairing a with $1000-a$ puts exactly half below 500, with no fixed coprime point.

3. The answer is 200.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2004-I-Q08] | Part B (Q6-Q10)

Answer: 199

1. Equal turning and equal lengths put the vertices on a circle with a constant step d among the regular n-gon vertices. Visiting all vertices once requires $\gcd(d, n) = 1$. Crossing excludes the ordinary polygon step 1.

2. Selecting the counterclockwise orientation gives $1 < d < n/2$; equivalently pair the coprime steps $d, n - d$ and remove the pair $1, n - 1$.

3. $\varphi(1000) = 1000(1 - 1/2)(1 - 1/5) = 400$. Therefore the count is $400/2 - 1 = 199$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2004-II-Q08] | Part B (Q6-Q10)

Answer: 054

1. Factor $2004 = 2^2 \cdot 3 \cdot 167$. A candidate is $2^i 3^j 167^k$, whose divisor count is $(i+1)(j+1)(k+1)$. We require this product to equal 2004.

2. Each factor is at most 2004, so all resulting exponents fit the original power bounds. Distribute the two factors of 2 among three slots in $\binom{4}{2} = 6$ ways; each of $3, 167$ chooses one of three slots.

3. The count is $6 \cdot 3 \cdot 3 = 54$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2019-I-Q09] | Part B (Q6-Q10)

Answer: 540

1. Except $n = 1$, both divisor counts are at least two. Thus the pair must be $(2, 5)$ or $(3, 4)$ in either order. One number is therefore a prime square or prime fourth power.

2. Test neighbors of these powers in increasing order. Through the sixth successful value, the candidates are prime squares $4, 9, 25, 49, 121, 169, 289, 361$ and fourth powers $16, 81$. Only $n = 8, 9, 16, 25, 121, 361$ qualify; the corresponding neighbors have counts four or are prime as required.

3. Their sum is $8 + 9 + 16 + 25 + 121 + 361 = 540$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2019-II-Q09] | Part B (Q6-Q10)

Answer: 472

1. Write $n = 2^a 5^b m$ with $a \geq 2, b \geq 1, \gcd(m, 10) = 1$. Divisor counts require $(a+1)(b+1)\tau(m) = 20$. The possible $a+1 \geq 3$ are $4, 5, 10, 20$.

2. The cases four and ten yield $5000, 2560$, too large; twenty is impossible. For $a+1 = 5$, either $b = 3, m = 1$, giving 2000, or $b = 1, m = p$ prime other than two,five. The bound gives $p \in \{3, 7, 11, 13, 17, 19, 23\}$.

3. Hence $S = 2000 + 80(93) = 9440$, and $S/20 = 472$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2026-II-Q09] | Part B (Q6-Q10)

Answer: 669

1. Expand each summand geometrically and regroup nonnegative terms: $S = \sum_{n\geq 1} d(n)10^{-n}$, since exponent n occurs once for each divisor of n.

2. After multiplying by 10^{100}, terms with $n \leq 100$ are integral. The remaining tail starts with $2/10 + 8/100 + 2/1000 + 8/10000 = .2828$; for later terms use $d(100+j) \leq 100 + j$ to bound the rest by $.002$. Hence the whole tail is less than one and causes no carry.

3. Only the last three integral terms matter modulo 1000: $100d(98) + 10d(99) + d(100) = 100 \cdot 6 + 10 \cdot 6 + 9 = 669$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q12] | Part C (Q11-Q15)

Answer: 025

1. Divisors pair except at squares, so $\tau(j)$ is odd exactly for square j. Hence $S(n)$ has parity $\lfloor\sqrt{n}\rfloor$.

2. Odd blocks correspond to $k = 1, 3, \ldots, 43$, each containing $2k+1$ integers from k^2 through $(k+1)^2 - 1$. They are all complete below 2005. Thus $a = 2(1 + 3 + \cdots + 43) + 22 = 990$.

3. Then $b = 1015$, and the absolute difference is 25.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2006-II-Q13] | Part C (Q11-Q15)

Answer: 015

1. Such a representation is $N = m^2 - k^2 = (m-k)(m+k)$ with integers $m > k \geq 0$. Thus it corresponds to a factor pair $a \leq b$ of equal parity, with length $j = a$.

2. For odd N, this is five unrestricted divisor pairs, requiring divisor count nine or ten. Below one thousand, the odd possibilities are $225, 441, 405, 567, 891$, giving five.

3. For even N, both factors are even, so $N = 4M$ and $M < 250$ must have nine or ten divisors. The possibilities are $36, 100, 196, 225$ and $48, 80, 112, 176, 208, 162$, giving ten. Total 15.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2016-II-Q11] | Part C (Q11-Q15)

Answer: 749

1. Prime exponents e_i of a give divisor count $\prod(ke_i+1) \equiv 1 \pmod{k}$. Conversely every $N = ke+1$ occurs using $a = 2^e$, including $N = 1$ via $a = 1$. Thus k-nice means exactly $N \equiv 1 \pmod{k}$.

2. From1 through999, the counts congruent to1 modulo7 and8 are143 and125. Their intersection is1 modulo56, with18 values.

3. Inclusion-exclusion leaves $999 - (143 + 125 - 18) = 749$ integers of neither kind.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2021-I-Q14] | Part C (Q11-Q15)

Answer: 125

1. Factor $2021 = 43 \cdot 47$. For a prime power factor of a, its divisor sum is $1 + p + \cdots + p^{en}$. It suffices that the nonconstant part vanish modulo each $q = 43, 47$.

2. If $p \equiv 0 \pmod{q}$ it vanishes automatically; if $p \equiv 1$, it requires $q \mid en$; otherwise $p^{en} = 1$ suffices. Thus $q(q-1) \mid n$ is sufficient.

3. The q factors are necessary by taking prime $a = 173 \equiv 1 \pmod{43}$ and $a = 283 \equiv 1 \pmod{47}$. For $q - 1$, take primes $a = 3$ of order 42 modulo 43 and $a = 5$ of order 46 modulo 47.

4. Hence $n = \operatorname{lcm}(43 \cdot 42, 47 \cdot 46) = 2 \cdot 3 \cdot 7 \cdot 23 \cdot 43 \cdot 47$. Its prime-factor sum is 125.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2003-II-Q03] | Part A (Q1-Q5)

Answer: 192

1. The first position has three choices. After any letter, exactly one successor is forbidden and two are allowed.

2. The count is $3 \cdot 2^6 = 192$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2004-II-Q04] | Part A (Q1-Q5)

Answer: 927

1. All 99 one- or two-digit numbers qualify. At length $L \geq 3$, either all digits agree, giving nine numbers, or choose a nonzero leading digit a and a distinct digit b: $9 \cdot 9$ choices.

2. The later $L-1$ positions use a, b, with at least one b, giving $2^{L-1}-1$ patterns. Hence the three- and four-digit counts are $9+81 \cdot 3=252$ and $9+81 \cdot 7=576$.

3. Total $99+252+576=927$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2005-I-Q05] | Part A (Q1-Q5)

Answer: 630

1. Read orientations bottom to top as U, D. The forbidden adjacency is UD, so every valid string consists of some down-facing coins followed by up-facing coins: nine choices for the split.

2. Independently choose the four gold positions in $\binom{8}{4} = 70$ ways. Total $9 \cdot 70 = 630$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2006-II-Q04] | Part A (Q1-Q5)

Answer: 462

1. The valley value a_6 is less than every other entry, so it is one. Choose the five numbers before it from the other eleven.

2. Their decreasing order and the remaining six numbers increasing order are forced. The number is $\binom{11}{5} = 462$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2007-II-Q01] | Part A (Q1-Q5)

Answer: 372

1. Strings with at most one zero use five distinct symbols from seven, giving $(7)_5 = 2520$.

2. With two zeroes, choose their positions in $\binom{5}{2}$ ways and order three distinct remaining symbols in $(6)_3$ ways, giving 1200. Thus $N = 3720$, so $N/10 = 372$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2011-I-Q05] | Part A (Q1-Q5)

Answer: 144

1. Subtract successive three-label sums: labels three places apart have the same residue modulo 3. Thus each of the three position classes modulo 3 carries one residue class.

2. The residue classes may be assigned in $3!$ ways, and their digits arranged in $(3!)^3$ ways. Every resulting triple has residues $0, 1, 2$, so all $6^4 = 1296$ labelings qualify.

3. Distinct digits prevent nonidentity rotational symmetries, so each orbit has size nine. The answer is $1296/9 = 144$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2012-I-Q03] | Part A (Q1-Q5)

Answer: 216

1. Choose the unique correctly served diner in nine ways. Suppose this is a beef orderer. Two beef meals remain for three chicken and three fish orderers, with the remaining two beef orderers receiving no beef.

2. If both leftover beef meals go to one group, choose their recipients in $2\binom{3}{2} = 6$ ways; all other types are forced. If they split between groups, there are $3 \cdot 3 = 9$ recipient choices, followed by two ways to assign the final chicken and fish meals to the two beef orderers.

3. The count is $9(6 + 9 \cdot 2) = 216$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2019-I-Q04] | Part A (Q1-Q5)

Answer: 122

1. There are always eleven choices of outgoing player. Incoming choices decrease from eleven to ten to nine. Thus the count is $1 + 11 \cdot 11 + (11 \cdot 11)(11 \cdot 10) + (11 \cdot 11)(11 \cdot 10)(11 \cdot 9)$.

2. This is 1331122, with remainder 122.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2019-II-Q05] | Part A (Q1-Q5)

Answer: 520

1. View the six even chairs as a cycle, with odd chairs between them. For a block of j occupied even chairs, advisors can choose adjacent odd chairs in $j+1$ ways: a string of left choices followed by right choices; a right-then-left boundary would collide.

2. Classify the two empty even chairs: adjacent (six choices) leaves one block of four, weight five; one chair between (six choices) gives blocks one and three, weight eight; opposite (three choices) gives blocks two and two, weight nine.

3. Total group-seat patterns $6 \cdot 5 + 6 \cdot 8 + 3 \cdot 9 = 105$. Assign the ambassador-advisor pairs in $4!$ ways, obtaining 2520, remainder 520.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2020-I-Q05] | Part A (Q1-Q5)

Answer: 052

1. For increasing remainders, insert a chosen exceptional label into the increasing order of the other five: $6 \cdot 6 = 36$ constructions.

2. The increasing row is counted six times, and each of the five adjacent swaps twice. No other duplication is possible: two possible deleted labels would have to form the only inverted pair, hence be adjacent. The distinct count is $36 - 5 - 5 = 26$.

3. Reversal gives another 26. An increasing and a decreasing subsequence of length five in six distinct labels would overlap in at least four labels, impossible since two such subsequences share at most one label. Total 52.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2021-II-Q03] | Part A (Q1-Q5)

Answer: 080

1. Fix the position i occupied by 3. Exactly two summands avoid it; modulo 3 their sum is $x_{i+2}x_{i+3}(x_{i+1}+x_{i+4})$.

2. The product factor is nonzero, so the two cyclic neighbors of 3 must lie in opposite nonzero residue classes. There are two choices per class and two orders, giving 8 neighbor assignments.

3. The other two labels have 2 orders, and 3 has 5 positions. Total $5\cdot 8\cdot 2=80$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2025-I-Q03] | Part A (Q1-Q5)

Answer: 016

1. Positive strictly decreasing counts summing to 9 are $(6,2,1),(5,3,1),(4,3,2)$.

2. They contribute multinomial counts $9!/(6!2!)=252$, $9!/(5!3!)=504$, and $9!/(4!3!2!)=1260$. The total is 2016, with remainder 16.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2004-I-Q06] | Part B (Q6-Q10)

Answer: 882

1. For any four distinct ordered digit values, five permutations have this alternating pattern. Place the largest in position two: three choices for position one, with the last pair forced. Or place it in position four: two choices for the remaining low positions.

2. If zero is absent, all five work, contributing $5\binom{9}{4} = 630$. If zero is present, it must be in position one or three; only three survive with zero in position three, contributing $3\binom{9}{3} = 252$.

3. Total 882.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2023-II-Q07] | Part B (Q6-Q10)

Answer: 928

1. Any inscribed rectangle has two diameters as its diagonals. Group the vertices into six opposite pairs; a forbidden rectangle occurs precisely when two pairs are both solid red or two pairs are both solid blue.

2. So there may be at most one all-red pair and at most one all-blue pair. Every remaining pair is mixed, with two orientations.

3. The four cases (zero solids, red only, blue only, both) give $2^6 + 6 \cdot 2^5 + 6 \cdot 2^5 + 6 \cdot 5 \cdot 2^4 = 928$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2025-I-Q10] | Part B (Q6-Q10)

Answer: 081

1. Fill the first block in $9!$ ways and label its row-digit sets R_1, R_2, R_3. The middle-block row sets partition the digits and row i must avoid R_i.

2. If its first row uses j members of R_2 and $3-j$ of R_3, the choices for the first and second row contents total $\binom{3}{j}^3$. Summing over $j=0,1,2,3$ gives 56. All third-block row contents are then forced.

3. The six row-triples in the latter two blocks may each be internally ordered in $3!$ ways. Thus the number is $9! \cdot 56 \cdot 6^6 = 2^{16}3^{10}5 \cdot 7^2$, giving $32+30+5+14=81$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2026-I-Q07] | Part B (Q6-Q10)

Answer: 396

1. Every onto self-map of a finite set is a permutation. The sixth iterate is identity precisely when all cycle lengths divide six, so only lengths four and five are forbidden.

2. The excluded cycle types $4+1+1, 4+2, 5+1$ have counts $6!/(4 \cdot 2!) = 90$, $6!/(4 \cdot 2) = 90$, and $6!/5 = 144$. Thus the count is $720 - 324 = 396$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2005-II-Q01] | Part A (Q1-Q5)

Answer: 013

1. The equation $\binom{n}{6} = 6\binom{n}{3}$ simplifies to $(n-3)(n-4)(n-5) = 720$.

2. The product increases for $n \geq 6$, and $10 \cdot 9 \cdot 8 = 720$, so the unique solution is $n = 13$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2012-II-Q03] | Part A (Q1-Q5)

Answer: 088

1. If every department contributes one man and one woman, there are $4^3 = 64$ committees.

2. Otherwise the gender splits by department must be two men, two women, and mixed. Choose those department roles in $3 \cdot 2$ ways and the mixed pair in four ways, adding twenty-four.

3. The total is 88.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q01] | Part A (Q1-Q5)

Answer: 390

1. Choose any three of15 points, giving $\binom{15}{3} = 455$ triples. A line not coinciding with a triangle side meets the convex boundary in at most two points, so every degenerate triple lies on one side.

2. Including endpoints, the side counts are $5, 6, 7$. Their collinear triples are disjoint and total $\binom{5}{3} + \binom{6}{3} + \binom{7}{3} = 65$. Thus the answer is $455 - 65 = 390$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2000-II-Q07] | Part B (Q6-Q10)

Answer: 137

1. Multiplying by $19!$ gives $19N = \sum_{k=2}^{9} \binom{19}{k}$. By symmetry the sum from $k = 0$ to 9 is half of 2^{19}, namely 2^{18}.

2. Subtract $\binom{19}{0} + \binom{19}{1} = 20$ to get $19N = 262124$, so $N = 13796$. The requested integer is 137.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2002-II-Q09] | Part B (Q6-Q10)

Answer: 501

1. For an ordered disjoint pair, every element is in the first subset, the second, or neither. There are 3^{10} assignments.

2. Exclude empty first and empty second subsets, giving $3^{10} - 2 \cdot 2^{10} + 1 = 57002$. The two nonempty disjoint subsets cannot be equal, so divide by two: $N = 28501$. Remainder 501.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2007-I-Q10] | Part B (Q6-Q10)

Answer: 860

1. Choose the first column in $\binom{6}{3} = 20$ ways. If the second overlaps it in j rows, its choices number $\binom{3}{j}\binom{3}{3-j}$. There are then j completed rows, j empty rows and $6-2j$ singly shaded rows.

2. The last two columns must both use the empty rows, and column three chooses $3-j$ of the singly shaded rows; the final column is forced. Thus $N = 20\sum_{j=0}^{3}\binom{3}{j}^2\binom{6-2j}{3-j} = 20(20+54+18+1) = 1860$.

3. The remainder is 860.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2009-II-Q06] | Part B (Q6-Q10)

Answer: 750

1. For the complement, write selected numbers $a_1 < \cdots < a_5$ with gaps at least two. Set $b_i = a_i - (i-1)$, a bijection to ordinary five-subsets of $\{1, \ldots, 10\}$.

2. Thus $m = \binom{14}{5} - \binom{10}{5} = 2002 - 252 = 1750$, remainder 750.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2010-I-Q07] | Part B (Q6-Q10)

Answer: 760

1. The unique elements in $A \cap B, B \cap C, C \cap A$ must be distinct. Assign them in $7 \cdot 6 \cdot 5 = 210$ ways.

2. Each of the four remaining elements may belong only to A, only to B, only to C, or none. These 4^4 choices introduce no new intersections.

3. The count is $210 \cdot 256 = 53760$, with remainder 760.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2010-II-Q08] | Part B (Q6-Q10)

Answer: 772

1. For $|A| = k$, $1 \le k \le 11$, the conditions force $k \in B$ and $12 - k \in A$. At $k = 6$ this is impossible.

2. For every other k, choose the other $k - 1$ elements of A from ten unrestricted elements: $\binom{10}{k-1}$. The complement fixes B.

3. The sum is $\sum_{j=0}^{10} \binom{10}{j} - \binom{10}{5} = 1024 - 252 = 772$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q07] | Part B (Q6-Q10)

Answer: 564

1. Set $c = 6 - a - b$ and use binomial symmetry. The sum becomes $\sum_{a+b+c=6} \binom{6}{a}\binom{6}{b}\binom{6}{c}$.

2. This counts choosing six objects from three disjoint six-object groups, classified by the number chosen from each. Thus $S = \binom{18}{6} = 18564$, with remainder564.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2020-I-Q07] | Part B (Q6-Q10)

Answer: 081

1. Choosing k men and $k+1$ women gives $N=\sum_{k=0}^{11}\binom{11}{k}\binom{12}{11-k}=\binom{23}{11}$. The second equality counts eleven selections from two disjoint groups.

2. In $23!/(11!12!)$, the nonzero prime exponents are those of $2,7,13,17,19,23$, each equal to one. For example $v_2=19-8-10=1$, while $v_3=9-4-5=0$.

3. The sum of these primes is $2+7+13+17+19+23=81$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2021-II-Q06] | Part B (Q6-Q10)

Answer: 454

1. Let $a = |A|, b = |B|, i = |A \cap B|$. The equation factors as $(a - i)(b - i) = 0$, so one set contains the other.

2. For $A \subseteq B$, each element is in neither, in B only, or in both: 3^5 choices. Reverse containment contributes another 3^5, while equality contributes 2^5 to both. The answer is $2 \cdot 3^5 - 2^5 = 454$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2022-II-Q10] | Part B (Q6-Q10)

Answer: 004

1. Expand the nested binomial: $\binom{\binom{k}{2}}{2} = (k+1)k(k-1)(k-2)/8 = 3\binom{k+1}{4}$.

2. Pascal telescoping (the hockey-stick identity) gives $\sum_{k=3}^{40}\binom{k+1}{4} = \binom{42}{5}$.

3. The sum is $3 \cdot 850668 = 2552004$, with remainder 4, displayed as AIME answer 004.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2024-II-Q06] | Part B (Q6-Q10)

Answer: 055

1. There are 2^{a-1} sets with maximum a: include a and choose any subset of $1, \ldots, a-1$. Hence $\sum_{a \in A} 2^{a-1} = 2024$.

2. Binary uniqueness and $2024 = 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^3$ give $A = \{11, 10, 9, 8, 7, 6, 4\}$. Their sum is 55.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2007-II-Q13] | Part C (Q11-Q15)

Answer: 640

1. If bottom values are $x_0, \ldots, x_{10}$, the top is $\sum_i \binom{10}{i} x_i$. Modulo three the only nonzero binomial weights occur at $i = 0, 1, 9, 10$, all equal one.

2. These four binary entries must sum to zero or three, giving $1 + 4 = 5$ choices. The seven others are free, so total $5 \cdot 2^7 = 640$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q12] | Part C (Q11-Q15)

Answer: 252

1. The number1 is forbidden by $1 \cdot 1 = 1$. If2 and3 are both absent, every subset of $\{4, \ldots, 10\}$ works because products exceed10. This contributes128, including the empty set.

2. If3 is the least element, exclude9 and choose freely from $4, 5, 6, 7, 8, 10$, giving64 sets. If2 is the least, exclude4; the free elements7,8 give a factor4.

3. In the latter case, $\{3, 6, 9\}$ has five valid choices: empty, each singleton, or $\{6, 9\}$. The pair $\{5, 10\}$ has three choices, excluding their joint selection. All remaining product constraints have now been covered, giving $4 \cdot 5 \cdot 3 = 60$ sets.

4. Total $128 + 64 + 60 = 252$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2023-II-Q11] | Part C (Q11-Q15)

Answer: 081

1. The 32 subsets form 16 complementary pairs. Exactly one member of each pair must be chosen. The full set is chosen and the empty set is not.

2. If a singleton is chosen, the entire family consists of all sets containing its element. This gives five families. Otherwise all four-element sets are chosen, and the only further restriction is that the selected two-element sets pairwise intersect; three-element choices are then fixed by complementation.

3. View two-element sets as edges of K_5. A pairwise-intersecting edge family is empty, a nonempty subfamily of a star, or a triangle. To see completeness, after edges ab, ac are chosen, any edge avoiding a must be bc, which prevents any fourth edge outside that triangle.

4. The counts are 1, $5(2^4 - 1) - 10 = 65$ (single edges counted twice), and $\binom{5}{3} = 10$. Add the five singleton families: $1 + 65 + 10 + 5 = 81$.

Source / 来源

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2026-I-Q13] | Part C (Q11-Q15)

Answer: 039

1. In the polynomial ring modulo both 503 and $x^{502}-1$, the reduced coefficients of $(1+x)^{10000}$ are precisely S_r. The prime-binomial identity gives $(1+x)^{503}=1+x^{503}=1+x$ in this ring.

2. As $10000=19\cdot 503+443$, reduce the power to $(1+x)^{462}$. Its coefficients $\binom{462}{r}$ for $r\leq 462$ are nonzero modulo 503, since none of the factorial factors contains 503.

3. Only exponents $r=463,\ldots,501$ have zero coefficient. Their number is 39.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2000-II-Q05] | Part A (Q1-Q5)

Answer: 376

1. Choose five rings in $\binom{8}{5}$ ways and order them in $5!$ ways, reading fingers in a fixed order.

2. The four finger lengths form a weak composition of 5, with $\binom{8}{3}$ choices. Thus $N = 56 \cdot 120 \cdot 56 = 376320$, whose requested digits are 376.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2021-I-Q04] | Part A (Q1-Q5)

Answer: 331

1. There are $\binom{65}{2} = 2080$ ordered positive triples. One has all entries equal.

2. For exactly two equal entries, use the multiset $\{u, u, 66 - 2u\}$ with $1 \leq u \leq 32$ and $u \neq 22$. The 31 multisets have three orderings each, giving 93.

3. The remaining $2080 - 1 - 93 = 1986$ triples have distinct entries. Each increasing triple has six orderings, so the answer is $1986/6 = 331$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2003-I-Q09] | Part B (Q6-Q10)

Answer: 615

1. For common digit sum $1 \leq s \leq 9$, the first pair has s possibilities because its first digit is nonzero; the last pair has $s+1$. For $10 \leq s \leq 18$, both pairs have $19-s$ possibilities.

2. The count is $\sum_{s=1}^{9} s(s+1) + \sum_{s=10}^{18} (19-s)^2 = 2\sum_{k=1}^{9} k^2 + \sum_{k=1}^{9} k = 615$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2009-I-Q09] | Part B (Q6-Q10)

Answer: 420

1. Arrange the seven digits in $7!/(4!3!) = 35$ ways. Then insert two cuts to form three nonempty blocks, each length at most four.

2. Allowed length patterns are permutations of $(1, 2, 4), (1, 3, 3), (2, 2, 3)$, with counts $6, 3, 3$. No leading-zero issue occurs. Thus $35 \cdot 12 = 420$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2009-I-Q10] | Part B (Q6-Q10)

Answer: 346

1. Ignoring individuals, group patterns consist of k ordered triples of nonempty M, V, E runs, for $1 \leq k \leq 5$. The seat-one/seat-fifteen condition fixes both endpoints at run boundaries.

2. For each group, split its five occurrences into k ordered positive run lengths, giving $\binom{4}{k-1}$. The groups are independent, so $N = \sum_{j=0}^{4} \binom{4}{j}^3 = 1 + 64 + 216 + 64 + 1 = 346$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2011-II-Q07] | Part B (Q6-Q10)

Answer: 003

1. The five greens make at most five green runs and therefore at most six red runs. Thus there are at most ten color changes. With m reds there are $m+4$ adjacent pairs, so equality forces $m+4 \leq 20$ and $m \leq 16$.

2. At $m=16$, all eleven runs must be present: six nonempty red runs separated by the five singleton greens. Any positive composition of sixteen into six parts attains the condition.

3. Stars and bars gives $N=\binom{15}{5}=3003$, with remainder 3.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2025-II-Q10] | Part B (Q6-Q10)

Answer: 907

1. Occupied runs have length one or two. If there are j runs, then $8-j$ are doubletons and $2j-8$ are singletons, so $4 \le j \le 8$. Their order has $\binom{j}{8-j}$ choices.

2. The eight empty chairs create nine gaps, of which choose j for these runs. Thus $N = \sum_{j=4}^{8} \binom{j}{8-j}\binom{9}{j} = 126 + 1260 + 1260 + 252 + 9 = 2907$. The remainder is 907.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2008-II-Q12] | Part C (Q11-Q15)

Answer: 310

1. If pole one has b blues and g greens, its arrangements number $\binom{b+1}{g}$ using distinct gaps around the blues. Pole two contributes $\binom{11-b}{9-g}$.

2. Allow empty poles temporarily and sum: Vandermonde gives $\sum_{b=0}^{10}\sum_{g}\binom{b+1}{g}\binom{11-b}{9-g} = 11\binom{12}{9} = 2420$.

3. An empty pole leaves all flags on the other in $\binom{11}{9} = 55$ ways, for each of two choices. Thus $N = 2420 - 110 = 2310$, remainder 310.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2018-II-Q15] | Part C (Q11-Q15)

Answer: 185

1. Let $d_i = f(i) - f(i-1) \in \{\pm 1, \pm 2, \pm 3\}$. Their sum is twelve. Two negative steps would make the sum at most $18 - 8 = 10$, so at most one is negative.

2. With no negatives, all pair inequalities follow immediately from steps one through three. The counts of ones, twos, threes can be $(0, 6, 0), (1, 4, 1), (2, 2, 2), (3, 0, 3)$, contributing $1 + 30 + 90 + 20 = 141$.

3. A negative two cannot adjoin any allowed positive step without violating the distance-two lower bound. A negative three would force all five other steps to be three to achieve total twelve, again violating that bound. Thus a single negative must be minus one, and its existing neighbors must be three. These local conditions suffice for every longer interval: a two-step interval containing it has sum at least two, and extending by positive steps preserves the lower bound; intervals containing it internally are larger still.

4. At either endpoint, the remaining four unfixed positive steps sum to ten, giving $4 + 6 = 10$ orders each. At each of four interior positions, the remaining three positive steps sum to seven, giving $3 + 3 = 6$ orders each. Total $141 + 20 + 24 = 185$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2022-II-Q13] | Part C (Q11-Q15)

Answer: 220

1. The numerator has constant term 1 and no other terms below degree 2310. Expanding the four geometric series, the desired coefficient counts $105u + 70v + 42w + 30t = 2022$ with all variables nonnegative integers.

2. Reduce modulo $2, 3, 5, 7$: $u = 2u_0, v = 3v_0, w = 5w_0 + 1, t = 7t_0 + 3$. These substitutions are forced, and conversely preserve nonnegativity.

3. The equation becomes $210(u_0 + v_0 + w_0 + t_0) + 132 = 2022$, or sum 9. Stars and bars gives $\binom{12}{3} = 220$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2001-II-Q02] | Part A (Q1-Q5)

Answer: 298

1. The integer enrollment ranges are $1601 \leq s \leq 1700$ and $601 \leq f \leq 800$. Since the union has size 2001, the overlap is $s + f - 2001$.

2. The extreme overlaps are 201 and 499. They are feasible because both overlaps are nonnegative and no larger than either enrollment; the remaining students can be assigned to the two single-language groups. Difference 298.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2003-I-Q03] | Part A (Q1-Q5)

Answer: 484

1. In increasing order, the element of rank j is the larger member of exactly $j-1$ pairs.

2. The weighted sum is $0\cdot 1+1\cdot 2+2\cdot 3+3\cdot 5+4\cdot 8+5\cdot 13+6\cdot 21+7\cdot 34=484$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2016-II-Q04] | Part A (Q1-Q5)

Answer: 180

1. Count total green cubes in two directions: $12a = 20b$, so $a : b = 5 : 3$. Counting yellow gives $a(bc - 21) = 25b = 15a$, hence $bc = 36$; then $ac = 60$.

2. Thus c divides both $36, 60$, so $c \leq 12$. The volume is $2160/c \geq 180$, with candidate dimensions $(a, b, c) = (5, 3, 12)$.

3. For attainment, at every first- and second-coordinate pair use a length12 row containing 3 reds, 4 greens and 5 yellows. Each first-direction layer has three such rows and each second-direction layer five, giving all required counts. Minimum volume 180 is attained.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2017-II-Q01] | Part A (Q1-Q5)

Answer: 196

1. There are $2^8 = 256$ subsets. Each forbidden containment class has $2^5 = 32$ members.

2. The overlap consists of subsets of the intersection $\{4, 5\}$, giving4. Inclusion-exclusion therefore leaves $256 - 32 - 32 + 4 = 196$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2023-I-Q03] | Part A (Q1-Q5)

Answer: 607

1. Count unordered pairs of lines, not intersection points: there are $\binom{40}{2} = 780$ pairs.

2. A point with k incident lines uses $\binom{k}{2}$ pairs. The specified higher intersections therefore use $3\binom{3}{2} + 4\binom{4}{2} + 5\binom{5}{2} + 6\binom{6}{2} = 173$.

3. Every unused pair gives a distinct ordinary double intersection. The count is $780 - 173 = 607$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2024-II-Q01] | Part A (Q1-Q5)

Answer: 073

1. The total number of ownership incidences is 2024. Give each resident one baseline incidence; the remaining 1124 incidences equal $437 + 2 \cdot 234 + 3x$.

2. Thus $3x = 1124 - 905 = 219$, so $x = 73$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2014-II-Q09] | Part B (Q6-Q10)

Answer: 581

1. Except for the full set, each maximal selected run of length at least three begins just after an unselected chair. Mark an unselected-selected-selected-selected pattern. There are ten starts and 2^6 assignments of the other chairs, giving 640 marked subsets.

2. Two such marked starts require disjoint four-chair patterns. The second start has three positions for each first start, so there are $10 \cdot 3/2 = 15$ unordered start pairs, each leaving two free chairs. This counts sixty subsets with two long runs. Three runs require at least twelve chairs and are impossible.

3. Subtract the double counts and add the full set: $640 - 60 + 1 = 581$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2018-I-Q09] | Part B (Q6-Q10)

Answer: 210

1. There are seven sixteen-pairs and eight twenty-four-pairs. Ten pair combinations share a center $a \in \{4, \ldots, 15\} \setminus \{8, 12\}$, leaving $56 - 10 = 46$ disjoint combinations. A disjoint combination determines its four-set uniquely, and no four-set has two such decompositions.

2. For overlapping pairs, the forced triple is $\{a, 16 - a, 24 - a\}$, with ten centers, and seventeen choices for the fourth element: 170 incidences. Exactly six sets have two centers: $\{1, 7, 9, 15\}, \{2, 6, 10, 14\}, \{3, 5, 11, 13\}, \{5, 11, 13, 19\}, \{6, 10, 14, 18\}, \{7, 9, 15, 17\}$. To justify the multiplicities, make a graph whose edges are the two allowed fixed-sum pairs. Every vertex has at most one edge of each sum, so degree is at most two and edges on a path alternate. An alternating cycle is impossible: composing the reflections $x \mapsto 16 - x$ and $x \mapsto 24 - x$ translates by 8, which cannot cycle. Thus any induced graph on four vertices is a disjoint union of paths and has at most two degree-two centers. Listing its alternating three-edge paths gives exactly the six sets above.

3. Thus overlapping-pair sets number $170 - 6 = 164$. They cannot also contain disjoint pairs: disjoint pairs force total sum forty, whereas a shared-center triple would then force its fourth element to repeat its center.

4. The total is $46 + 164 = 210$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2020-II-Q09] | Part B (Q6-Q10)

Answer: 090

1. Use inclusion-exclusion on the five forbidden edges $12, 23, 34, 45, 56$. A selected subset of k edges forming r nonempty runs forces r blocks, each orientable in two ways, leaving $(6-k)!$ block arrangements.

2. Summing $2^r(6-k)!$ over subsets of size $k = 0, 1, 2, 3, 4, 5$ gives $720, 1200, 768, 228, 32, 2$. These follow by grouping edge subsets into runs; for $k = 3$ the run counts are $3, 6, 1$ for $r = 1, 2, 3$.

3. Alternating the totals gives $720 - 1200 + 768 - 228 + 32 - 2 = 90$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2022-I-Q06] | Part B (Q6-Q10)

Answer: 228

1. Initially $6 \leq a < b \leq 29$, giving $\binom{24}{2} = 276$ pairs. A forbidden progression uses either one or both variable terms, because the fixed terms have no four-term progression.

2. With only one variable, the progression must extend $3, 4, 5$ by 6 or $30, 40, 50$ by 20. Thus exclude all pairs containing 6 or 20: $23 + 23 - 1 = 45$ pairs.

3. For a progression using both variables, test the six possible placements of two fixed terms in four equally spaced positions. The only new admissible pairs are $(7, 9), (12, 21), (16, 28)$; $(6, 7), (10, 20)$ were already excluded. Therefore $276 - 45 - 3 = 228$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2010-II-Q11] | Part C (Q11-Q15)

Answer: 068

1. There are $\binom{9}{5} = 126$ matrices. A bad matrix has at least two constant lines. Two zero lines require at least five zeros, impossible. A zero line and a one line must be disjoint, hence parallel rows or columns.

2. This mixed-color bad case has $6 \cdot 2 \cdot 3 = 36$ possibilities: choose the one line, the zero line, and the location of the lone zero in the remaining parallel line. Each has exactly two constant lines.

3. Two all-one lines must cross and use all five ones. Their pairs are row-column (9), row/column-diagonal (12), or the two diagonals (1). These 22 configurations have no further constant line and do not overlap the mixed case.

4. The valid count is $126 - 36 - 22 = 68$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2022-I-Q12] | Part C (Q11-Q15)

Answer: 245

1. Count triples (x, A, B) with $x \in A \cap B$ instead. For a fixed x and subset size k, there are $\binom{n-1}{k-1}^2$ choices.

2. Vandermonde summation yields $S_n = n\binom{2n-2}{n-1}$. Hence $S_{2022}/S_{2021} = 2 \cdot 2022 \cdot 4041/2021^2$.

3. As $2021 = 43 \cdot 47$ is coprime to $2 \cdot 2022 \cdot 4041$, the fraction is already reduced. Modulo 1000, $p = 804$ and $q = 441$, giving remainder 245.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2024-I-Q11] | Part C (Q11-Q15)

Answer: 371

1. Label positions modulo 8 and let B be the blue set, $b = |B|$. We need $B \cap (B + k) = \varnothing$ for some k. Summing overlap sizes over the seven nonzero k gives $b(b-1)$. For $b \le 3$ this is at most 6, so one overlap is zero. These cases contribute $\sum_{j=0}^{3} \binom{8}{j} = 93$.

2. For $b = 4$, the rotation must exchange the blue and red sets. Odd k permits only the two alternating colorings; $k = 2, 6$ gives four colorings by alternating on each parity cycle; $k = 4$ gives $2^4 = 16$ colorings by choosing one vertex from each opposite pair. The three families are disjoint. For $b > 4$ success is impossible.

3. There are $93 + 2 + 4 + 16 = 115$ successful colorings out of 256, so the requested sum is $115 + 256 = 371$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2007-I-Q06] | Part B (Q6-Q10)

Answer: 169

1. Split by whether the sequence visits 13 and 26. Between successive landmarks $0, 13, 26, 39$, there are five routes: climb successive multiples of three, then jump to the next landmark at one of five possible starting positions.

2. If the intermediate landmark is omitted, each corresponding skip segment has four routes. Thus paths visiting both landmarks number $5^3 = 125$, only either one contribute $5 \cdot 4 = 20$ each, and neither contributes four.

3. Total $125 + 20 + 20 + 4 = 169$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2007-II-Q06] | Part B (Q6-Q10)

Answer: 640

1. Build from right to left. A final digit d may be preceded by an odd digit below d or an even digit above d. For every $d = 0, \ldots, 9$, these possibilities total exactly four; zero is never an allowed predecessor.

2. The last digit has ten choices and each preceding digit four, with a nonzero leading digit automatic. Count $10 \cdot 4^3 = 640$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2013-II-Q09] | Part B (Q6-Q10)

Answer: 106

1. With k available colors, color the first tile in k ways. At each of six following cells, either continue the current tile or start a new tile in any of k colors. This gives $k(k+1)^6$.

2. Inclusion-exclusion for missing colors yields $N = 3 \cdot 4^6 - 3(2 \cdot 3^6) + 3(2^6) = 8106$.

3. The required remainder is 106.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2015-II-Q10] | Part B (Q6-Q10)

Answer: 486

1. Let T_n be the count. Insert the maximum n into an admissible permutation of $1, \ldots, n-1$. The entry immediately after it, if present, must be $n-1$ or $n-2$. Hence there are exactly three insertion positions: before those entries or at the end.

2. Every insertion preserves all other inequalities. Conversely, deleting n is safe: its following entry is at least $n-2$, while its preceding entry is at most $n-1$, so their newly formed adjacent inequality still holds.

3. Thus $T_n = 3T_{n-1}$ for $n \geq 3$, with $T_2 = 2$. Therefore $T_7 = 2 \cdot 3^5 = 486$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-II-Q08] | Part B (Q6-Q10)

Answer: 556

1. For either coordinate, a multiset of jumps totaling four is of type $(1,1,1,1)$, $(1,1,2)$, or $(2,2)$. Choose one type independently for horizontal and vertical jumps, and interleave their distinct directional steps.

2. The symmetric table of multinomial counts for these three types is $\begin{pmatrix} 70 & 105 & 15 \\ 105 & 180 & 30 \\ 15 & 30 & 6 \end{pmatrix}$. For example the middle entry is $6!/(2!2!) = 180$, since the two horizontal singles and two vertical singles are indistinguishable within each direction.

3. Summing all entries gives 556.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2024-I-Q06] | Part B (Q6-Q10)

Answer: 294

1. Four turns mean five nonempty runs. A path beginning right has run directions R, U, R, U, R.

2. The three horizontal run lengths are positive integers summing to 8, giving $\binom{7}{2}$ choices; the two vertical lengths give $\binom{7}{1}$.

3. Interchanging horizontal and vertical accounts for paths starting up. Hence $2\binom{7}{2}\binom{7}{1} = 294$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2000-I-Q15] | Part C (Q11-Q15)

Answer: 927

1. Assign the original positions $1, \ldots, 2000$ to a queue. Repeatedly remove the front position and rotate the next position to the back. The removed positions receive labels in increasing order.

2. For a reproducible recurrence, let $L = [1, \ldots, 2000]$ and repeat $p = L.pop(0)$, followed, when nonempty, by $L.append(L.pop(0))$. Recording the last two removals gives $928, 1952$. This finite procedure tracks positions exactly and uses no assumptions about card labels.

3. Thus label 1999 began at position 928, with 927 cards above it.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2001-I-Q14] | Part C (Q11-Q15)

Answer: 351

1. Encode receipt by 1 and nonreceipt by 0. Track counts A_n, B_n, C_n of strings ending respectively in 1, exactly one trailing zero, or exactly two trailing zeros.

2. The updates are $(A_{n+1}, B_{n+1}, C_{n+1}) = (B_n + C_n, A_n, B_n)$, starting from $(1, 1, 0)$ at length one.

3. The total counts for lengths 13 through 19 are $65, 86, 114, 151, 200, 265, 351$. Thus there are 351 patterns.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2004-I-Q15] | Part C (Q11-Q15)

Answer: 511

1. Build exact first-hit levels backwards. Start $T_1 = \{1\}$. Each z has predecessor $10z$, and also $z-1$ precisely when $z > 2$ and $z \not\equiv 1 \pmod{10}$. The latter exclusions prevent an incorrect divide-by-ten predecessor and a return from the absorbing state one.

2. These predecessor sets are disjoint because f has a unique output. Iterating this explicit rule gives level sizes at $n = 10, \ldots, 20$:
$256, 511, 1022, 2043, 4084, 8164, 16320, 32624, 65216, 130368, 260608$.

3. Thus $M = 260608 = 2^9 \cdot 509$. Trial division through $\sqrt{509} < 23$ proves 509 prime, and the required sum is 511.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2005-I-Q13] | Part C (Q11-Q15)

Answer: 083

1. Track counts by the last step, $R(x,y), U(x,y), D(x,y)$. Then $R(x,y) = R(x-1,y) + D(x-1,y)$, $U(x,y) = U(x,y-1) + D(x,y-1)$, and $D(x,y)$ is the sum of all three states at $(x-1,y-1)$.

2. Use one empty starting state and allow all three first steps; along each axis the only path uses that axis direction. Filling the recurrence gives $(D,R,U)(5,5) = (27,28,28)$.

3. The desired total is $27+28+28 = 83$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2006-I-Q11] | Part C (Q11-Q15)

Answer: 458

1. In a legal tower using $1, \ldots, n$, the new largest cube $n+1$ can be inserted only at bottom, directly above n, or directly above $n-1$: exactly three choices for $n \geq 2$. Each insertion stays legal.

2. Conversely deleting $n+1$ preserves legality because its supporting cube had side at least $n-1$, and the cube above it has side at most n. Thus $T_{n+1} = 3T_n$, with $T_2 = 2$.

3. Therefore $T_8 = 2 \cdot 3^6 = 1458$, remainder 458.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2008-I-Q11] | Part C (Q11-Q15)

Answer: 172

1. Let a_n, b_n count valid words starting with A, B. Removing the initial AA gives $a_n = a_{n-2} + b_{n-2}$ for $n \geq 3$. For a B start, either remove a single initial B before an A run, or remove BB from a longer odd B run; hence $b_n = a_{n-1} + b_{n-2}$.

2. Initialize $(a_1, b_1) = (0, 1)$ and $(a_2, b_2) = (1, 0)$. Iteration gives $(a_{10}, b_{10}) = (16, 21)$, $(a_{12}, b_{12}) = (37, 43)$, and $(a_{14}, b_{14}) = (80, 92)$.

3. The desired total is 172.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2015-II-Q12] | Part C (Q11-Q15)

Answer: 548

1. Fix the first letter. A valid string is uniquely specified by its successive run lengths, each in $\{1, 2, 3\}$ and summing to 10. The letters alternate automatically.

2. Let C_n count these compositions, with $C_0 = 1$ and $C_n = 0$ for $n < 0$. Removing the last part gives $C_n = C_{n-1} + C_{n-2} + C_{n-3}$.

3. The values for $n = 1, \ldots, 10$ are $1, 2, 4, 7, 13, 24, 44, 81, 149, 274$. Either first letter is possible, so the answer is $2 \cdot 274 = 548$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2016-II-Q12] | Part C (Q11-Q15)

Answer: 732

1. Let R_n count proper colorings of an n-cycle. A proper row of n positions has $4 \cdot 3^{n-1}$ colorings. Rows with different end colors correspond to R_n; those with equal ends merge to a properly colored $(n-1)$-cycle.

2. Thus $R_n + R_{n-1} = 4 \cdot 3^{n-1}$. From $R_3 = 24$, obtain $R_4 = 84$, $R_5 = 240$, $R_6 = 732$. The six-region answer is 732.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-I-Q14] | Part C (Q11-Q15)

Answer: 351

1. Group nonabsorbing vertices into $A = \{S, P_1\}, B = \{P_2, P_5\}, C = \{P_3, P_4\}$. Let (a_n, b_n, c_n) count still-active paths in these classes after n steps.

2. The transitions give $(a_{n+1}, b_{n+1}, c_{n+1}) = (a_n + b_n, a_n + c_n, b_n)$, initially $(1, 0, 0)$. Exactly c_n paths first reach E next.

3. For $n = 0, \ldots, 11$, the c_n values are $0, 0, 1, 1, 3, 4, 9, 14, 28, 47, 89, 155$. Their sum is 351.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-II-Q11] | Part C (Q11-Q15)

Answer: 461

1. Failure means a proper prefix contains exactly the first k positive integers. Let c_n count permutations of $1, \ldots, n$ with no such proper prefix.

2. Every permutation has a unique earliest closed prefix, of some size k. Its internal order has c_k choices, and the rest are arbitrary, so $n! = \sum_{k=1}^{n} c_k (n-k)!$.

3. Starting with $c_1 = 1$ yields $c_2 = 1, c_3 = 3, c_4 = 13, c_5 = 71$. Then $c_6 = 720 - (120 + 24 + 18 + 26 + 71) = 461$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2019-II-Q12] | Part C (Q11-Q15)

Answer: 047

1. Let $g(s)$ count such sequences of sum s with first term at least two. Set $g(1)=0$. For $s \geq 2$, the single-term sequence contributes one. If first term is a proper divisor $d \geq 2$ of s, divide the remaining sequence by d; it has sum $s/d-1$ and first term at least two. Thus $g(s)=1+\sum_{d \mid s,\, 2 \leq d<s} g(s/d-1)$.

2. The desired count is $1+\sum_{d \mid 360,\, d<360} g(360/d-1)$. There are twenty-three arguments. Fifteen have value one, one argument is one and contributes zero, while arguments $119, 44, 39, 35, 14, 9, 8$ have values $6, 8, 6, 4, 3, 2, 2$.

3. The sum is $1+15+6+8+6+4+3+2+2=47$. The recursion only calls smaller positive sums, so it gives a finite exhaustive count.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2023-I-Q11] | Part C (Q11-Q15)

Answer: 235

1. Let T_m count subsets of a block of m consecutive integers with no adjacent pair. Choosing or omitting its last element gives $T_m = T_{m-1} + T_{m-2}$, with $T_0 = 1, T_1 = 2$. Thus $T_m = F_{m+2}$.

2. If the unique pair is $k, k+1$, forbid $k-1, k+2$. The blocks on either side contribute $F_k F_{10-k}$; at an endpoint, the empty block contributes 1.

3. For $k = 1, \ldots, 9$, these contributions are $34, 21, 26, 24, 25, 24, 26, 21, 34$. Their sum is 235.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2026-I-Q15] | Part C (Q11-Q15)

Answer: 083

1. Enumerate every rectangle $[r, r+a) \times [c, c+b)$, keeping its boundary-cell bitmask. There are $(10 \cdot 9/2)^2 = 2025$ candidate loops. A partition is an exact cover of the 100 cells by five masks.

2. Use the following exact recurrence: $C(U, j) = \sum C(U \cup L, j-1)$, where v is the first uncovered cell and the sum is over candidate loops L containing v with $L \cap U = \varnothing$. Set $C(U, 0) = 1$ exactly for the full grid, otherwise zero. This counts each partition once, because its loop covering v is unique.

3. Evaluating the recurrence gives $C(\varnothing, 5) = 83$. The count splits into 27 partitions with a full 10×10 outer loop, 28 with multiple vertical outer slabs, and 28 with multiple horizontal outer slabs. The companion exact-cover certificate records these masks and totals.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2026-II-Q14] | Part C (Q11-Q15)

Answer: 157

1. The running-value parity equals the ordinary prefix-sum parity, since adding and subtracting agree modulo two. Hence the final value is 12 minus twice the sum of even terms placed after an odd prefix; those subtracted terms must sum to six.

2. For a completely explicit count, define $D(s, v)$ as the number of positive-part sequences of total s with running value v, starting at $D(0, 0) = 1$. For every $1 \le t \le 12 - s$, add $D(s, v)$ to $D(s + t, v - t)$ if v is odd and t even, and otherwise to $D(s + t, v + t)$. This is an exhaustive disjoint extension recurrence.

3. Evaluating by total gives $D(12, 0) = 157$. Equivalently, grouping by the number $2m$ of odd terms and their total A, the six nonzero contributions $(m, A) = (1, 2), (1, 4), (1, 6), (2, 4), (2, 6), (3, 6)$ are $20, 16, 12, 36, 48, 25$, summing to 157.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2002-I-Q05] | Part A (Q1-Q5)

Answer: 183

1. Each pair among twelve vertices determines three squares: two with the pair as a side and one with it as a diagonal. This gives $3\binom{12}{2} = 198$ incidences.

2. If three square vertices lie on the dodecagon circumcircle, the fourth lies on the same circle. Successive square vertices are $90^\circ = 3 \cdot 30^\circ$ apart, so the fourth is also a dodecagon vertex. There are exactly three such squares, obtained by taking every third vertex.

3. Each of those is counted six times, rather than once. Subtract $3(6-1)$ to get 183.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2004-I-Q03] | Part A (Q1-Q5)

Answer: 241

1. All vertex pairs number $\binom{26}{2} = 325$. Remove the 60 edges.

2. Triangular faces have no face diagonals; the twelve quadrilaterals have two each. Convexity prevents one such diagonal from belonging to two distinct faces. Thus remove another 24, leaving 241.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2007-II-Q05] | Part A (Q1-Q5)

Answer: 888

1. The intercept rectangle is 223×9, containing 2007 unit squares. Since $\gcd(223, 9) = 1$, the segment meets no interior lattice point, and it crosses $223 + 9 - 1 = 231$ square interiors.

2. The remaining 1776 squares split equally above and below by half-turn symmetry about the rectangle center. Thus the count is 888.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2014-I-Q05] | Part A (Q1-Q5)

Answer: 134

1. A separating circle intersects the common circumcircle in at most two points, unless the circles coincide; coincidence cannot strictly separate a proper nonempty subset. Thus any such subset is one consecutive run around the polygon.

2. Conversely any proper consecutive run can be strictly separated by a chord line placed in the two adjoining gaps. A sufficiently large circle approximating the appropriate half-plane separates the same finite vertex sets.

3. For each size one through eleven there are twelve runs. Add the empty and full subsets: $12 \cdot 11 + 2 = 134$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2016-I-Q03] | Part A (Q1-Q5)

Answer: 810

1. A path enters the upper ring once, moves horizontally there, descends once to the lower ring, moves horizontally again, and then reaches B. It can never return to an earlier level.

2. On a five-cycle, a simple path from a fixed start has 1 zero-step option and two directions for each length $1, 2, 3, 4$, totaling 9. Changing horizontal direction would revisit a vertex.

3. There are 5 first upper vertices, 9 upper-ring paths, 2 descending edges, and 9 lower-ring paths. These choices are independent, giving $5 \cdot 9 \cdot 2 \cdot 9 = 810$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2022-II-Q05] | Part A (Q1-Q5)

Answer: 072

1. For labels $i < j < k$, put $u = j - i, v = k - j$. All of $u, v, u + v$ must be prime. Parity forces exactly one of u, v to be 2, and the other must be p with $p, p + 2$ twin primes.

2. The possible p are $3, 5, 11, 17$. For each, the span is $p + 2$, allowing $20 - (p + 2)$ starting labels and two orders of the gaps.

3. Total $2(15 + 13 + 7 + 1) = 72$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2025-II-Q03] | Part A (Q1-Q5)

Answer: 082

1. Choose the four central-cross colors first. Each small square has two internal edges and two independent outer edges; if its internal colors differ, its outer edges have two choices, otherwise only one.

2. For the central cross, monochromatic cases number 2, each weighted 1; cases with one odd-colored arm number 8, each weighted 4; two adjacent red arms number 4, each weighted 4; two opposite red arms number 2, each weighted 16.

3. Total $2+8\cdot 4+4\cdot 4+2\cdot 16=82$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2008-II-Q10] | Part B (Q6-Q10)

Answer: 240

1. The nine possible squared distances are $1, 2, 4, 5, 8, 9, 10, 13, 18$, so at most ten points occur. A ten-point path must use these in that order.

2. Count backwards. The last two points are opposite corners, with four ordered choices. The preceding squared distance thirteen allows two symmetric choices. Fix the last points as $P_{10} = (0, 0), P_9 = (3, 3), P_8 = (0, 1)$. Distances ten,nine,eight,five,four,two successively force $P_7 = (3, 2), P_6 = (0, 2), P_5 = (2, 0), P_4 = (1, 2), P_3 = (1, 0), P_2 = (2, 1)$; the alternative ten-distance choice $(3, 0)$ fails at the next step because its distance-three neighbors were used.

3. The first point has three unused unit-neighbors of $(2, 1)$: $(1, 1), (2, 2), (3, 1)$. Thus ten-point paths exist and $r = 4 \cdot 2 \cdot 3 = 24$, giving $mr = 240$.

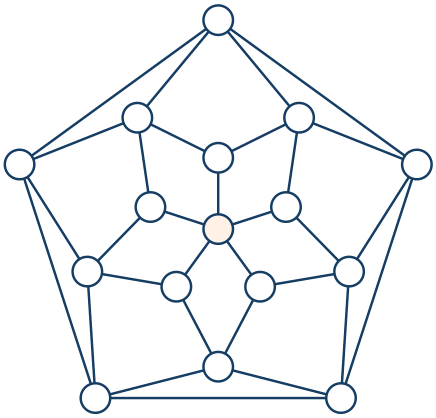
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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2012-I-Q07] | Part B (Q6-Q10)

Answer: 280



1. Let c_v be the coins at vertex v and d_v its degree. Fixed-state transfer gives $c_v = \sum_{u \sim v} c_u/d_u$. Thus c_v/d_v is the average of these same ratios at neighboring vertices.

2. A maximal ratio must equal all neighboring ratios; connectedness propagates this equality throughout the graph. Consequently c_v is proportional to d_v.

3. The central degree is five; the five inner degrees are three, and the ten other degrees are four. Total degree is $5 + 5 \cdot 3 + 10 \cdot 4 = 60$. The center receives $3360(5/60) = 280$ coins.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2018-I-Q07] | Part B (Q6-Q10)

Answer: 052

1. Each single base contributes six triangles with sides $1, 1, \sqrt{3}$ and two equilateral triangles, or eight. Both bases give sixteen.

2. For a mixed triangle choose which base contains the pair, in two ways. A cross-base distance is at least two, while base-pair lengths are $1, \sqrt{3}, 2$. Adjacent base pairs cannot be isosceles: no opposite-base vertex is horizontally equidistant from their endpoints.

3. For the six base pairs of length $\sqrt{3}$, two opposite-base vertices lie over their perpendicular bisector, giving twelve. For the three diameters of length two, a top vertex directly over either endpoint makes one cross-base side equal two, giving six; there are no equidistant top vertices for these diameters.

4. Total $16 + 2(12 + 6) = 52$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2022-II-Q09] | Part B (Q6-Q10)

Answer: 244

1. Choosing two points on each line determines exactly one pair of crossing segments. Hence the interior crossing count is $X = \binom{7}{2}\binom{5}{2} = 210$.

2. Trim both full lines beyond the marked points; this does not alter bounded faces. For general counts m, n, the connected graph has $V = m + n + X + 4$ vertices and $E = (m + 1) + (n + 1) + mn + 2X$ edges.

3. Euler gives bounded faces $E - V + 1 = mn + X - 1$. Substituting $m = 7, n = 5$ gives $35 + 210 - 1 = 244$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2012-II-Q14] | Part C (Q11-Q15)

Answer: 016

1. A simple graph with all degrees two is a disjoint union of cycles of length at least three. On nine vertices, the possible cycle sizes are $3+3+3, 3+6, 4+5, 9$.

2. An undirected cycle on k fixed labels has $(k-1)!/2$ arrangements. The four cases give $9!/[(3!)^3 3!] = 280$, $\binom{9}{3}(5!/2) = 5040$, $\binom{9}{4}(3!/2)(4!/2) = 4536$, and $8!/2 = 20160$.

3. The total is 30016, leaving remainder 16.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q11] | Part C (Q11-Q15)

Answer: 544

1. For five vertices, failure of strong connectivity forces an all-outgoing source or all-incoming sink. Indeed, a proper set of vertices reachable from a vertex has all cross edges entering it. A reachable set of size1 or2 contains a global sink; if its size is3 or4, the complementary set of size2 or1 contains a global source.

2. An all-outgoing vertex can be chosen5 ways, with the other6 edges free:320 orientations. There is at most one such vertex. Similarly there are320 with an all-incoming vertex.

3. The intersection has5 choices of source,4 of sink and3 free remaining edges, giving160. Out of $2^{10} = 1024$ orientations, the valid count is $1024 - 320 - 320 + 160 = 544$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q13] | Part C (Q11-Q15)

Answer: 245

1. At each apex there are $\lfloor (n-1)/2 \rfloor$ symmetric vertex pairs. Every nonequilateral isosceles triangle has one apex, but an equilateral is counted three times. Thus $f(n) = n\lfloor (n-1)/2 \rfloor - (2n/3)\mathbf{1}_{3|n}$.

2. Write $n = 6k + j$ for $j = 0, 1, 2, 3, 4, 5$. Substituting into this formula gives successive differences $13k, 3k, 5k + 1, 7k + 3, 9k + 6, -k - 2$, respectively.

3. Equaling78 yields only $k = 6, j = 0$; $k = 26, j = 1$; and $k = 8, j = 4$. Therefore $n = 36, 157, 52$, with sum245.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q14] | Part C (Q11-Q15)

Answer: 168

1. Choose a primitive integral direction vector. Eight points span seven primitive steps, so every component has absolute value at most1. Axis-parallel lines contain10 points and do not qualify.

2. If exactly one component is zero, the line lies in one of30 coordinate-parallel 10×10 planes. In each plane, two diagonals of each slope $+1, -1$ have exactly8 points (offset magnitude2). This contributes120.

3. If all components are nonzero, there are four directions up to sign, equivalent by cube symmetry. For $(1, 1, 1)$, normalize a point on the line so its least coordinate is1. Exactly8 points means its greatest coordinate is3. The count of triples in $\{1, 2, 3\}^3$ using both1 and3 is $27 - 8 - 8 + 1 = 12$.

4. The space-diagonal contribution is $4 \cdot 12 = 48$, for total $120 + 48 = 168$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2023-I-Q14] | Part C (Q11-Q15)

Answer: 608

1. Label states $(a, b) \in \mathbb{Z}_{12}^2$, and let $h(a, b)$ be 1 when the next move increases a, or 0 when it increases b. A Hamiltonian cycle must give every state one incoming edge. Thus $h(a-1, b) + [1 - h(a, b-1)] = 1$, so $h(a-1, b) = h(a, b-1)$.

2. Therefore h is constant on each diagonal $a + b \pmod{12}$. A valid transition rule is a binary word of length 12, read repeatedly as the diagonal sum increases one per move.

3. If the word contains c moves of the first hand, every twelve moves change the state by $(c, -c)$. Return occurs after $12/\gcd(c, 12)$ such blocks; all 144 states form one cycle exactly when $\gcd(c, 12) = 1$. This condition is sufficient as well as necessary.

4. The possible counts are $c = 1, 5, 7, 11$, giving $\binom{12}{1} + \binom{12}{5} + \binom{12}{7} + \binom{12}{11} = 1608$ sequences. The remainder is 608.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2024-II-Q15] | Part C (Q11-Q15)

Answer: 315

1. On the unit circumcircle, chord directions form twelve classes separated by 15°. There are six perpendicular direction pairs, three of each of two types. In a direction class of the first type, chord offsets are $0, \pm 1/2, \pm\sqrt{3}/2$; in the second they are $\pm\cos 75^\circ, \pm\cos 45^\circ, \pm\cos 15^\circ$.

2. Choose two chords in each perpendicular class. If D_1, D_2 are the maximum absolute offsets of each selected pair, their intersections lie on the chord segments exactly when $D_1^2 + D_2^2 \le 1$. For the five-chord class, selected pairs have $D = 1/2, \sqrt{3}/2$ with counts $3, 7$. The allowed ordered combinations number $3^2 + 2 \cdot 3 \cdot 7 = 51$.

3. For the six-chord class, the three possible D values have pair counts $1, 5, 9$. Every combination with the smallest D is allowed, as is the middle-middle combination. Thus the count is $1 + 2 \cdot 5 + 2 \cdot 9 + 25 = 54$. Total $3(51 + 54) = 315$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2025-II-Q11] | Part C (Q11-Q15)

Answer: 113

1. Fix the common chord step $k = 1, \ldots, 12$. For $k < 12$, edges joining i to $i \pm k$ form $g = \gcd(24, k)$ cycles of length $24/g$. An even cycle has two perfect matchings and an odd cycle none.

2. For $k = 1, 5, 7, 11$ contributions are 2 each; $k = 2, 10$ give 4 each; $k = 3, 9$ give 8 each; $k = 4$ gives 16; $k = 6$ gives 64; $k = 8$ gives zero. The diameter case $k = 12$ contributes one.

3. Summing yields $8 + 8 + 16 + 16 + 64 + 1 = 113$. Different steps have different lengths, so no matching is duplicated.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2026-II-Q15] | Part C (Q11-Q15)

Answer: 393

1. Work over residues $0, 1, -1$. The triples $\{i, i+1, i+3\}$ are the seven Fano-plane lines. Let k be the number of zero positions; only lines avoiding those positions contribute to the cubic sum.

2. For $k = 7, 6, 5, 4, 3$, the respective counts are $1, 0, 42, 56, 42$: two nonzero entries must have opposite signs; three must be equal and not form a line; four must split two and two and the zero triple must itself be a line.

3. For $k = 2$, the five signs split four and one; the two surviving lines meet once, and their products cancel exactly when the minority sign is at one of four nonintersection positions. Count $\binom{7}{2} \cdot 2 \cdot 4 = 168$. For $k = 1$, the six positions correspond to edges of K_4 on the four surviving lines. Three negative signs must form a three-edge path to make exactly two line products negative; there are $4!/2 = 12$ paths per zero position, giving 84. With no zero entries the linear condition forces two or five negative signs, giving respectively four or three negative line products, never a zero cubic sum.

4. Total $1 + 42 + 56 + 42 + 168 + 84 = 393$. An independent exhaustive check can inspect all $3^7 = 2187$ residue tuples using the two displayed sums.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2026-II-Q02] | Part A (Q1-Q5)

Answer: 243

1. Each square strip is crossed exactly once, using its bottom edge, top edge, or rising diagonal. These are three independent choices per strip.

2. The ten choices determine all vertical moves: at each vertical line use its edge if and only if arrival and departure heights differ, including the starting and ending lines. This never repeats a segment and yields a bijection with the 3^{10} choice strings.

3. Thus $\sqrt{N} = 3^5 = 243$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2005-II-Q06] | Part B (Q6-Q10)

Answer: 392

1. A card $i \le n$ moves to top-based position $2(n-i)+1$, while $i > n$ moves to $2(2n-i)+2$. Thus an odd fixed position must belong to A.

2. For $i = 131$, solve $131 = 2(n-131)+1$, giving $n = 196$. Card 262 from B is also fixed because $2(392-262)+2 = 262$. Thus the required total is 392.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2006-II-Q08] | Part B (Q6-Q10)

Answer: 336

1. The central triangle stays fixed under every symmetry, while the three corner triangles may undergo all six permutations. Hence only the center color and multiset of corner colors matter.

2. Three-color multisets selected with repetition from six colors number $\binom{8}{3} = 56$. Multiply by six center choices to obtain 336.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2010-I-Q10] | Part B (Q6-Q10)

Answer: 202

1. Write $a_i = 10b_i + c_i$ with digits b_i, c_i. Then the equation is $10B + C = 2010$, where $B = \sum b_i 10^i, C = \sum c_i 10^i$.

2. Conversely, every choice $B = 0, \ldots, 201$ gives nonnegative $C = 2010 - 10B$. Both are at most four-digit numbers, so padding their decimal digits with zeros uniquely recovers the coefficients.

3. This is a bijection with 202 choices of B, so the count is 202.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2010-II-Q10] | Part B (Q6-Q10)

Answer: 163

1. Write $f(x) = a(x-r)(x-s)$, so $ars = 2010 = 2 \cdot 3 \cdot 5 \cdot 67$. Allocate each prime to one of $|a|, |r|, |s|$: there are $3^4 = 81$ ordered magnitude triples.

2. For each allocation, four sign patterns have positive product. There are therefore 324 signed ordered triples (a, r, s). Swapping r, s identifies polynomials.

3. Only $r = s = 1$ and $r = s = -1$ are fixed under the swap, both with $a = 2010$. The number of orbits is $(324 + 2)/2 = 163$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2011-II-Q06] | Part B (Q6-Q10)

Answer: 080

1. The reflection $x \mapsto 11 - x$ exchanges $d - c > b - a$ and $d - c < b - a$. Thus the desired count is one half of the difference between $\binom{10}{4}$ and the equality count.

2. Equality has outer gaps $k = b - a = d - c$ and middle gap $j = c - b$. For $k = 1, 2, 3, 4$, positive integers (a, j) satisfy $a + j \leq 10 - 2k$. Their counts are $\binom{8}{2}, \binom{6}{2}, \binom{4}{2}, \binom{2}{2}$, totaling 50.

3. Therefore the answer is $(210 - 50)/2 = 80$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2018-I-Q10] | Part B (Q6-Q10)

Answer: 004

1. Code counterclockwise or inward as X, clockwise or outward as Y. From any state there is exactly one move of each type, so binary words describe paths bijectively. A step ends on the inner ring exactly when its code is X.

2. If the final code is X, spoke moves balance inward and outward, so the net angular displacement equals $\#X - \#Y$ modulo five. A return therefore requires $2\#X - 15 \equiv 0 \pmod 5$, with $\#X = 5, 10, 15$.

3. Fixing the last X, the count is $\binom{14}{4} + \binom{14}{9} + \binom{14}{14} = 1001 + 2002 + 1 = 3004$, giving remainder 4.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2023-II-Q10] | Part B (Q6-Q10)

Answer: 144

1. First fill with residues $0, 1, 2$, each appearing four times; then permute the four actual integers within each class, contributing $(4!)^3$. Adjacent residues must differ.

2. For a column (a, b) with remaining residue c, the next column has exactly the three possibilities $(b, a), (b, c), (c, a)$. Thus each choice of its unordered residue pair uniquely determines the orientation once the previous column is known.

3. For each residue to occur four times in six columns, the three unordered pair types must each occur twice. There are $6!/(2!)^3 = 90$ pair sequences and two orientations of the first column, yielding 180 residue fillings.

4. Hence $N = 180 \cdot 24^3 = 2^{11}3^5 \cdot 5$. Its divisor count is $(11 + 1)(5 + 1)(1 + 1) = 144$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2024-II-Q09] | Part B (Q6-Q10)

Answer: 902

1. A maximal placement has no empty row or column: an empty row allows a chip matching some column color to be added. Assign each row and column its occupied color.

2. A cell is filled exactly when its row and column colors agree; otherwise it must remain empty. Thus rows and columns must use the same set of colors. This gives a bijection with pairs of row/column colorings using either one common color or both colors.

3. There are two monochromatic possibilities and $(2^5 - 2)^2$ mixed possibilities. Total $2 + 30^2 = 902$.

Source / 来源

Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2026-II-Q13] | Part C (Q11-Q15)

Answer: 107

1. A three-element consecutive block makes any pairing impossible. Every maximal consecutive block therefore has size one or two. A two-block is forced to map outward; a singleton can move one unit left or right. Only blocks separated by one missing integer can compete for a target, so group them into chains with such gaps.

2. For fixed disjoint S, T, the pairing is unique if it exists: the allowed edges are edges of the integer line, whose finite components are paths, and an even path has only one perfect matching. Thus counting legal shifts counts image sets without multiplicity.

3. A chain of k singleton blocks permits $k+1$ images: an initial run shifts left and the remainder shifts right, since a right-then-left pattern collides. A chain with one two-block has one image; with two two-blocks it has none. Different chains are independent, so positive counts factor as $\prod(k_i+1)$; two-blocks cannot help minimize size.

4. To minimize $\sum k_i$ for product $4040 = 2^3 \cdot 5 \cdot 101$, split every composite factor gh into g, h, since $(g-1)+(h-1) < gh-1$. Cost $1+1+1+4+100 = 107$, realized by separated alternating-integer chains of those five lengths.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2000-I-Q05] | Part A (Q1-Q5)

Answer: 026

1. Let box sizes be a, b and black counts p, q. Then $a + b = 25$ and $50pq = 27ab$, so $50 \mid ab$. Up to exchange, only $(a, b) = (5, 20), (10, 15)$ fit.

2. For sizes $5, 20$, $pq = 54$ with $p < 5, q < 20$ forces black counts $3, 18$, leaving whites $2, 2$. For sizes $10, 15$, $pq = 81$ forces $9, 9$, leaving $1, 6$.

3. Both cases give white probability $1/25$, so the answer is 26.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2000-II-Q03] | Part A (Q1-Q5)

Answer: 758

1. The remaining multiplicities are nine groups of four and one group of two, so there are $9\binom{4}{2}+\binom{2}{2}=55$ matching pairs.

2. There are $\binom{38}{2}=703$ possible pairs. Since $55/703$ is reduced, the answer is 758.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2002-I-Q01] | Part A (Q1-Q5)

Answer: 059

1. In a three-symbol block, being palindromic means the last symbol equals the first. The two block probabilities are $1/26, 1/10$, independently.

2. The union probability is $1/26 + 1/10 - 1/260 = 7/52$, so the sum is 59.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2005-II-Q02] | Part A (Q1-Q5)

Answer: 079

1. For the first bag, after any initial roll, the next differs in type with probability $6/8$ and the third supplies the missing type with probability $3/7$. Thus the chance is $9/28$.

2. Two of each type remain. The second bag succeeds with chance $(4/5)(2/4) = 2/5$, and the last bag is then forced. Probability $9/70$, giving 79.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2010-I-Q01] | Part A (Q1-Q5)

Answer: 107

1. $2010^2 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 67^2$ has $3^4 = 81$ positive divisors. The square divisors choose each exponent from $0, 2$, so there are16 squares and65 nonsquares.

2. The probability is $16 \cdot 65 / \binom{81}{2} = 26/81$, giving 107.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2010-II-Q04] | Part A (Q1-Q5)

Answer: 052

1. There are $12 \cdot 11$ ordered distinct gate pairs. For a separation of d positions, there are $2(12-d)$ pairs.

2. Add $d = 1, 2, 3, 4$: favorable pairs total $2(11+10+9+8) = 76$. The probability is $76/132 = 19/33$.

3. Thus $m + n = 52$.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2013-I-Q04] | Part A (Q1-Q5)

Answer: 429

1. Each four-cell orbit must be monochromatic. Since the total red count is eight, the central cell is blue and exactly two of the three orbits are red.

2. There are three favorable choices and $\binom{13}{8} = 1287$ total colorings. Thus the probability is $3/1287 = 1/429$, giving 429.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2015-I-Q02] | Part A (Q1-Q5)

Answer: 139

1. There are $\binom{9}{3} = 84$ possible subsets. Choosing the nationality of the repeated pair gives favorable count $\binom{2}{2} \cdot 7 + \binom{3}{2} \cdot 6 + \binom{4}{2} \cdot 5 = 55$.

2. The cases are disjoint because a three-person subset cannot have two different repeated nationalities. Thus the probability is $55/84$, and $m + n = 139$.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2015-II-Q05] | Part A (Q1-Q5)

Answer: 090

1. The grid has $n(n-1)$ horizontal and the same number of vertical neighboring pairs. Among $\binom{n^2}{2}$ unordered pairs, the probability is $2n(n-1)/\binom{n^2}{2} = 4/[n(n+1)]$.

2. The strict inequality becomes $n(n+1) > 8060$. Since $89 \cdot 90 = 8010 < 8060 < 90 \cdot 91 = 8190$, the least possible n is 90.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2018-I-Q03] | Part A (Q1-Q5)

Answer: 157

1. Treat individual cards as distinct, giving $10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30240$ equally likely layouts. For a fixed color word with r reds, the number of fillings is $(5!/(5-r)!)(5!/r!)$.

2. The monochromatic words give 120 each. For $r = 1, 2, 3, 4$, two block orders qualify, with respective fillings $600, 1200, 1200, 600$.

3. Favorable layouts total $240 + 2(3600) = 7440$, so probability $31/126$ and answer 157.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2019-I-Q02] | Part A (Q1-Q5)

Answer: 029

1. The $20 \cdot 19 = 380$ ordered pairs are equally likely. Favorable choices for $J = 1, \ldots, 18$ number $19 - J$.

2. Sum $18 + \cdots + 1 = 171$, so probability $171/380 = 9/20$, giving 29.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2019-I-Q05] | Part A (Q1-Q5)

Answer: 252

1. Success requires reaching $(1,1)$ and then taking a diagonal. If the preceding path has d diagonals, it has $3-d$ steps of each other type and length $6-d$. Their multinomial counts for $d=0,1,2,3$ are $20,30,12,1$.

2. Weight paths by their lengths and the final diagonal: probability $\frac{1}{3}(20/3^6+30/3^5+12/3^4+1/3^3)=245/3^7$. Thus $m+n=252$.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2023-I-Q01] | Part A (Q1-Q5)

Answer: 191

1. Treat the seven diameters as pairs of positions. The five male positions are uniformly distributed among $\binom{14}{5} = 2002$ subsets.

2. Choose five distinct diameter pairs, then one end of each: $\binom{7}{5}2^5 = 672$ choices.

3. Thus the probability is $672/2002 = 48/143$, and $m + n = 191$.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2026-II-Q05] | Part A (Q1-Q5)

Answer: 190

1. Equal hypergeometric counts give $\binom{r}{4}\binom{b}{3}=\binom{r}{5}\binom{b}{2}$. Cancel to obtain $(b-2)/3=(r-4)/5$, or $5b=3r-2$.

2. Thus $r\equiv 4 \pmod 5$. The bounds make the first five r values $14,19,24,29,34$, with $b=8,11,14,17,20$.

3. Totals are $22,30,38,46,54$, whose sum is 190.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q06] | Part B (Q6-Q10)

Answer: 079

1. Each multiset of four values from six labels has exactly one nondecreasing ordering. The favorable count is $\binom{9}{4} = 126$ by weak compositions of four into six multiplicities.

2. Of $6^4 = 1296$ equally likely outcomes, the probability is $126/1296 = 7/72$, giving 79.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q10] | Part B (Q6-Q10)

Answer: 200

1. The midpoint automatically stays in the geometric box; it is a lattice point exactly when the coordinate parities agree. Allowing equal ordered points, the same-parity coordinate pair counts are $2^2+1^2=5$, $2^2+2^2=8$, $3^2+2^2=13$.

2. Of their 520 product pairs, 60 are equal-point pairs. Therefore the desired probability is $460/(60\cdot 59)=23/177$, giving 200.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-II-Q09] | Part B (Q6-Q10)

Answer: 929

1. For the four block events, a single block forces 4 red cells. Of six block pairs, four force 6 cells and two force 7. Each triple forces 8 cells; all four force 9.

2. In units of $1/512$, the count containing a red block is $4 \cdot 32 - (4 \cdot 8 + 2 \cdot 4) + 4 \cdot 2 - 1 = 95$.

3. The desired probability is $(512 - 95)/512 = 417/512$, giving 929.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2004-II-Q10] | Part B (Q6-Q10)

Answer: 913

1. Each number is $2^a + 2^b$ for $0 \le a < b \le 39$, giving $\binom{40}{2} = 780$ choices.

2. Divisibility by nine is equivalent to $2^{b-a} \equiv -1 \pmod 9$, hence $b - a \equiv 3 \pmod 6$. Allowed differences $3, 9, \ldots, 39$ contribute $37 + 31 + 25 + 19 + 13 + 7 + 1 = 133$ pairs.

3. The probability $133/780$ is reduced, giving 913.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2005-I-Q09] | Part B (Q6-Q10)

Answer: 074

1. The edge joining the two unpainted faces is uniformly one of twelve edges. At a large-cube corner, edge position, face center, the fractions placing both bad faces inward are $3/12 = 1/4$, $5/12$, and $8/12 = 2/3$, respectively.

2. With eight corner, twelve edge and six face-center cubes, multiply to get $(1/4)^8(5/12)^{12}(2/3)^6 = 5^{12}/(2^{34}3^{18})$. The central cube imposes no condition.

3. The requested sum is $12 + 34 + 18 + 5 + 2 + 3 = 74$.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2007-II-Q10] | Part B (Q6-Q10)

Answer: 710

1. Given $|A| = j$, allowed B number $2^j + 2^{6-j} - 1$, subtracting the empty-set overlap.

2. Summing over A gives favorable pairs $\sum_j \binom{6}{j}(2^j + 2^{6-j} - 1) = 2 \cdot 3^6 - 2^6$. Divide by 2^{12} to get $697/2^{11}$, so the answer is 710.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2011-I-Q10] | Part B (Q6-Q10)

Answer: 503

1. An inscribed triangle is obtuse exactly when all three vertices lie in an open semicircle. Its unique first clockwise vertex determines the other two within that half-circle, so this counts each obtuse triangle once.

2. For $n = 2k$, there are $n\binom{k-1}{2}$ obtuse triples. Dividing by $\binom{2k}{3}$ gives $3(k-2)/[2(2k-1)] = 93/125$, whose root is $k = 188$, or $n = 376$.

3. For $n = 2k+1$, the probability is $3(k-1)/[2(2k-1)] = 93/125$, giving $k = 63$, or $n = 127$.

4. Both satisfy the count formulas; the required sum is 503.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2013-I-Q06] | Part B (Q6-Q10)

Answer: 047

1. The three math-book positions form a uniformly random three-subset of twelve slots. A box of capacity k contains them in $\binom{k}{3}$ possible slot subsets.

2. The favorable counts add because boxes are disjoint: $\binom{3}{3}+\binom{4}{3}+\binom{5}{3}=15$. Thus probability $15/\binom{12}{3}=3/44$, giving 47.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2020-I-Q09] | Part B (Q6-Q10)

Answer: 077

1. Every divisor is uniquely 2^u5^v, with $u \in \{0, \ldots, 18\}$ and $v \in \{0, \ldots, 9\}$. Both exponent coordinates are independent uniform choices.

2. For k possible exponents, a nondecreasing triple has $\binom{k+2}{3}$ possibilities. Consequently the probability is $\binom{21}{3}\binom{12}{3}/190^3$.

3. This equals $(70/361)(11/50) = 77/1805$. As 77 and 1805 are coprime, the numerator is 77.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2022-I-Q09] | Part B (Q6-Q10)

Answer: 247

1. For positions $i < j$, the gap count $j - i - 1$ is even exactly when i and j have opposite parity. Thus every color must occur once in the six odd slots and once in the six even slots.

2. There are $(6!)^2$ favorable color rows and $12!/2^6$ total color rows. Their ratio is $2^6(6!)^2/12! = 16/231$.

3. The numerator and denominator are coprime, giving $m + n = 247$.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2025-I-Q07] | Part B (Q6-Q10)

Answer: 821

1. The total pairings are $11!! = 10395$. The last word has the largest smaller letter among all pairs.

2. If G is the first letter, it has five later partners. Each remaining later letter must pair with a distinct member of $A, \ldots, F$, giving $5 \cdot (6 \cdot 5 \cdot 4 \cdot 3) = 1800$ pairings.

3. If G is the second letter, its partner must be F: otherwise too many later letters would require earlier partners. The five remaining late letters biject with $A, \ldots, E$, giving 120 pairings. Probability $(1800 + 120)/10395 = 128/693$, so the answer is 821.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q15] | Part C (Q11-Q15)

Answer: 085

1. The dual graph is a cube. Encode its vertices by binary triples. Two successive labels are allowed exactly when their vertices differ in two or three bits.

2. Fix label 1 at vertex 000. Let $H(S, v)$ count paths starting there, using exactly subset S, ending at v, with every step allowed. Start $H(\{000\}, 000) = 1$, and append unused w when the bit-distance from v to w is at least two.

3. The sums of these path counts at lengths 1 through 8 are $1, 4, 12, 30, 66, 126, 156, 102$. Of the full paths, 60 have last vertex also allowed adjacent to 000, closing the label cycle. The conditional favorable count is 60 out of $7! = 5040$.

4. The probability is $1/84$, and the requested sum is 85.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2011-II-Q12] | Part C (Q11-Q15)

Answer: 097

1. Country patterns are equally likely, numbering $9!/(3!)^3 = 1680$. Failure occurs exactly when a country occupies three consecutive seats, since its middle delegate then has two compatriot neighbors.

2. For one specified country, there are $9\binom{6}{3} = 180$ such patterns. For two specified countries, place the first block in nine ways and the second in one of four triples along the remaining six-seat arc: 36 patterns. For all three, there are three cyclic partitions into triples and $3!$ country assignments, or 18.

3. Inclusion-exclusion gives $3(180) - 3(36) + 18 = 450$ failures, hence probability $(1680 - 450)/1680 = 41/56$.

4. The answer is 97.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2014-I-Q12] | Part C (Q11-Q15)

Answer: 453

1. If f has image size k, exactly $(4-k)^4$ functions g avoid it. Count f by image size: 4 for $k=1$; $\binom{4}{2}(2^4-2)=84$ for $k=2$; $\binom{4}{3}(3^4-3\cdot 2^4+3)=144$ for $k=3$; and 24 for $k=4$.

2. Favorable ordered pairs number $4\cdot 81+84\cdot 16+144=1812$, out of $4^8=65536$.

3. The reduced probability is $453/16384$, giving $m=453$.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2014-II-Q13] | Part C (Q11-Q15)

Answer: 028

1. Pairings are permutations of owners: left shoe j goes to right shoe $\pi(j)$. A collection using exactly its owners full pairs is an invariant subset of π, hence a union of cycles.

2. Thus there may be no cycle shorter than five. The only partitions of ten into allowable cycle lengths are ten or five plus five.

3. The favorable counts are $9!$ and $\binom{10}{5}(4!)^2/2 = 9!/5$. Dividing their sum by $10!$ gives $3/25$.

4. The requested sum is 28.

Source / 来源

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2018-I-Q12] | Part C (Q11-Q15)

Answer: 683

1. Six elements have each residue modulo three. Multiples of three may be chosen freely and cancel from the probability. Let a, b be counts chosen from residues one and two; the condition is $a \equiv b \pmod 3$.

2. The cases $a - b = 0, \pm 3, \pm 6$ contribute $\binom{12}{6} = 924$, $2\binom{12}{9} = 440$, and two choices respectively, by Vandermonde convolution.

3. The probability is $1366/2^{12} = 683/2048$, so $m = 683$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2004-II-Q02] | Part A (Q1-Q5)

Answer: 441

1. The matching cases are RR/RR, BB/BB, and mixed/mixed. Each single-color case has probability $[\binom{10}{2}/\binom{20}{2}][\binom{8}{2}/\binom{18}{2}] = 14/323$.

2. The mixed case has probability $[100/\binom{20}{2}][81/\binom{18}{2}] = 90/323$.

3. Adding gives $118/323$, already reduced, so the answer is 441.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2006-II-Q05] | Part A (Q1-Q5)

Answer: 029

1. Write exceptional probabilities $1/6 + t$ and $1/6 - t$ with $t > 0$. The sum-seven probability is $4/36 + 2[(1/6)^2 - t^2] = 1/6 - 2t^2$.

2. Equating gives $2t^2 = 1/288$, so $t = 1/24$ and $P(F) = 5/24$. The answer is 29.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-I-Q02] | Part A (Q1-Q5)

Answer: 144

1. The two same-color events are disjoint, giving probability $[4 \cdot 16 + 6N]/[10(16 + N)] = 29/50$.

2. Clearing denominators gives $320 + 30N = 464 + 29N$, hence $N = 144$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-II-Q02] | Part A (Q1-Q5)

Answer: 076

1. Let the triple-intersection probability be t. Then $t/(0.14+t)=1/3$, giving $t=0.07$.

2. The union has probability $3(0.1)+3(0.14)+0.07=0.79$, while A has probability $0.1+2(0.14)+0.07=0.45$.

3. The desired probability is $0.21/0.55=21/55$, giving 76.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2015-I-Q05] | Part A (Q1-Q5)

Answer: 341

1. Deal the ten distinct socks into five labeled daily pairs. Permuting day labels preserves uniformity, so the pattern mismatch, mismatch, match has the same probability as match, mismatch, mismatch.

2. The latter pattern has probability $(1/9)(6/7)(13/15)$. After the first match, four complete pairs remain among eight socks; after a mismatch, only two complete pairs remain among six socks.

3. The product is $26/315$, already reduced. The requested sum is 341.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2015-II-Q02] | Part A (Q1-Q5)

Answer: 025

1. Using a population scale of 100, the Latin groups have sizes $40, 24, 10, 2$, for a total of 76. These are proportional counts, so no assumption about actual school size is needed.

2. The desired conditional probability is $24/76 = 6/19$. Hence $m + n = 25$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2016-I-Q02] | Part A (Q1-Q5)

Answer: 071

1. Normalization gives $\Pr(k) = k/21$. By independence, outcome $(k, 7-k)$ has probability $k(7-k)/21^2$.

2. Summing the six possibilities gives $(6+10+12+12+10+6)/441 = 56/441 = 8/63$. Therefore $m+n = 71$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2016-II-Q02] | Part A (Q1-Q5)

Answer: 107

1. Let $p = \Pr(\text{Sunday rain} \mid \text{Saturday dry})$. Total probability gives $(2/5)(2p) + (3/5)p = 3/10$, hence $p = 3/14$.

2. The probability both days are dry is $(3/5)(11/14) = 33/70$. Its complement is $37/70$, giving $a + b = 107$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2017-II-Q02] | Part A (Q1-Q5)

Answer: 781

1. Team4 reaches the final with probability $4/5$. Its opponent is team2 with probability $2/5$ and team3 with probability $3/5$.

2. Conditioning on that opponent, the title probability is $(4/5)[(2/5)(2/3)+(3/5)(4/7)]=256/525$. The numerator is a power of2 and the denominator odd, so the reduced sum is 781.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2021-I-Q01] | Part A (Q1-Q5)

Answer: 097

1. Exactly one of races 2 through 6 must be a loss. If the loss is last, the path has four repeats and one switch, with probability $16/243$.

2. At any of the other four positions, there are two switches and three repeats, each path having probability $8/243$. Thus the total is $(16 + 4 \cdot 8)/243 = 16/81$, giving 97.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2022-II-Q02] | Part A (Q1-Q5)

Answer: 125

1. With probability $1/3$, C faces A immediately. Conditional on that pairing, the title chance is $(1/3)(3/4) = 1/4$.

2. For either other pairing, C wins the semifinal with probability $3/4$. In the final the opponent is A with probability $3/4$, so the conditional title chance is $(3/4)[(3/4)(1/3) + (1/4)(3/4)] = 21/64$.

3. Average over pairings: $(1/3)(1/4) + (2/3)(21/64) = 29/96$. Thus $p + q = 125$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2024-I-Q04] | Part A (Q1-Q5)

Answer: 116

1. Within the prize event, draws with exactly $2, 3, 4$ matches number $\binom{4}{2}\binom{6}{2} = 90$, $\binom{4}{3}\binom{6}{1} = 24$, and 1.

2. Exactly one of these 115 equally likely draws is the top prize, so the conditional probability is $1/115$ and $m + n = 116$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-II-Q06] | Part B (Q6-Q10)

Answer: 167

1. Bayes gives posterior odds biased:fair equal to $(2/3)^2 : (1/6)^2 = 16 : 1$. Thus their posterior probabilities are $16/17, 1/17$.

2. The predictive probability is $(16/17)(2/3) + (1/17)(1/6) = 65/102$, already reduced.

3. The requested sum is 167.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2017-II-Q09] | Part B (Q6-Q10)

Answer: 013

1. Complete coverage by eight cards means exactly one repeated number and exactly one repeated color. A discard is possible exactly when their intersection card is present, in which case that card is uniquely discardable.

2. A favorable hand comes uniquely from a seven-card number-color matching (7! choices) plus one of the42 unused cards, yielding211680 hands.

3. For an unfavorable hand, choose the repeated number7 ways and its two colors $\binom{7}{2}$ ways. The repeated color must be among the other5; choose its two numbers from the remaining6 in $\binom{6}{2}$ ways, then match the four leftover numbers and colors in $4!$ ways. Count $7\binom{7}{2} \cdot 5\binom{6}{2} \cdot 4! = 264600$.

4. The conditional probability is $211680/(211680 + 264600) = 4/9$, giving $p + q = 13$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2026-I-Q09] | Part B (Q6-Q10)

Answer: 029

1. Let f_j denote the face on roll j. Visibility requires $f_3, f_4, f_5, f_6 \neq f_2$ and $f_5, f_6 \neq f_4$. The conditioned sample space therefore has $6 \cdot 6 \cdot 5 \cdot 5 \cdot 4 \cdot 4 = 14400$ equiprobable sequences.

2. One blank face means five distinct rolled faces, so exactly one pair of roll indices coincides. Of the fifteen index pairs, four involving index two and later indices, and two involving four and later indices, are forbidden. Thus nine repeated-pair choices remain.

3. Assign five distinct faces in $(6)_5 = 720$ ways, giving 6480 favorable sequences. Probability $6480/14400 = 9/20$, and the requested sum is 29.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2010-II-Q13] | Part C (Q11-Q15)

Answer: 263

1. The other two cards must both lie below a or both above $a+9$. Thus $p(a) = [\binom{a-1}{2} + \binom{43-a}{2}]/\binom{50}{2}$, with $1 \le a \le 43$.

2. The numerator equals $(a-22)^2 + 420$. The condition is $(a-22)^2 \ge 385/2$, or $|a-22| \ge 14$ for integers.

3. The least admissible probability occurs at $a = 8, 36$, giving $616/1225 = 88/175$. Its numerator-denominator sum is 263.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2011-I-Q12] | Part C (Q11-Q15)

Answer: 594

1. The possible ordered male-block sizes under the conditioning are $(2,2,2),(2,4),(4,2),(3,3),(6)$. A pattern with j blocks occupies j of the $n+1$ female gaps.

2. The denominator count is $\binom{n+1}{3}+3\binom{n+1}{2}+n+1$. Favorable blocks are $(2,4),(4,2),(6)$, giving $2\binom{n+1}{2}+n+1$. Their ratio is $6(n+1)/(n^2+8n+6)$.

3. The one-percent condition is $n^2-592n-594\geq 0$. Its positive root lies strictly between 593 (value -1) and 594 (value 594); the other root is negative.

4. The least nonnegative integer number of women is therefore 594.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2010-II-Q02] | Part A (Q1-Q5)

Answer: 281

1. The points with nearest-side distance at least t form a central square of side $1-2t$.

2. The desired area, hence probability, is $(3/5)^2-(1/3)^2=56/225$. Boundaries have area zero.

3. The answer is 281.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2017-II-Q03] | Part A (Q1-Q5)

Answer: 409

1. Comparing squared distances gives the half-planes $x > 6$ and $y < (2/5)x + 1$. Intersecting these with the triangle produces vertices $(6, 0), (12, 0), (10, 5), (6, 17/5)$.

2. The quadrilateral area is $15 + 34/5 = 109/5$, by splitting along the diagonal from $(6, 0)$ to $(10, 5)$. The original triangle area is60.

3. The probability is $(109/5)/60 = 109/300$, giving $p + q = 409$. Boundary equalities have area zero.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2020-II-Q02] | Part A (Q1-Q5)

Answer: 171

1. The threshold line is $y = x/2 + 1/16$. To its right of $x = 5/8$ the desired points lie above it; to its left they lie below it. The vertical line itself has area zero.

2. The right trapezoid has area $\frac{3}{8}\frac{5/8+7/16}{2} = 51/256$; the left one has area $\frac{5}{8}\frac{1/16+3/8}{2} = 35/256$.

3. The total area, hence probability, is $86/256 = 43/128$. The answer is 171.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2004-I-Q10] | Part B (Q6-Q10)

Answer: 817

1. With the diagonal from $(0,0)$ to $(36,15)$, centers lie in $[1,35]\times[1,14]$, area 442. The diagonal line is $5x-12y=0$, so avoidance requires $|5x-12y|>13$.

2. One favorable triangle has vertices $(5,1)$, $(35,1)$, $(35,27/2)$, area $375/2$. A half-turn produces an equal region on the other side.

3. Thus probability $375/442$ is reduced, giving 817.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2017-I-Q06] | Part B (Q6-Q10)

Answer: 048

1. The three consecutive arcs between triangle vertices have measures $2x, 2x, 360 - 4x$. A chord misses the triangle exactly when both endpoints lie on the same such arc, apart from zero-probability boundary cases.

2. With $u = x/180$, the miss probability is $2u^2 + (1-2u)^2 = 11/25$. This reduces to $75u^2 - 50u + 7 = 0$, whose roots are $1/5, 7/15$.

3. The corresponding angles $36^\circ, 84^\circ$ both lie in $(0, 90^\circ)$ and form nondegenerate triangles. Their difference is48.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2017-I-Q08] | Part B (Q6-Q10)

Answer: 041

1. Thales theorem places Q, R on the same semicircle with diameter OP and radius100. The chord length is $QR = 200 \sin |a - b|$. Since the angle difference is below75 degrees, the event is equivalent to $|a - b| \leq 30$.

2. In the uniform 75×75 parameter square, the excluded region consists of two triangles of leg45. Its total area is 45^2, so the probability is $1 - 45^2/75^2 = 16/25$. Hence $m + n = 41$.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2023-II-Q06] | Part B (Q6-Q10)

Answer: 035

1. The midpoint always stays in $[0,2]^2$. Failure means its two coordinates both exceed 1, placing it in the missing upper-right square.

2. This requires one point in the upper arm and one in the right arm, probability $2/9$. Conditional on that choice, each coordinate-sum exceeds 2 with probability $1/2$; the two coordinate events are independent.

3. Thus failure has probability $(2/9)(1/2)^2 = 1/18$, and success is $17/18$, yielding 35.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2003-I-Q11] | Part C (Q11-Q15)

Answer: 092

1. Symmetry about 45° permits restricting to $0 < x \leq 45^\circ$. Here the largest length is $\cos^2 x$, so failure means $\cos^2 x \geq \sin^2 x + \sin x \cos x$.

2. Using double angles, this is $\cos 2x \geq \frac{1}{2} \sin 2x$, or $x \leq \frac{1}{2} \arctan 2$.

3. The two symmetric failure intervals have total degree length $\arctan 2$, so $p = \arctan 2 / 90^\circ$. Thus the requested parameters give $m + n = 2 + 90 = 92$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2019-II-Q02] | Part A (Q1-Q5)

Answer: 107

1. Let u_k be the visit probability. The last step to k comes uniquely from $k-1$ by one or $k-2$ by two, so $u_k = (u_{k-1} + u_{k-2})/2$, with $u_1 = 1, u_2 = 1/2$.

2. Iteration gives $u_3 = 3/4, u_4 = 5/8, u_5 = 11/16, u_6 = 21/32, u_7 = 43/64$. Thus $p + q = 107$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2023-I-Q06] | Part B (Q6-Q10)

Answer: 051

1. The prediction cannot affect the remaining deck. Therefore maximizing each immediate success probability is optimal: always choose a color with the larger remaining count.

2. For $E(r,b)$, the expected future successes, use $E(r,0)=r, E(0,b)=b$ and $E(r,b)=[\max(r,b)+rE(r-1,b)+bE(r,b-1)]/(r+b)$.

3. By symmetry, the necessary values are $E(1,1)=3/2, E(2,1)=7/3, E(2,2)=17/6, E(3,1)=13/4, E(3,2)=18/5$.

4. Thus $E(3,3)=1/2+18/5=41/10$, and $m+n=51$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2026-II-Q07] | Part B (Q6-Q10)

Answer: 754

1. Condition on exactly k Alice/Bob rolls before the first Carol roll. For $k \geq 4$, the good binary strings number $2^k - 2(k+1)$, since strings with at most one occurrence of either person are disjoint bad families.

2. Each such length-k string followed by Carol has probability $3^{-(k+1)}$. Thus $P = \frac{1}{3}\sum_{k\geq 4}[2^k - 2(k+1)]/3^k$. The two sums are $16/27$ and $11/54$, using geometric-series differentiation.

3. Hence $P = \frac{1}{3}(16/27 - 11/54) = 7/54$ and $100m + n = 754$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2015-I-Q12] | Part C (Q11-Q15)

Answer: 431

1. Write each chosen subset in increasing order. The unchosen integers form 1001 gaps: before the first chosen number, between consecutive chosen numbers, and after the last. Their sizes are nonnegative and sum to 1015.

2. The correspondence between subsets and these gap vectors is a bijection, so all weak compositions are equally likely. Permuting gap coordinates preserves their distribution, giving mean first gap $1015/1001$.

3. The smallest chosen number is one plus that gap. Its mean is $1 + 1015/1001 = 2016/1001 = 288/143$. Thus $p + q = 431$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2016-I-Q13] | Part C (Q11-Q15)

Answer: 273

1. Only height matters. Let T_j be the expected remaining jumps at height j. First-step analysis gives $T_{24} = 0$, $T_0 = 3 + T_1$, and $T_{j+1} - 2T_j + T_{j-1} = -4$ for $1 \leq j \leq 23$.

2. Write differences $d_j = T_j - T_{j-1}$. Then $d_1 = -3$ and $d_{j+1} = d_j - 4$, so $d_j = 1 - 4j$.

3. Thus $T_{21} = -\sum_{j=22}^{24} d_j = (87 + 91 + 95) = 273$. Finite expectation follows, for example, because every block of24 jumps has a fixed positive chance of all upward choices and hence absorption.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2016-II-Q13] | Part C (Q11-Q15)

Answer: 371

1. There are $6! = 720$ configurations. Every score is at least2 and at most7, the latter because the six occupied values sum to42. Therefore the total score is $2 \cdot 720 + \sum_{j=3}^{7} B_j$, where B_j counts scores at least j.

2. To have score at least j, row r must use column $c \geq j - r$. For the first $j - 2$ rows, allowed sets are nested, giving exactly $8 - j$ unused choices at each step; the remaining $8 - j$ rows can be permuted freely. Hence $B_j = (8 - j)^{j-2}(8 - j)!$.

3. The five tail counts are $600, 384, 162, 32, 1$. Mean score is $(1440 + 600 + 384 + 162 + 32 + 1)/720 = 2619/720 = 291/80$, so $p + q = 371$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2018-II-Q13] | Part C (Q11-Q15)

Answer: 647

1. Use prefix states zero (no match), one (last result one), and two (last two results one-two). Let u_j be the expected value of $(-1)^R$ for the number R of remaining rolls from each state. Stopping is almost sure, since independent three-roll blocks have a fixed positive chance of being one-two-three.

2. Each next roll changes parity, so conditioning gives $6u_0 = -u_1 - 5u_0$, $6u_1 = -u_1 - u_2 - 4u_0$, and $6u_2 = -1 - u_1 - 4u_0$. The constant term corresponds to immediate completion from state two.

3. The first two yield $u_1 = -11u_0, u_2 = 73u_0$. The third gives $431u_0 = -1$, so the odd-roll probability is $(1 - u_0)/2 = 216/431$.

4. The fraction is reduced and $m + n = 647$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2021-I-Q12] | Part C (Q11-Q15)

Answer: 019

1. The three cyclic gaps stay even. Before collision their unordered possibilities are $A = (4,4,4)$, $B = (2,4,6)$, $C = (2,2,8)$. Enumerating the eight jump-sign triples gives transition counts: $A \to (A,B) = (2,6)$; $B \to (A,B,C,\text{stop}) = (1,4,1,2)$; $C \to (B,C,\text{stop}) = (2,2,4)$.

2. With expected remaining times a, b, c, first-step equations are $a = 1 + a/4 + 3b/4$, $b = 1 + a/8 + b/2 + c/8$, $c = 1 + b/4 + c/4$.

3. The last gives $c = 4/3 + b/3$ and the first $a = 4/3 + b$. Substitution gives $b = 4$, hence $a = 16/3$ and the requested sum 19.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2025-I-Q13] | Part C (Q11-Q15)

Answer: 204

1. With probability one, all interior crossings are distinct except the prescribed diameter crossing. Region count equals $1 + 27+$ the number of crossing pairs. The diameters contribute one crossing, and each random chord crosses each diameter with probability $4/6 = 2/3$.

2. For two random chords there are 36 equally likely ordered quadrant-pair choices. Two combinations force crossing, four forbid it, and the remaining thirty cross with probability $1/2$ by endpoint-order symmetry. Hence the crossing probability is $17/36$.

3. Linearity of expectation gives $28 + 1 + 25 \cdot 4/3 + \binom{25}{2} \cdot 17/36 = 204$.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2009-I-Q03] | Part A (Q1-Q5)

Answer: 011

1. Equal binomial coefficients cancel in $\binom{8}{3}p^3(1-p)^5 = \binom{8}{5}p^5(1-p)^3/25$, leaving $(1-p)^2 = p^2/25$.

2. Positivity gives $1-p = p/5$, so $p = 5/6$ and the answer is 11.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2010-I-Q04] | Part A (Q1-Q5)

Answer: 515

1. The generating polynomial for one head count is $[(1+t)/2]^2(3+4t)/7 = (3+10t+11t^2+4t^3)/28$.

2. Independence makes the matching probability the sum of the squared coefficients: $(3^2+10^2+11^2+4^2)/28^2 = 123/392$.

3. The requested sum is 515.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2006-II-Q10] | Part B (Q6-Q10)

Answer: 831

1. Each team has five other games, giving $2^{10} = 1024$ equally likely joint outcomes. Let remaining-win counts be X, Y; A finishes ahead iff $X \geq Y$.

2. The equality count is $\sum_{j=0}^{5} \binom{5}{j}^2 = \binom{10}{5} = 252$. Symmetry splits the remaining outcomes evenly. Thus probability $(1024 + 252)/(2 \cdot 1024) = 319/512$, yielding 831.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2008-I-Q09] | Part B (Q6-Q10)

Answer: 190

1. If counts of the three heights are u, v, w, then $u+v+w=10$ and $3u+4v+6w=41$. Thus $v+3w=11$, giving $(u,v,w)=(1,8,1),(3,5,2),(5,2,3)$.

2. Multinomial counts are $90, 2520, 2520$, totaling 5130. Divide by 3^{10} and reduce: probability $190/3^7$, so $m=190$.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2009-II-Q08] | Part B (Q6-Q10)

Answer: 041

1. First-six probability at roll k is $p_k = (5/6)^{k-1}/6$. Equal counts have probability $\sum p_k^2 = (1/36)/(1 - 25/36) = 1/11$.

2. One specified player needing exactly one more roll contributes $(5/6)\sum p_k^2 = 5/66$. Both directions give total $1/11 + 10/66 = 8/33$, so the answer is 41.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2021-II-Q08] | Part B (Q6-Q10)

Answer: 049

1. All $3 \cdot 2^7 = 384$ allowed paths are equally likely. Count paths by states (B_h, B_v, T_h, T_v), recording bottom/top face and horizontal/vertical last edge.

2. The update is $(B_h + 2B_v, T_h, T_h + 2T_v, B_h)$, because a vertical edge must be followed horizontally, whereas a horizontal edge allows one horizontal and one vertical continuation.

3. Starting after one step at $(2, 0, 0, 1)$, successive states through steps 2 to 8 are $(2, 0, 2, 2)$, $(2, 2, 6, 2)$, $(6, 6, 10, 2)$, $(18, 10, 14, 6)$, $(38, 14, 26, 18)$, $(66, 26, 62, 38)$, $(118, 62, 138, 66)$.

4. The top count is 204, so the probability is $204/384 = 17/32$, giving 49.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2001-II-Q11] | Part C (Q11-Q15)

Answer: 341

1. Win-loss reversal is a symmetry, so the desired probability is half the probability of unequal counts.

2. For equal counts k wins and k losses, $k = 0, 1, 2, 3$, the sequence counts are $6!/[k!^2(6-2k)!] = 1, 30, 90, 20$, totaling 141.

3. Thus the probability is $(729 - 141)/(2 \cdot 729) = 98/243$, and the sum is 341.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2002-II-Q12] | Part C (Q11-Q15)

Answer: 660

1. Let $D(n, j)$ count admissible prefixes of length n with j hits. Set $D(0, 0) = 1$, update by a miss or hit, and retain only $j \leq \lfloor 2n/5 \rfloor$ for $n < 10$.

2. The counts at maximum permitted hit totals for $n = 3, 4, 5, 6, 7, 8, 9$ are $1, 2, 2, 5, 9, 9, 23$. The tenth shot must hit, giving 23 admissible four-hit sequences.

3. Each has probability $(2/5)^4(3/5)^6$. Thus the total is $2^4 3^6 \cdot 23/5^{10}$ and the requested value is $(2 + 3 + 23 + 5)(4 + 6 + 10) = 660$.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2003-II-Q13] | Part C (Q11-Q15)

Answer: 683

1. Let p_j be the probability of being at the start after j steps. Then $p_0 = 1$ and $p_{j+1} = (1 - p_j)/2$.

2. The fixed point is $1/3$, so $p_j - 1/3 = (2/3)(-1/2)^j$. Therefore $p_{10} = 171/512$, giving 683.

Source / 来源

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2014-I-Q11] | Part C (Q11-Q15)

Answer: 391

1. Use $u = x + y, v = x - y$. The four moves realize each pair of signs $(\Delta u, \Delta v) \in \{\pm 1\}^2$ with probability one quarter. Thus these coordinates are independent six-step sign walks.

2. The target is $u = 0$ or $v = 0$. Each event has probability $\binom{6}{3}/2^6 = 5/16$, and their intersection has the square of this probability.

3. Inclusion-exclusion gives $2(5/16) - (5/16)^2 = 135/256$, whose parameter sum is 391.

Source / 来源

Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2001-I-Q02] | Part A (Q1-Q5)

Answer: 651

1. Neither adjoined value was already present, because otherwise its union would not change the mean. If the set has n elements, then $(nM+1)/(n+1) = M-13$ and $(nM+2001)/(n+1) = M+27$.

2. These become $M-1 = 13(n+1)$ and $2001-M = 27(n+1)$. Adding gives $n+1 = 50$, and $M = 1+13\cdot 50 = 651$.

Source / 来源

Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2021-II-Q01] | Part A (Q1-Q5)

Answer: 550

1. Each palindrome is $101a + 10b$ for $1 \leq a \leq 9, 0 \leq b \leq 9$, with every pair occurring once.

2. The mean digits are 5 and $9/2$, so the mean number is $101 \cdot 5 + 10 \cdot 9/2 = 550$.

Source / 来源

Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2024-II-Q02] | Part A (Q1-Q5)

Answer: 236

1. The length is even, and 9 appears two or three times. If it appears three times, the remaining entries sum to 3; the possible even lengths either have median 9 or use three 1s, tying the mode. Hence exactly two 9s occur and all other entries are distinct.

2. For four entries, the other entries sum to 12. If one exceeds 9, the middle entries are both 9, which fails. Otherwise write $a < b < 9$; the median $(b + 9)/2$ must be integral, so $b = 7, a = 5$.

3. For six entries the four other distinct positives sum to 12, giving only $\{1, 2, 3, 6\}$ or $\{1, 2, 4, 5\}$; both yield median $9/2$. Eight or more entries require six distinct positives summing to at least $21 > 12$. Thus the list is $5, 7, 9, 9$ and the squared sum is 236.

Source / 来源

Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2017-I-Q11] | Part C (Q11-Q15)

Answer: 360

1. Write the four digits below5 as L and the four above5 as G. The row containing5 must have one L and one G. The remaining two row medians must straddle5.

2. The other row types are either LLL and GGG, or LLG and LGG. Assigning the three row types and ordering the L5G row gives $3! \cdot 3! = 36$ patterns. The two cases supply another factor1 or9, respectively, for360 letter patterns.

3. Fill the L and G slots independently in $4!$ ways each. Thus $Q = 360(4!)^2 = 207360$, with remainder360.

Source / 来源

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2024-I-Q03] | Part A (Q1-Q5)

Answer: 809

1. The losing positions for the player to move are exactly $n \equiv 0, 2 \pmod 5$. From either residue every legal move reaches $1, 3$, or 4; from $1, 3, 4$ one can respectively remove $1, 1, 4$ to reach a losing residue.

2. The base cases $0, 1, 2, 3, 4$ verify the claim, so induction applies. The required count is $\lfloor 2024/5 \rfloor + \lfloor (2024 - 2)/5 \rfloor + 1 = 404 + 405 = 809$.

Source / 来源

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2010-I-Q12] | Part C (Q11-Q15)

Answer: 243

1. For $m = 243$, suppose there is no monochromatic product, and color 3 red. Then 9 is blue, 81 red, and 243 blue. If 27 is red, $3 \cdot 27 = 81$ fails; if blue, $9 \cdot 27 = 243$ fails. This proves the upper bound.

2. For $m = 242$, color $[3, 8] \cup [81, 242]$ red and $[9, 80]$ blue. Products of two small red numbers lie in $[9, 64]$, hence blue; any product involving a large red number exceeds 242. Products of two blue numbers exceed 80, hence are not blue within the set.

3. This product-free coloring also restricts to every smaller initial set. Therefore the least value is 243.

Source / 来源

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2012-I-Q11] | Part C (Q11-Q15)

Answer: 373

1. Every move keeps $x+y$ a multiple of three and $x-y$ a multiple of five. Write $x+y=3j, x-y=5k$; integer coordinates require j,k of the same parity. The diamond bound is $\max(|3j|,|5k|)\le 100$, so $|j|\le 33, |k|\le 20$.

2. These constraints are sufficient. Repeating a suitable move reaches some point with each desired integer k. Two positive moves together shift by $(9,9)$; two negative moves shift by $(-15,-15)$. Combining them in counts $(2,1)$ or $(3,2)$ gives diagonal shifts $(3,3)$ or $(-3,-3)$.

3. These shift j by ± 2 without changing k, reaching every required same-parity value. Count even pairs $33\cdot 21$ and odd pairs $34\cdot 20$, for $M=1373$.

4. The remainder is 373.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2001-II-Q05] | Part A (Q1-Q5)

Answer: 253

1. If $a_1 < \cdots < a_{10}$ contains no triangle, then $a_{i+2} \geq a_i + a_{i+1}$. Starting with the least possible first terms $4, 5$, this forces the sequence at least $4, 5, 9, 14, 23, 37, 60, 97, 157, 254$.

2. Thus no triangle-free ten-set fits below 254. Conversely the displayed sequence has no triangle: any two entries before a_j sum to at most $a_{j-2} + a_{j-1} = a_j$.

3. Therefore $n = 253$ works and 254 fails, giving 253.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2004-I-Q05] | Part A (Q1-Q5)

Answer: 849

1. Alpha's ratios are $8/15 < 7/10$. Beta's ratio on each day is below $7/10$, so its total integer score is below 350, hence at most 349.

2. This is attainable: Beta scores 1 of 2 on day one and 348 of 498 on day two. These ratios are strictly below $8/15, 7/10$, and the first-day amount is not 300.

3. Thus the maximum ratio is $349/500$, giving 849.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2006-I-Q02] | Part A (Q1-Q5)

Answer: 901

1. The minimum is $1 + \cdots + 90 = 4095$, maximum $11 + \cdots + 100 = 4995$.

2. Unless the subset is the maximum final block, some selected integer $k < 100$ has unselected successor $k + 1$. Replacing it raises the sum by exactly one. Repetition constructs every intermediate sum.

3. Thus the number is $4995 - 4095 + 1 = 901$.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2007-II-Q08] | Part B (Q6-Q10)

Answer: 896

1. Let there be h length-four and v length-five lines. Then $4h + 5v = 2007$ and basic count $(h-1)(v-1)$. Put $x = h-1, y = v-1$, so $4x + 5y = 1998$.

2. The product $xy = x(1998 - 4x)/5$ is a concave quadratic, peaking at $x = 249.75$. Integer feasibility requires $x \equiv 2 \pmod 5$, whose nearest candidates are $247, 252$, yielding products $49894, 49896$.

3. The latter is maximal, with remainder 896.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2022-I-Q07] | Part B (Q6-Q10)

Answer: 289

1. The choice $(2 \cdot 3 \cdot 6 - 1 \cdot 5 \cdot 7)/(4 \cdot 8 \cdot 9) = 1/288$ provides an attainable bound.

2. Every positive numerator is an integer and every denominator is at most $7 \cdot 8 \cdot 9 = 504$. A numerator at least 2 gives a value at least $2/504 > 1/288$. A better value would therefore require numerator 1 and denominator greater than 288.

3. The denominator triples above 288 are $789, 689, 679, 589, 678, 579$ (each denotes a product). Their complementary digits always include both 2 and 4, so both triple products cannot have opposite parity unless one triple contains both; checking the ten unordered splits per complement gives minimum positive differences $6, 2, 2, 8, 6, 4$ respectively. None is 1. Thus the minimum is $1/288$, yielding 289.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2025-II-Q08] | Part B (Q6-Q10)

Answer: 610

1. An optimal choice has at most nine pennies and four dimes, since ten pennies can be replaced by a dime and five dimes by two quarters. Thus its non-quarter value is at most 49, so it uses either $q = \lfloor N/25 \rfloor$ or $q - 1$ quarters.

2. Let $r = N \bmod 25$ and $h(v) = \lfloor v/10 \rfloor + (v \bmod 10)$. Compare greedy count $q + h(r)$ with rival $q - 1 + h(r + 25)$. For $q \geq 1$, the rival wins exactly when $r \in \{5, 6, 7, 8, 9, 15, 16, 17, 18, 19\}$.

3. Each of these ten residue classes has forty representatives up to 1000, but its representative below 25 has no rival. Failures $= 10(40 - 1) = 390$, successes $= 610$.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2003-II-Q12] | Part C (Q11-Q15)

Answer: 134

1. With committee size t and candidate votes v, the condition is $100v/t \leq v - 1$, or $v(t - 100) \geq t$. Hence $t > 100$ and $v \geq \lceil t/(t - 100) \rceil$.

2. For $101 \leq t \leq 133$, the lower bound is at least five votes per candidate, forcing $t \geq 135$, a contradiction.

3. At $t = 134$, four votes suffice for each candidate. Give twenty-six candidates four votes and the last thirty; all satisfy the inequality. Thus the minimum is 134.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2009-II-Q12] | Part C (Q11-Q15)

Answer: 803

1. The sum of used elements is at least $1 + \cdots + 2k = k(2k+1)$. The distinct pair sums total at most $2009 + 2008 + \cdots + (2010 - k) = k(4019 - k)/2$. Thus $5k \leq 4017$, so $k \leq 803$.

2. Attain this with 401 pairs $(i, 1206 + i)$ for $1 \leq i \leq 401$, giving even sums $1208, \ldots, 2008$, and 402 pairs $(a, a + 403)$ for $402 \leq a \leq 803$, giving odd sums $1207, \ldots, 2009$.

3. Used elements are 1–803 and 805–1607, with no repeats, and all sums satisfy the rule. Maximum 803.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2022-II-Q14] | Part C (Q11-Q15)

Answer: 188

1. To form 1, necessarily $a = 1$. A collection of x ones, y stamps of value b, and z of value c covers a full interval through its total exactly when $x \geq b - 1$ and $x + yb \geq c - 1$; the total must also be at least 1000. These follow by preventing gaps at b and c.

2. If $c \leq 10$, even 97 stamps total at most 970, so $f \neq 97$. For $c = 11, b = 7$, six ones, one seven, and ninety elevens give 97 stamps. With only 96, at least six ones and one seven are required, and the total is at most $6 + 7 + 89 \cdot 11 = 992$. Hence $f = 97$.

3. For any $12 \leq c \leq 87$ and $2 \leq b < c$, use $b - 1$ ones and $c - b$ copies of b. These $c - 1$ stamps have value $b - 1 + b(c - b) \geq 2c - 3$. Adding $\lceil 1003/c \rceil - 2$ copies of c covers through 1000. The count is $c - 3 + \lceil 1003/c \rceil \leq 96$, since the concave quadratic $c(99 - c)$ is at least 1003 throughout this interval. Thus none of these c qualifies.

4. For $c = 88$ or 89, take $b = 87$, eighty-six ones, one 87, and ten copies of c: 97 stamps suffice. Only 96 stamps would total at most $86 + 87 + 9c < 1000$, because the smaller denominations are mandatory. Thus both have $f = 97$, and the answer is $11 + 88 + 89 = 188$.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2026-I-Q11] | Part C (Q11-Q15)

Answer: 896

1. Orient each edge from its lower to its higher value. Writing $M = \sum c_v a_v$, the coefficient satisfies $|c_v| \le d_v$ and $\sum c_v = 0$, where degree d_v is four at 36 interior cells, three at 24 edge cells, two at four corners.

2. Therefore $M \le \sum d_v |a_v - 65/2|$. Rearrangement pairs the 36 largest deviations (values 1–$18, 47$–64, sum 828) with degree four, the next 24 (sum 192) with degree three, and the last four (sum 4) with degree two. Bound $M \le 4 \cdot 828 + 3 \cdot 192 + 2 \cdot 4 = 3896$.

3. Attain the bound using a checkerboard: all low values 1–32 on one color, all high values 33–64 on the other, allocating each degree class the indicated extremes. Each color has the required 18 interior,12 edge,2 corner cells. Thus the maximum is 3896, remainder 896.

Source / 来源

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

[AIME-2021-II-Q11] | Part C (Q11-Q15)

Answer: 258

1. List all valid four-integer runs allowed by the announcement. A student can initially identify the run exactly when their number occurs in only one candidate run. Thus the unanimous failure removes every run containing such a unique number.

2. Enumerating the starts 10 through 96 leaves exactly the runs $[34, 37], [47, 50], [76, 79], [89, 92]$. Equivalently these have the consecutive 6- and 7-multiples in their two middle positions.

3. The four surviving runs are disjoint, so every number now identifies its run, meeting the second condition. Their largest values sum to $37 + 50 + 79 + 92 = 258$.

Source / 来源